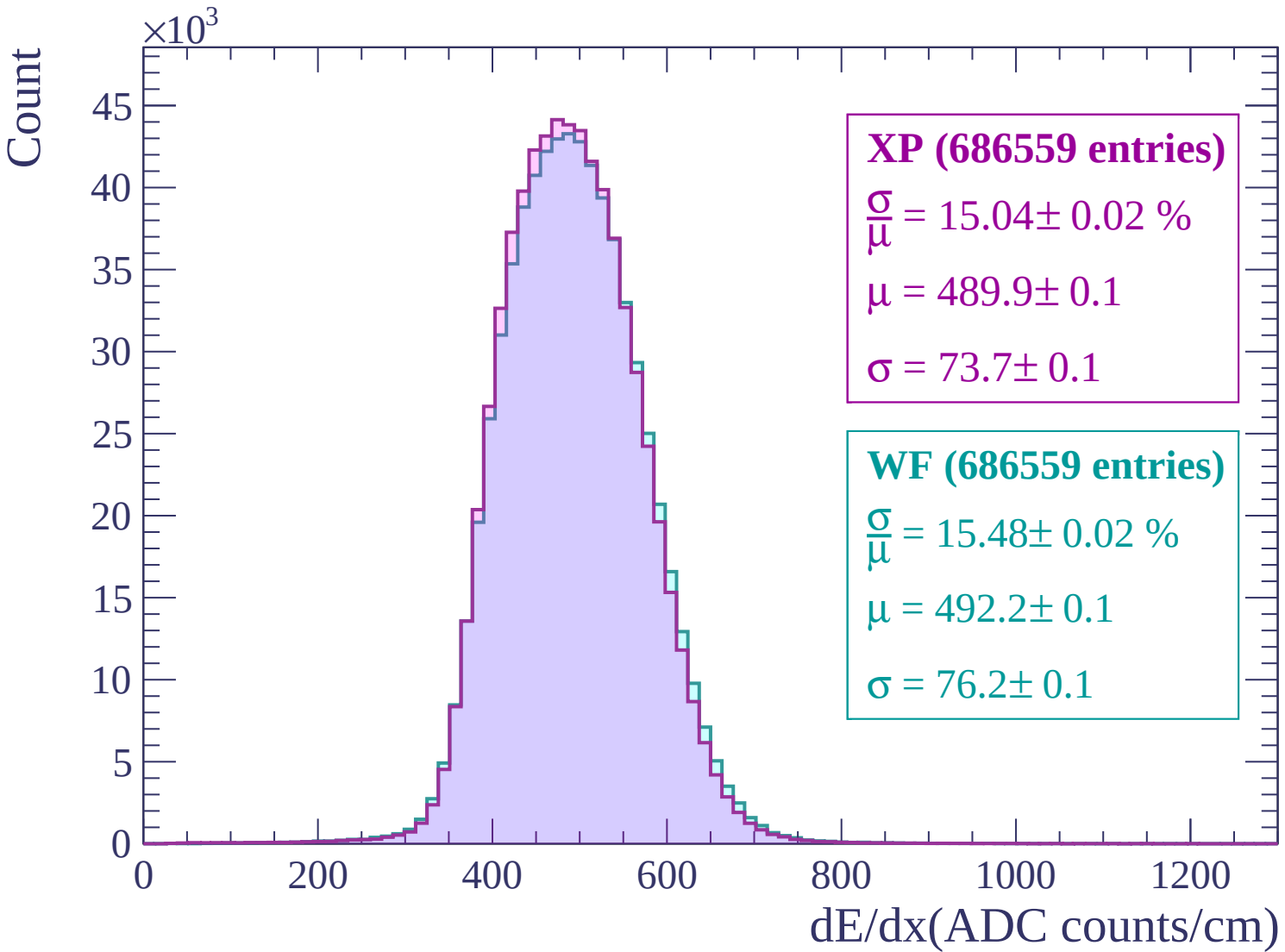
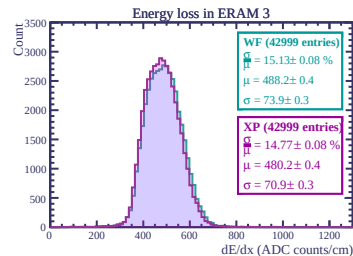
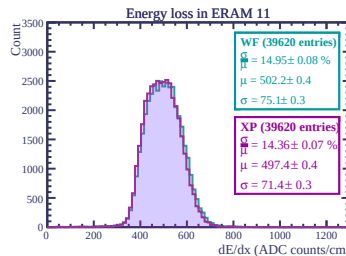
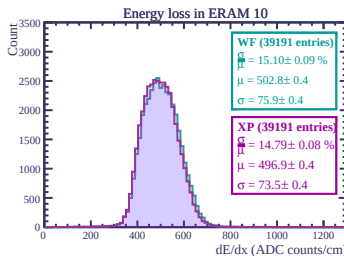
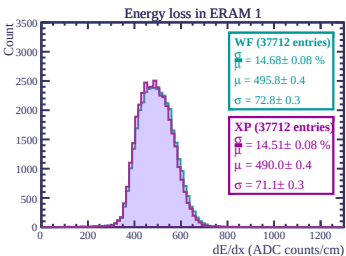
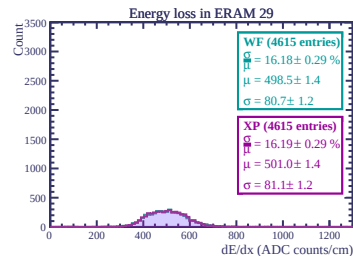
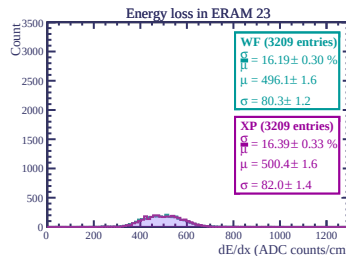
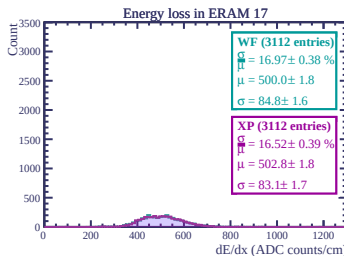
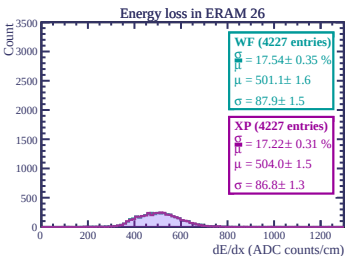
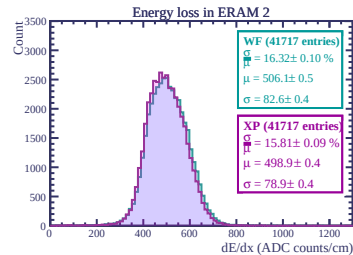
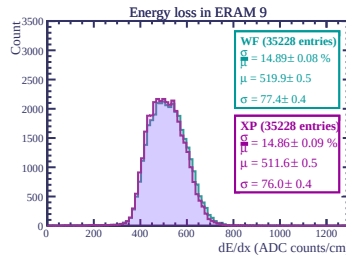
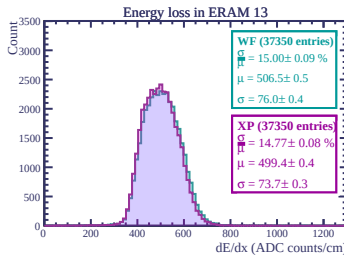
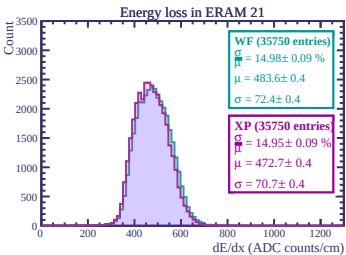
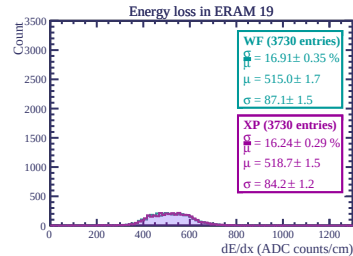
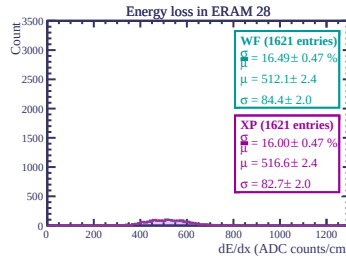
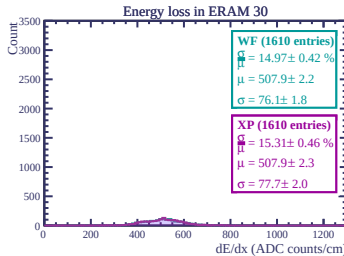
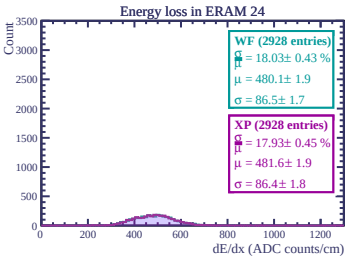
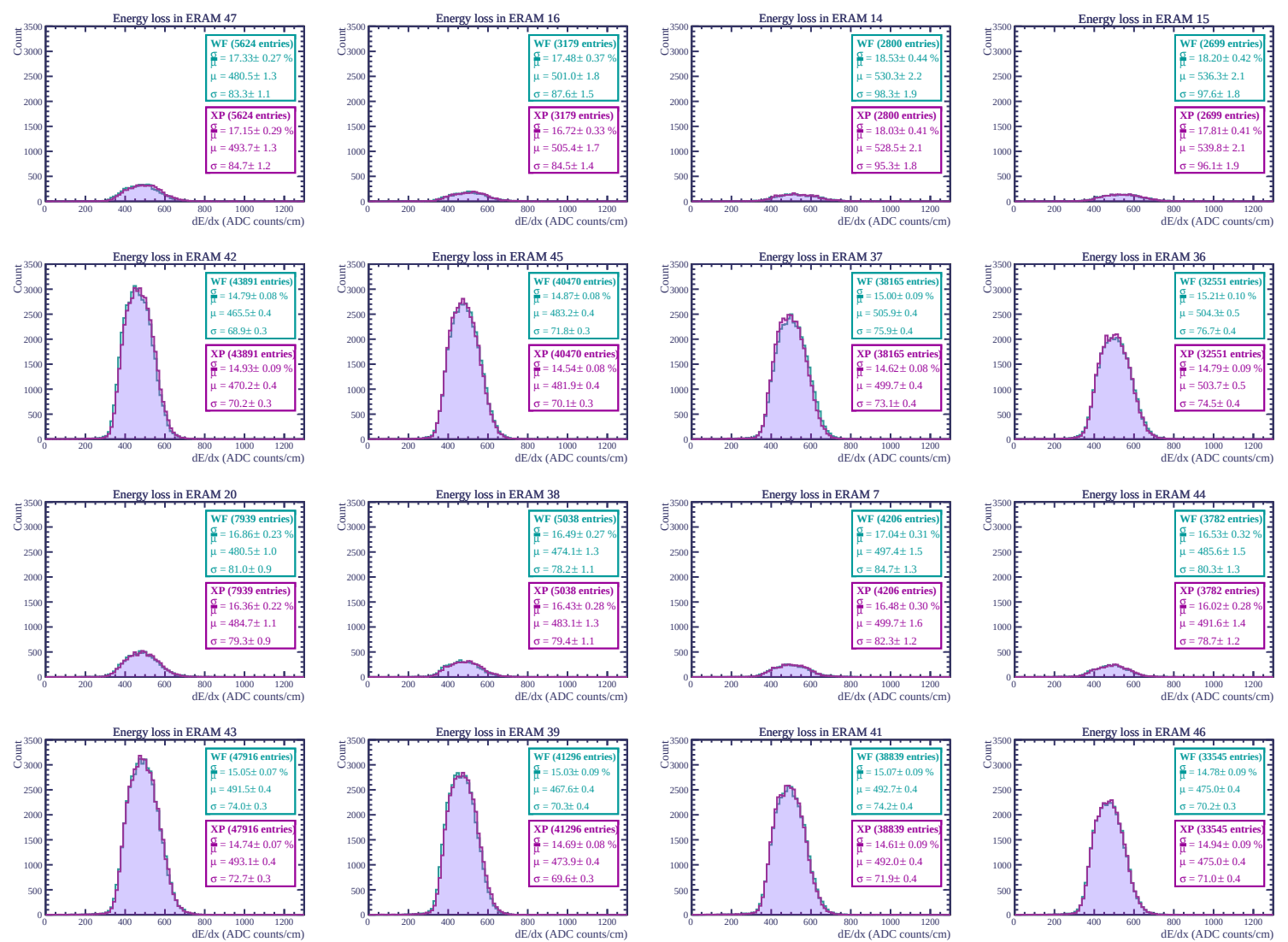
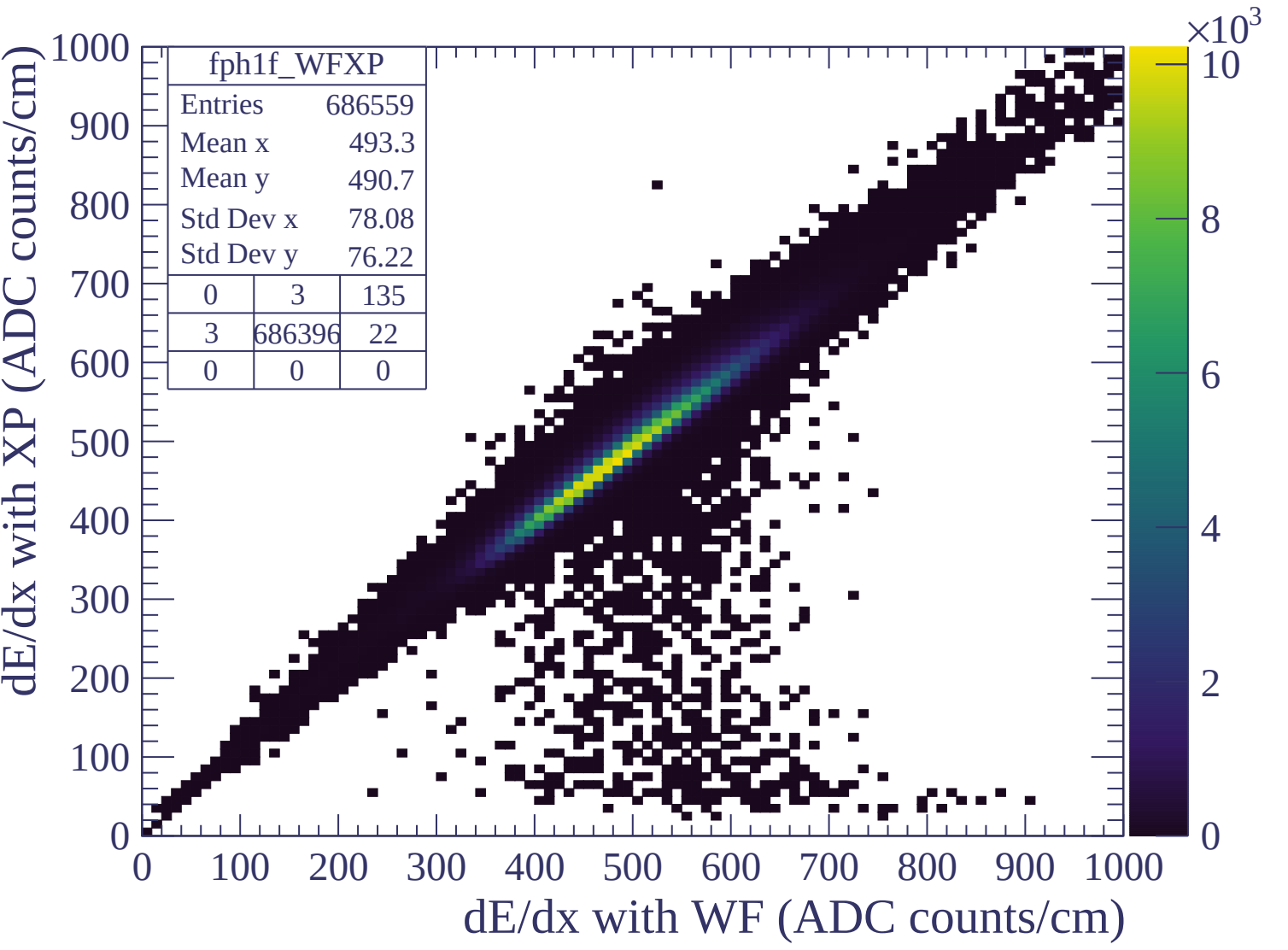
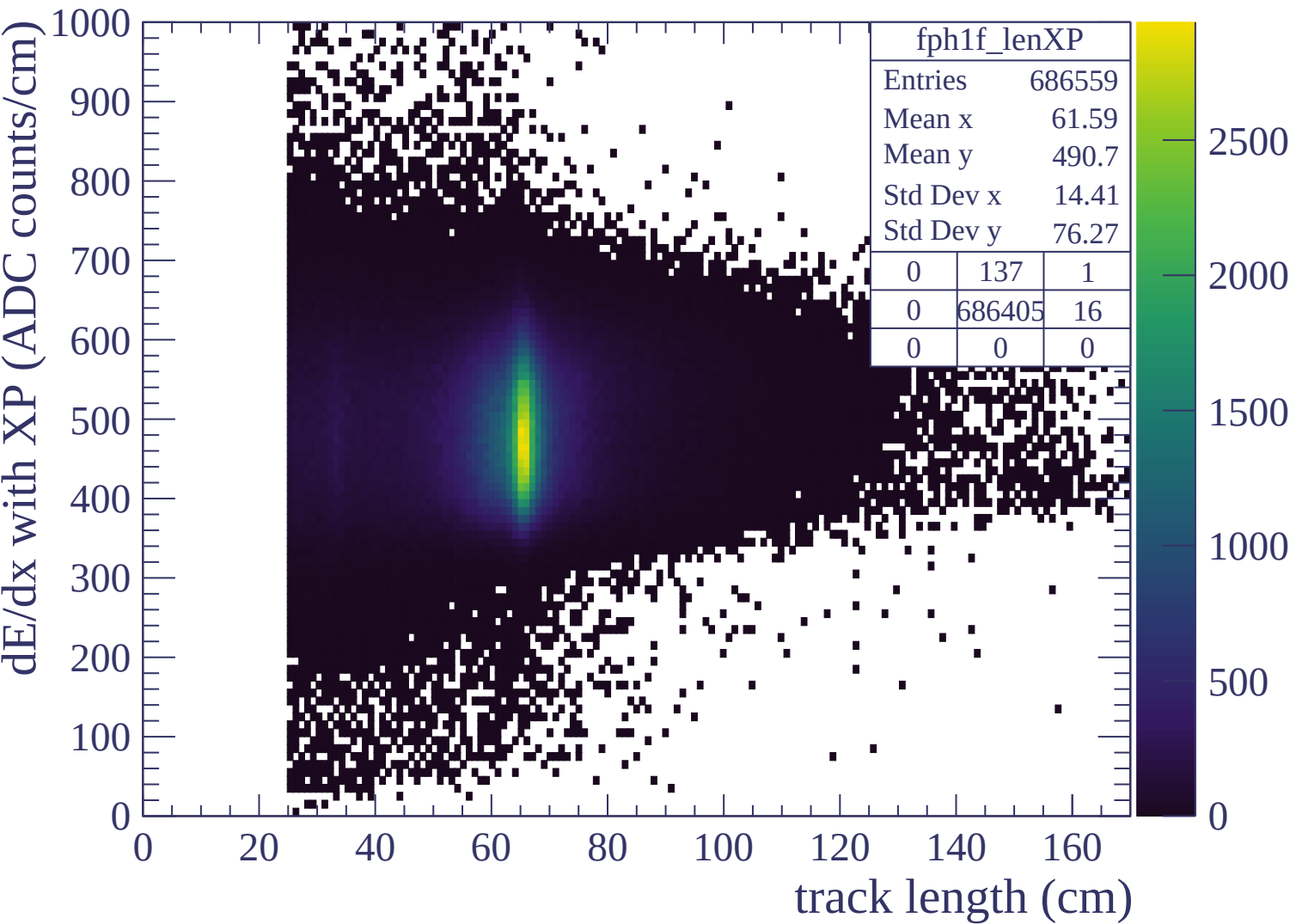


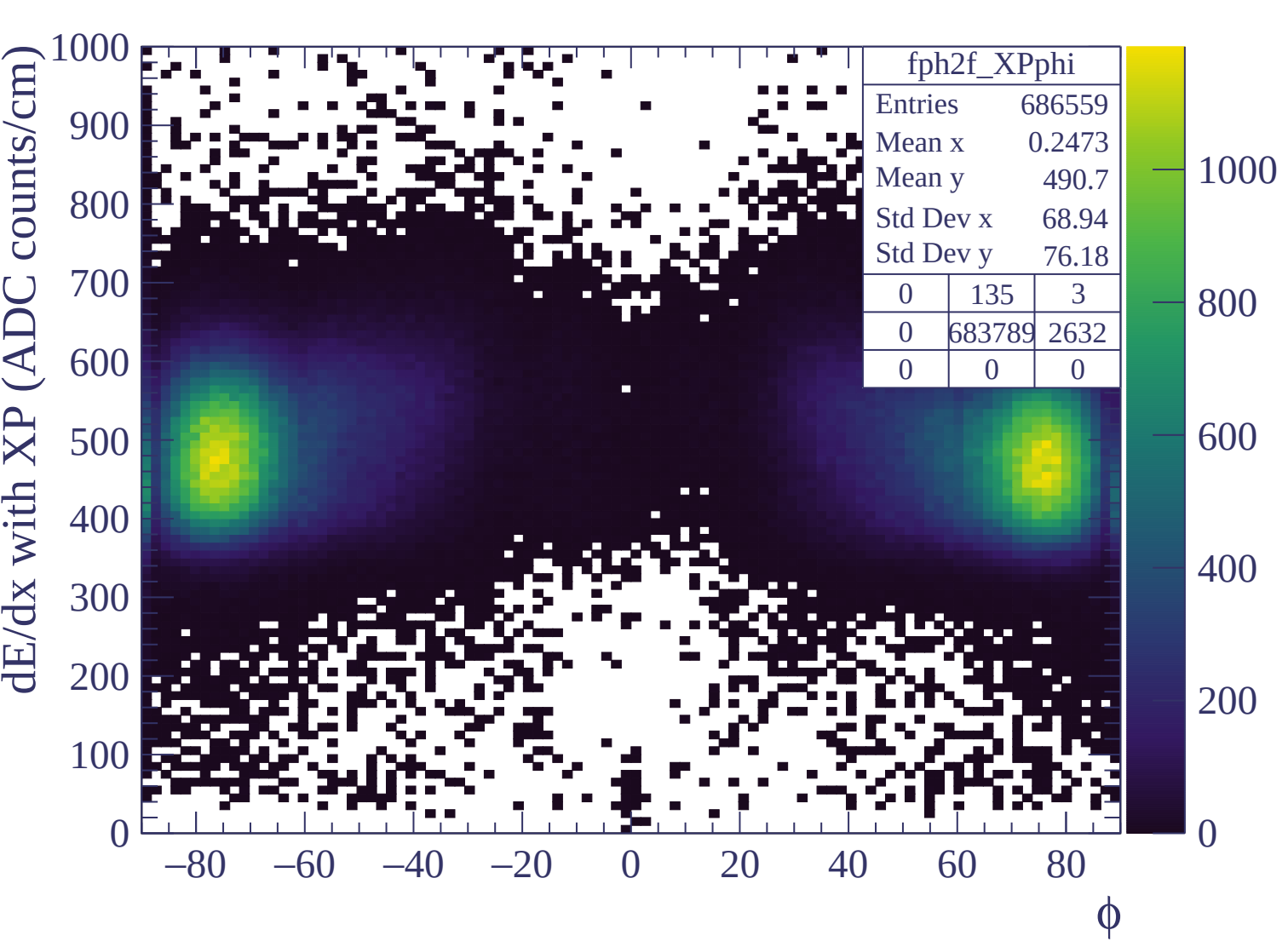


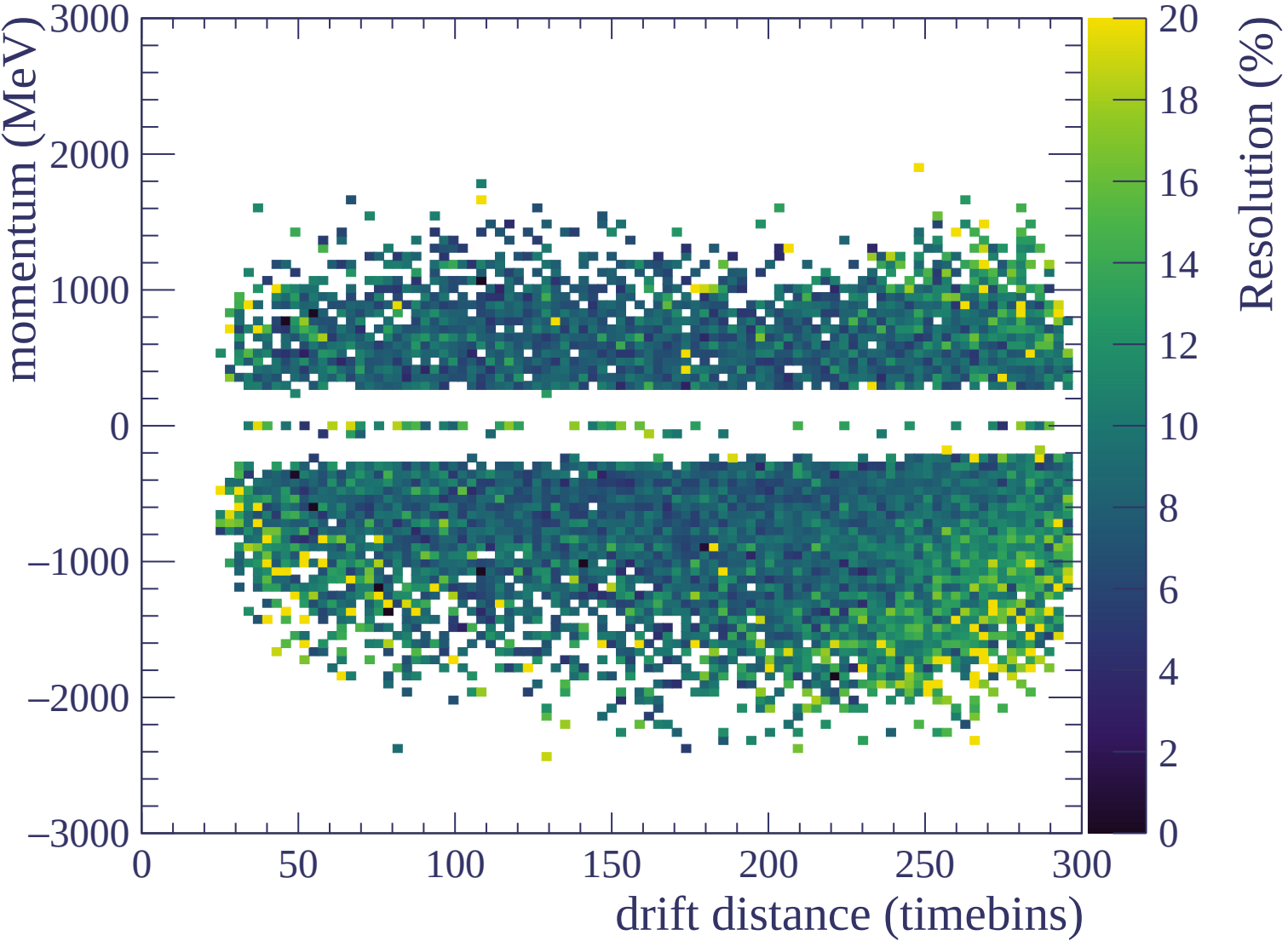


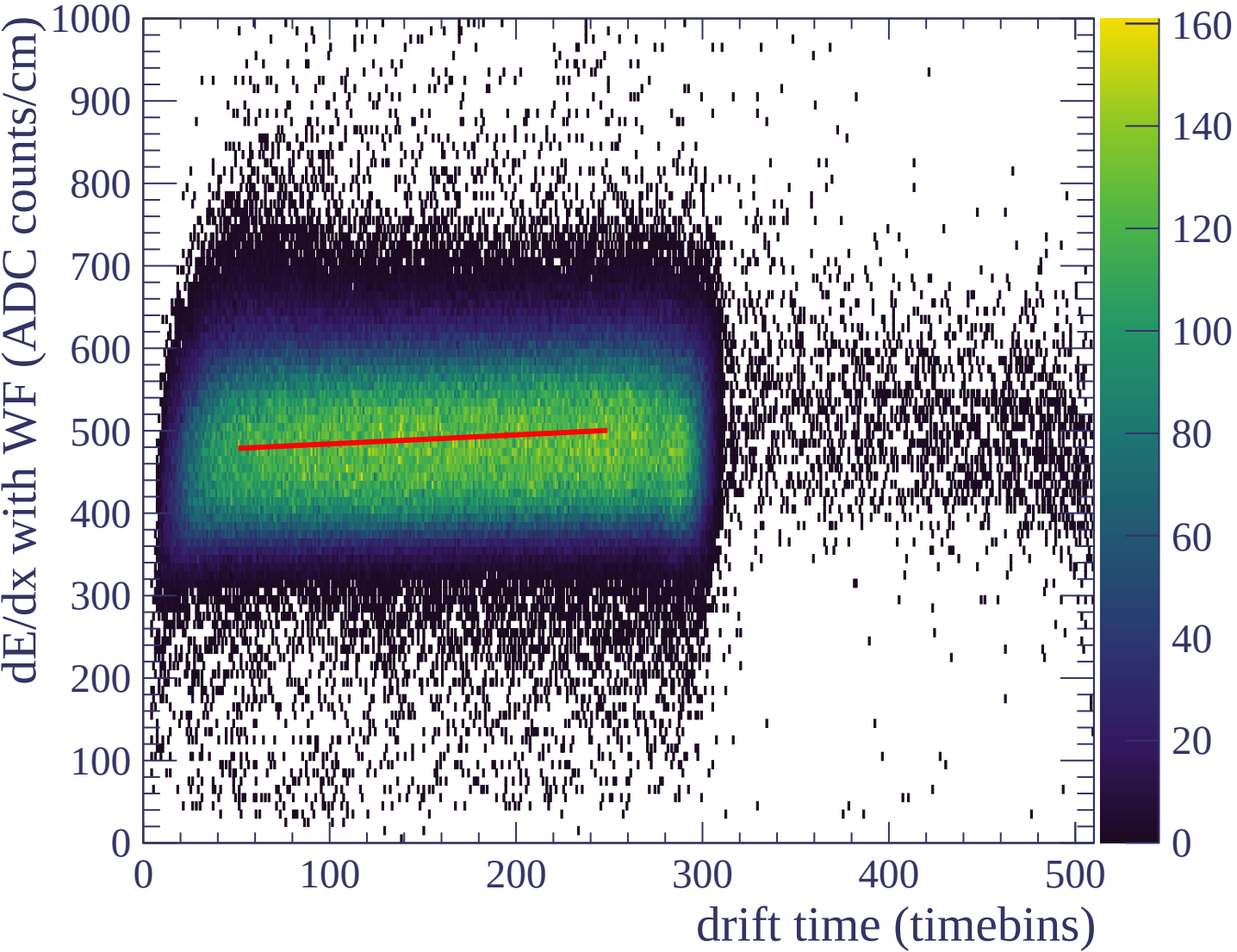


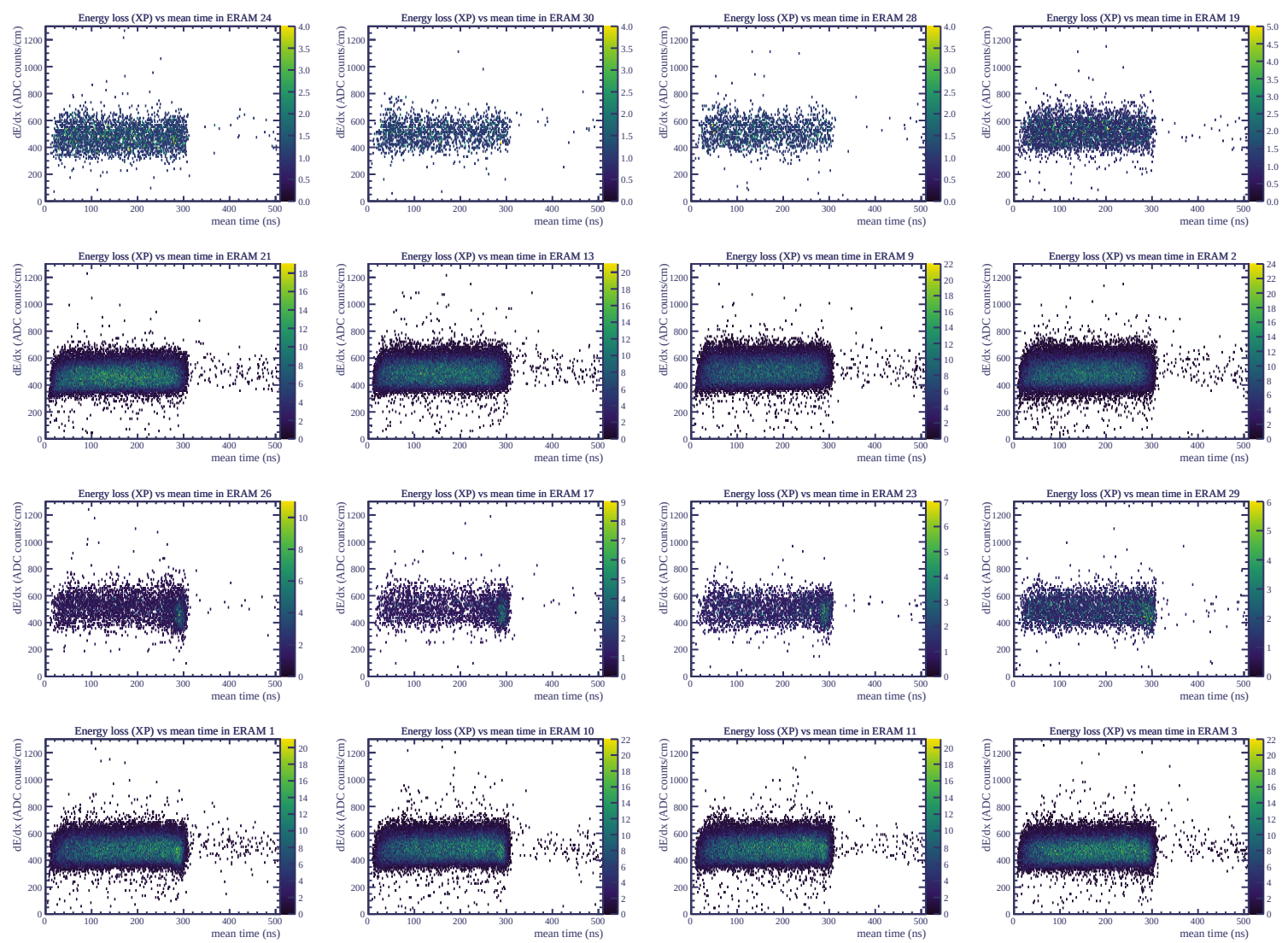


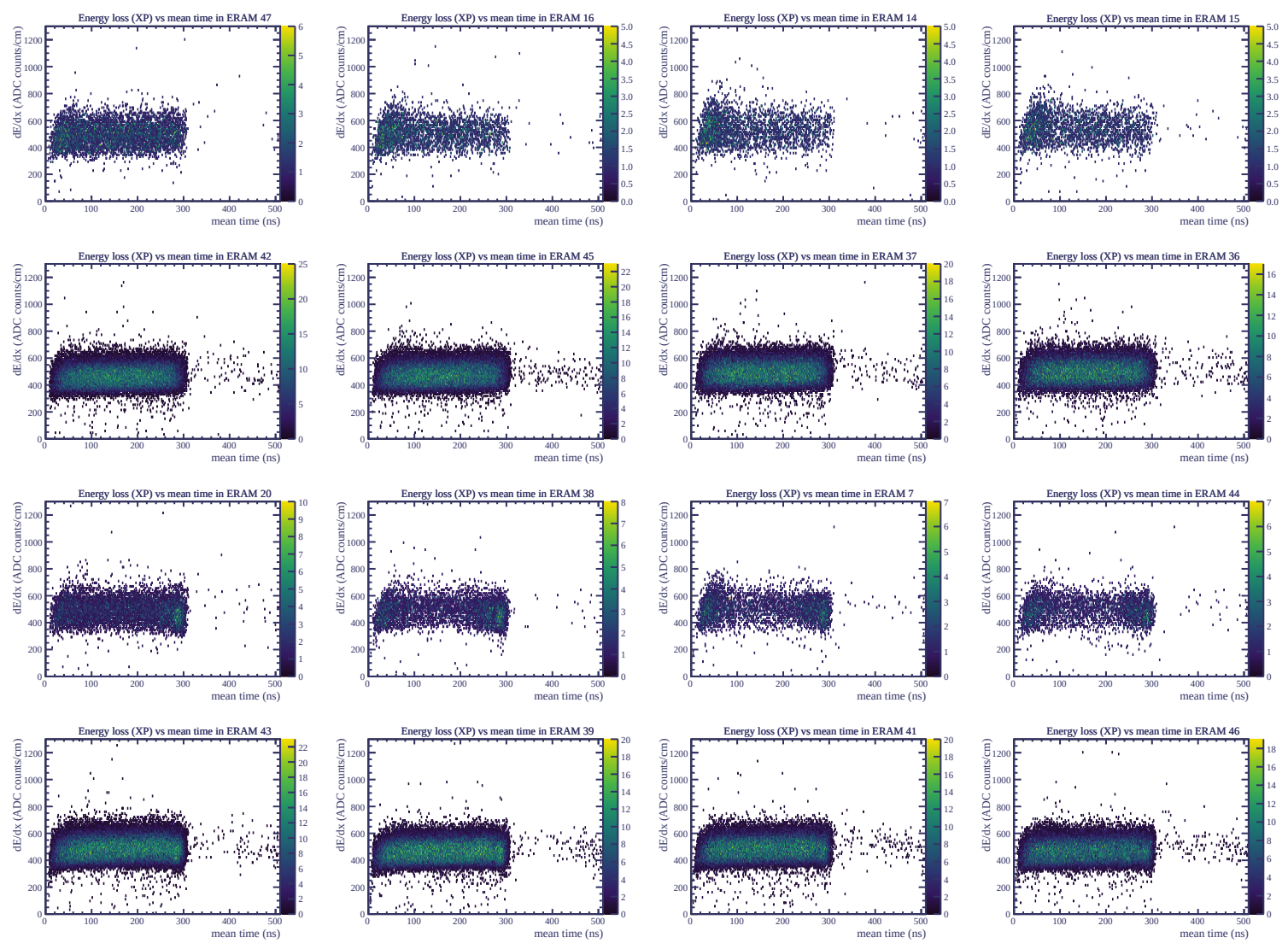


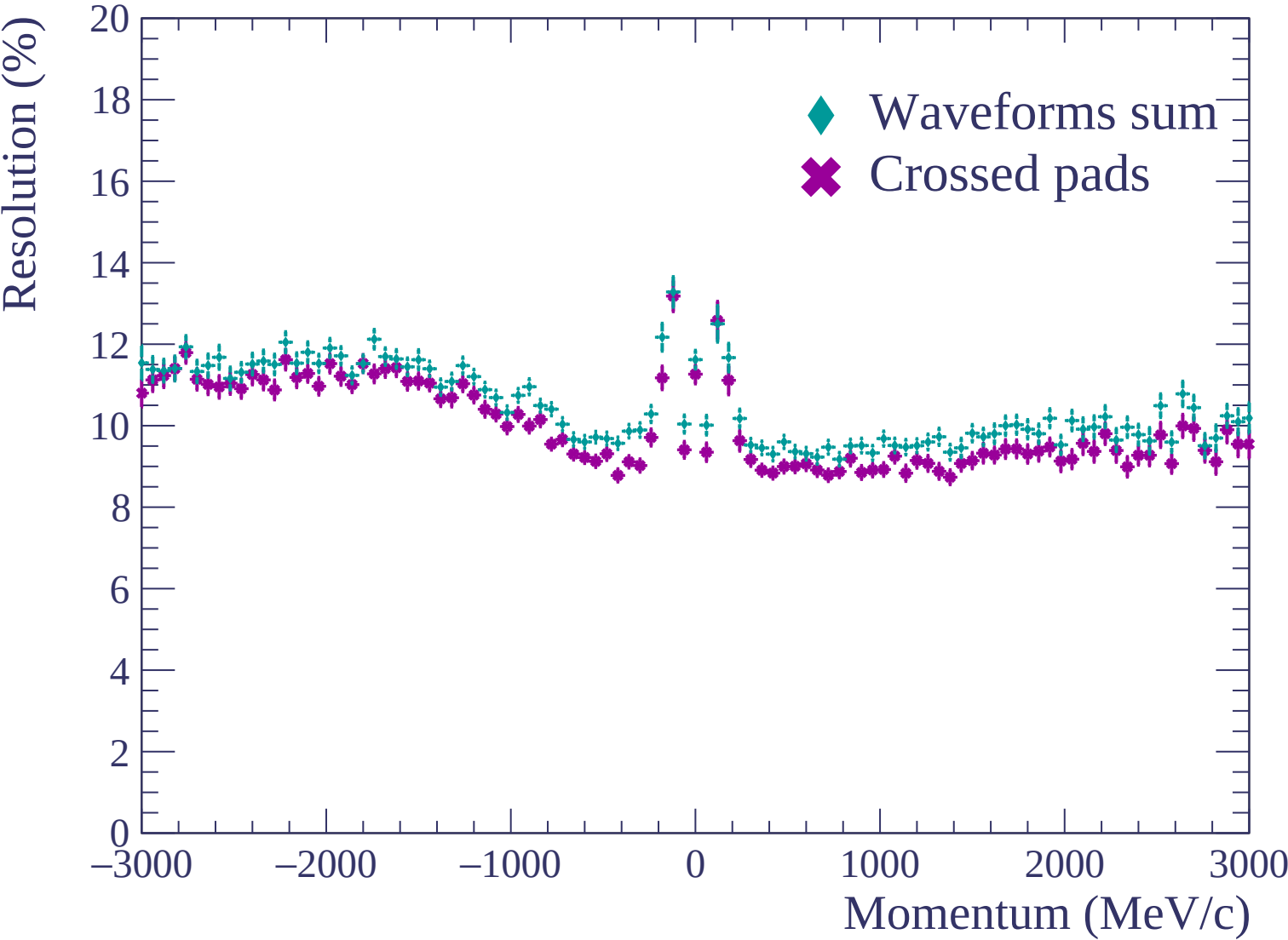


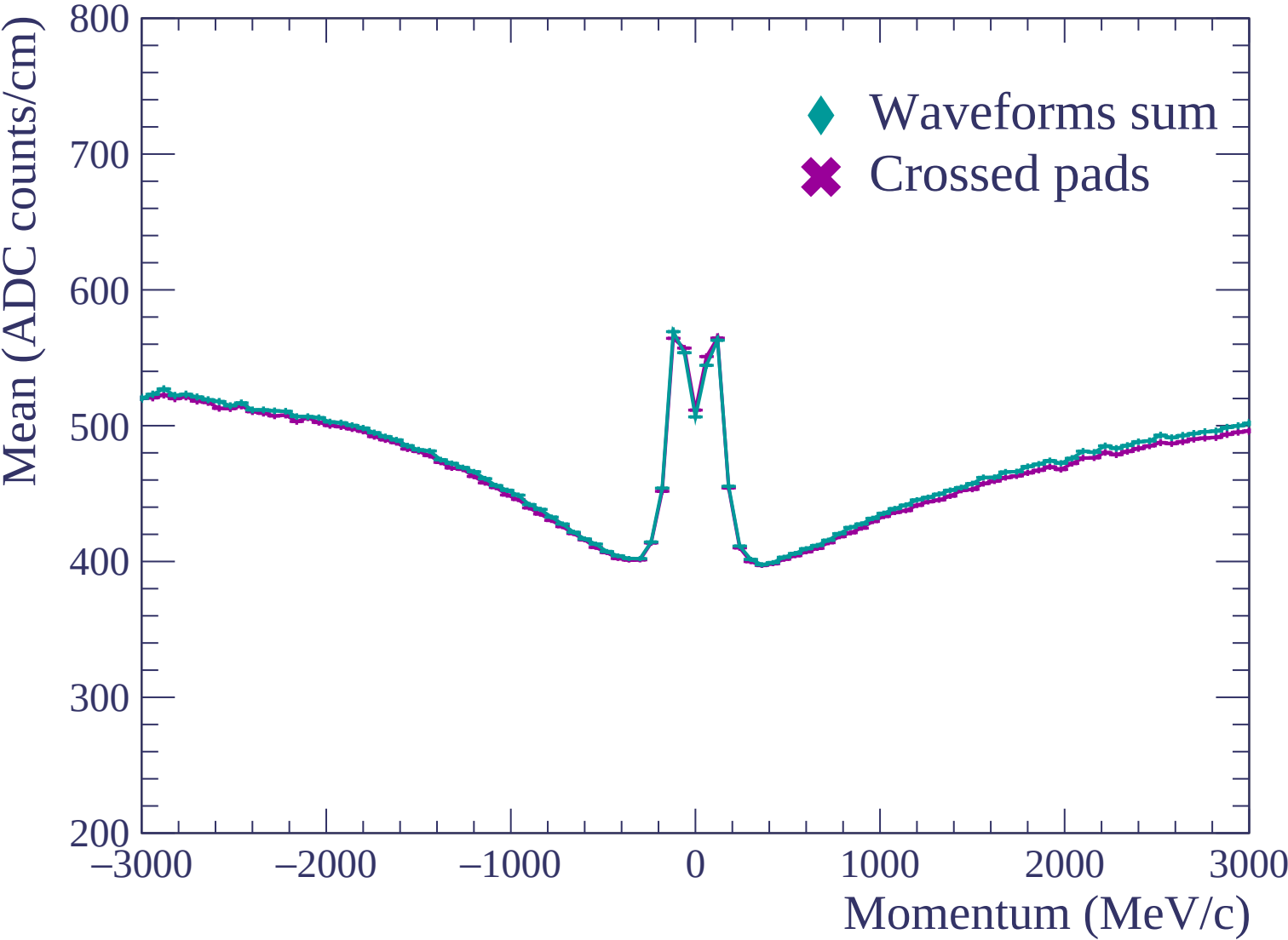


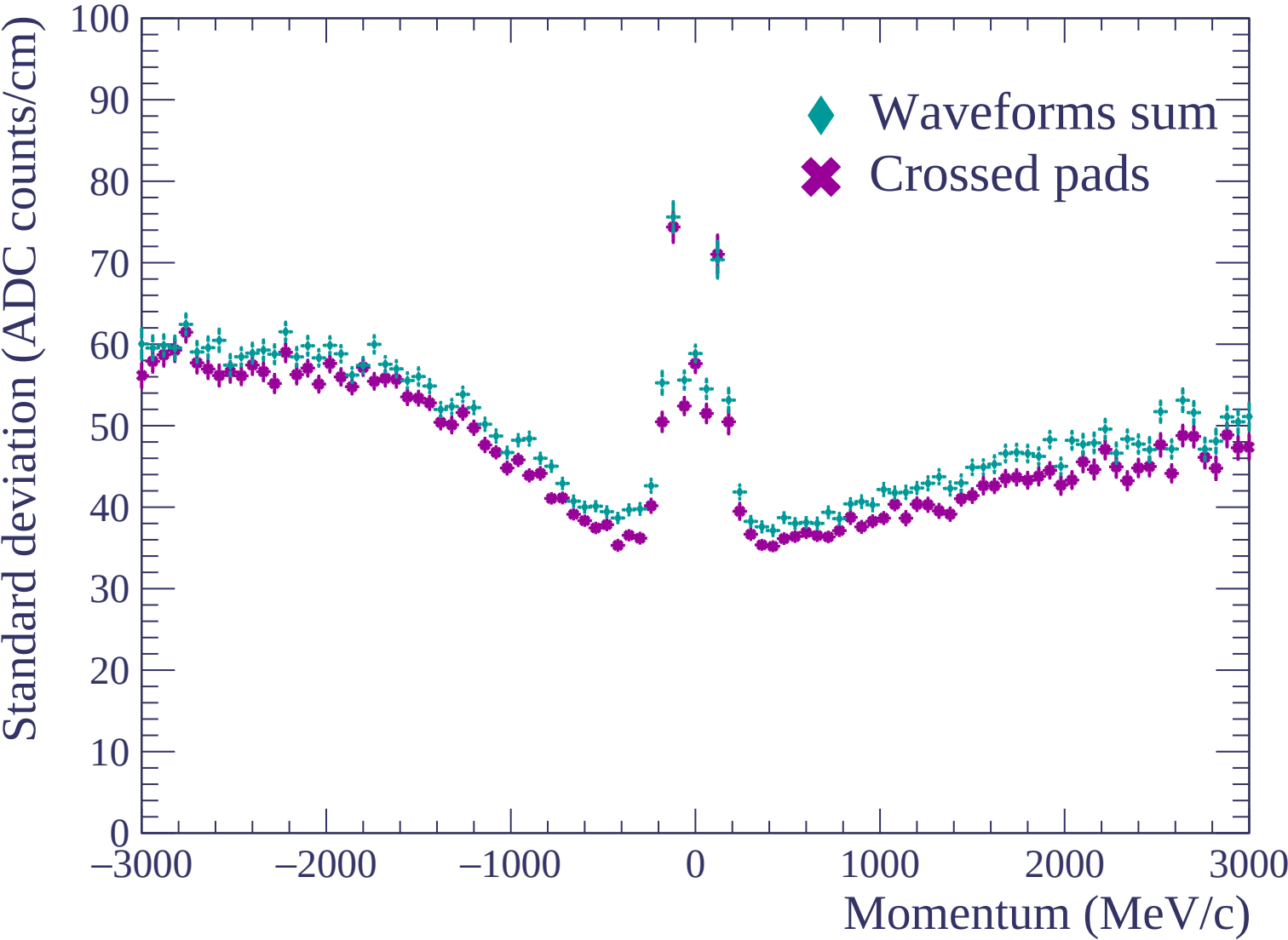


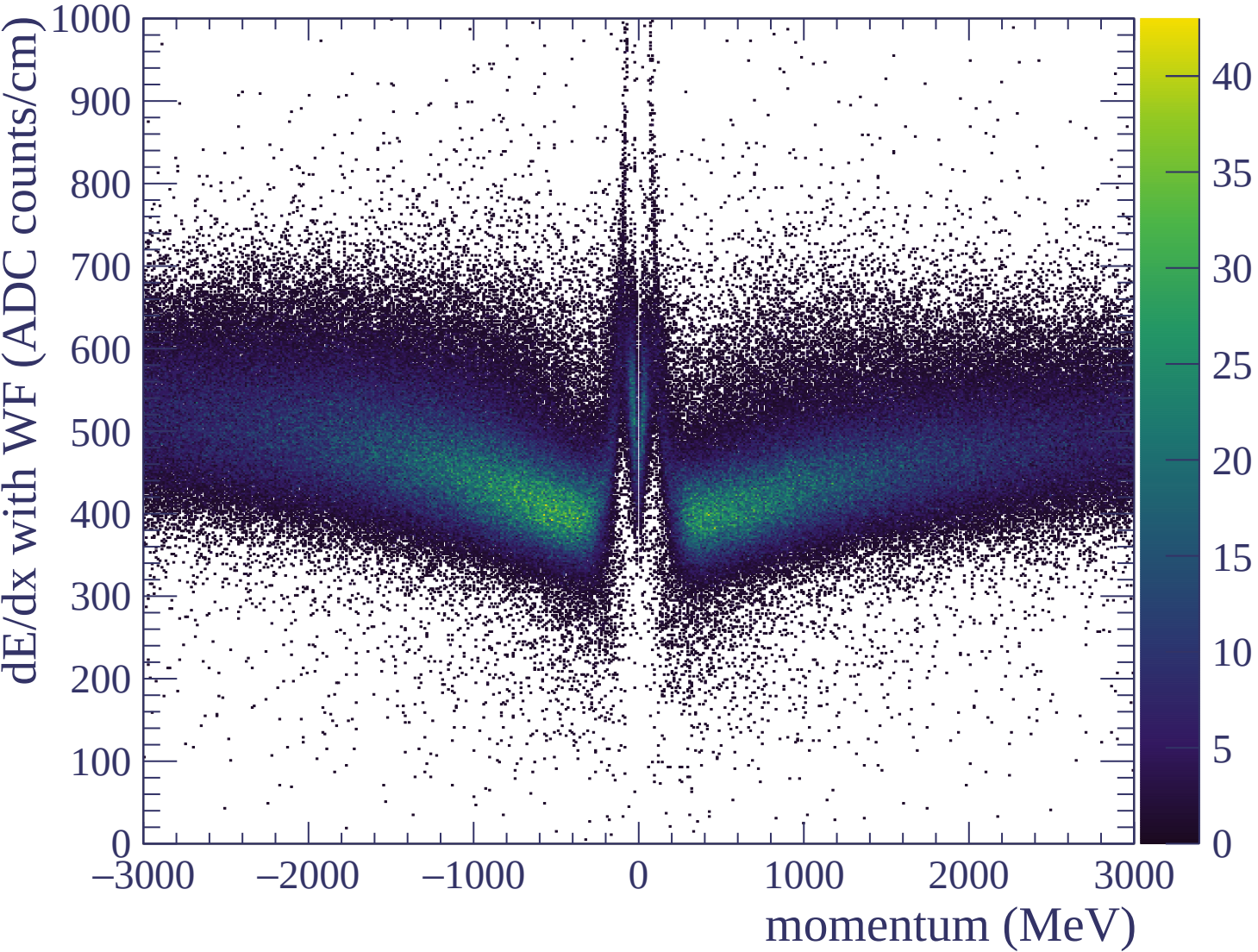


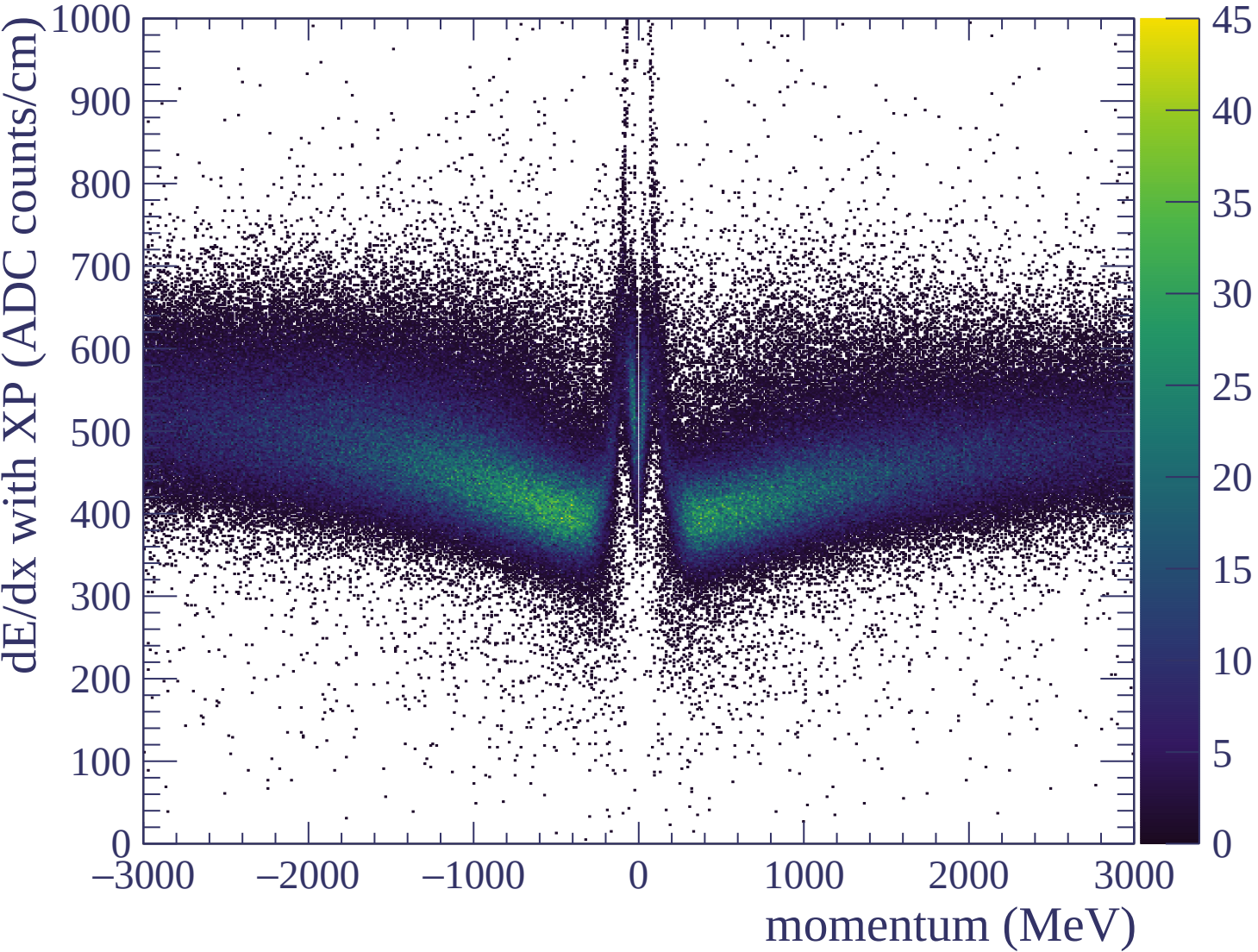


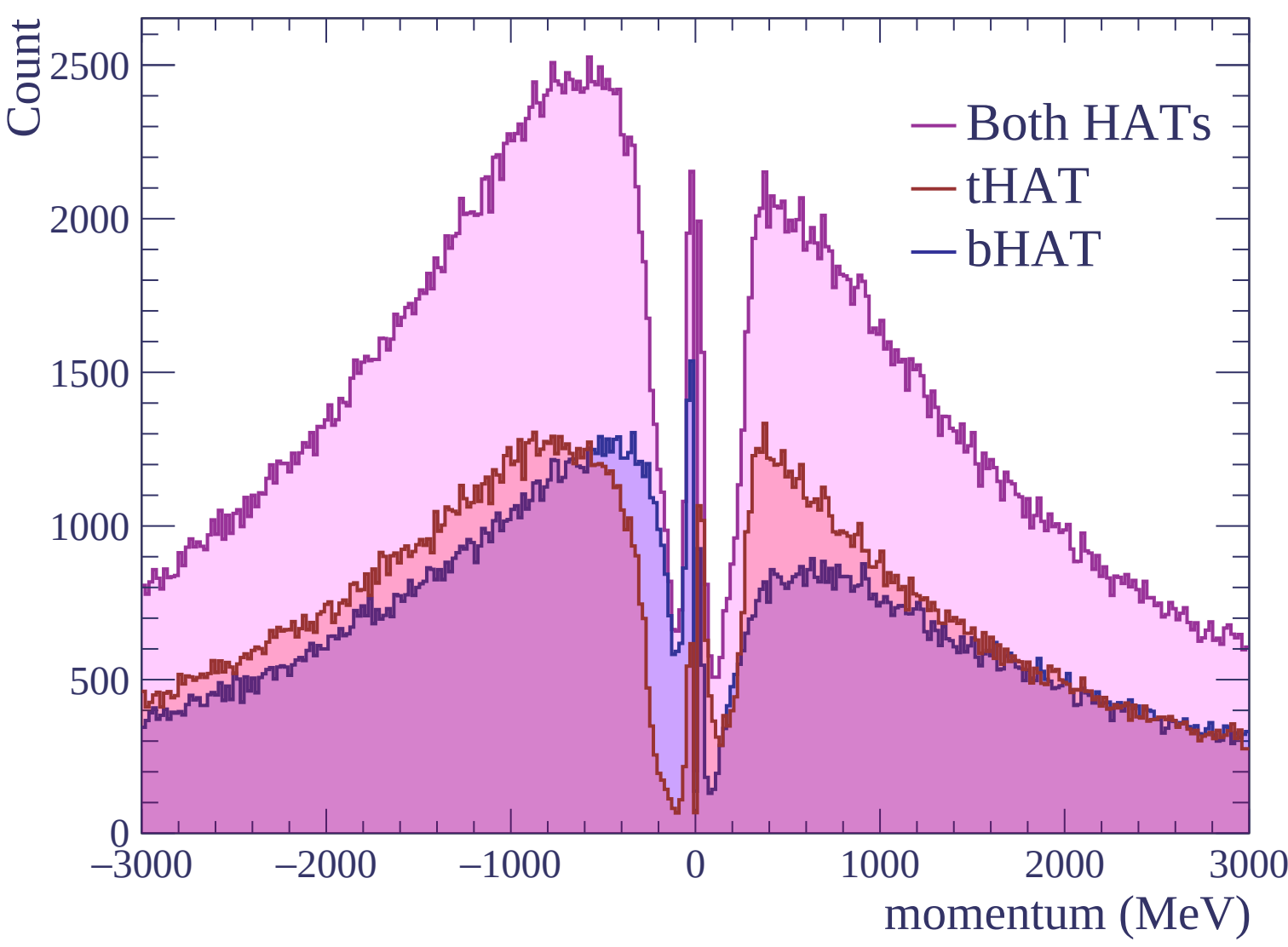


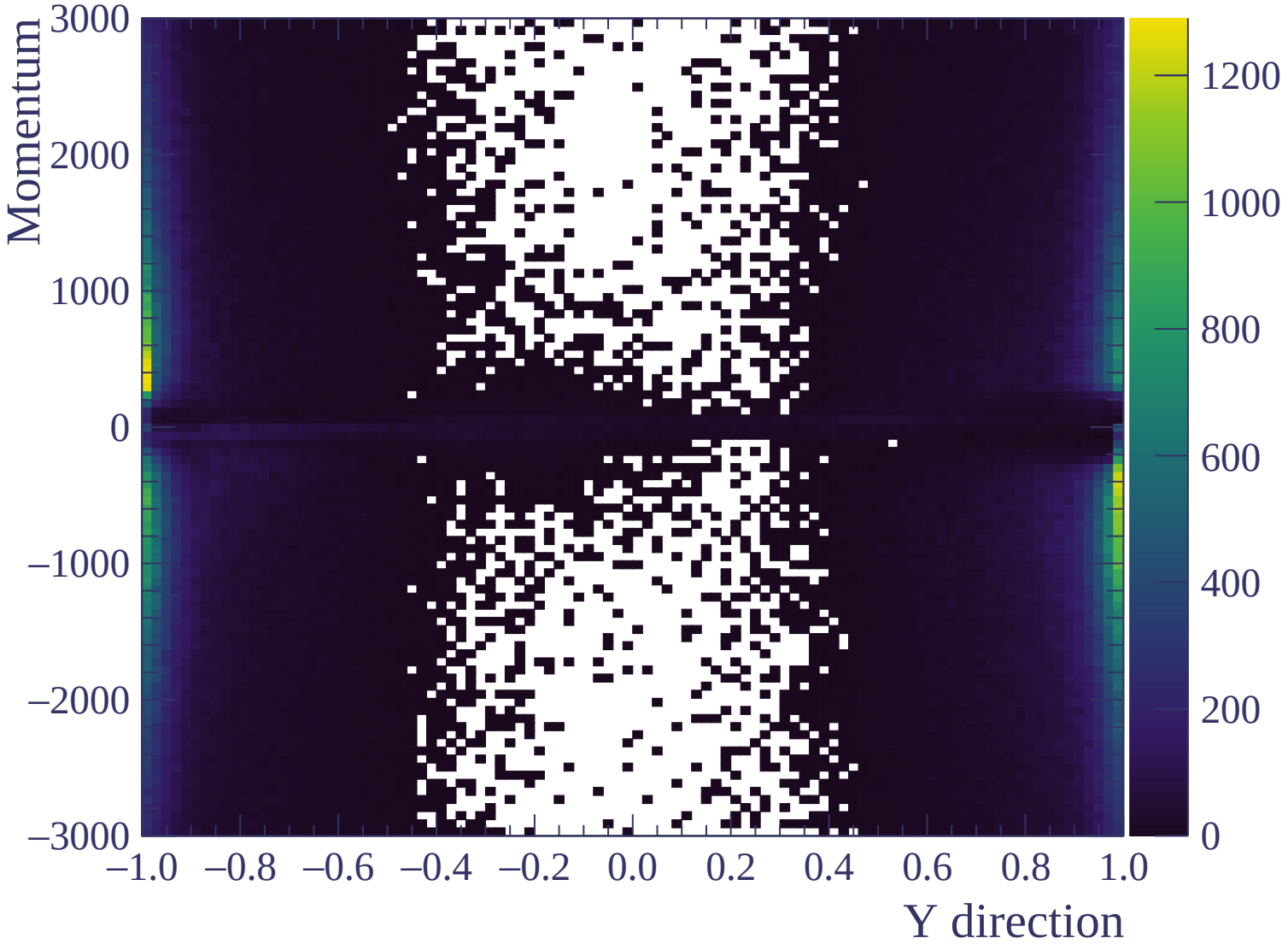


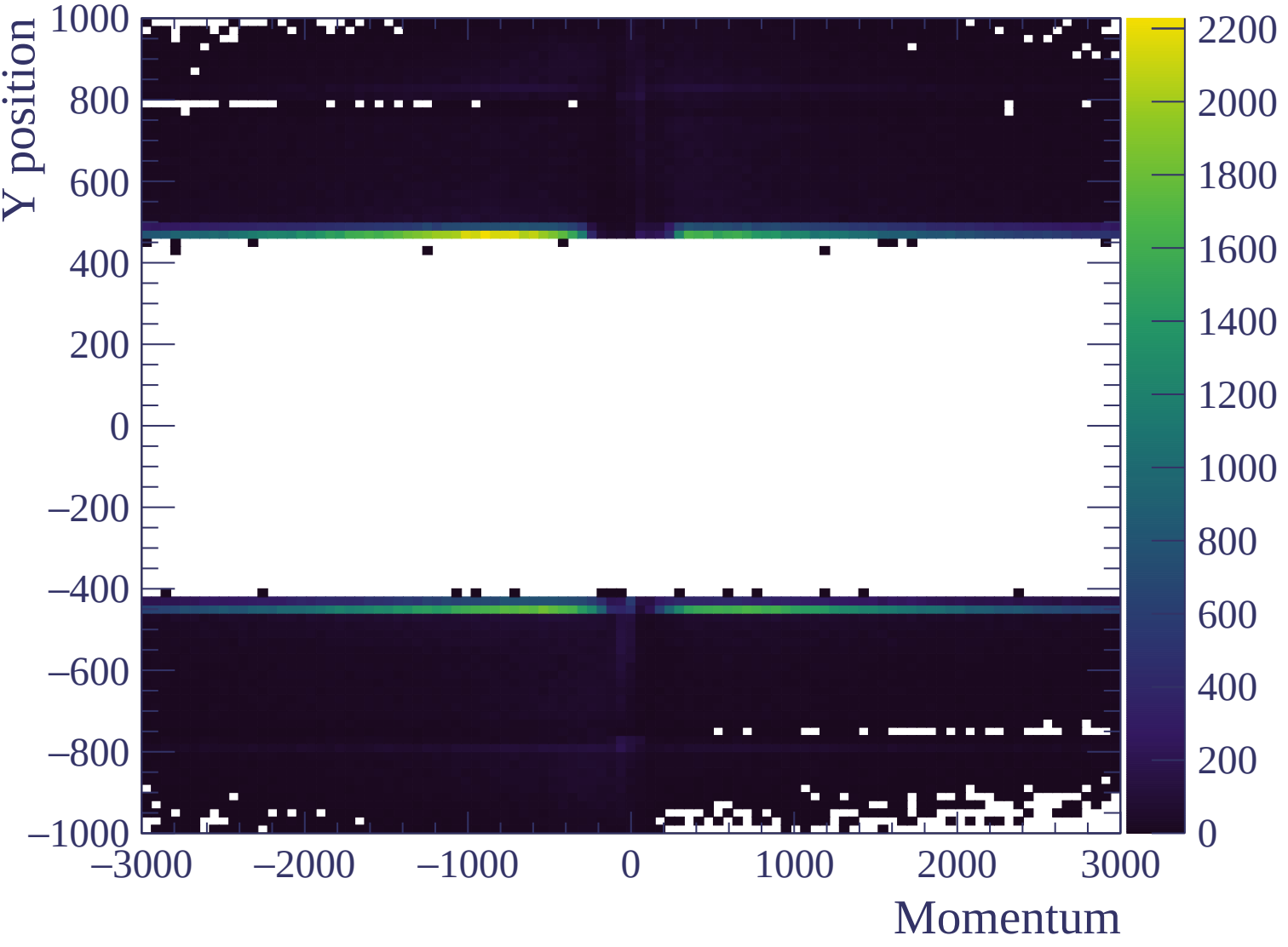


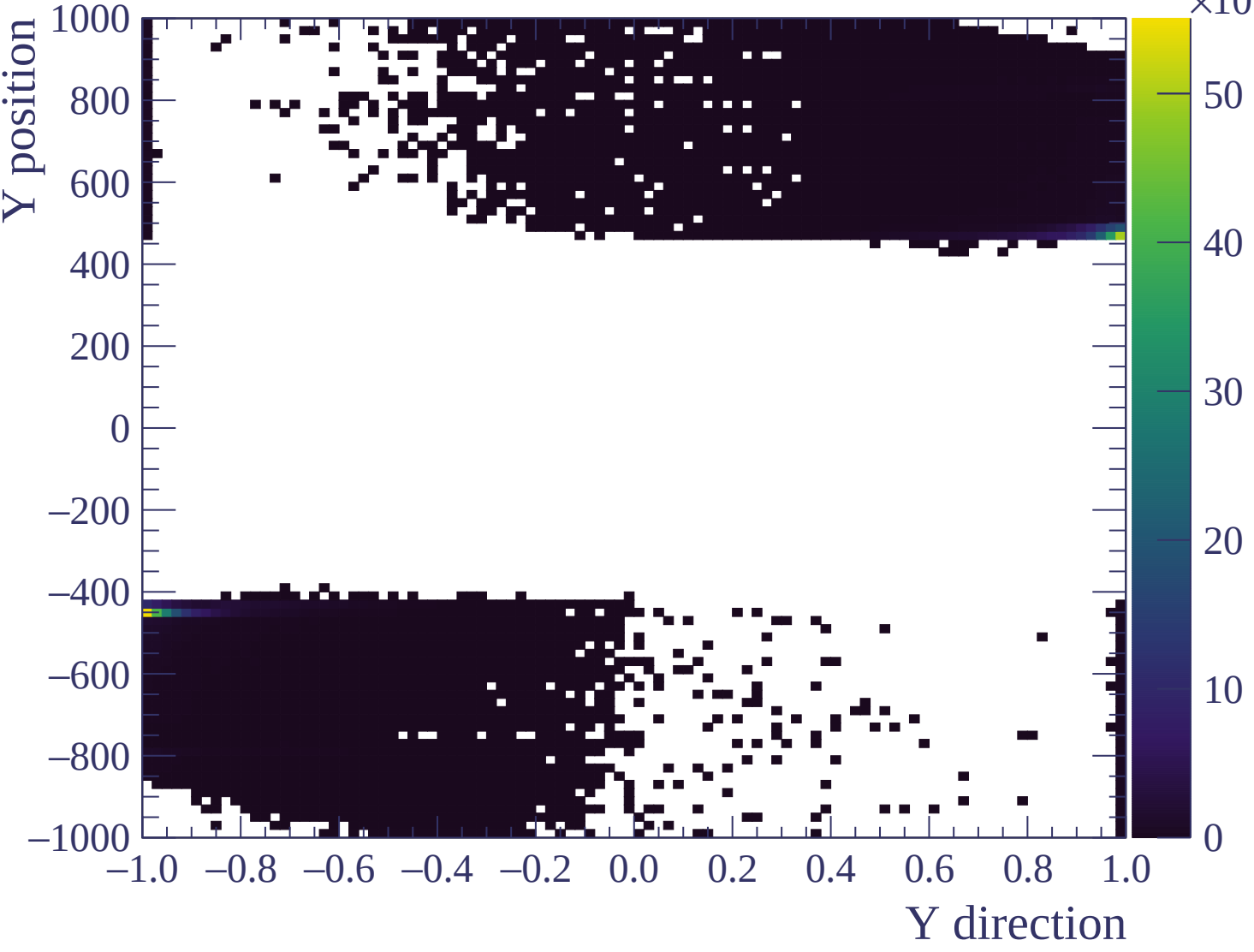


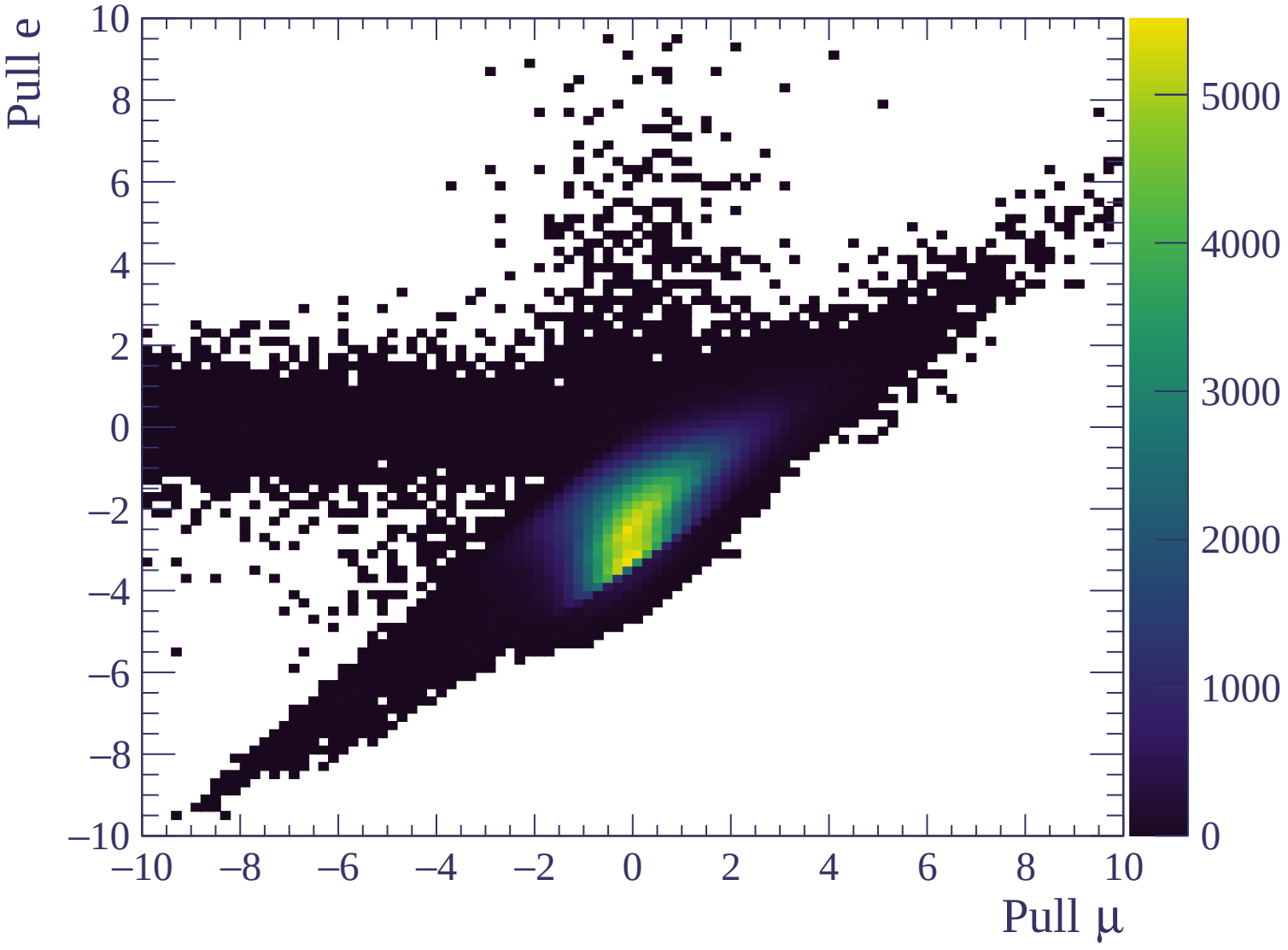






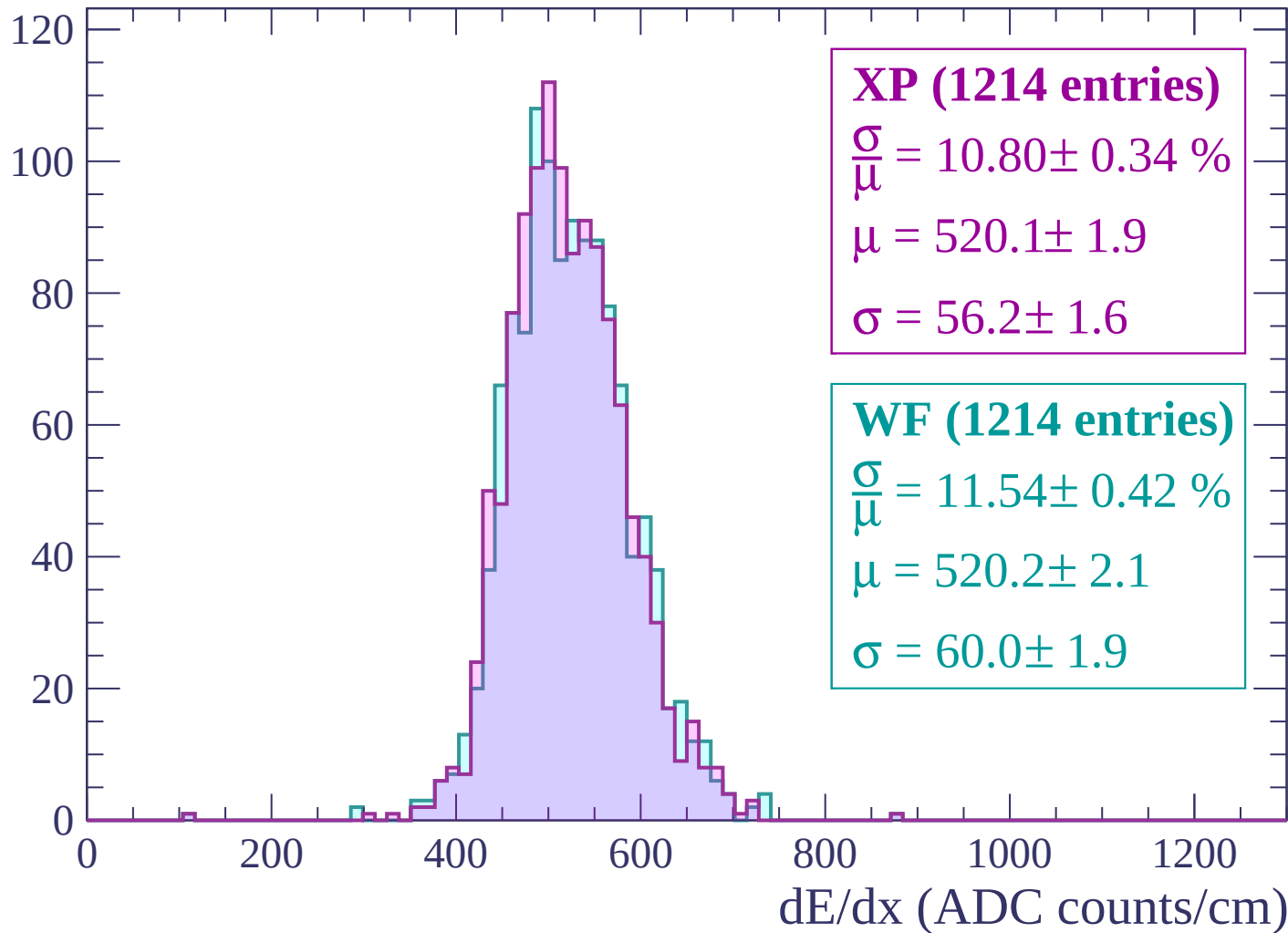






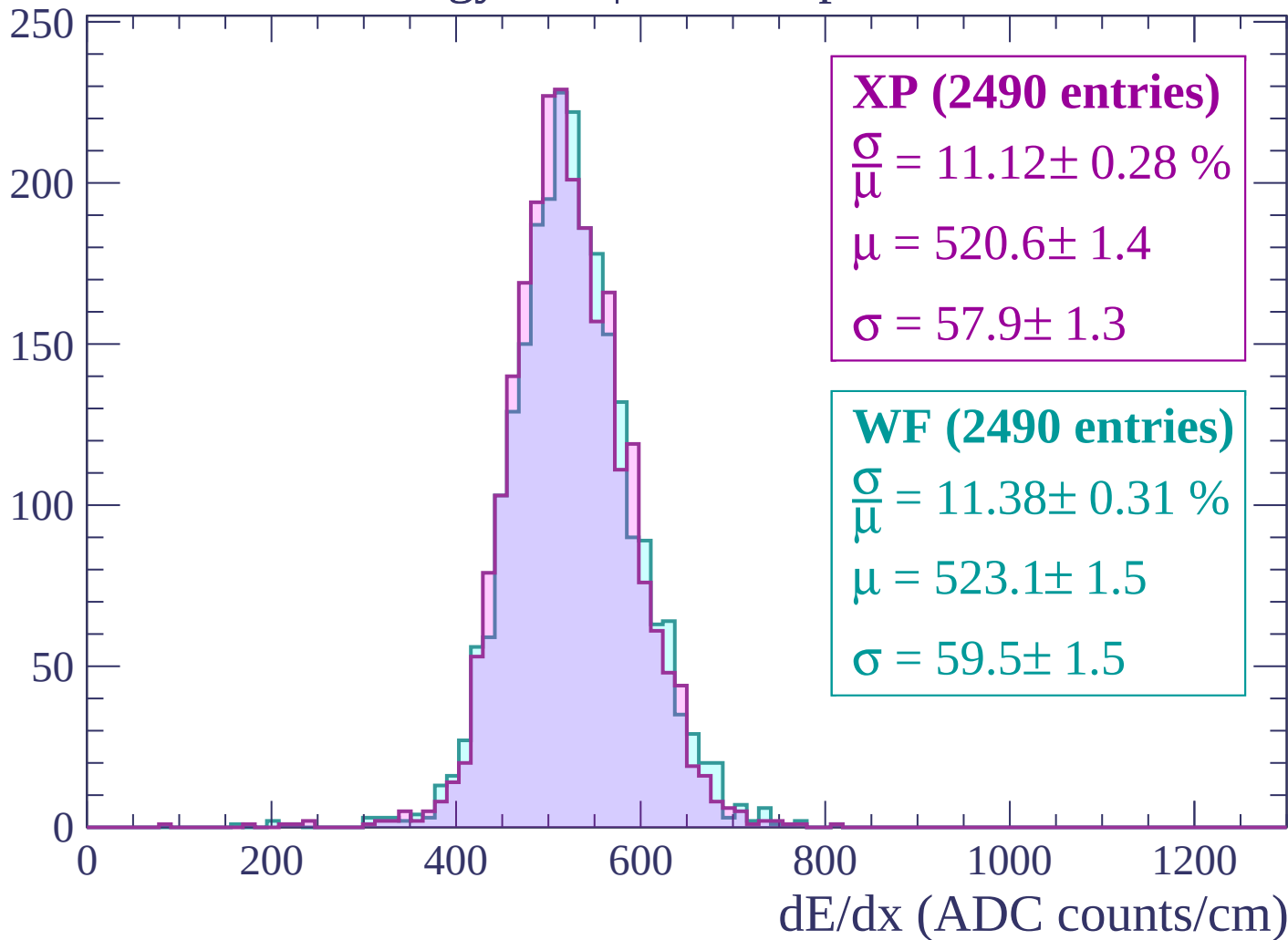
Energy loss | $-3000 < p < -2940$

Count



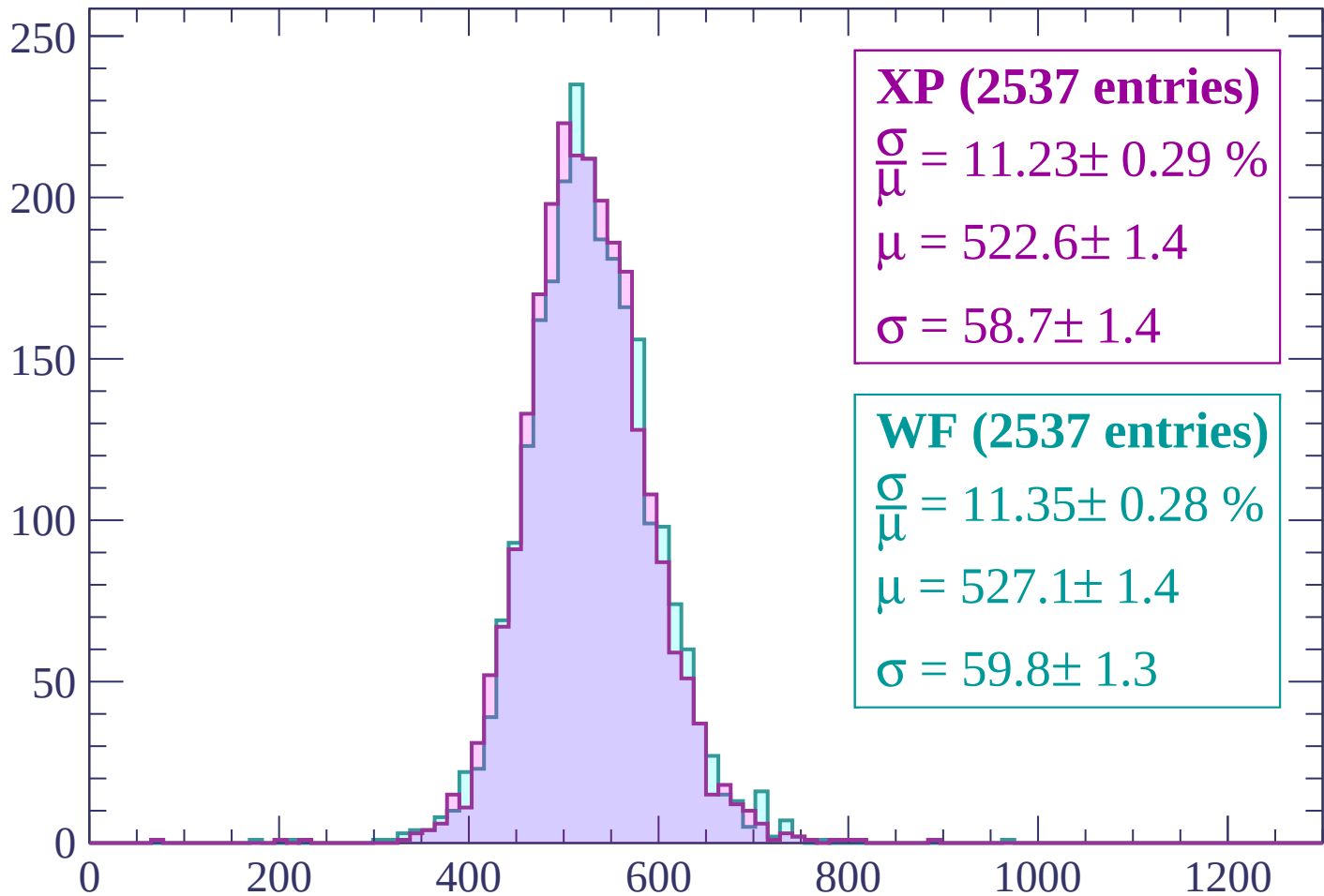
Energy loss | -2940 < p < -2880

Count



Energy loss | $-2880 < p < -2820$

Count



XP (2537 entries)

$\frac{\sigma}{\mu} = 11.23 \pm 0.29 \%$

$\mu = 522.6 \pm 1.4$

$\sigma = 58.7 \pm 1.4$

WF (2537 entries)

$\frac{\sigma}{\mu} = 11.35 \pm 0.28 \%$

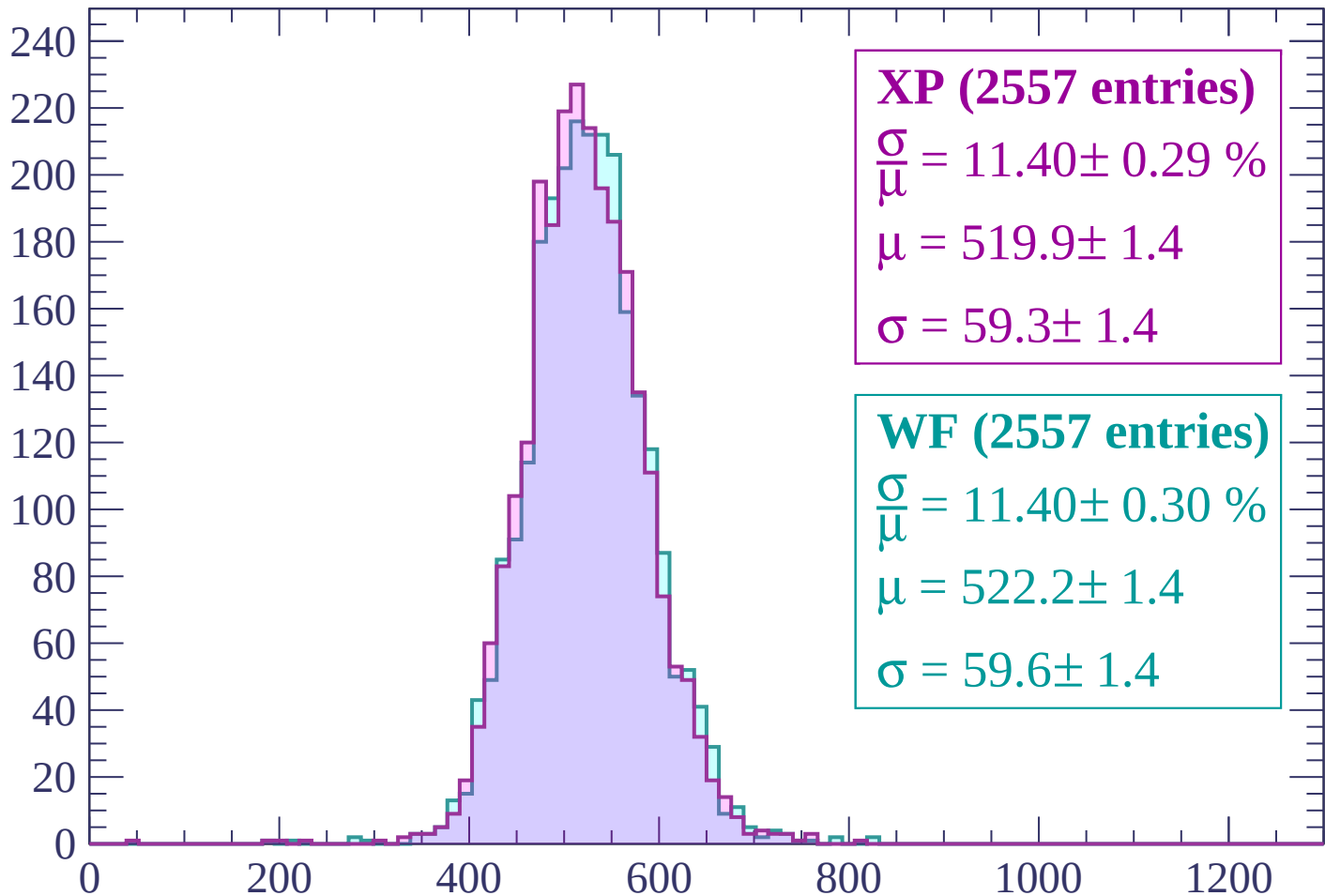
$\mu = 527.1 \pm 1.4$

$\sigma = 59.8 \pm 1.3$

dE/dx (ADC counts/cm)

Energy loss | $-2820 < p < -2760$

Count



XP (2557 entries)

$\frac{\sigma}{\mu} = 11.40 \pm 0.29 \%$

$\mu = 519.9 \pm 1.4$

$\sigma = 59.3 \pm 1.4$

WF (2557 entries)

$\frac{\sigma}{\mu} = 11.40 \pm 0.30 \%$

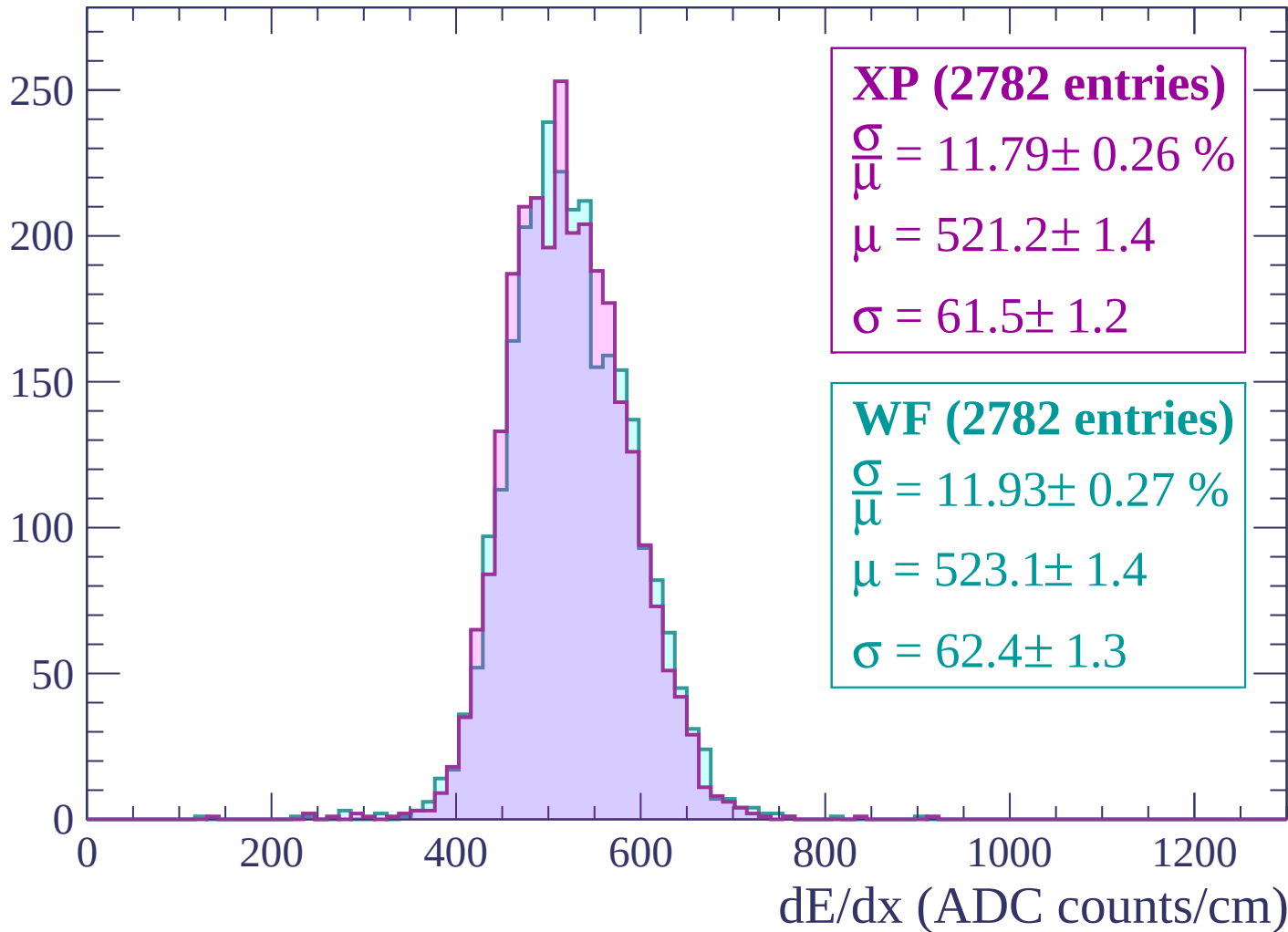
$\mu = 522.2 \pm 1.4$

$\sigma = 59.6 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (2849 entries)

$\frac{\sigma}{\mu} = 11.14 \pm 0.26 \%$

$\mu = 518.1 \pm 1.3$

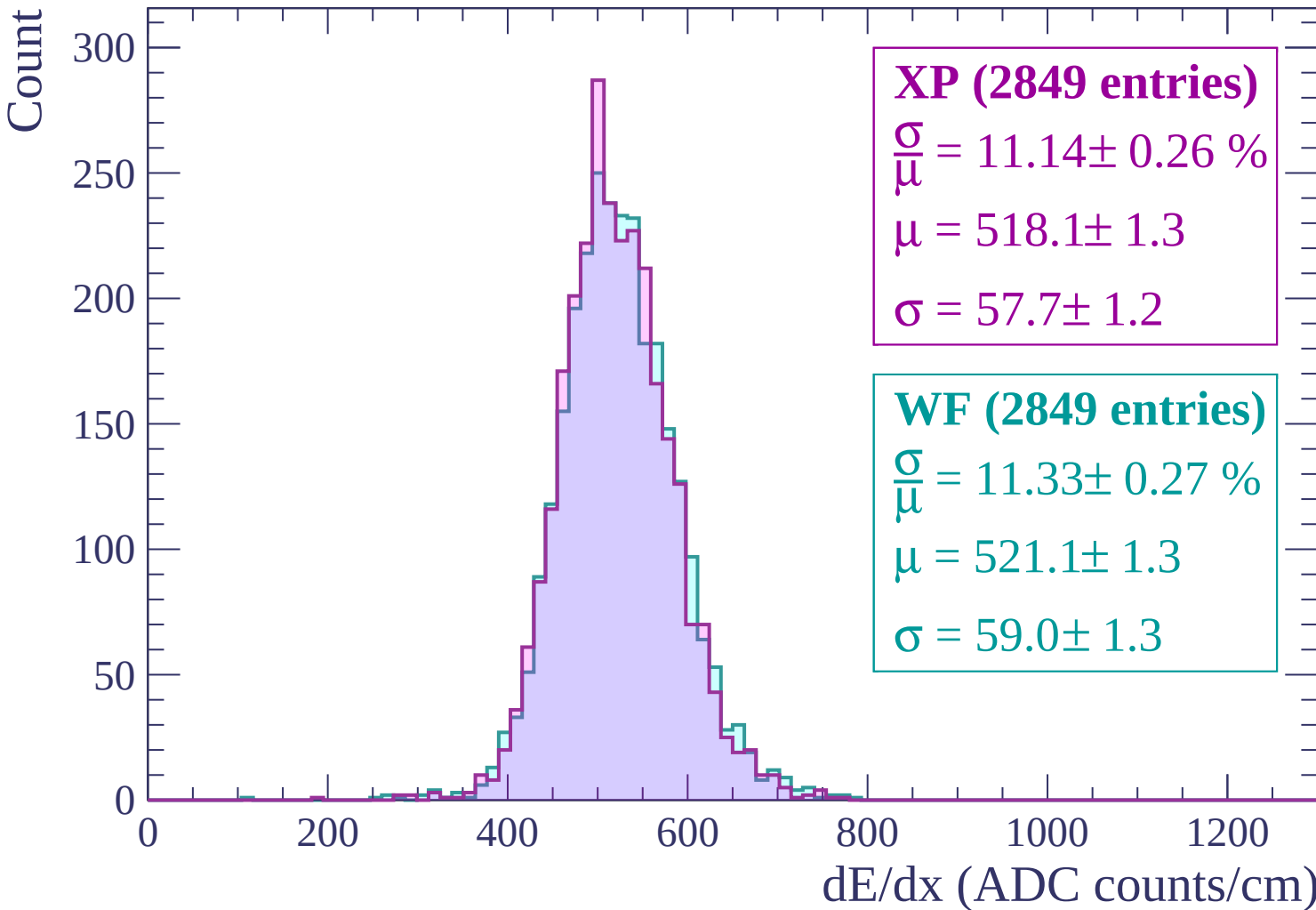
$\sigma = 57.7 \pm 1.2$

WF (2849 entries)

$\frac{\sigma}{\mu} = 11.33 \pm 0.27 \%$

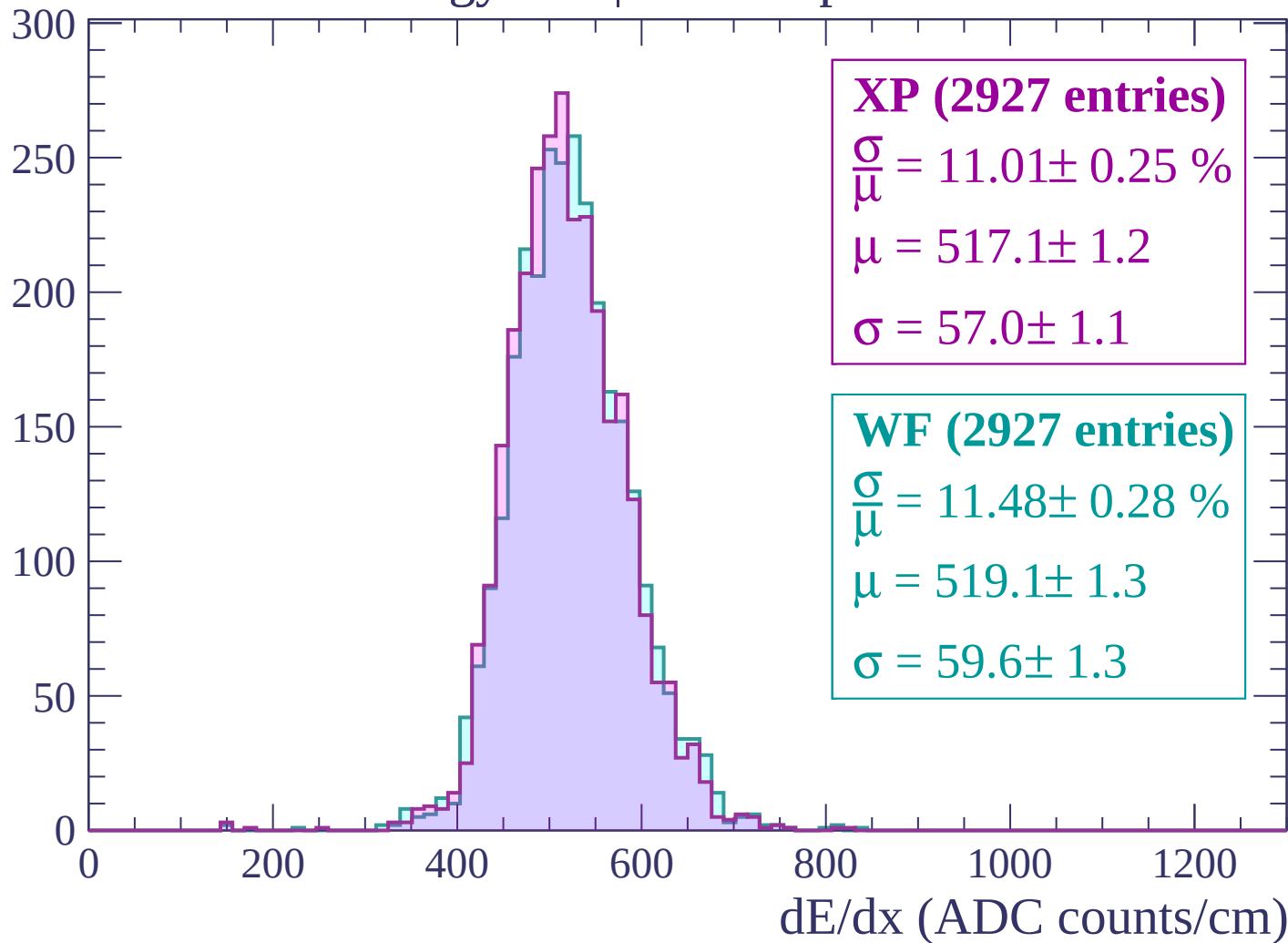
$\mu = 521.1 \pm 1.3$

$\sigma = 59.0 \pm 1.3$



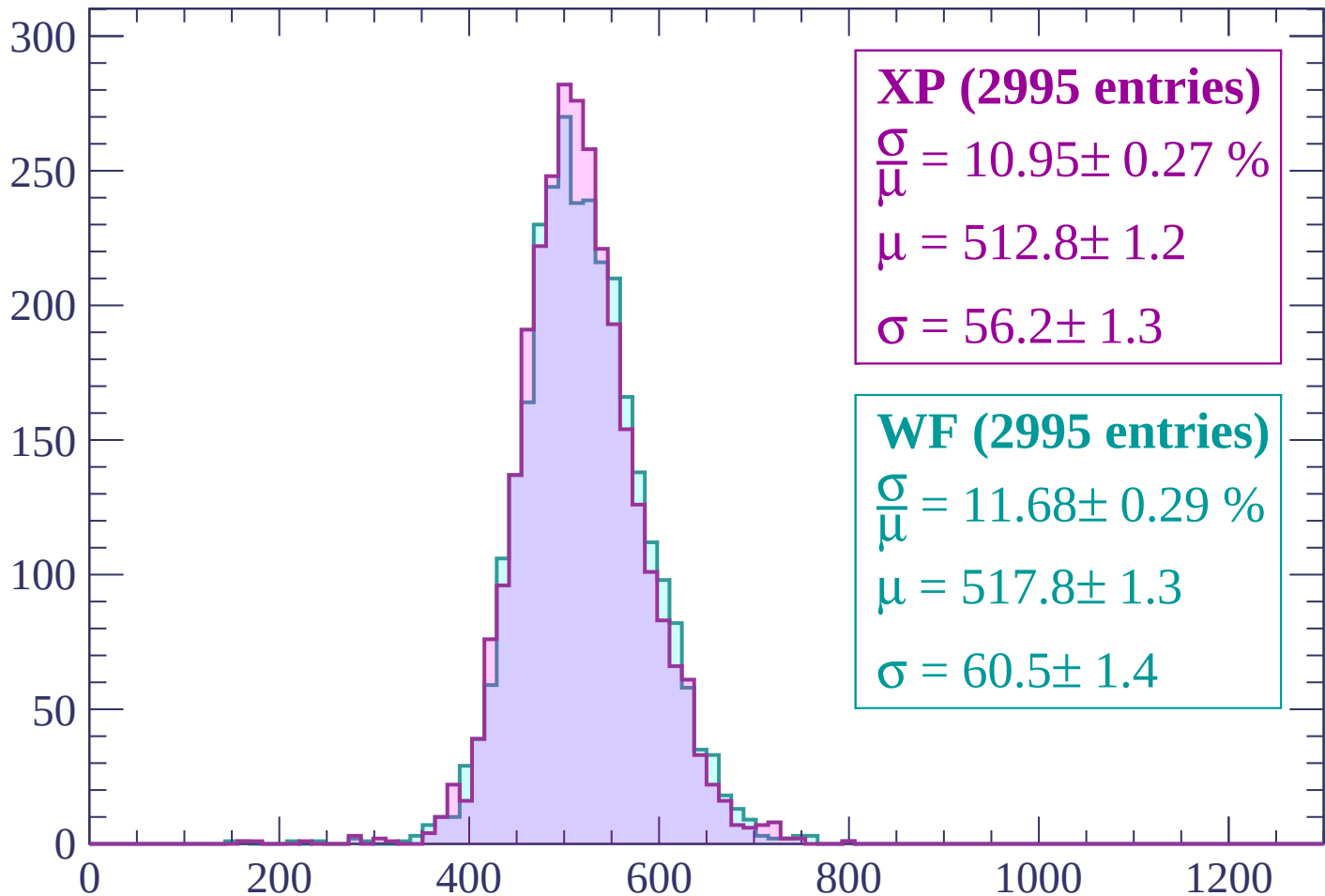
Energy loss | $-2640 < p < -2580$

Count



Energy loss | $-2580 < p < -2520$

Count



XP (2995 entries)

$$\frac{\sigma}{\mu} = 10.95 \pm 0.27 \%$$

$$\mu = 512.8 \pm 1.2$$

$$\sigma = 56.2 \pm 1.3$$

WF (2995 entries)

$$\frac{\sigma}{\mu} = 11.68 \pm 0.29 \%$$

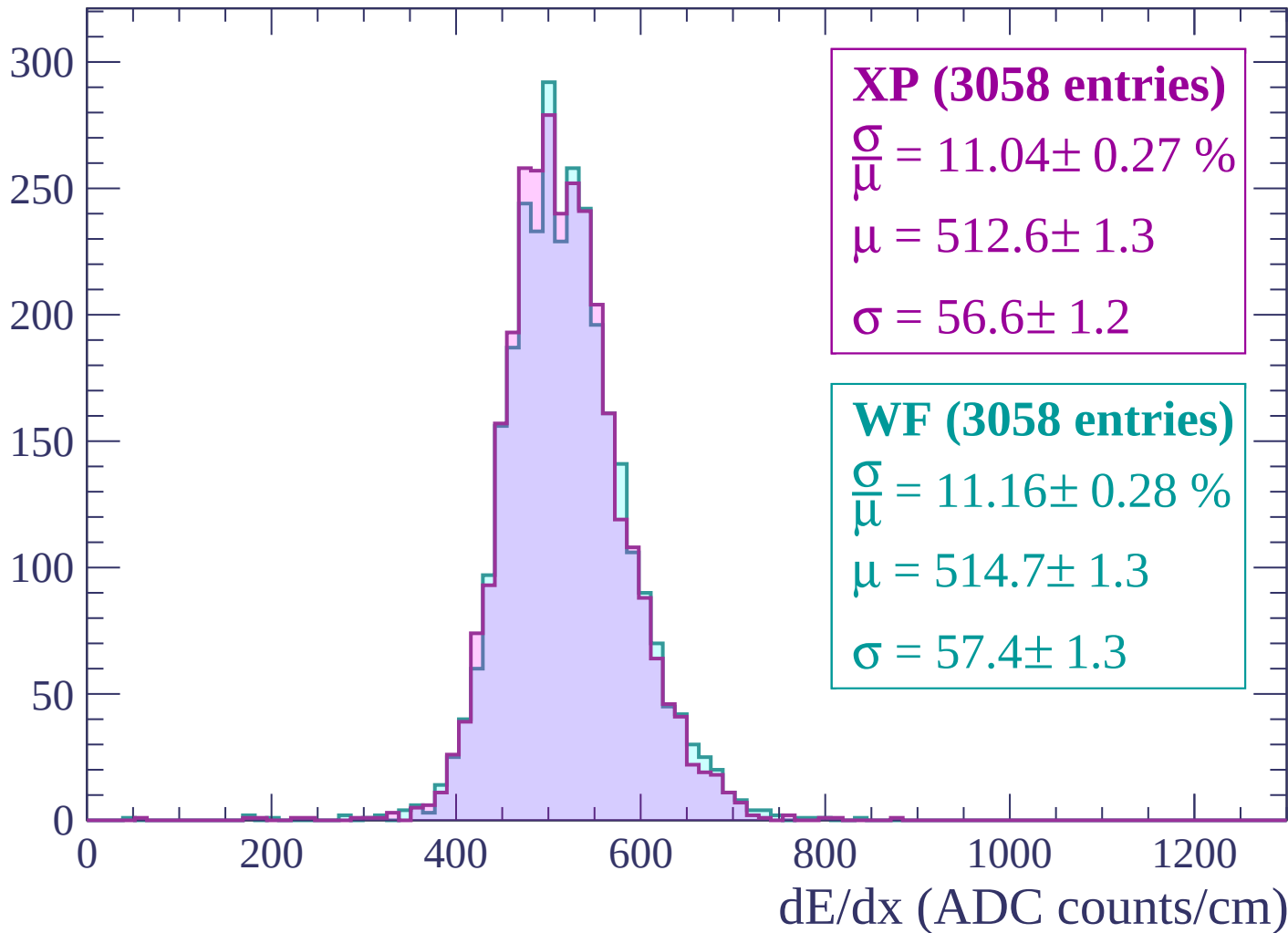
$$\mu = 517.8 \pm 1.3$$

$$\sigma = 60.5 \pm 1.4$$

dE/dx (ADC counts/cm)

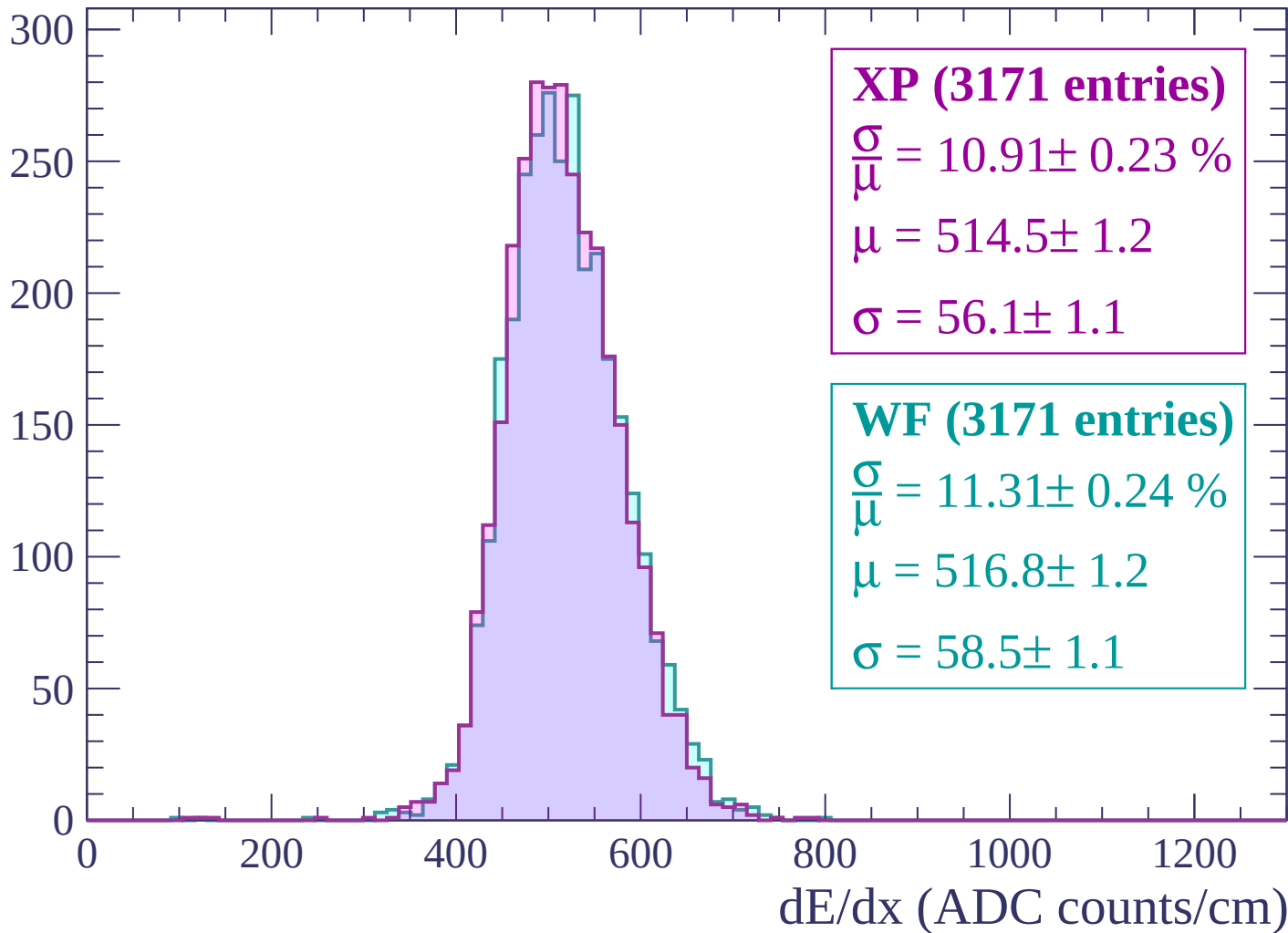
Energy loss | -2520 < p < -2460

Count



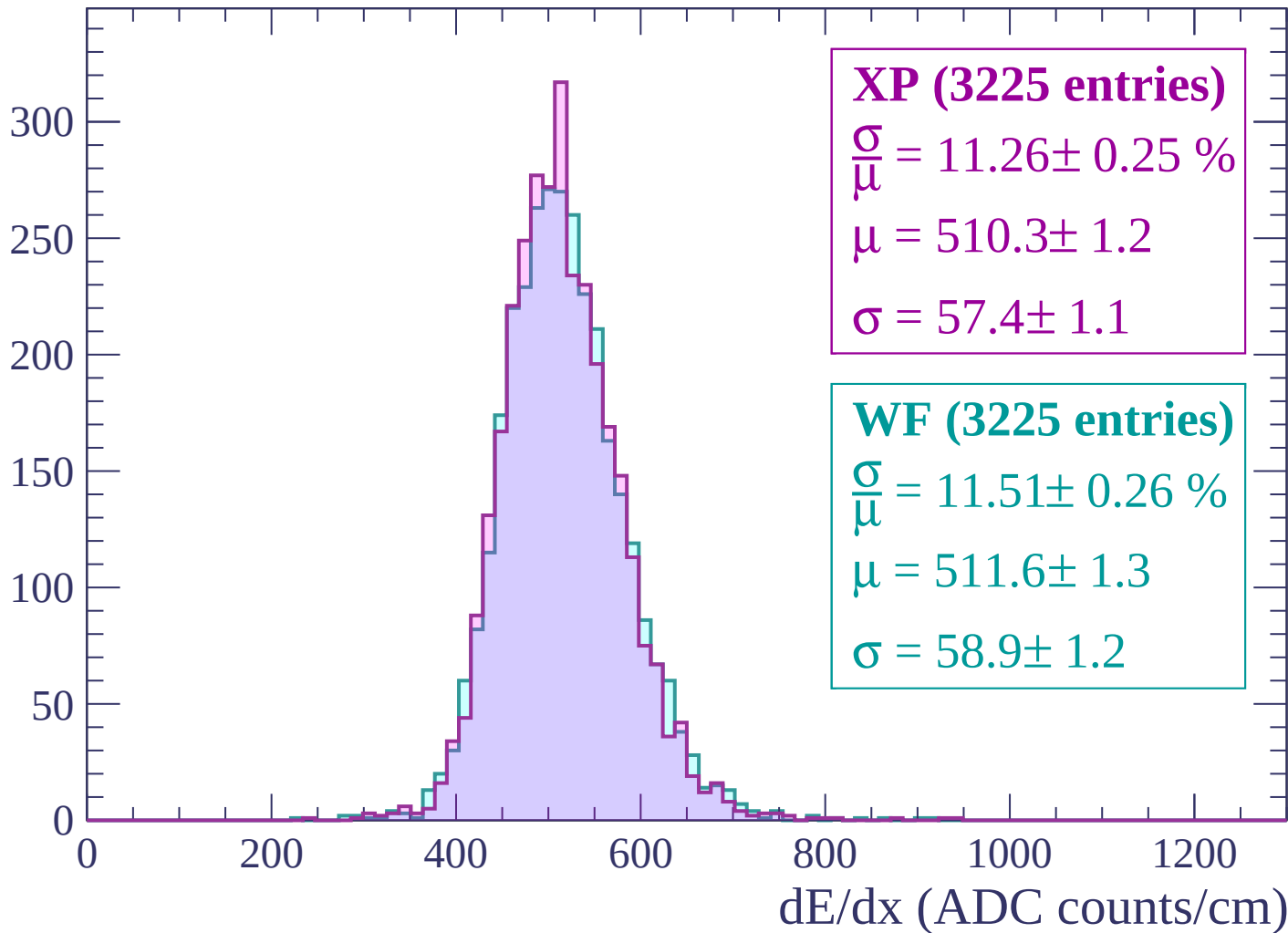
Energy loss | $-2460 < p < -2400$

Count



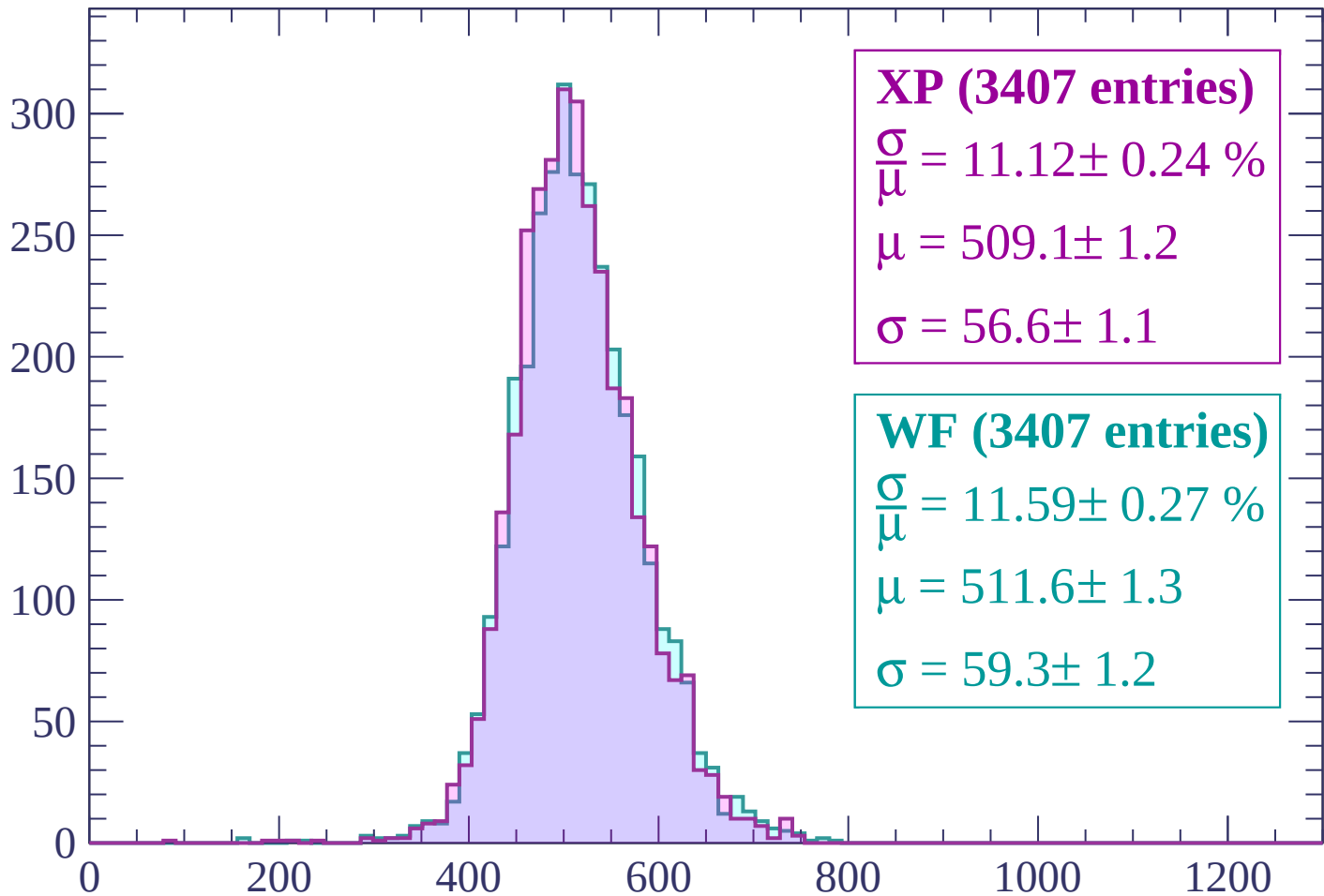
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (3407 entries)

$$\frac{\sigma}{\mu} = 11.12 \pm 0.24 \%$$

$$\mu = 509.1 \pm 1.2$$

$$\sigma = 56.6 \pm 1.1$$

WF (3407 entries)

$$\frac{\sigma}{\mu} = 11.59 \pm 0.27 \%$$

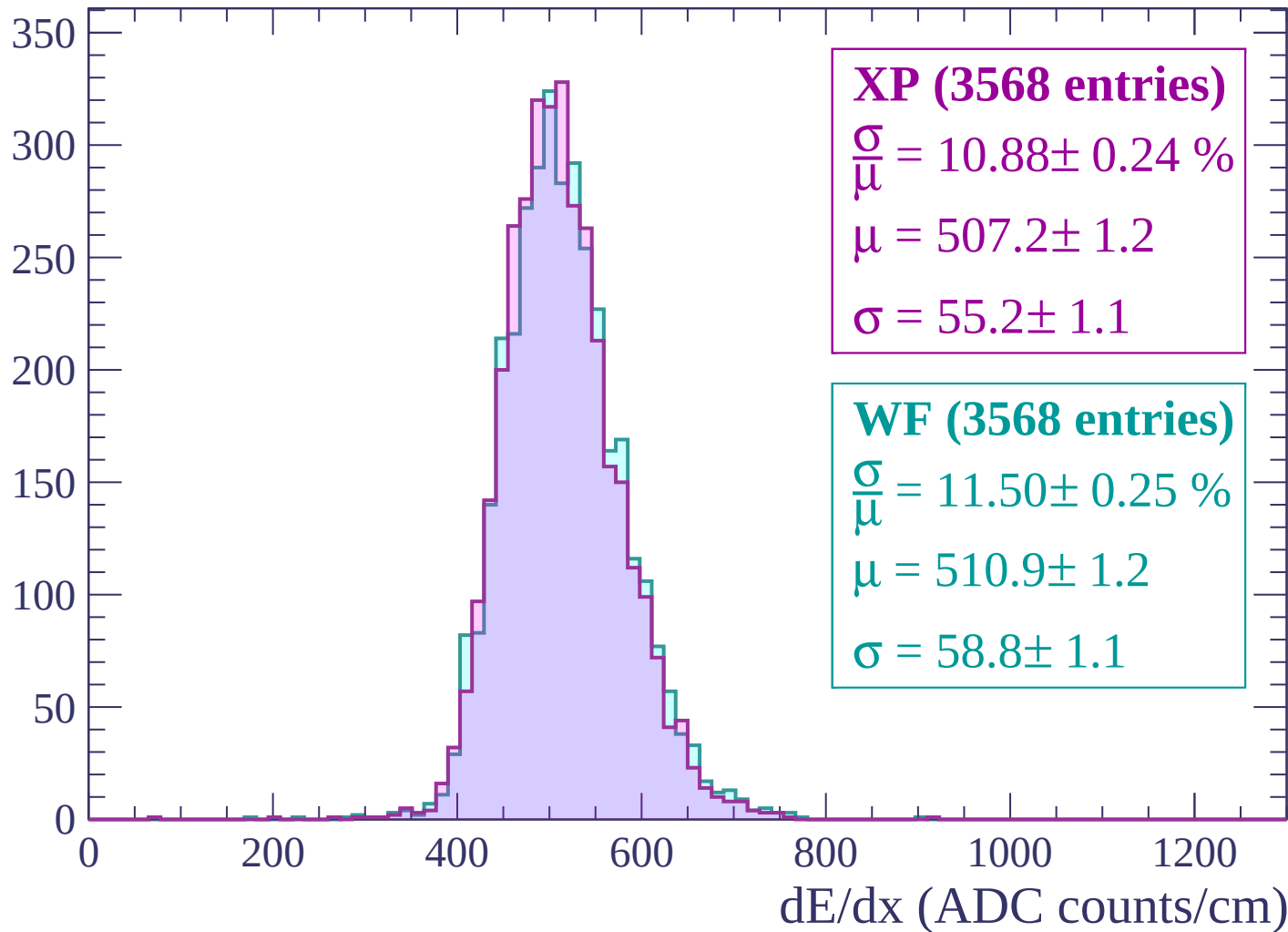
$$\mu = 511.6 \pm 1.3$$

$$\sigma = 59.3 \pm 1.2$$

dE/dx (ADC counts/cm)

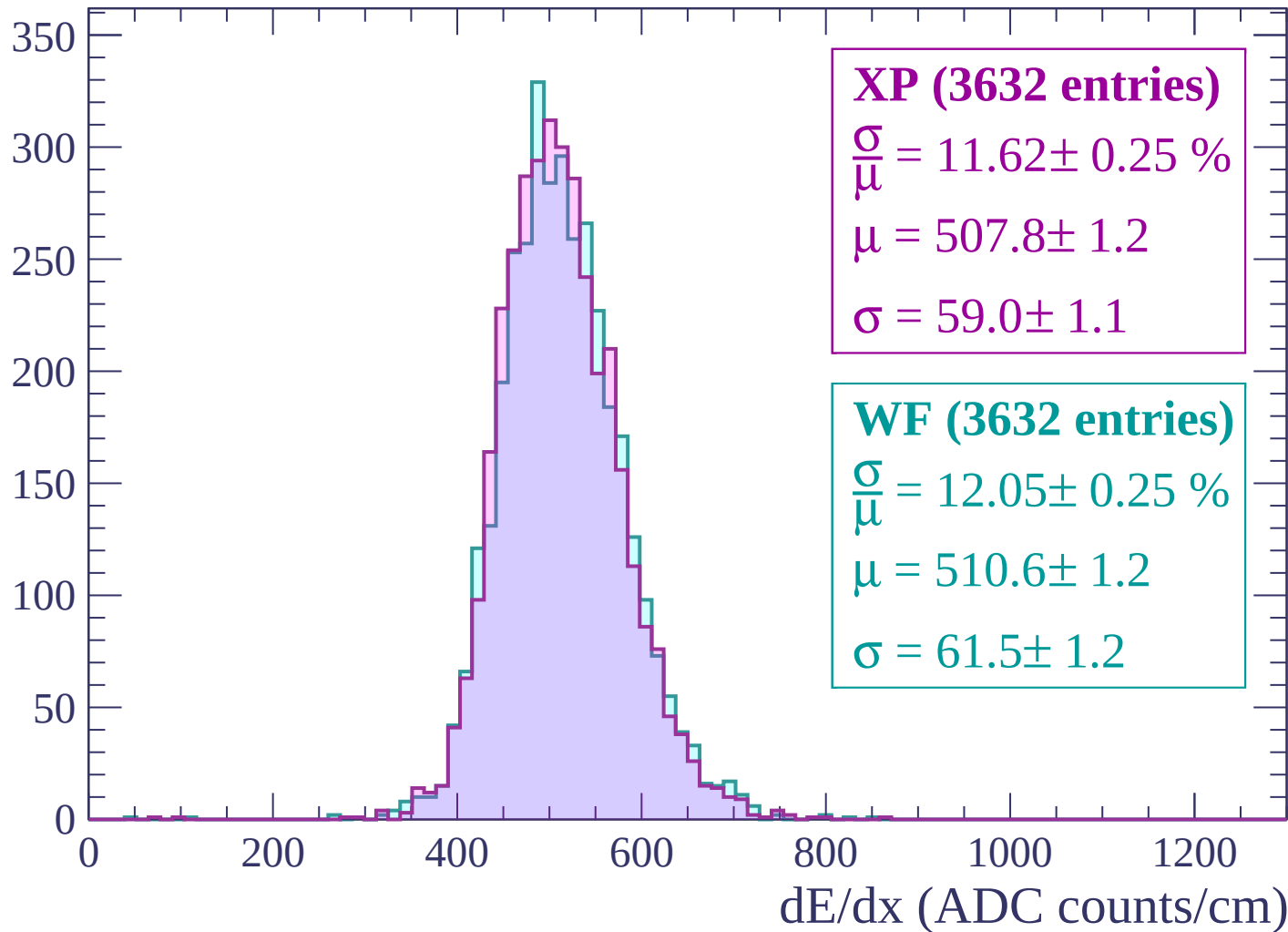
Energy loss | $-2280 < p < -2220$

Count



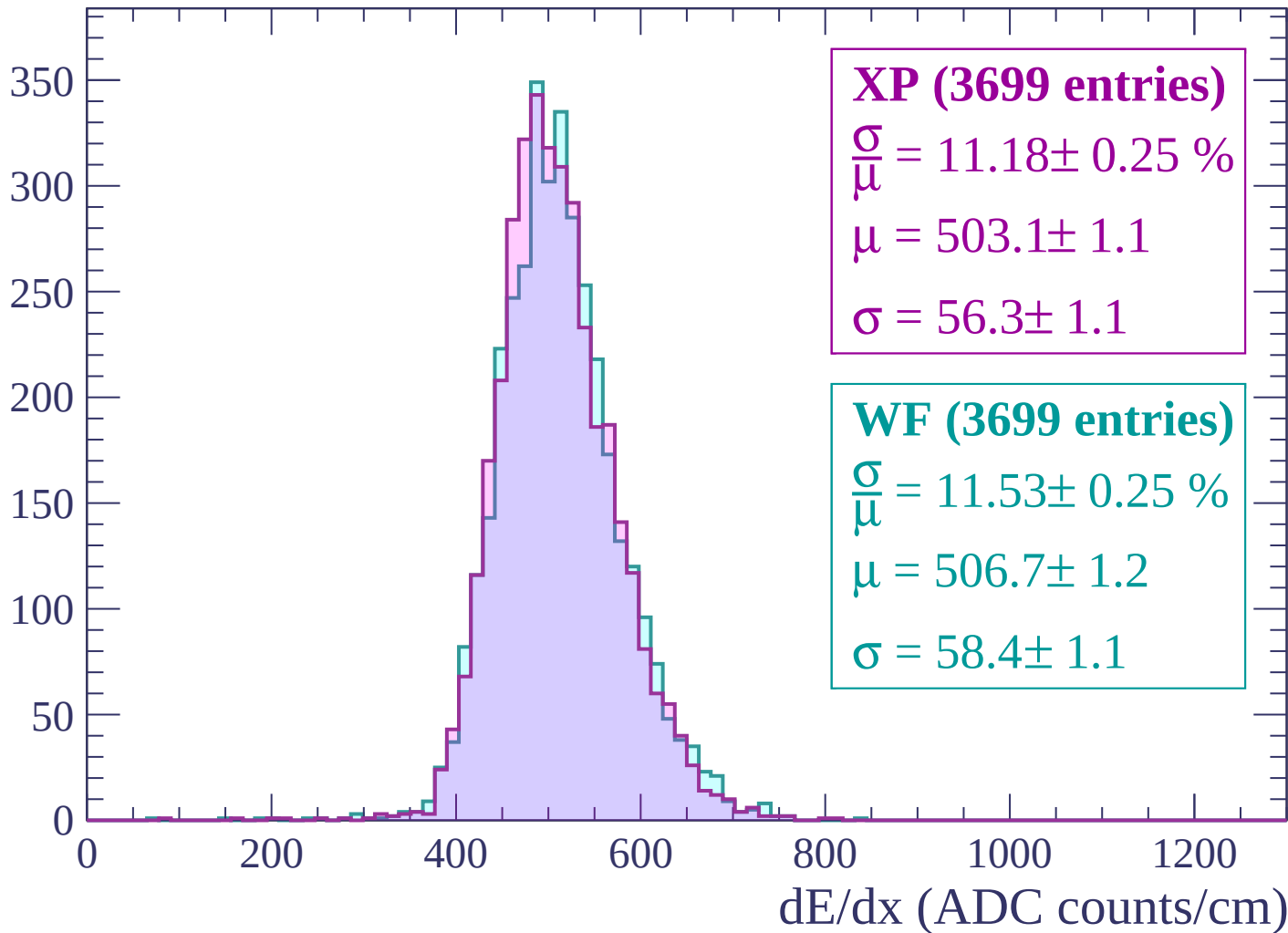
Energy loss | -2220 < p < -2160

Count



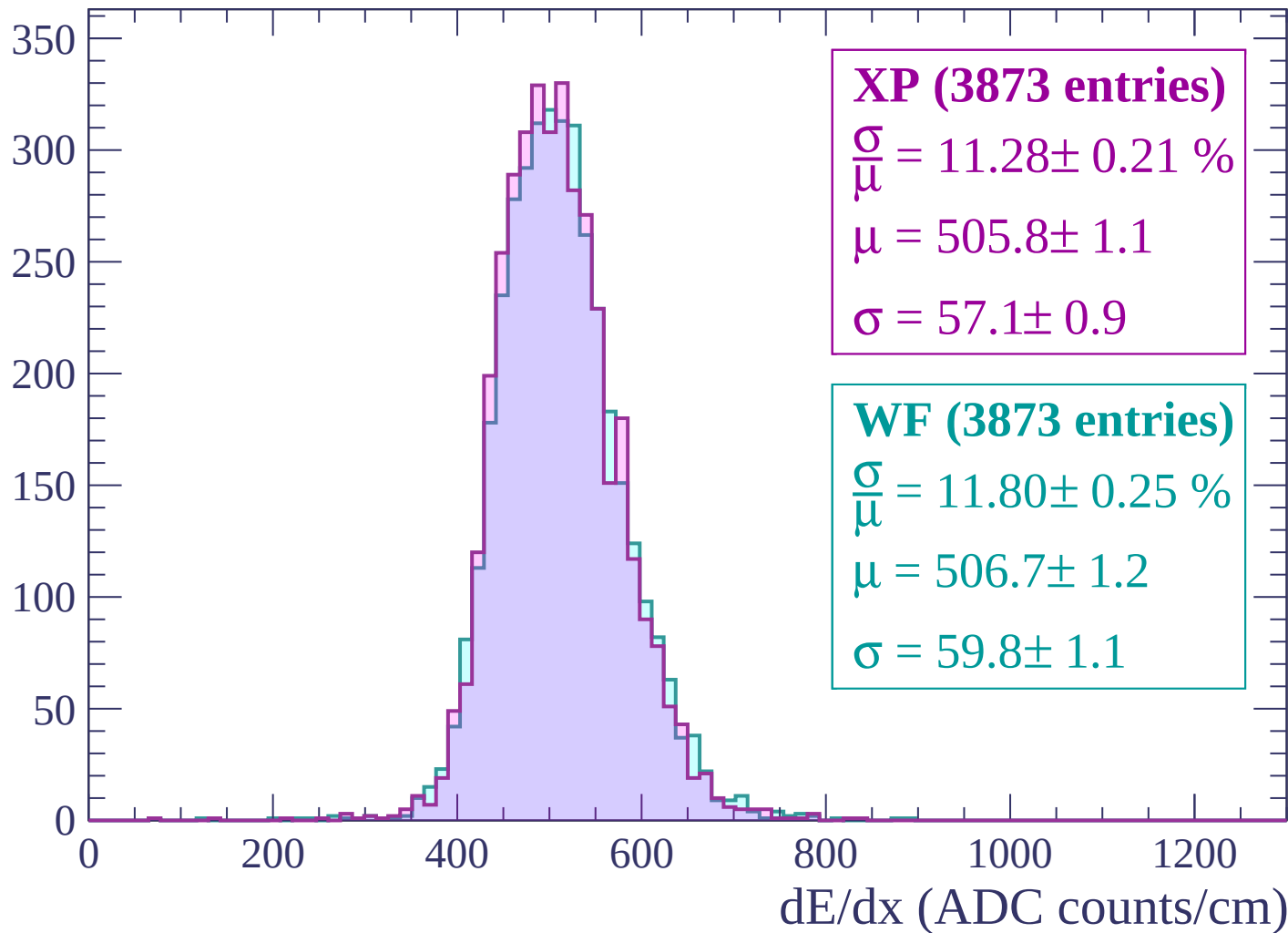
Energy loss | -2160 < p < -2100

Count



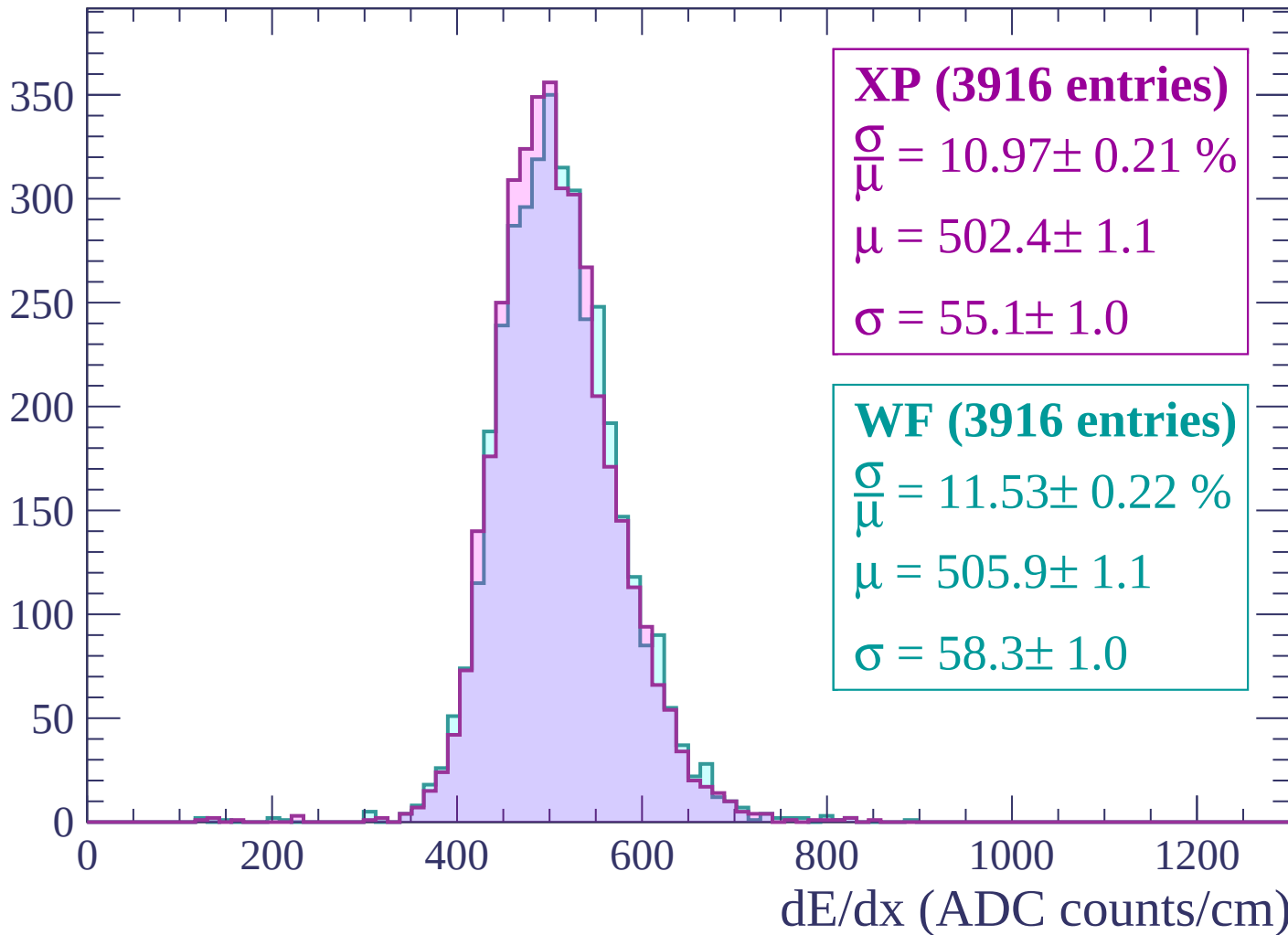
Energy loss | -2100 < p < -2040

Count



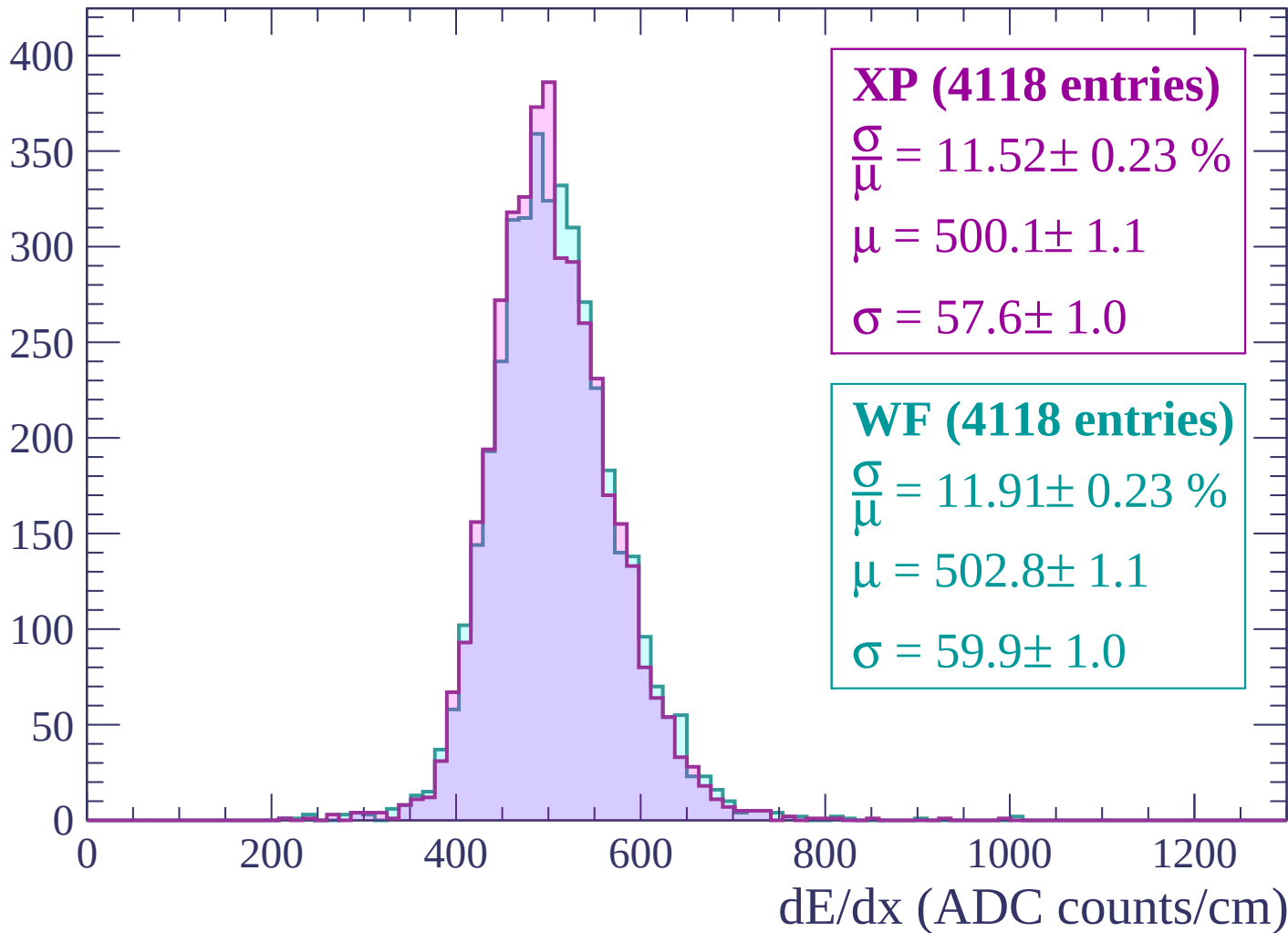
Energy loss | -2040 < p < -1980

Count



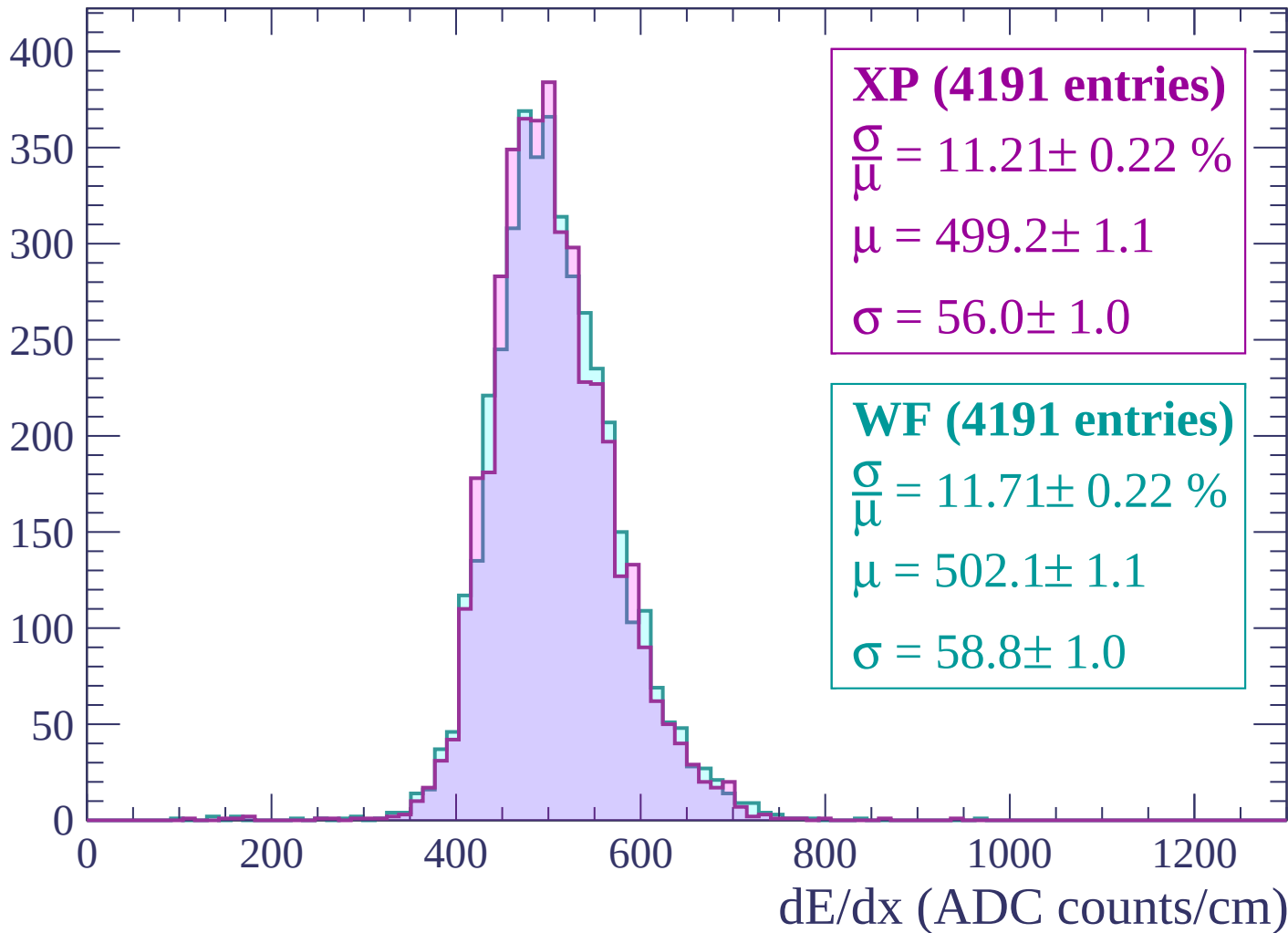
Energy loss | -1980 < p < -1920

Count

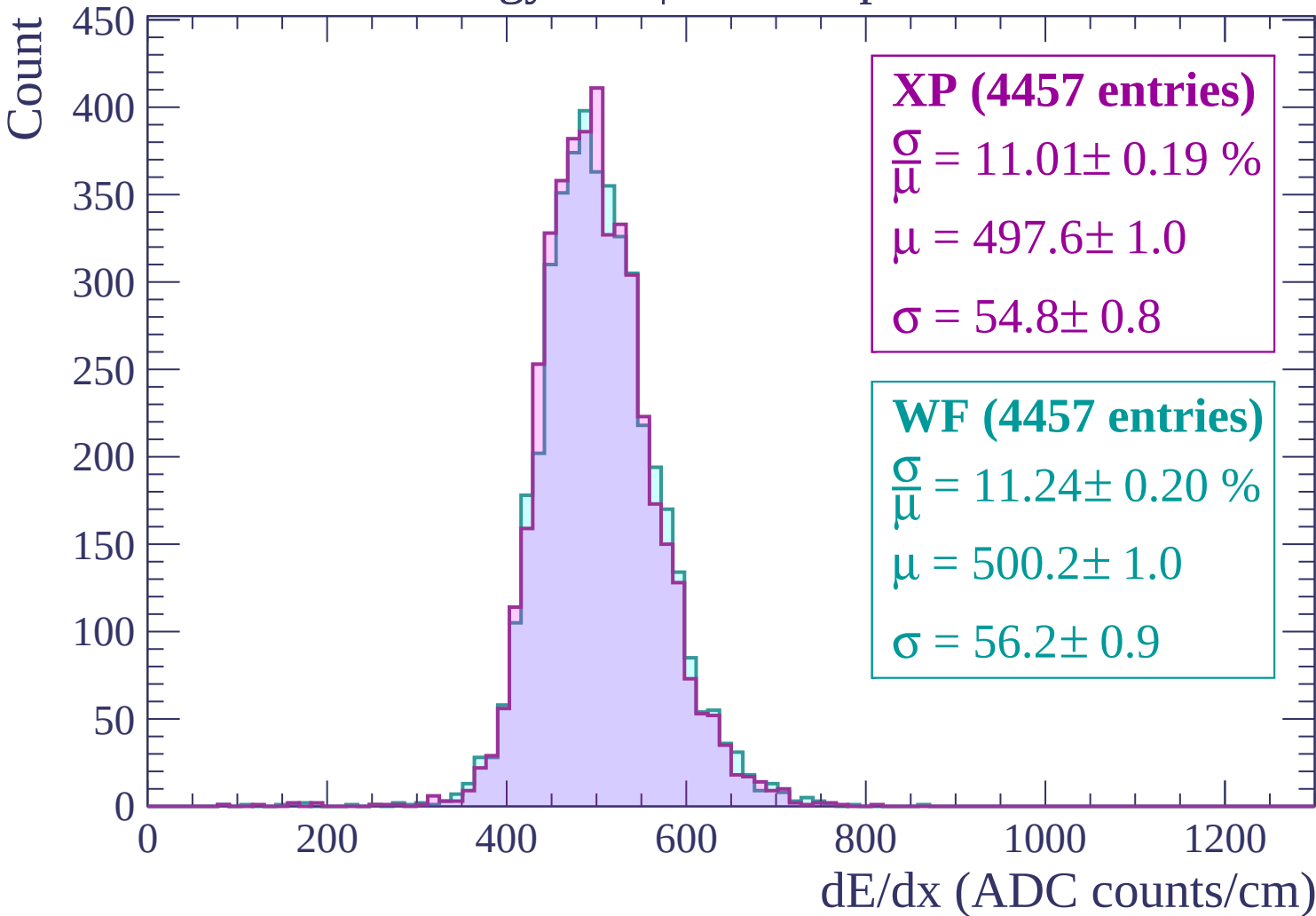


Energy loss | -1920 < p < -1860

Count

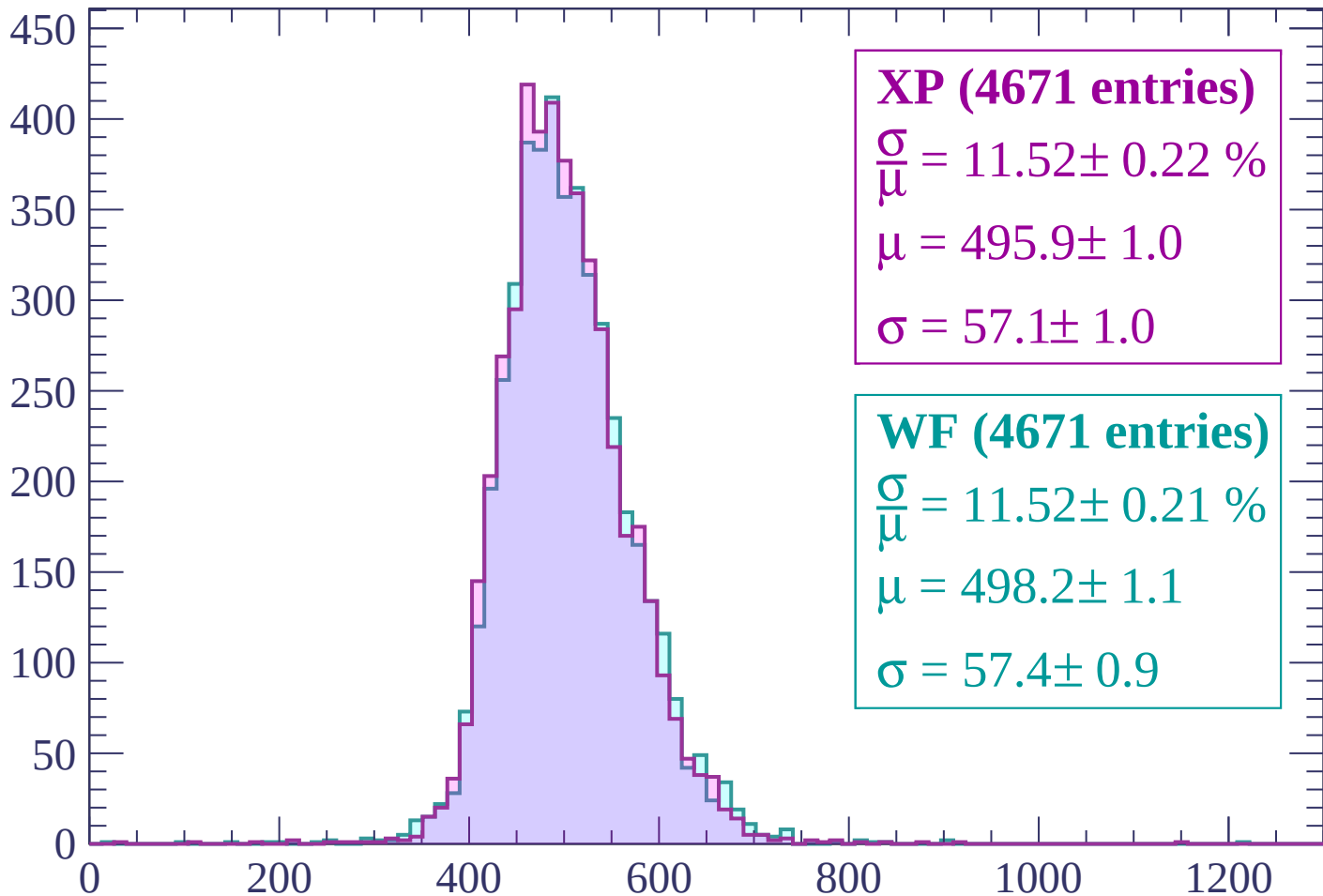


Energy loss | -1860 < p < -1800



Energy loss | -1800 < p < -1740

Count



XP (4671 entries)

$$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$$

$$\mu = 495.9 \pm 1.0$$

$$\sigma = 57.1 \pm 1.0$$

WF (4671 entries)

$$\frac{\sigma}{\mu} = 11.52 \pm 0.21 \%$$

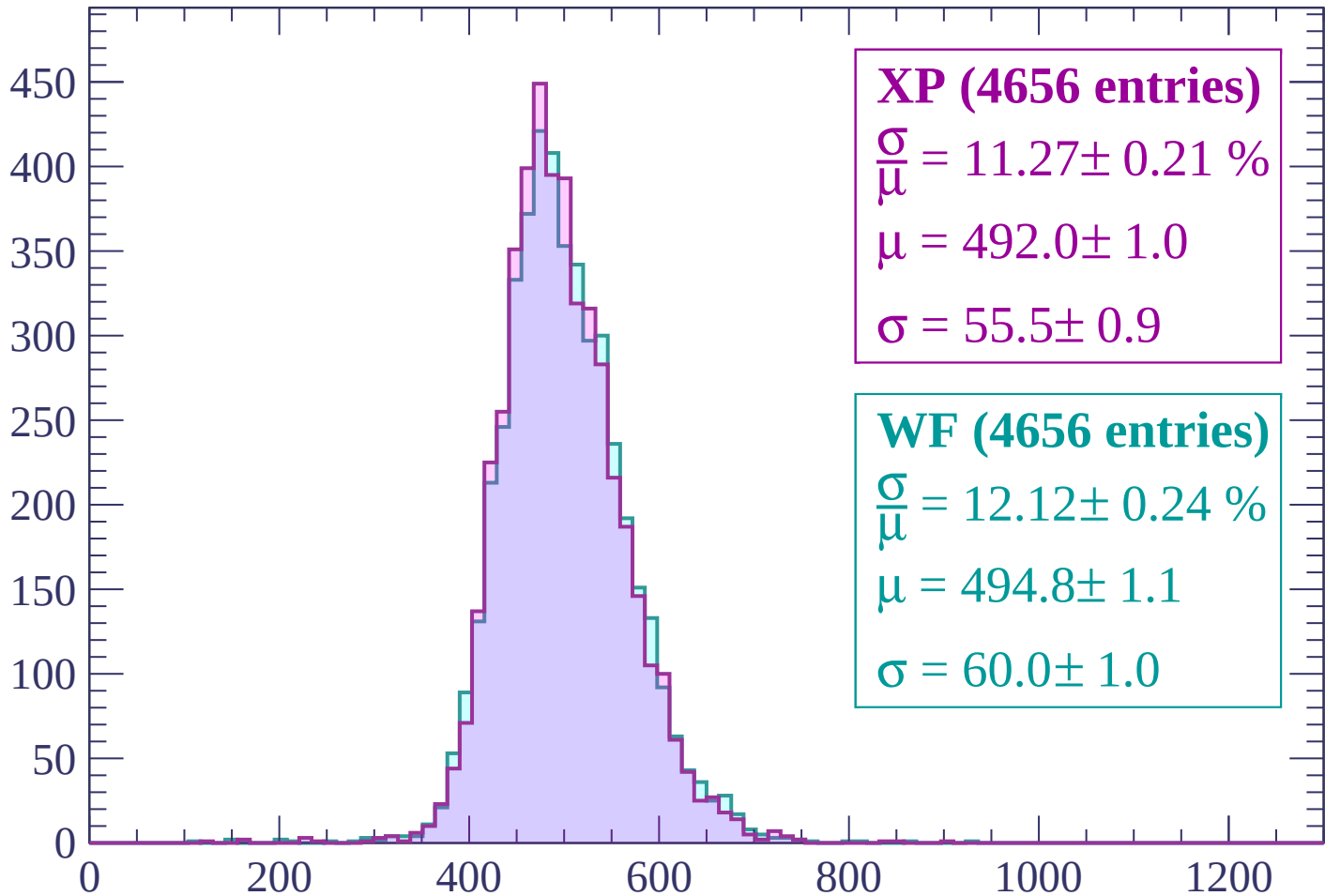
$$\mu = 498.2 \pm 1.1$$

$$\sigma = 57.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (4656 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.21 \%$$

$$\mu = 492.0 \pm 1.0$$

$$\sigma = 55.5 \pm 0.9$$

WF (4656 entries)

$$\frac{\sigma}{\mu} = 12.12 \pm 0.24 \%$$

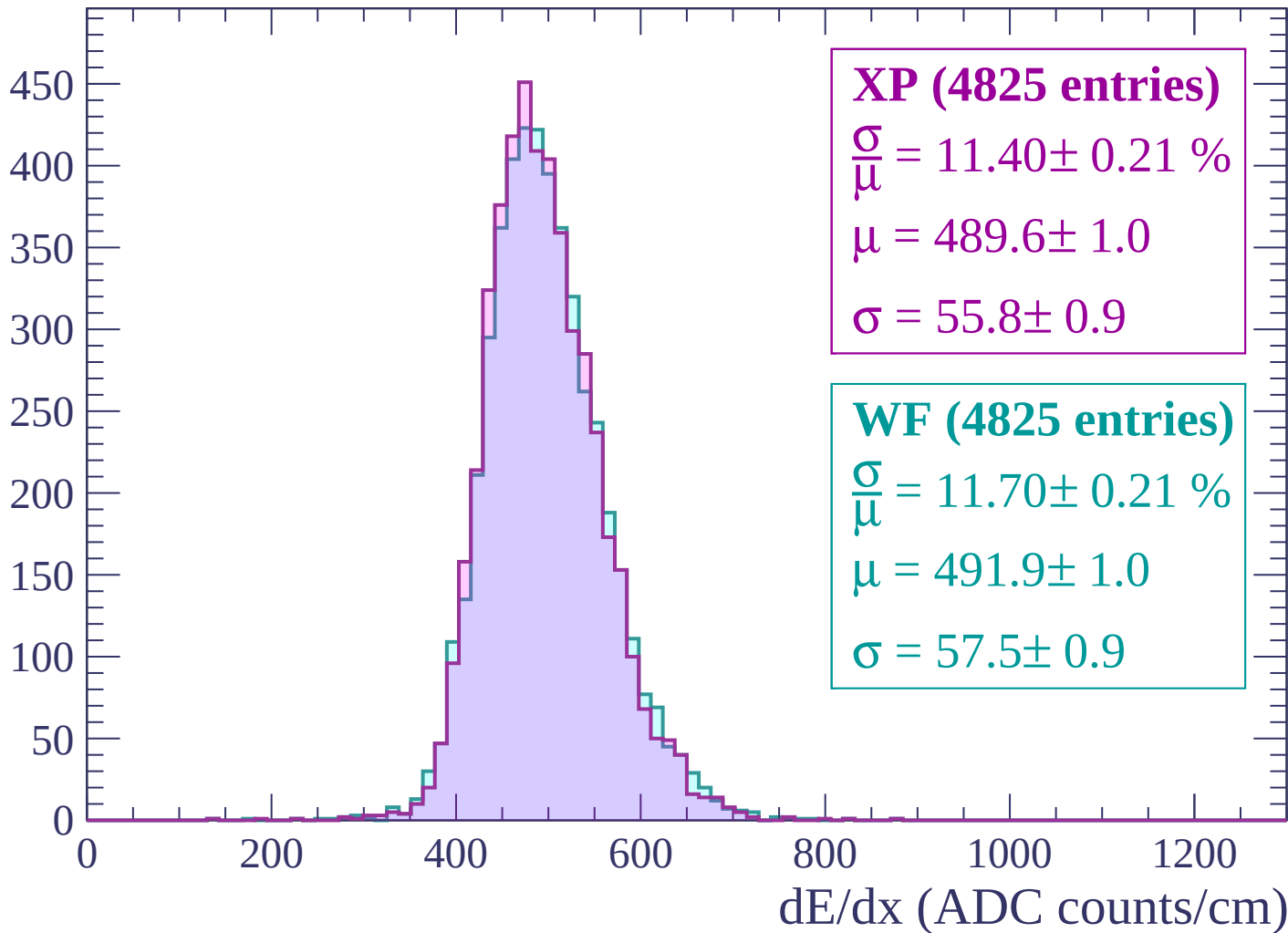
$$\mu = 494.8 \pm 1.1$$

$$\sigma = 60.0 \pm 1.0$$

dE/dx (ADC counts/cm)

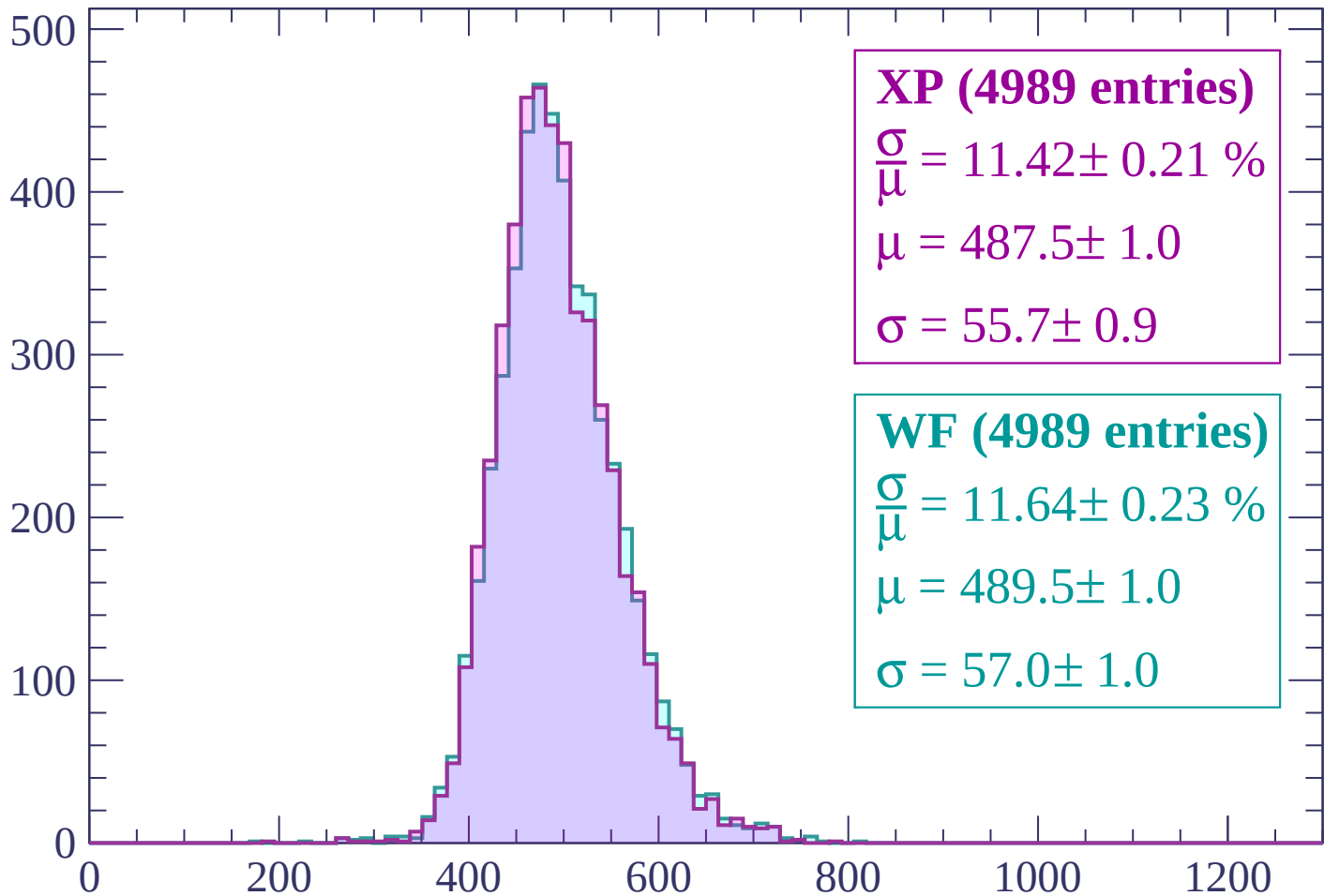
Energy loss | -1680 < p < -1620

Count



Energy loss | -1620 < p < -1560

Count



XP (4989 entries)

$$\frac{\sigma}{\mu} = 11.42 \pm 0.21 \%$$

$$\mu = 487.5 \pm 1.0$$

$$\sigma = 55.7 \pm 0.9$$

WF (4989 entries)

$$\frac{\sigma}{\mu} = 11.64 \pm 0.23 \%$$

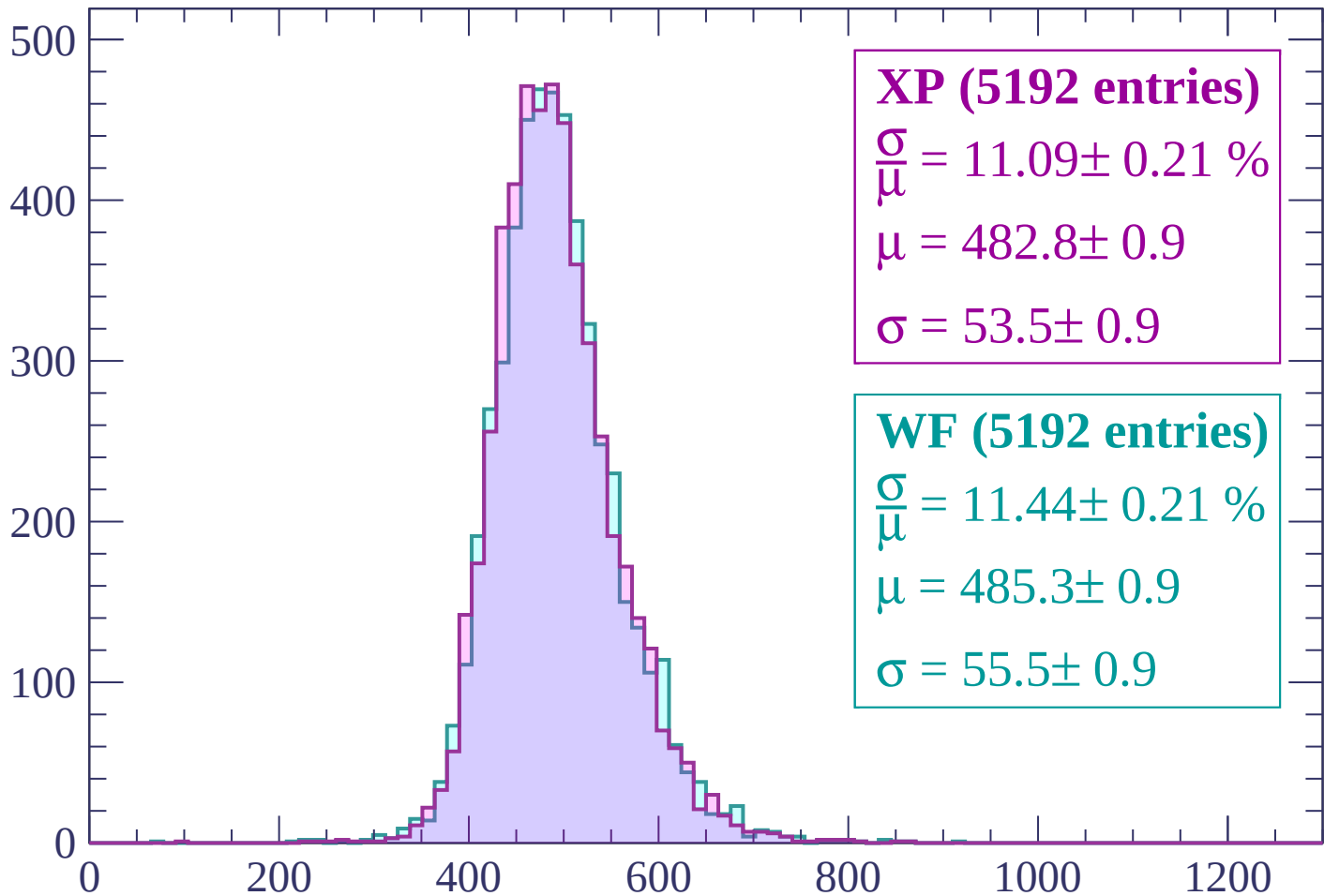
$$\mu = 489.5 \pm 1.0$$

$$\sigma = 57.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (5192 entries)

$$\frac{\sigma}{\mu} = 11.09 \pm 0.21 \%$$

$$\mu = 482.8 \pm 0.9$$

$$\sigma = 53.5 \pm 0.9$$

WF (5192 entries)

$$\frac{\sigma}{\mu} = 11.44 \pm 0.21 \%$$

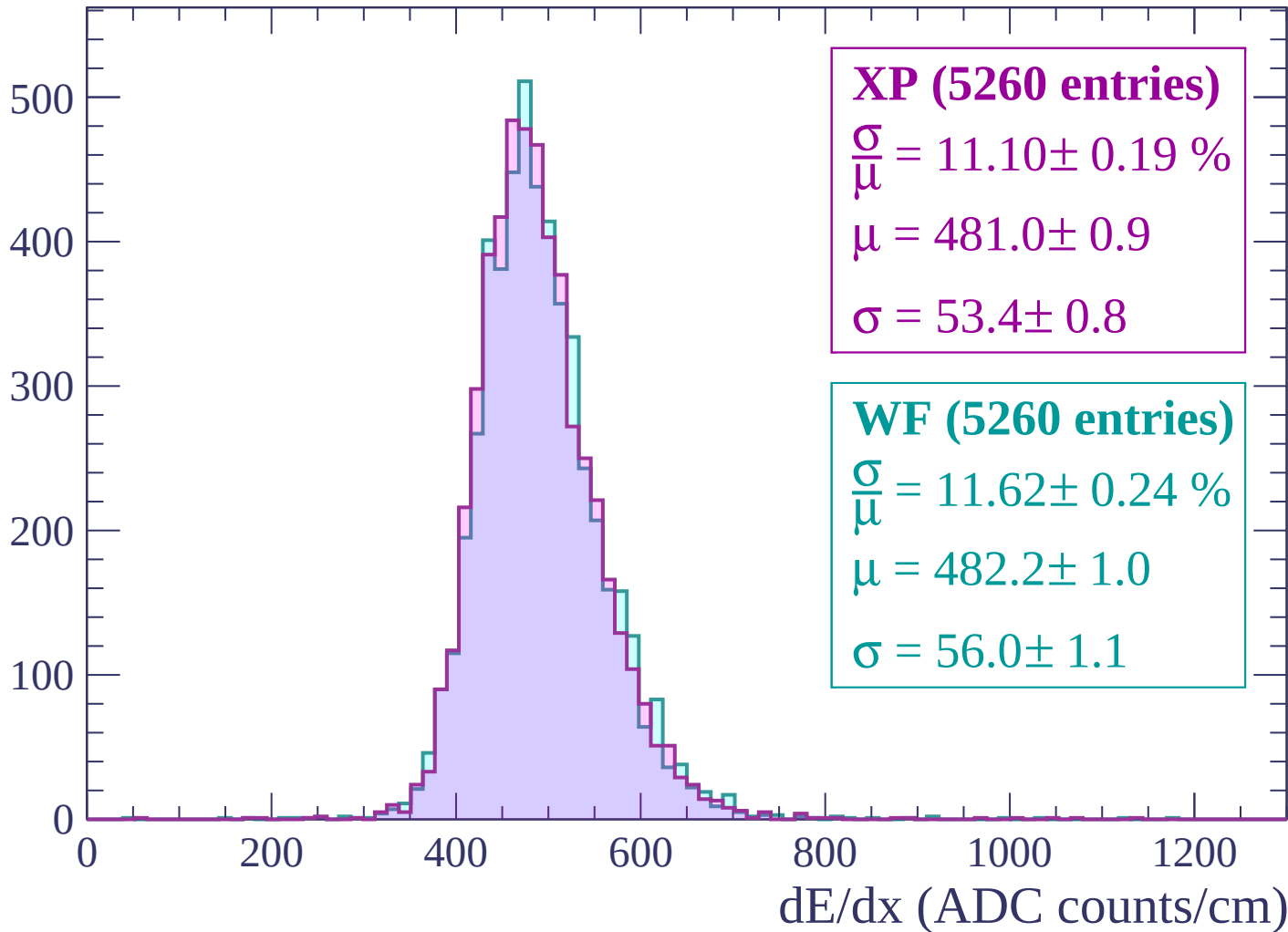
$$\mu = 485.3 \pm 0.9$$

$$\sigma = 55.5 \pm 0.9$$

dE/dx (ADC counts/cm)

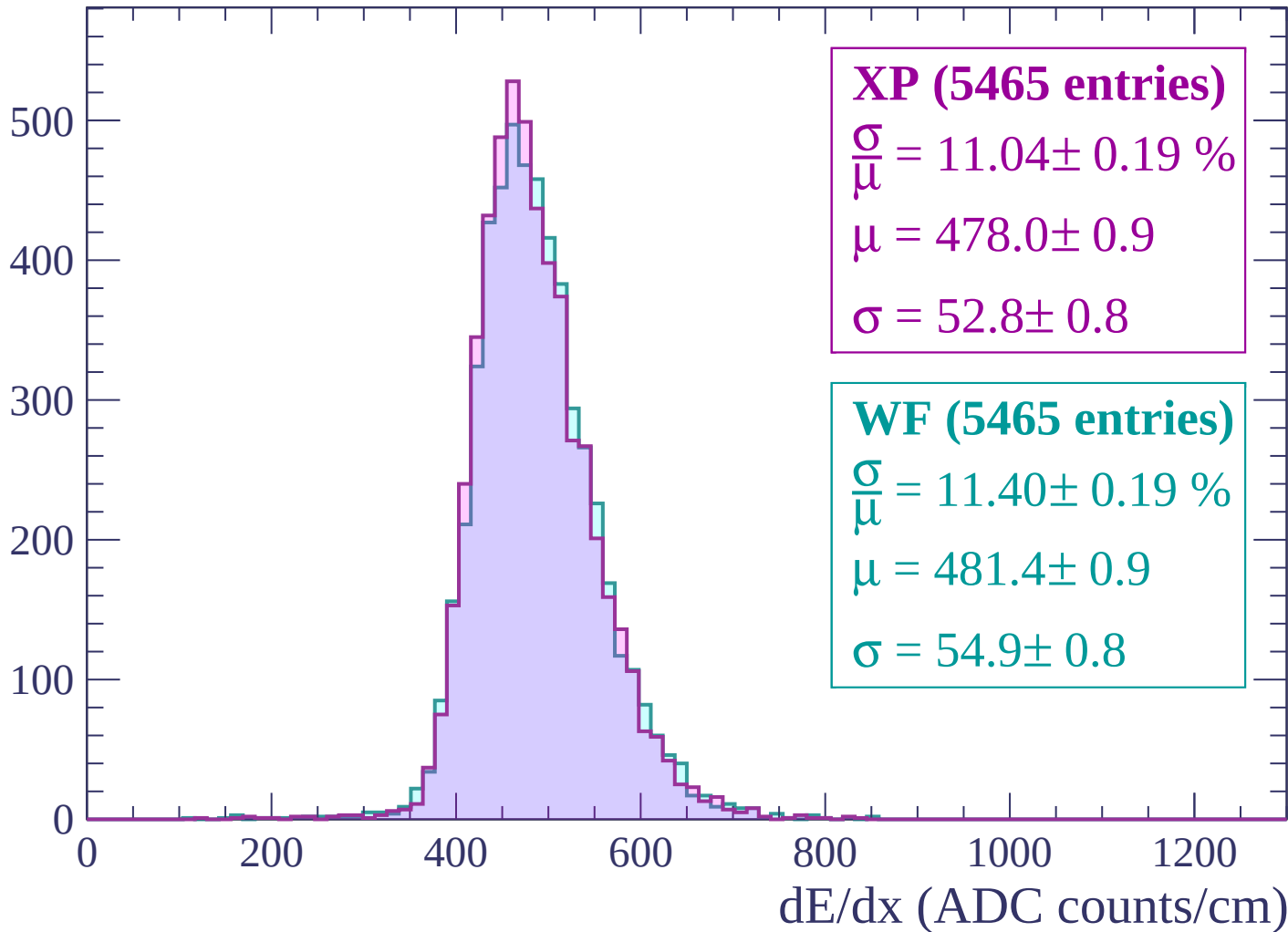
Energy loss | -1500 < p < -1440

Count



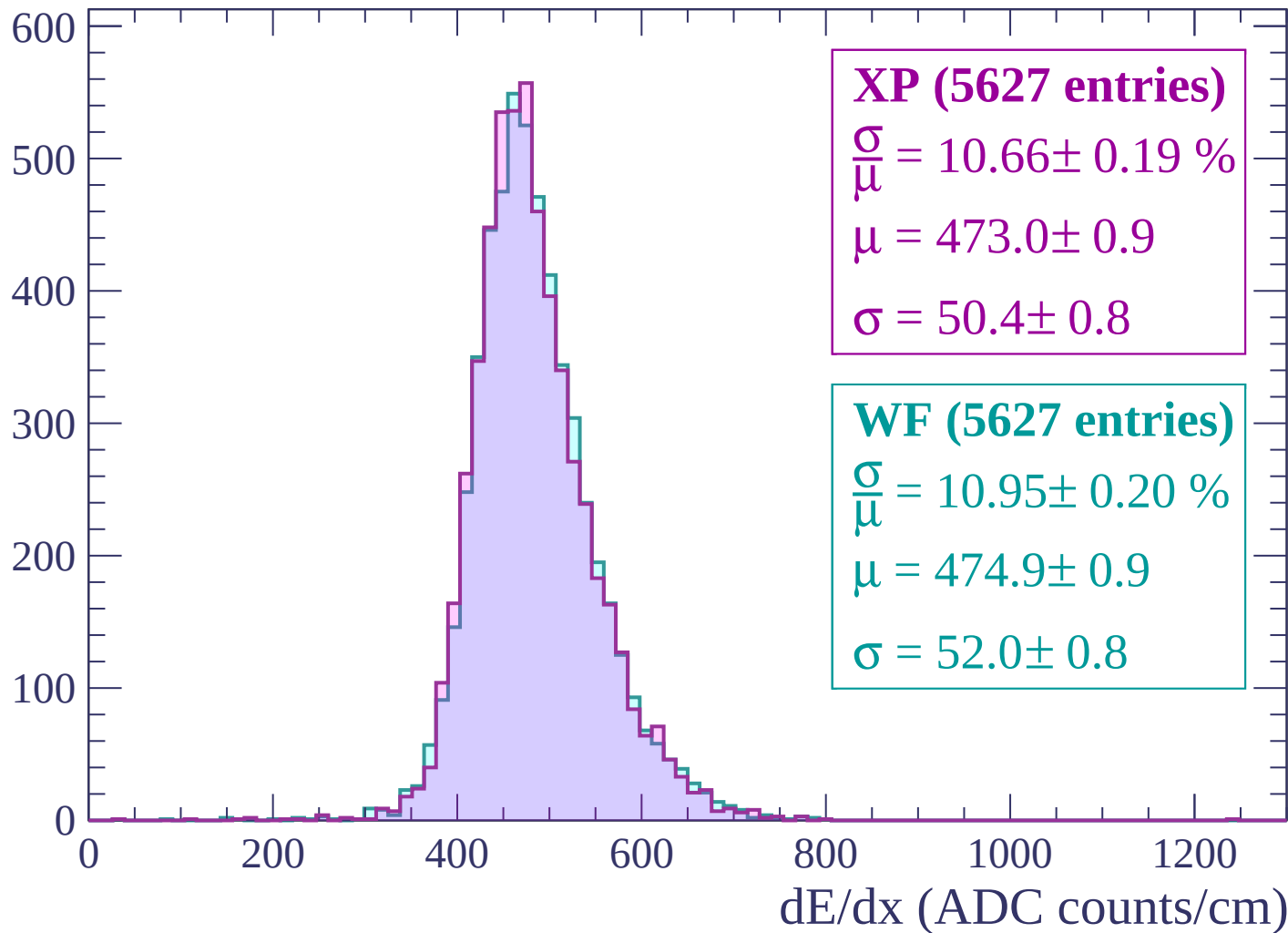
Energy loss | $-1440 < p < -1380$

Count

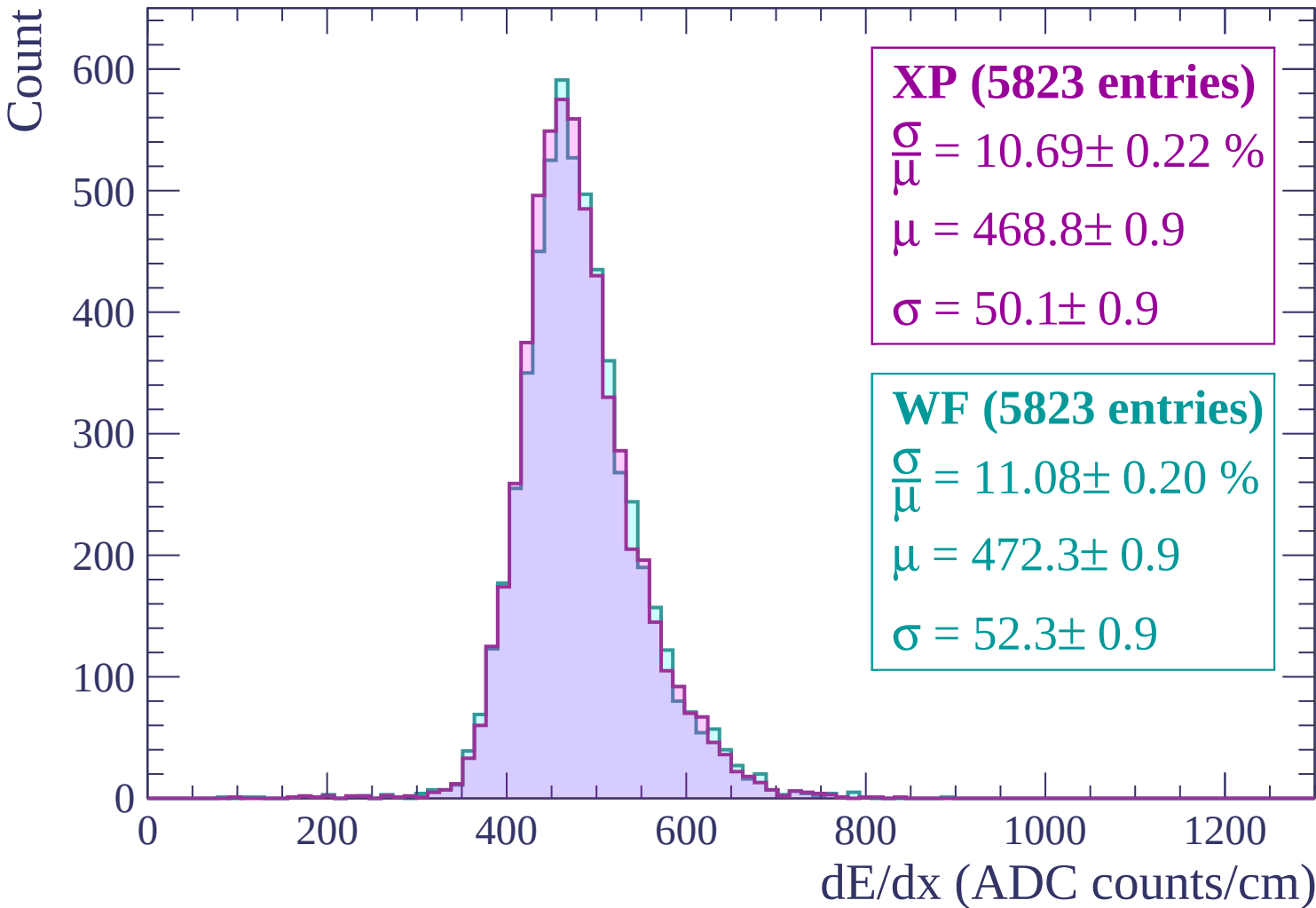


Energy loss | -1380 < p < -1320

Count

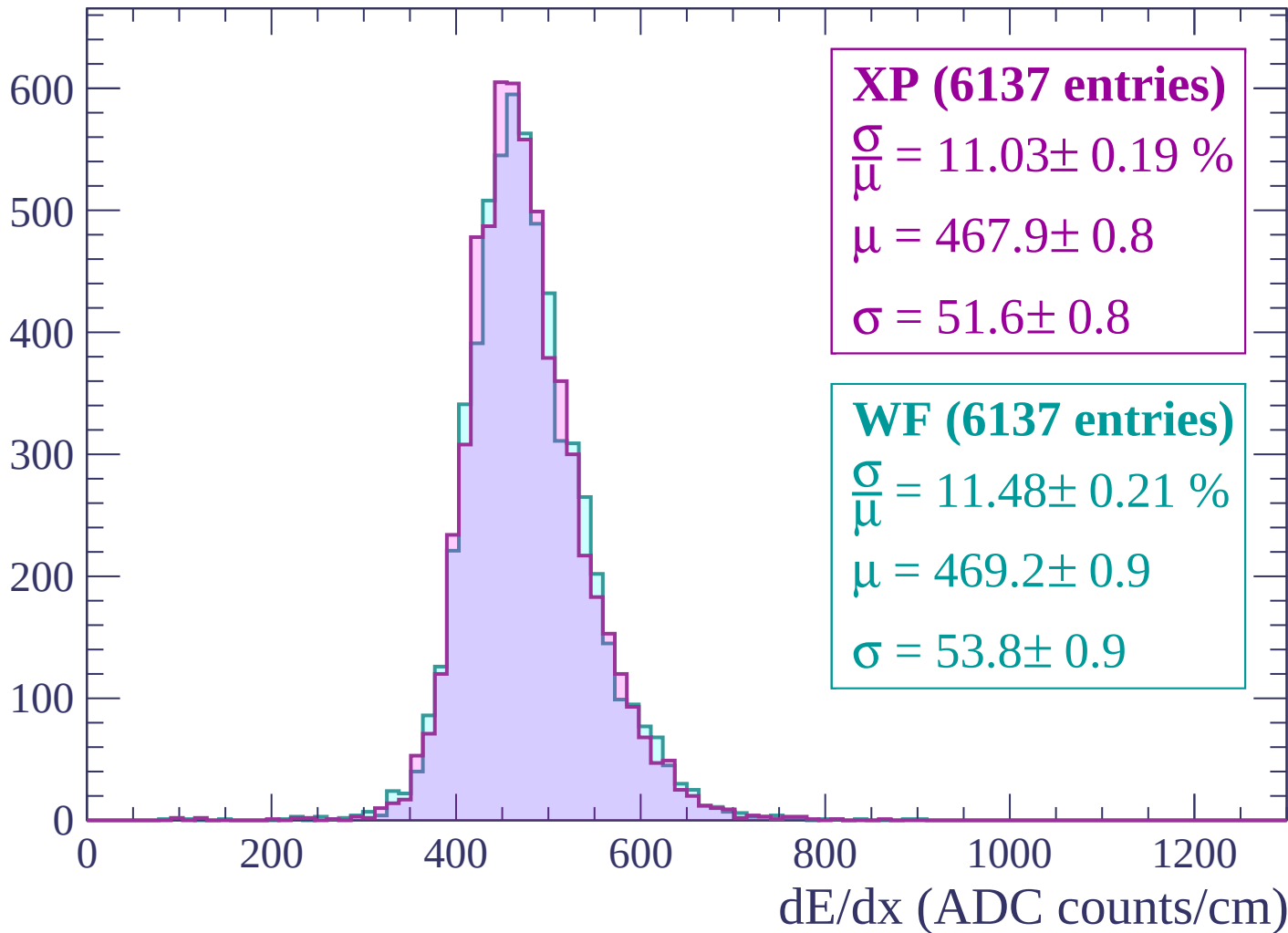


Energy loss | -1320 < p < -1260



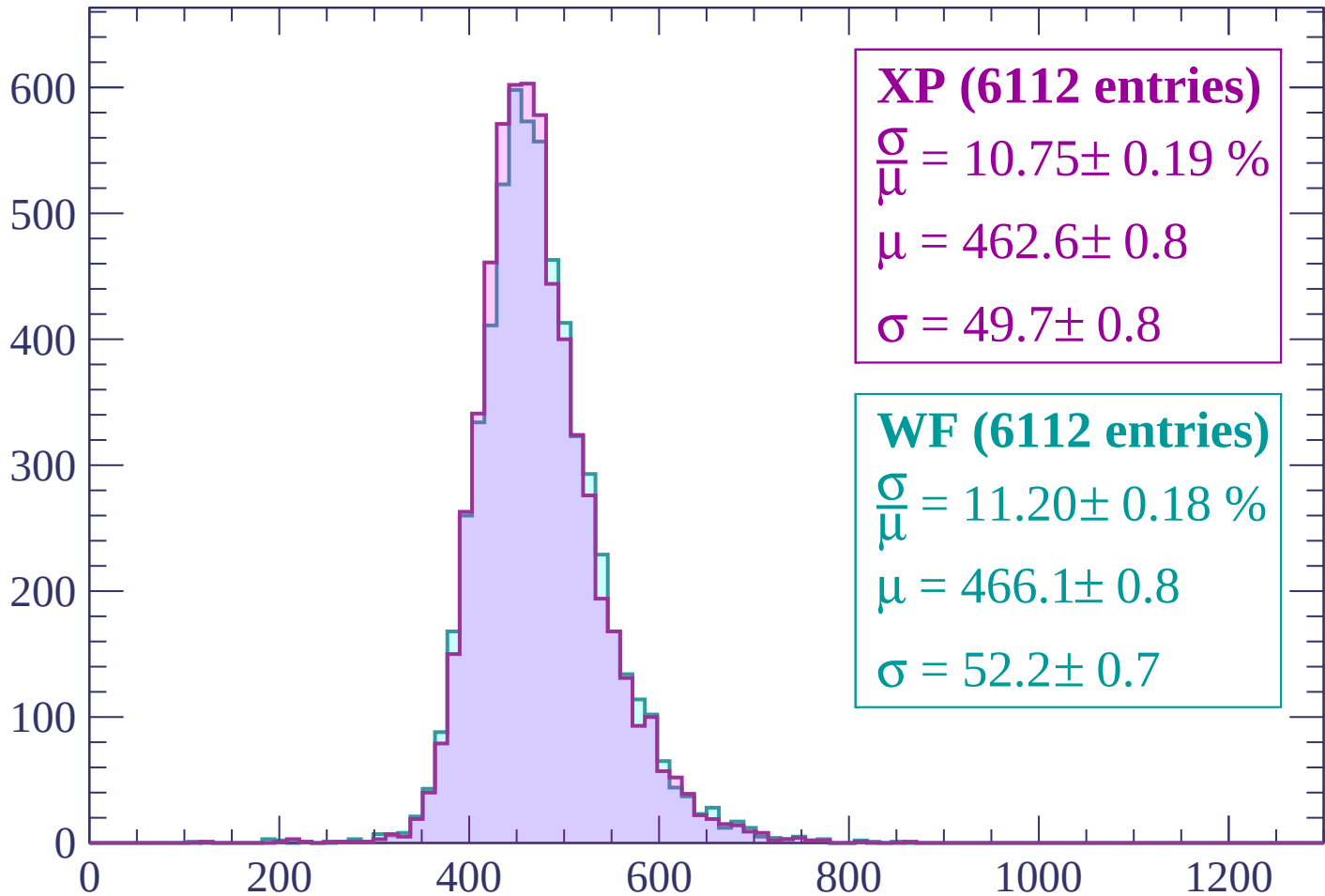
Energy loss | $-1260 < p < -1200$

Count



Energy loss | -1200 < p < -1140

Count



XP (6112 entries)

$$\frac{\sigma}{\mu} = 10.75 \pm 0.19 \%$$

$$\mu = 462.6 \pm 0.8$$

$$\sigma = 49.7 \pm 0.8$$

WF (6112 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 0.18 \%$$

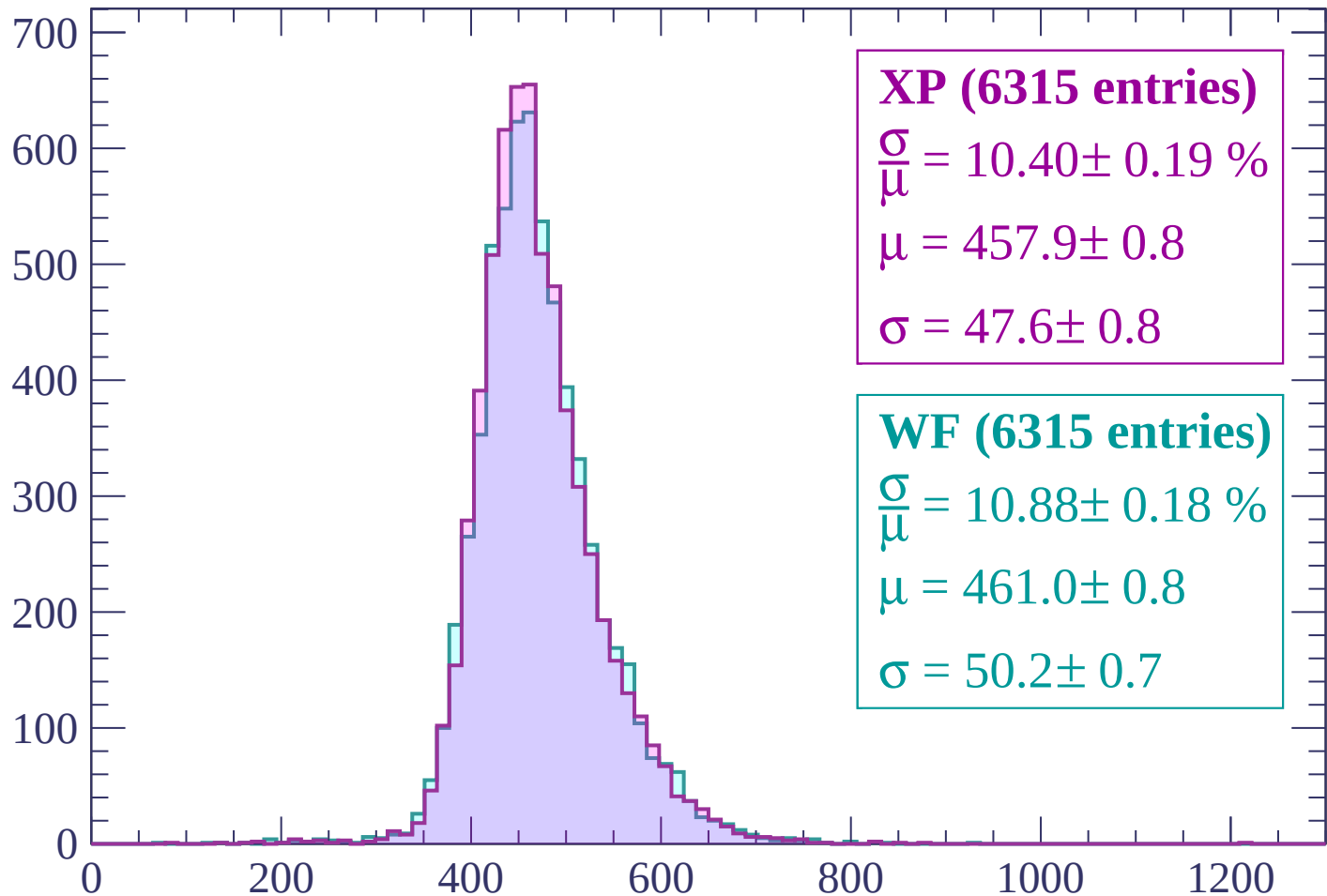
$$\mu = 466.1 \pm 0.8$$

$$\sigma = 52.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count



XP (6315 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.19 \%$$

$$\mu = 457.9 \pm 0.8$$

$$\sigma = 47.6 \pm 0.8$$

WF (6315 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.18 \%$$

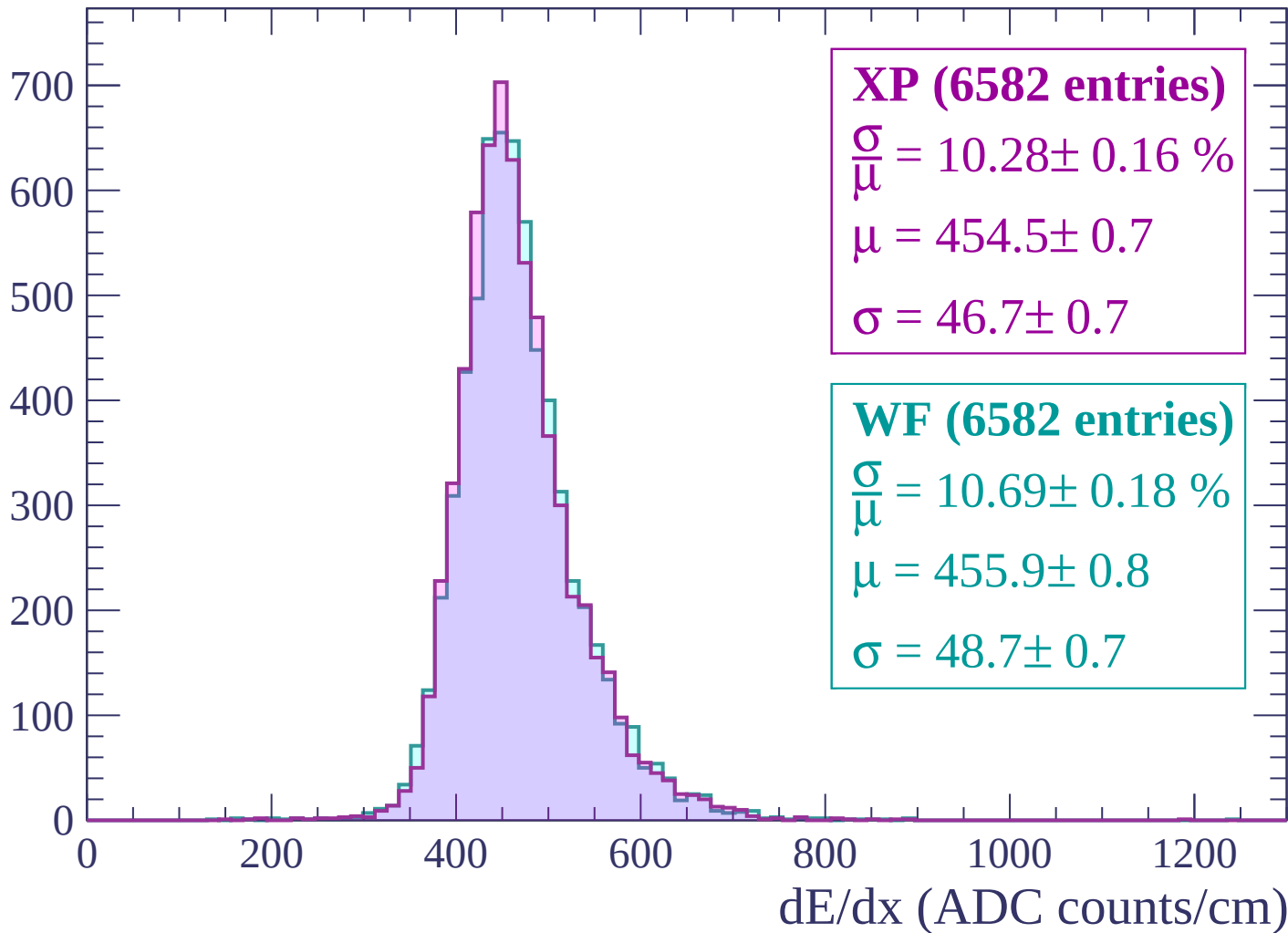
$$\mu = 461.0 \pm 0.8$$

$$\sigma = 50.2 \pm 0.7$$

dE/dx (ADC counts/cm)

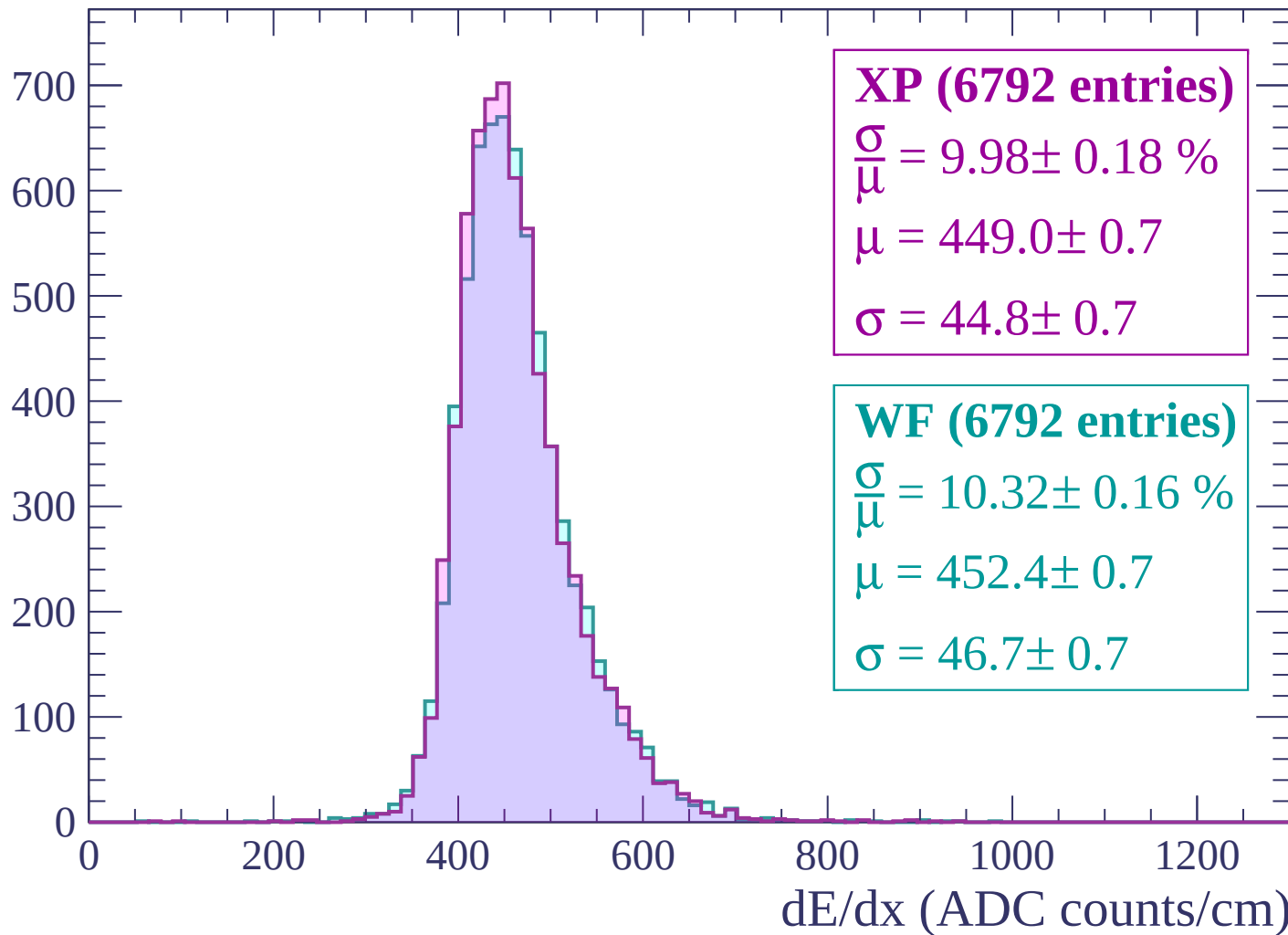
Energy loss | $-1080 < p < -1020$

Count



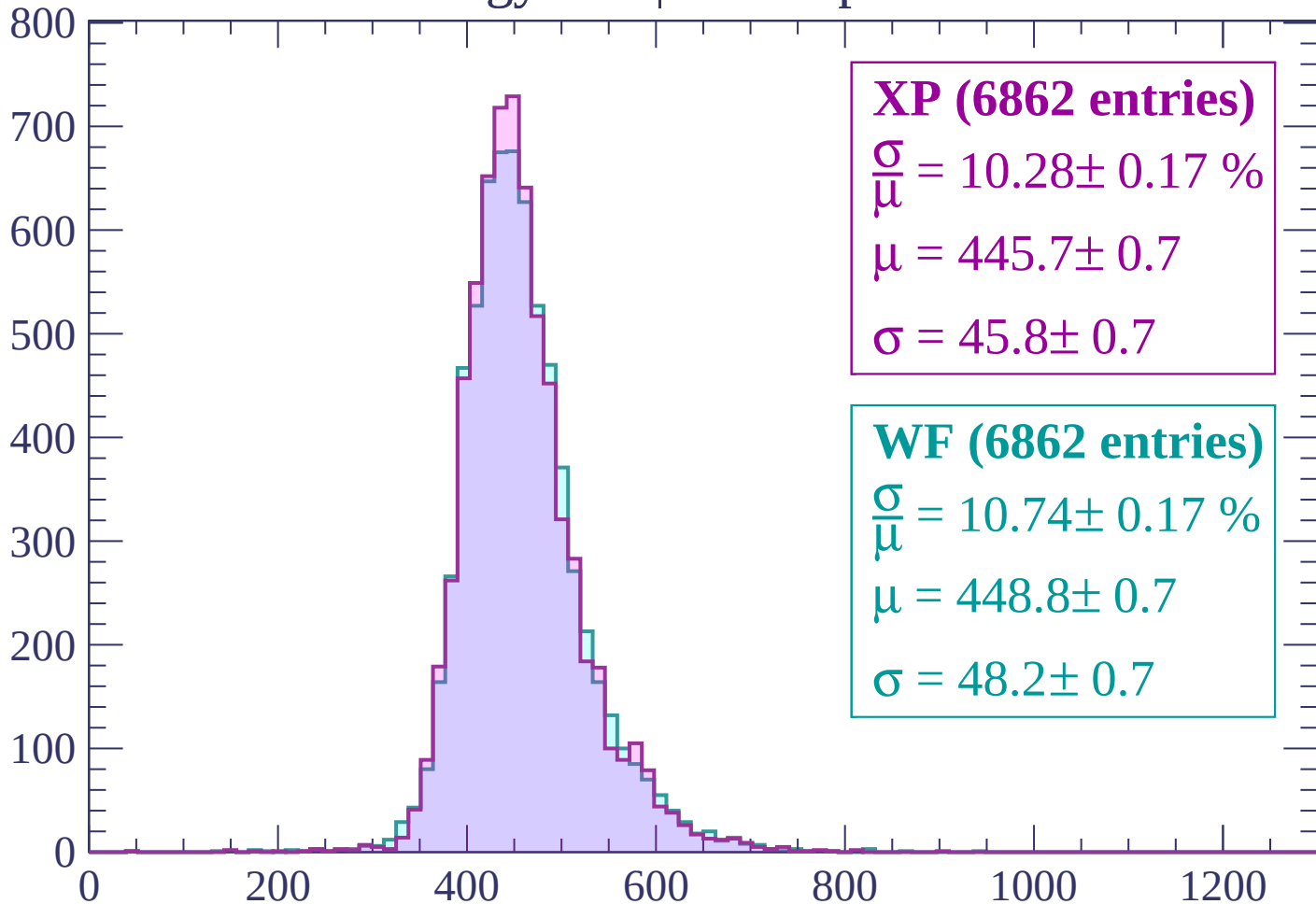
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (6862 entries)

$\frac{\sigma}{\mu} = 10.28 \pm 0.17 \%$

$\mu = 445.7 \pm 0.7$

$\sigma = 45.8 \pm 0.7$

WF (6862 entries)

$\frac{\sigma}{\mu} = 10.74 \pm 0.17 \%$

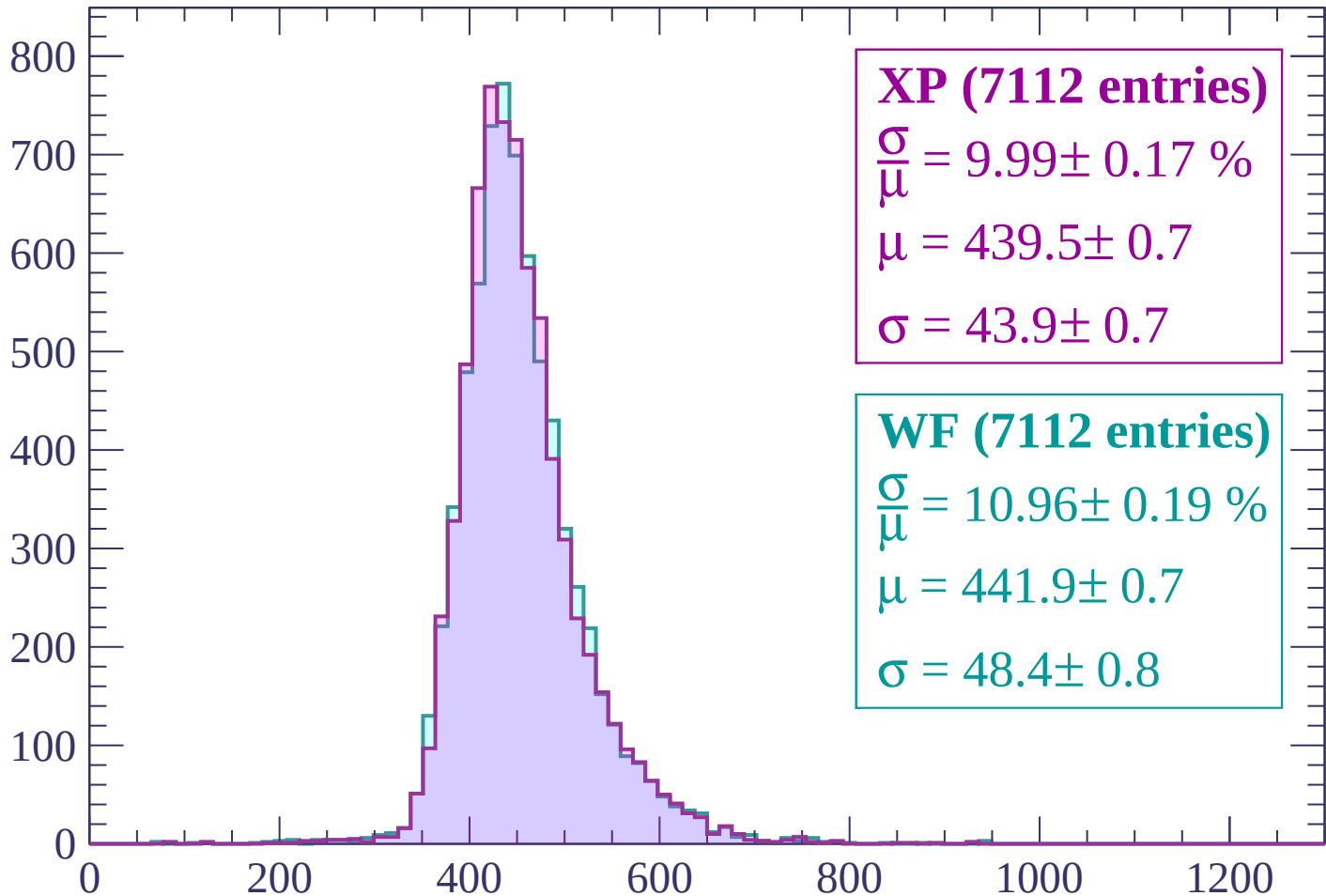
$\mu = 448.8 \pm 0.7$

$\sigma = 48.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (7112 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.17 \%$$

$$\mu = 439.5 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

WF (7112 entries)

$$\frac{\sigma}{\mu} = 10.96 \pm 0.19 \%$$

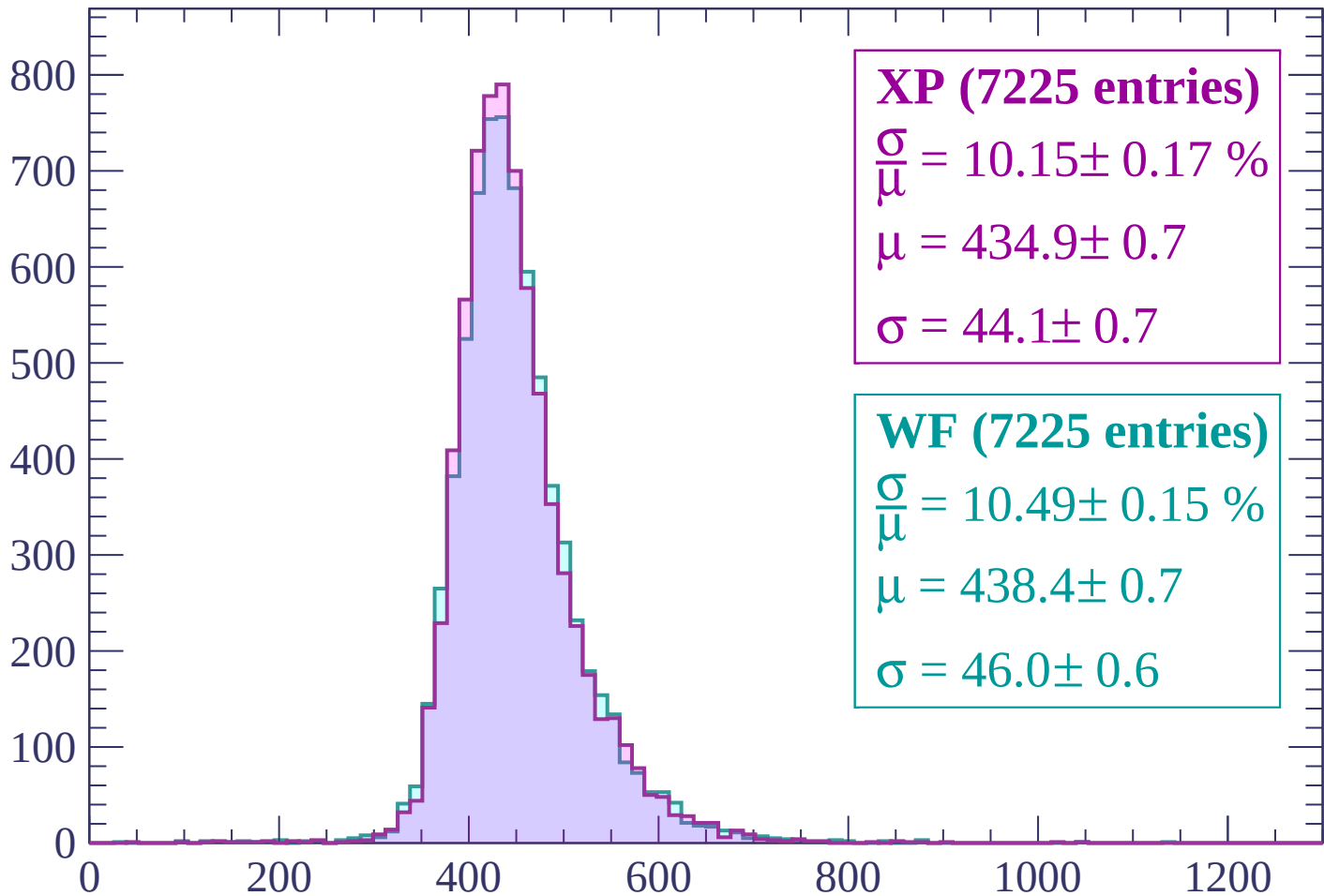
$$\mu = 441.9 \pm 0.7$$

$$\sigma = 48.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (7225 entries)

$$\frac{\sigma}{\mu} = 10.15 \pm 0.17 \%$$

$$\mu = 434.9 \pm 0.7$$

$$\sigma = 44.1 \pm 0.7$$

WF (7225 entries)

$$\frac{\sigma}{\mu} = 10.49 \pm 0.15 \%$$

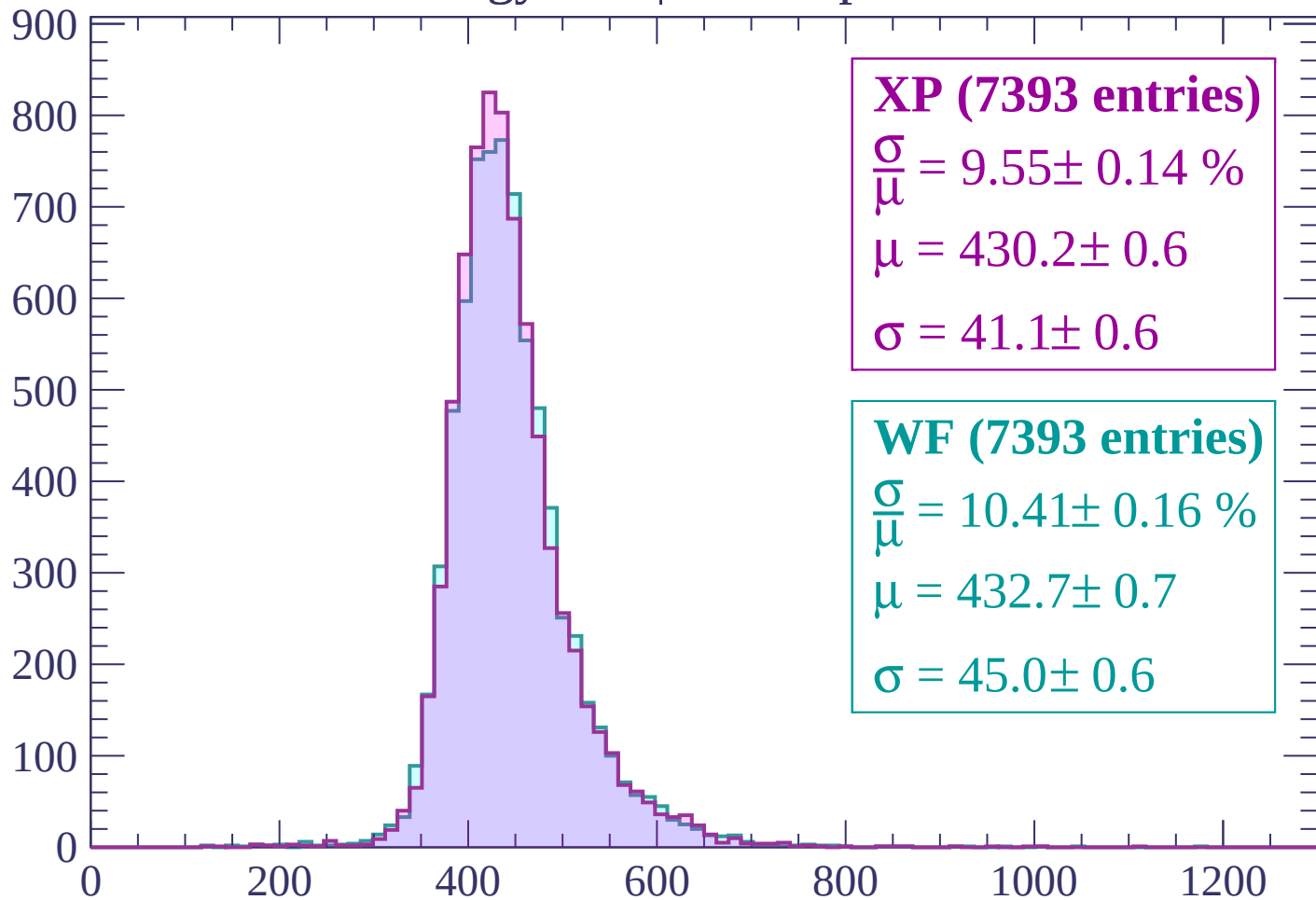
$$\mu = 438.4 \pm 0.7$$

$$\sigma = 46.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (7393 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.14 \%$$

$$\mu = 430.2 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

WF (7393 entries)

$$\frac{\sigma}{\mu} = 10.41 \pm 0.16 \%$$

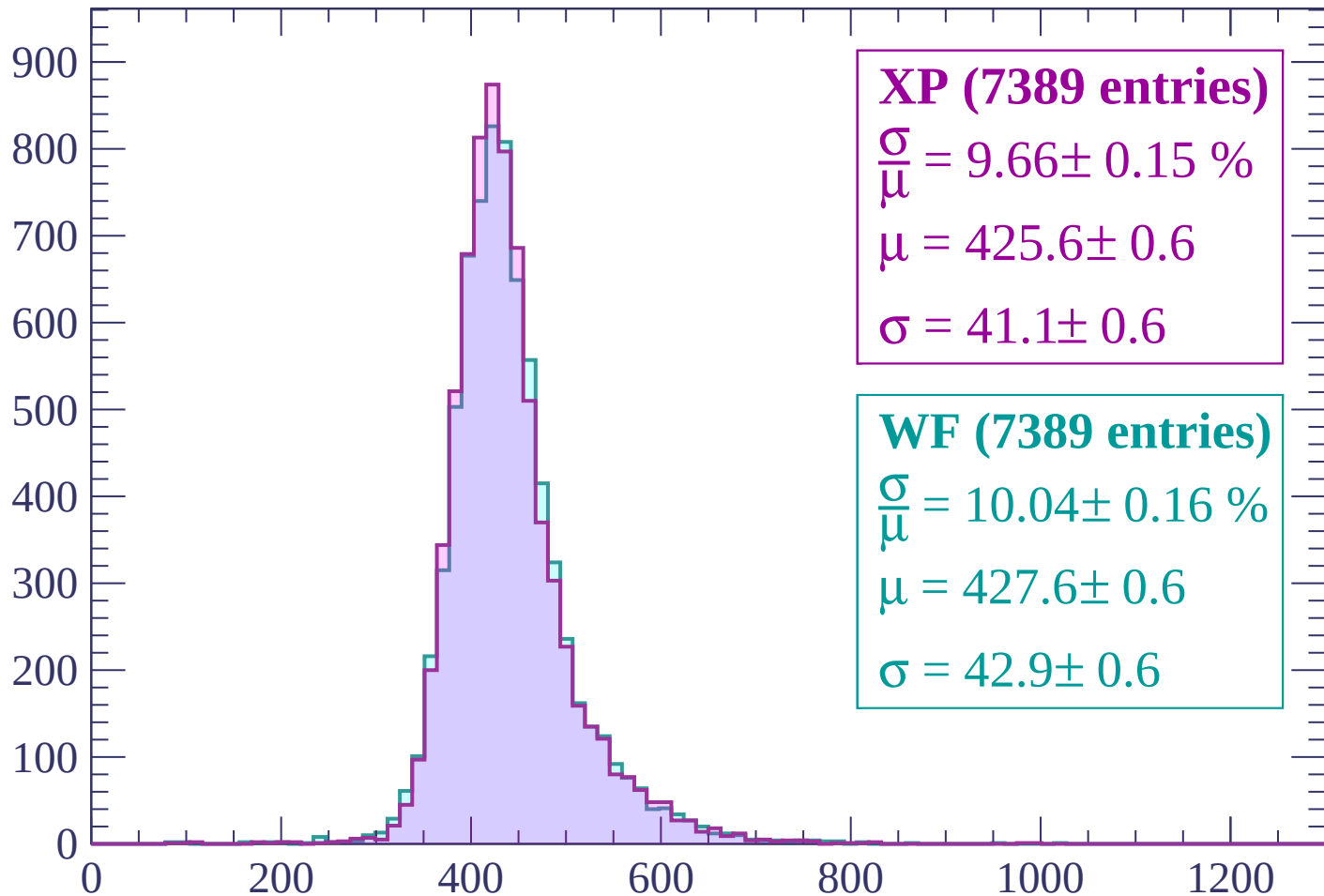
$$\mu = 432.7 \pm 0.7$$

$$\sigma = 45.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (7389 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.15 \%$$

$$\mu = 425.6 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

WF (7389 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

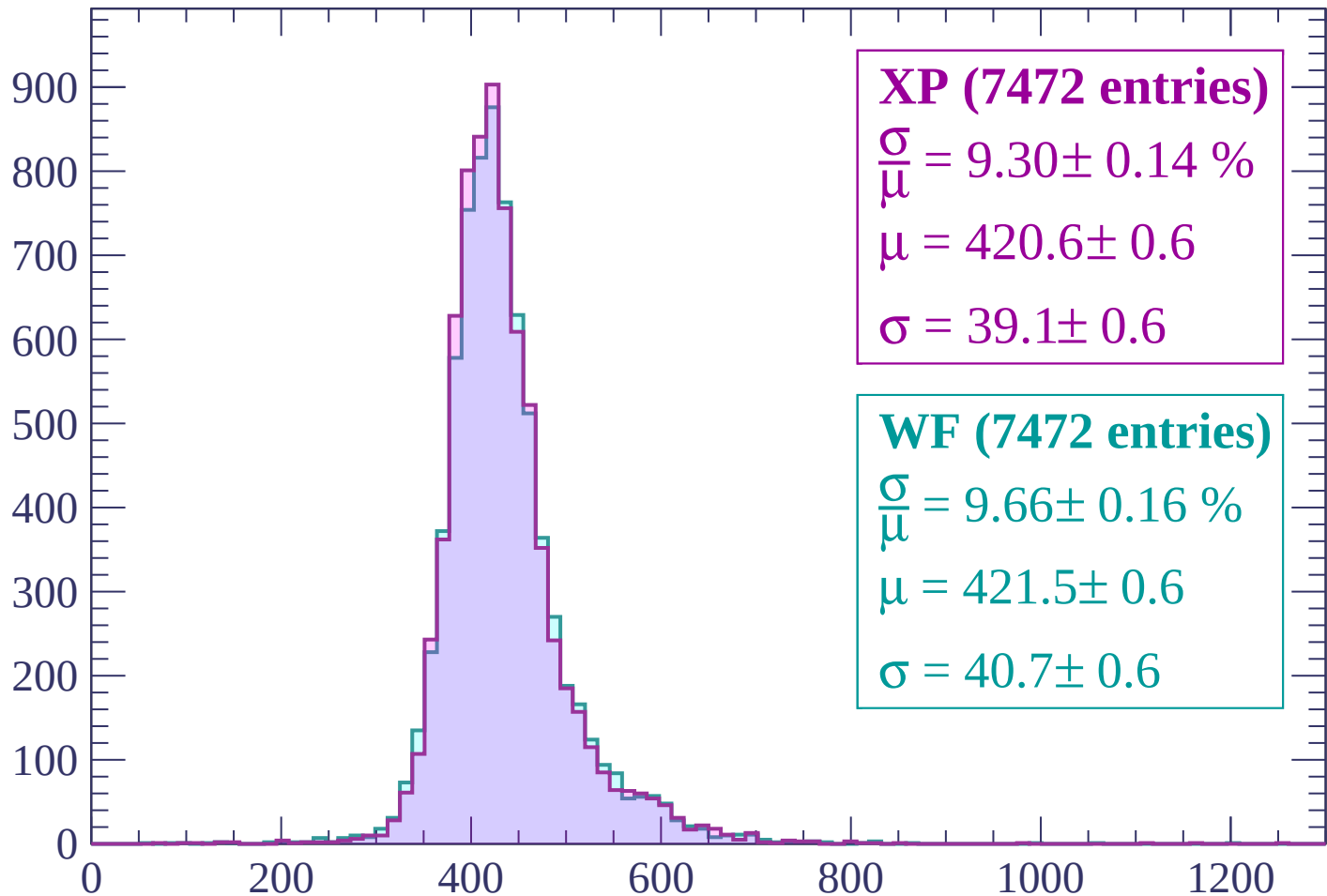
$$\mu = 427.6 \pm 0.6$$

$$\sigma = 42.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (7472 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.14 \%$$

$$\mu = 420.6 \pm 0.6$$

$$\sigma = 39.1 \pm 0.6$$

WF (7472 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.16 \%$$

$$\mu = 421.5 \pm 0.6$$

$$\sigma = 40.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count

1000

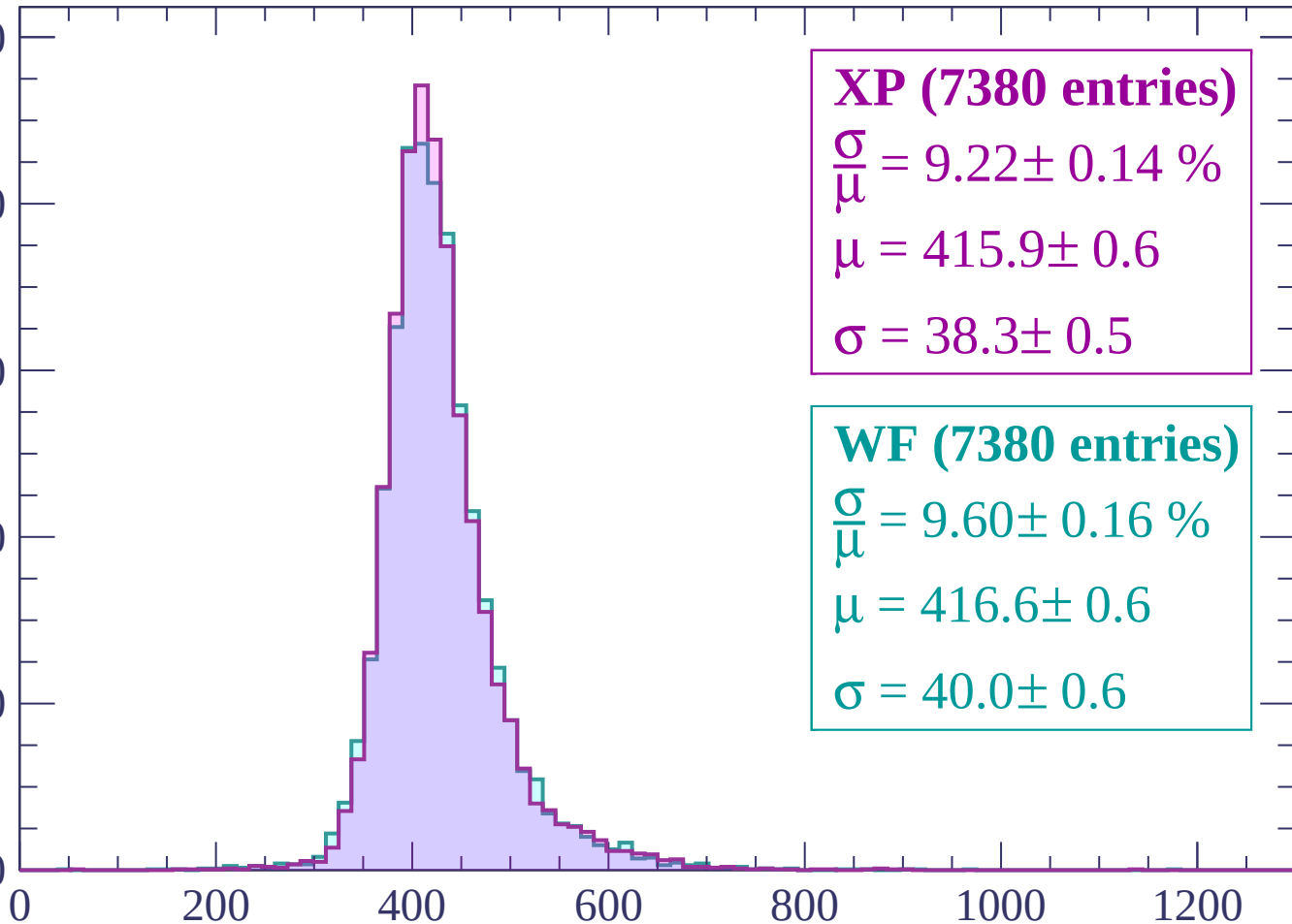
800

600

400

200

0



XP (7380 entries)

$\frac{\sigma}{\mu} = 9.22 \pm 0.14 \%$

$\mu = 415.9 \pm 0.6$

$\sigma = 38.3 \pm 0.5$

WF (7380 entries)

$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$

$\mu = 416.6 \pm 0.6$

$\sigma = 40.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count

1000

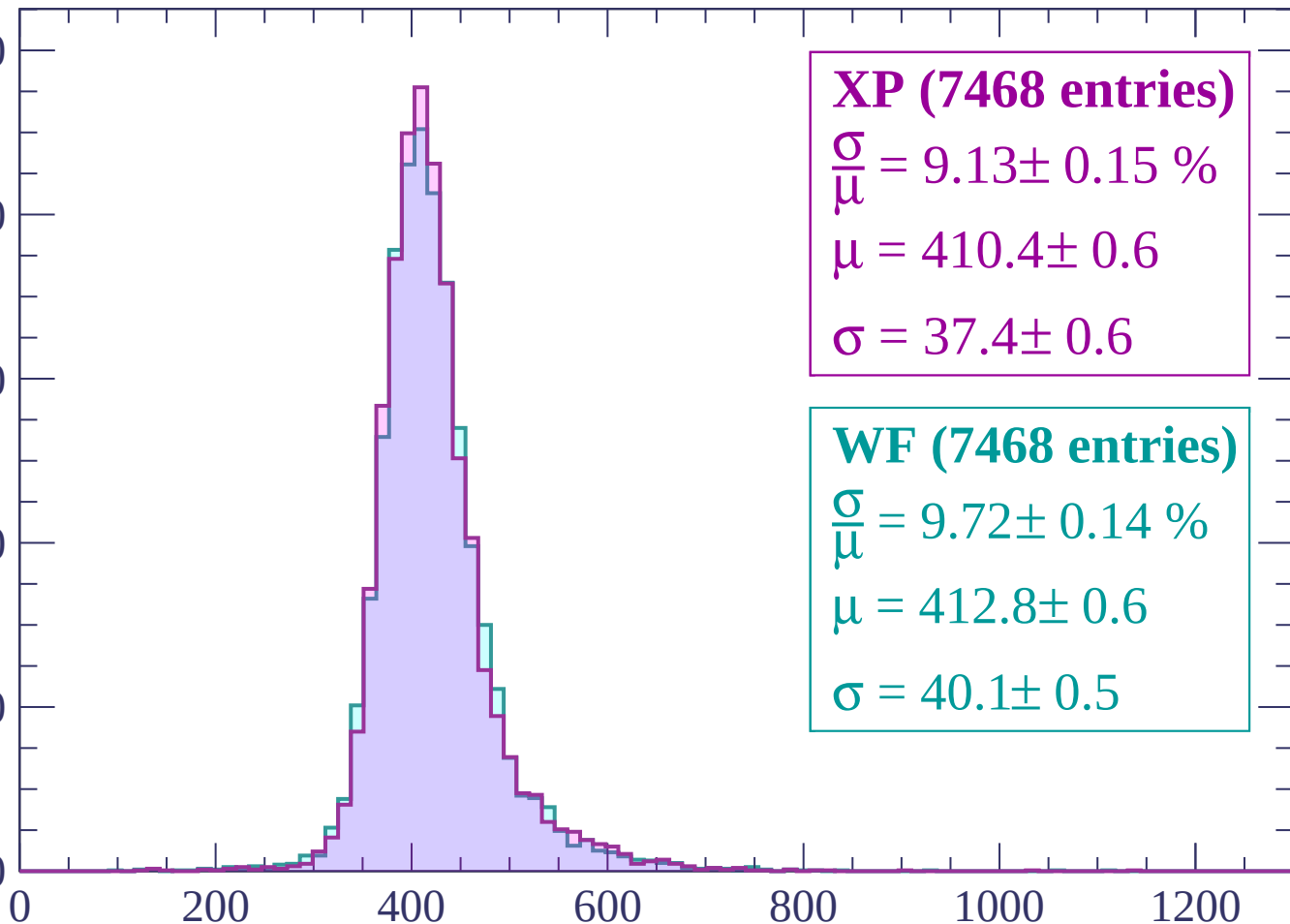
800

600

400

200

0



XP (7468 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.15 \%$

$\mu = 410.4 \pm 0.6$

$\sigma = 37.4 \pm 0.6$

WF (7468 entries)

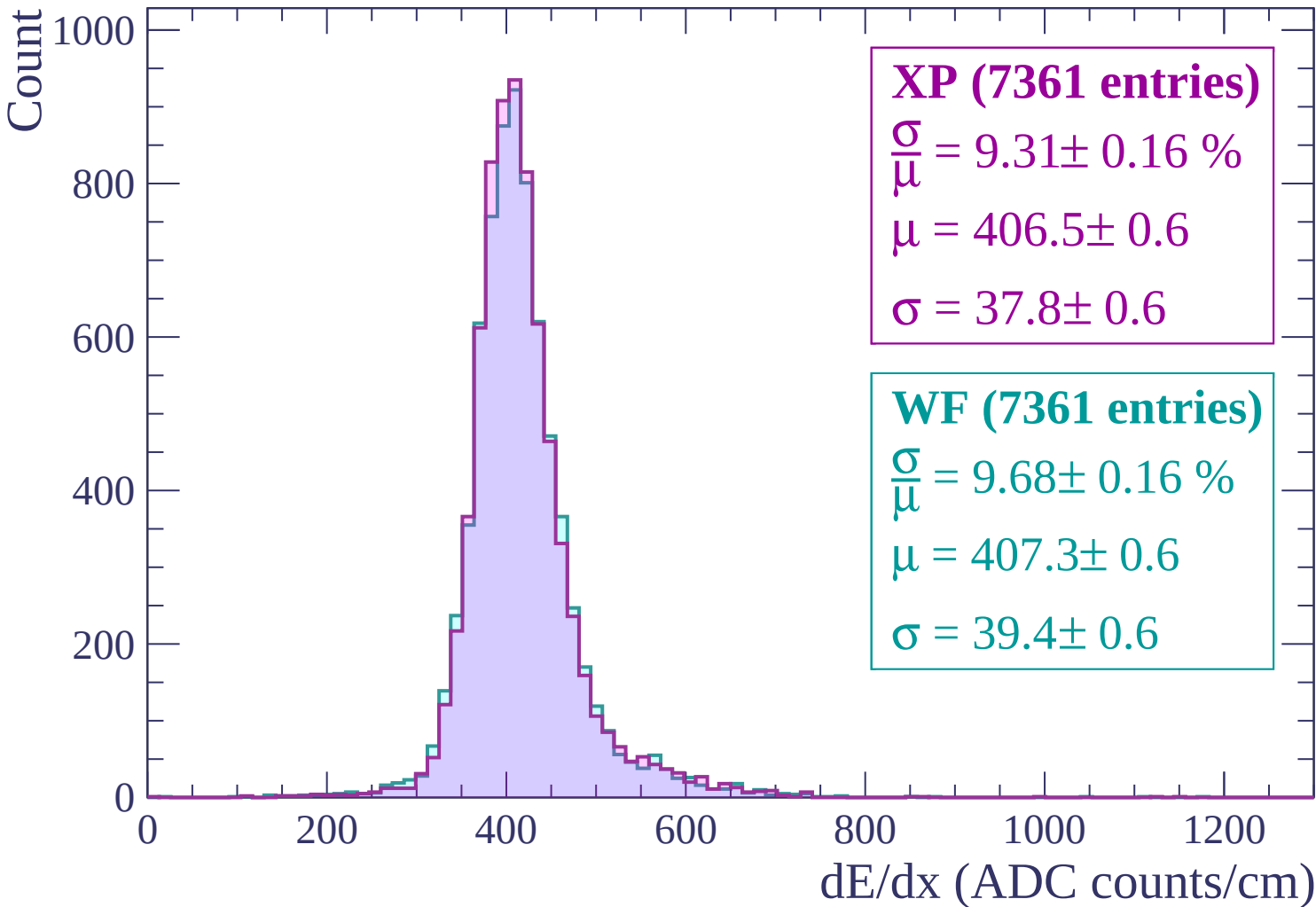
$\frac{\sigma}{\mu} = 9.72 \pm 0.14 \%$

$\mu = 412.8 \pm 0.6$

$\sigma = 40.1 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$



Energy loss | $-420 < p < -360$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (7237 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.16 \%$

$\mu = 402.2 \pm 0.5$

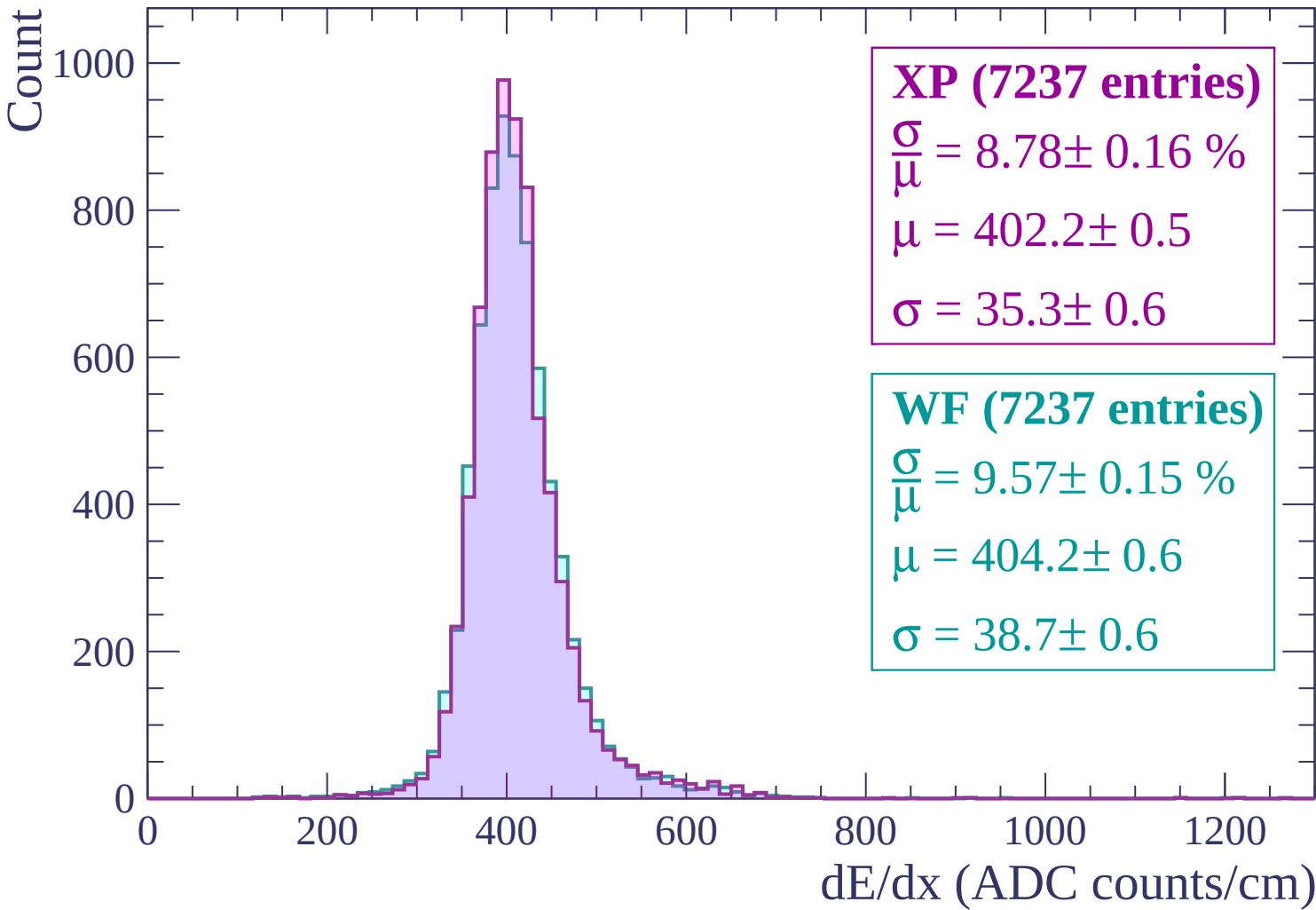
$\sigma = 35.3 \pm 0.6$

WF (7237 entries)

$\frac{\sigma}{\mu} = 9.57 \pm 0.15 \%$

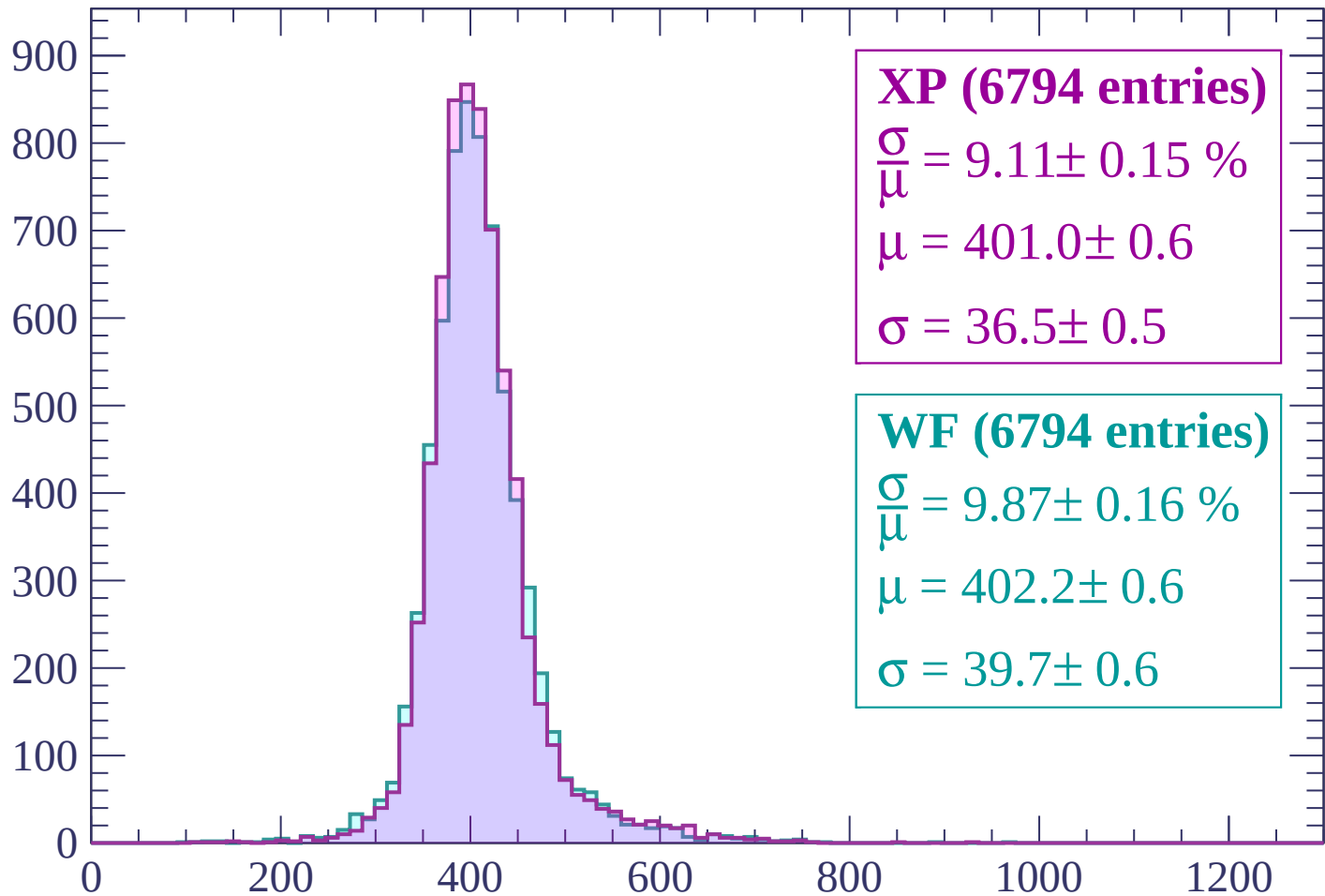
$\mu = 404.2 \pm 0.6$

$\sigma = 38.7 \pm 0.6$



Energy loss | $-360 < p < -300$

Count



XP (6794 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.15 \%$$

$$\mu = 401.0 \pm 0.6$$

$$\sigma = 36.5 \pm 0.5$$

WF (6794 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.16 \%$$

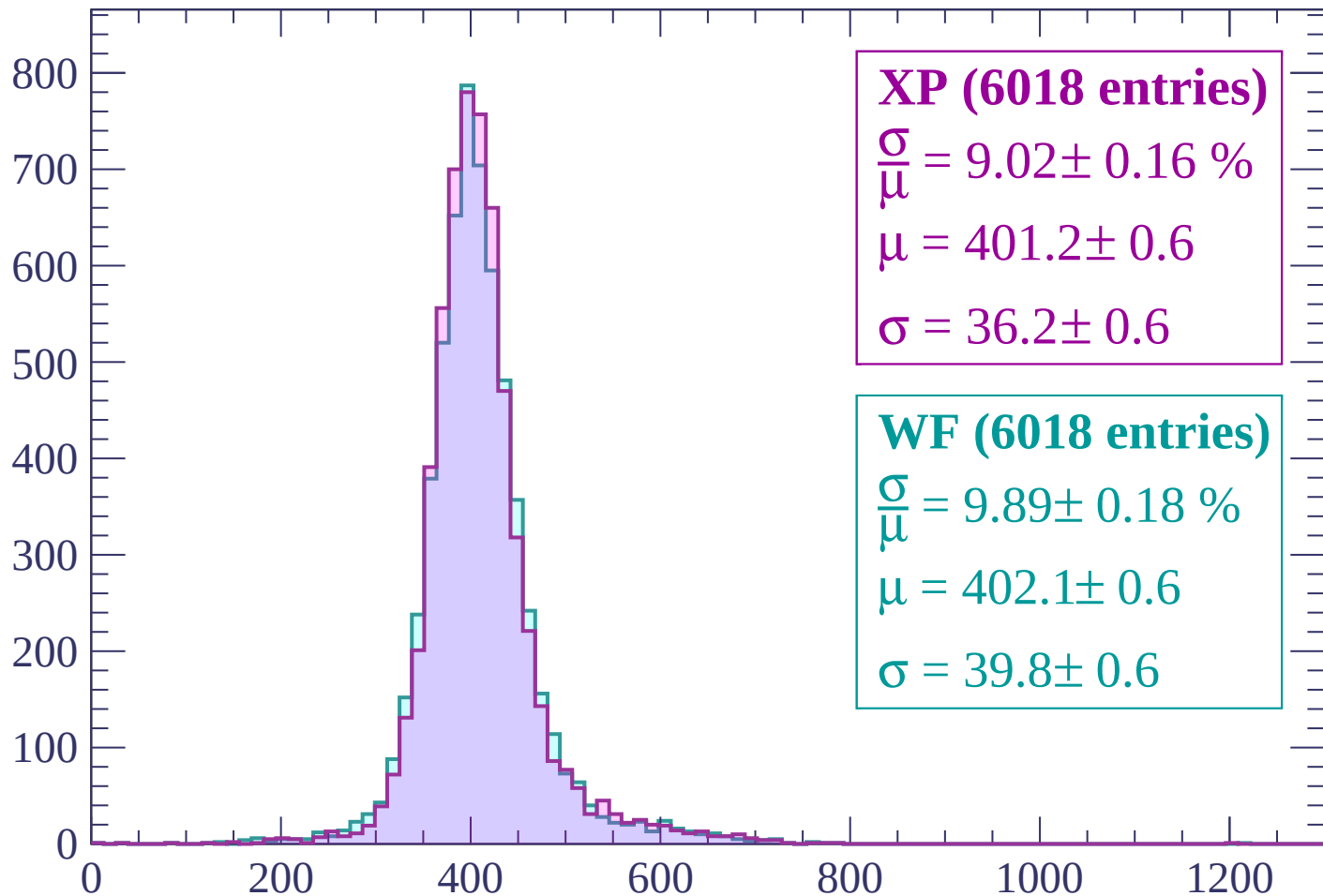
$$\mu = 402.2 \pm 0.6$$

$$\sigma = 39.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (6018 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$$

$$\mu = 401.2 \pm 0.6$$

$$\sigma = 36.2 \pm 0.6$$

WF (6018 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.18 \%$$

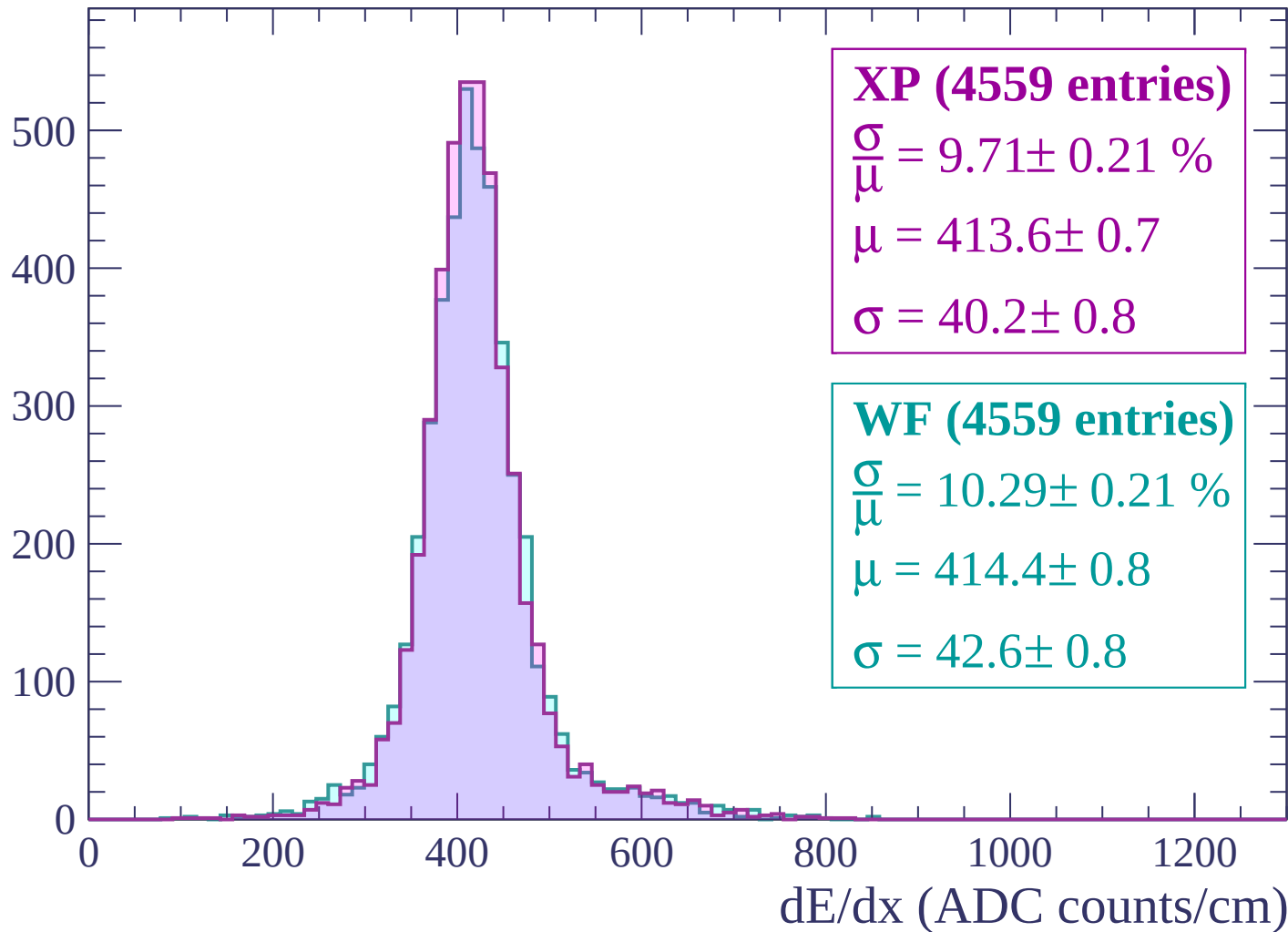
$$\mu = 402.1 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

dE/dx (ADC counts/cm)

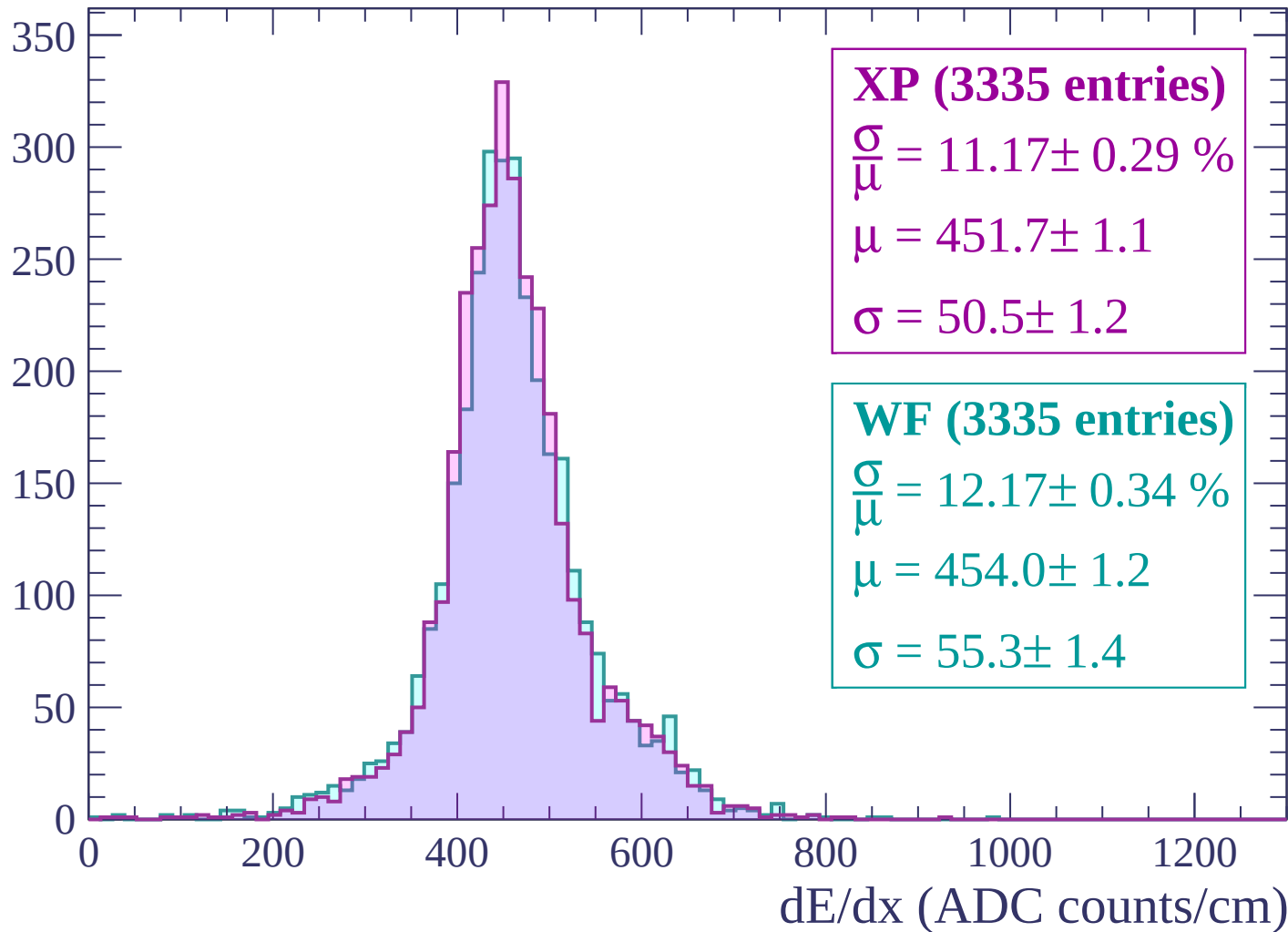
Energy loss | $-240 < p < -180$

Count



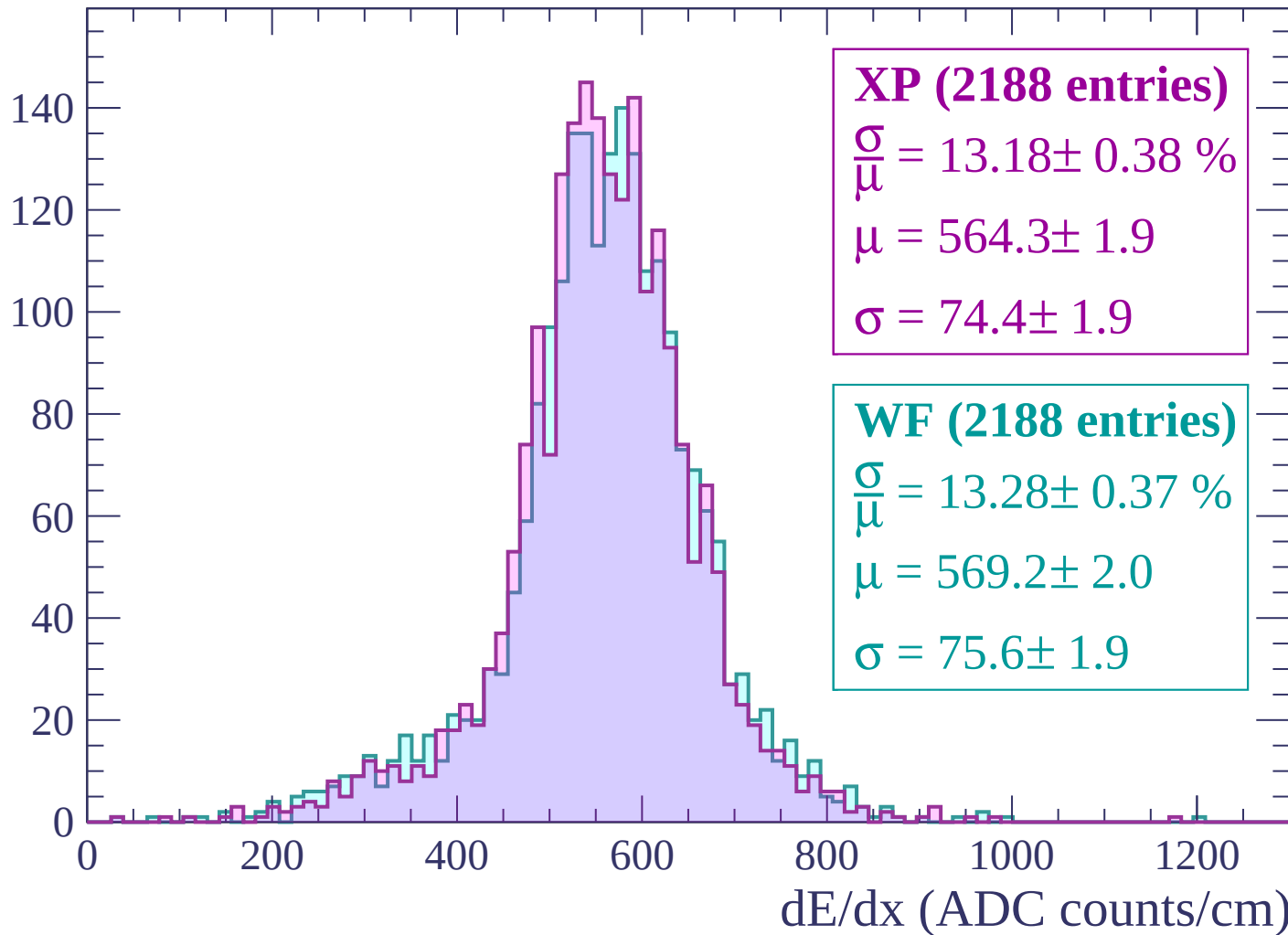
Energy loss | $-180 < p < -120$

Count



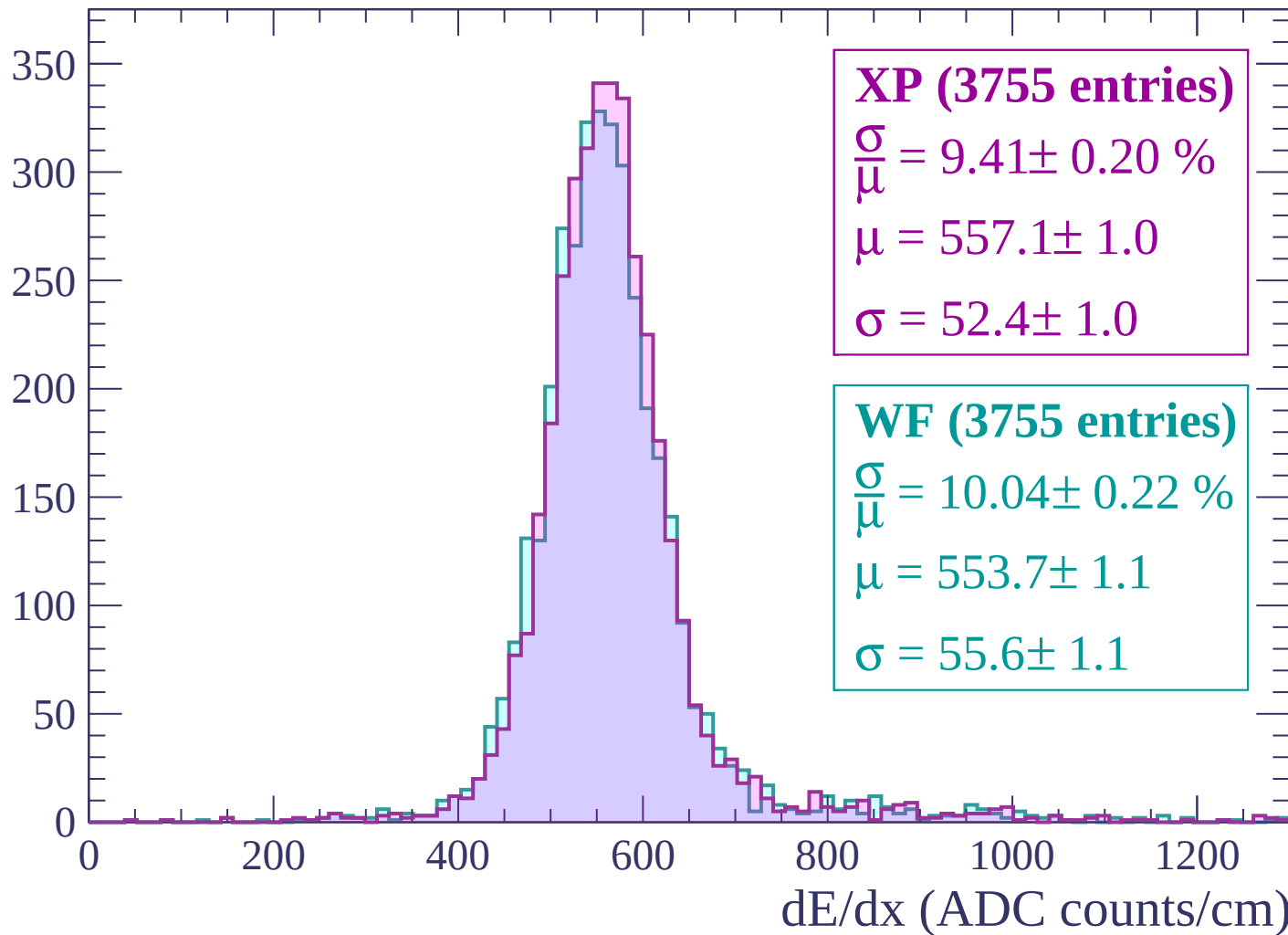
Energy loss | $-120 < p < -60$

Count



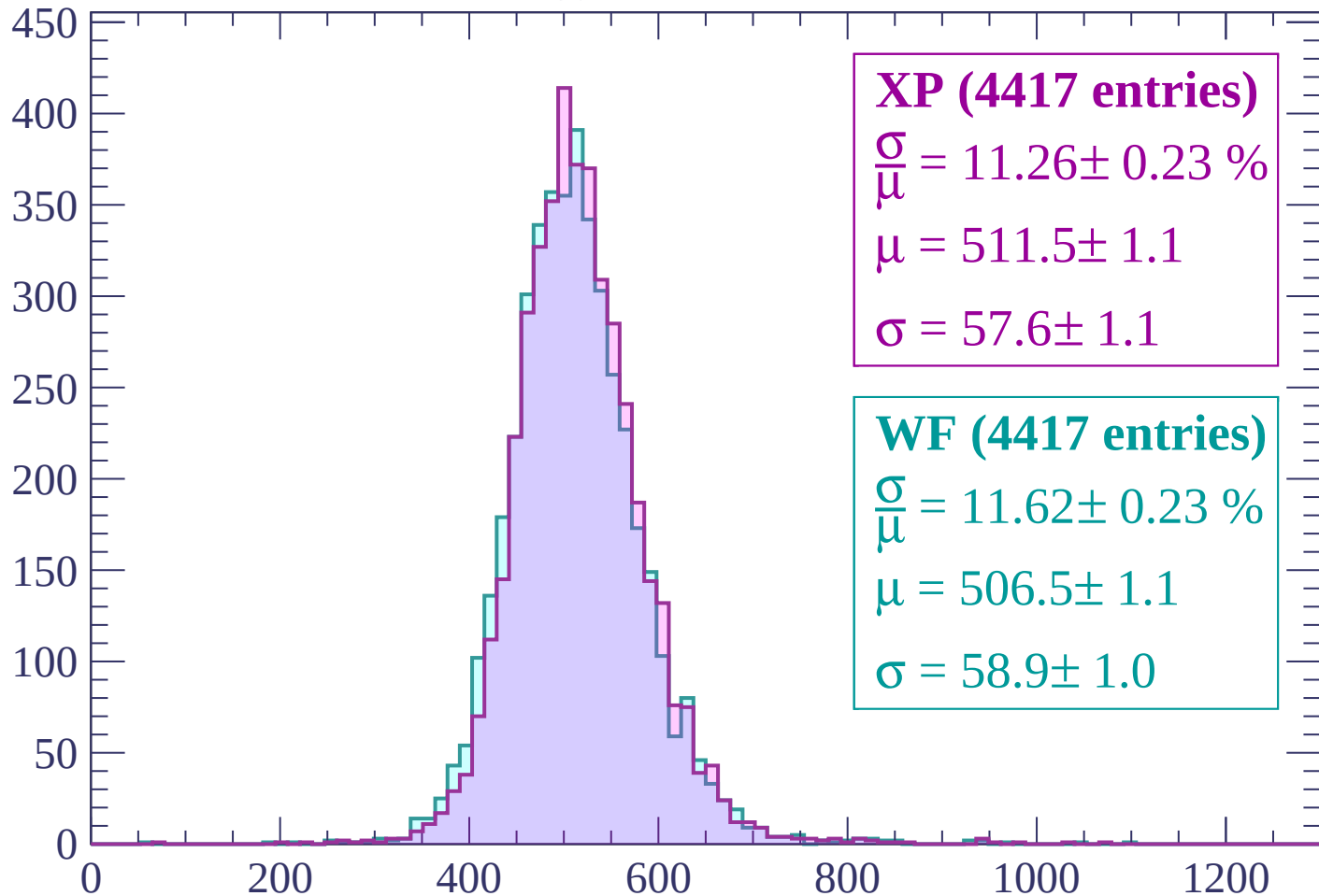
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP (4417 entries)

$\frac{\sigma}{\mu} = 11.26 \pm 0.23 \%$

$\mu = 511.5 \pm 1.1$

$\sigma = 57.6 \pm 1.1$

WF (4417 entries)

$\frac{\sigma}{\mu} = 11.62 \pm 0.23 \%$

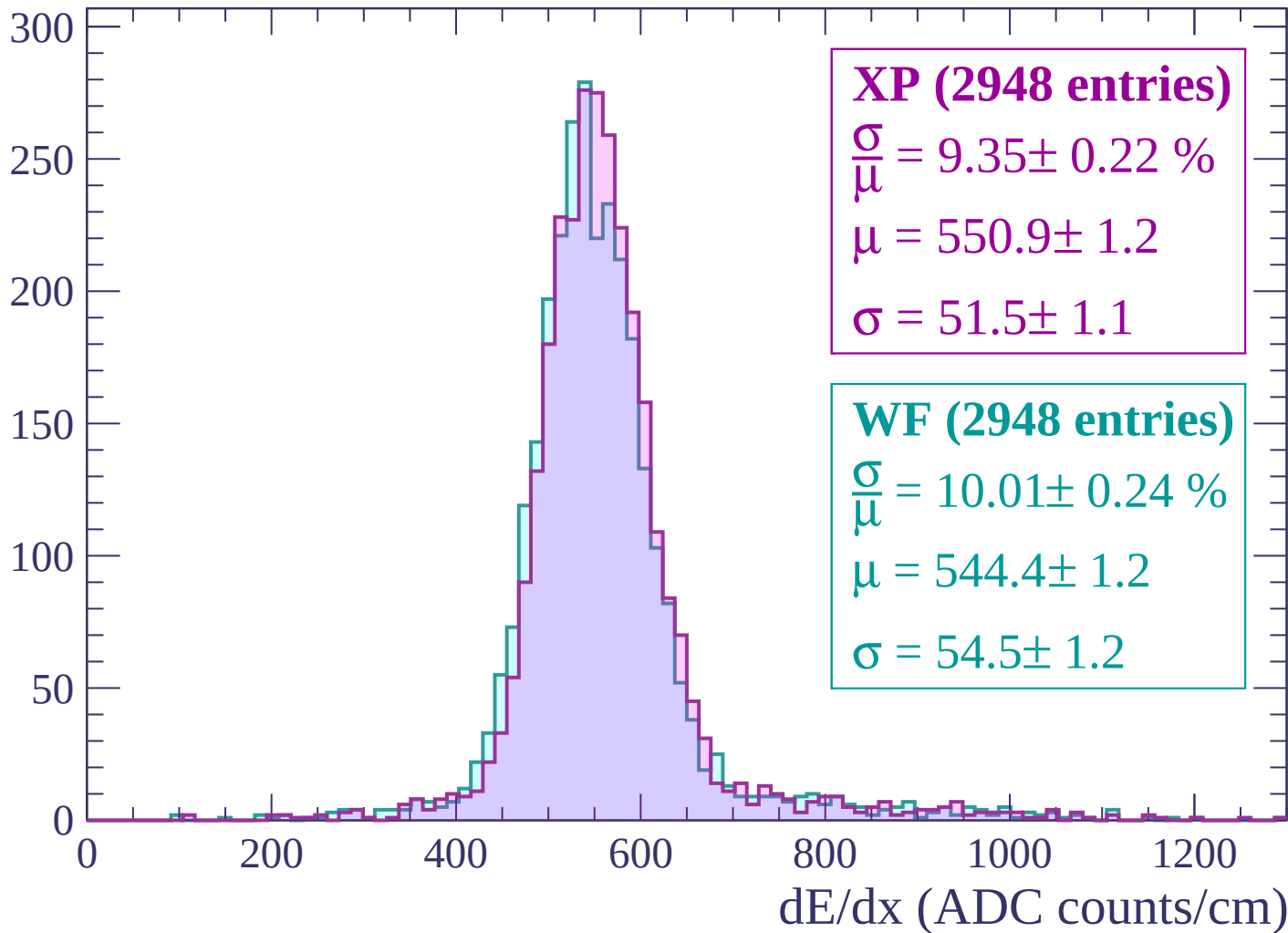
$\mu = 506.5 \pm 1.1$

$\sigma = 58.9 \pm 1.0$

dE/dx (ADC counts/cm)

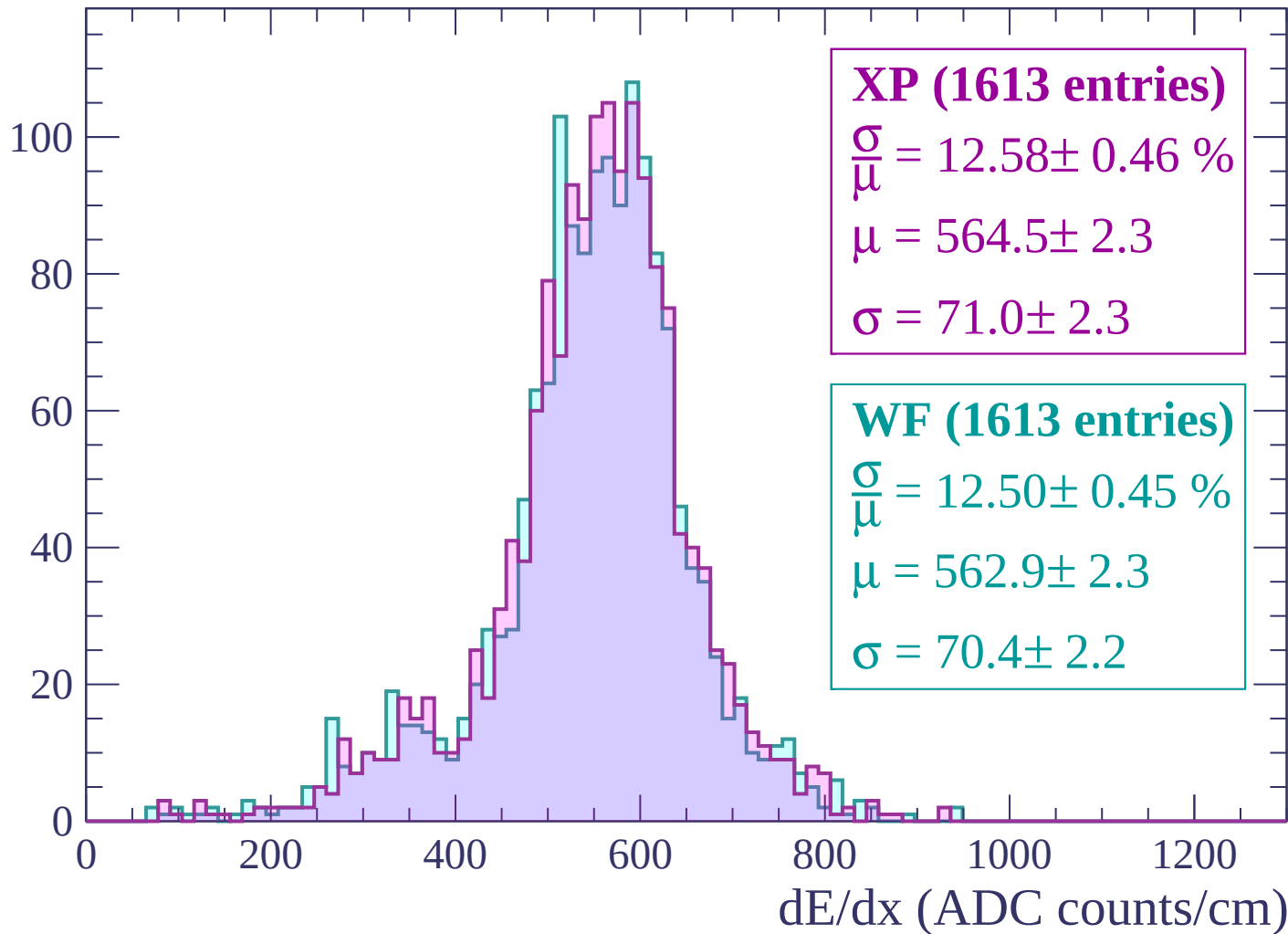
Energy loss | $60 < p < 120$

Count



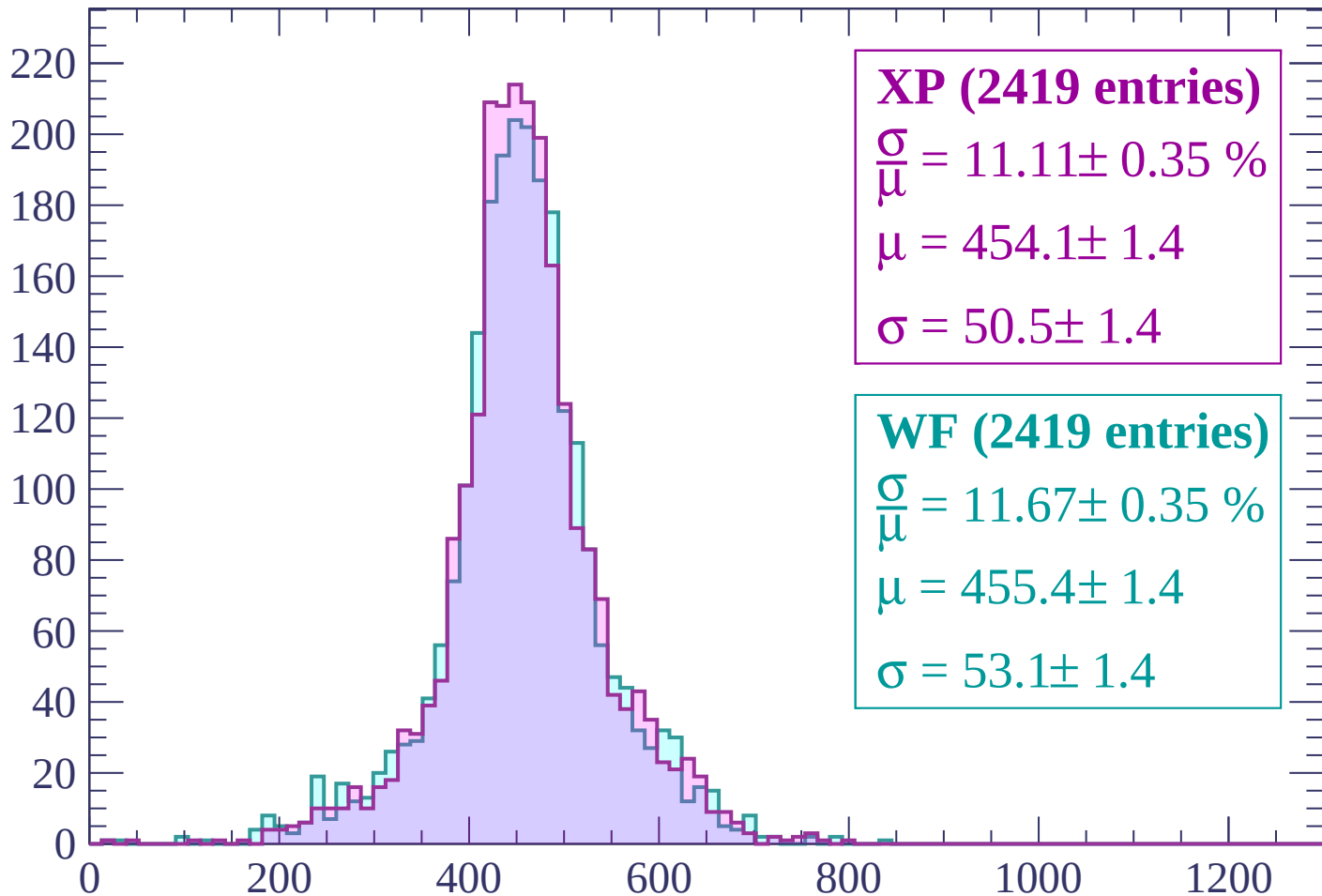
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (2419 entries)

$\frac{\sigma}{\mu} = 11.11 \pm 0.35 \%$

$\mu = 454.1 \pm 1.4$

$\sigma = 50.5 \pm 1.4$

WF (2419 entries)

$\frac{\sigma}{\mu} = 11.67 \pm 0.35 \%$

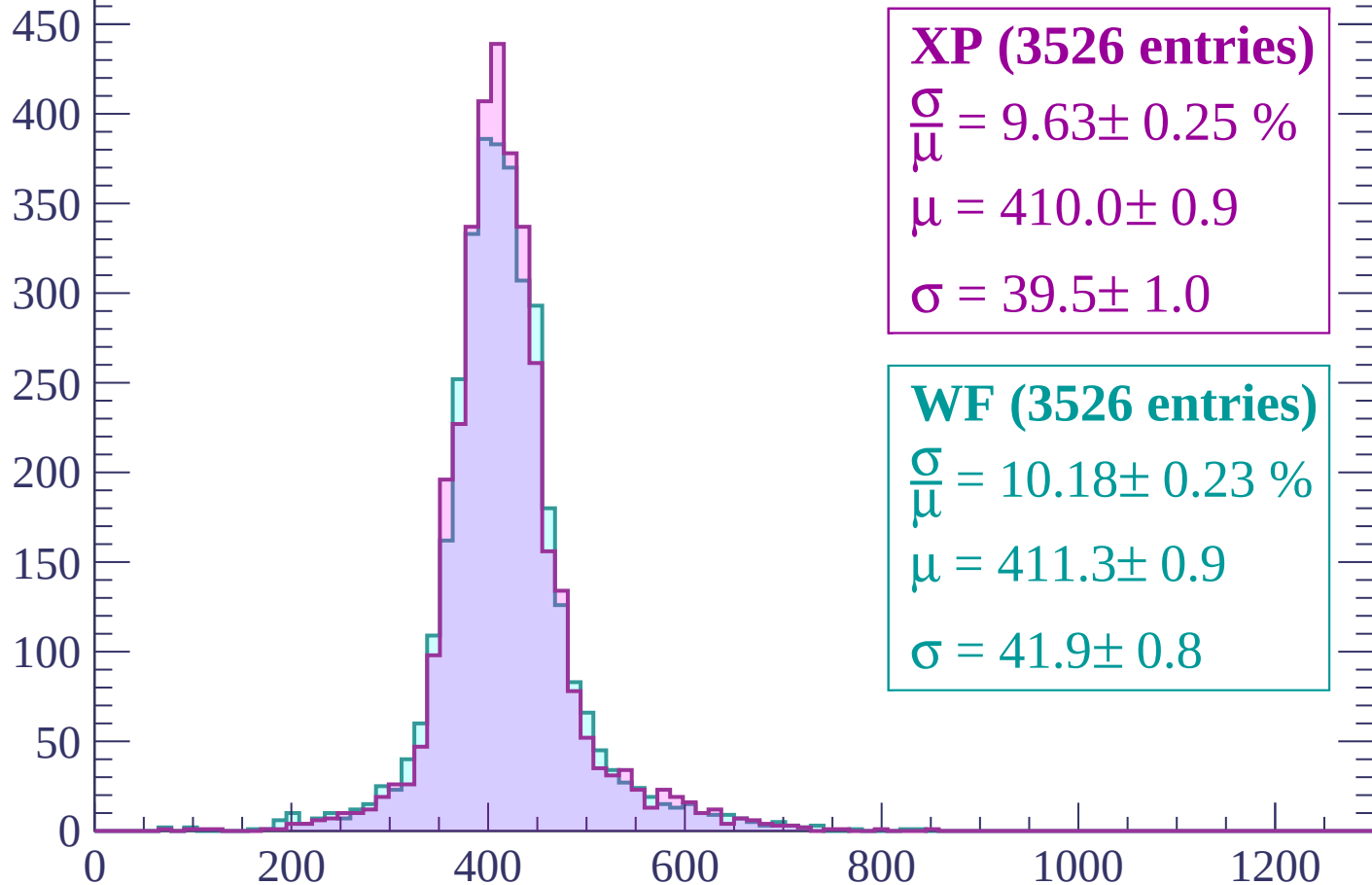
$\mu = 455.4 \pm 1.4$

$\sigma = 53.1 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (3526 entries)

$$\frac{\sigma}{\mu} = 9.63 \pm 0.25 \%$$

$$\mu = 410.0 \pm 0.9$$

$$\sigma = 39.5 \pm 1.0$$

WF (3526 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.23 \%$$

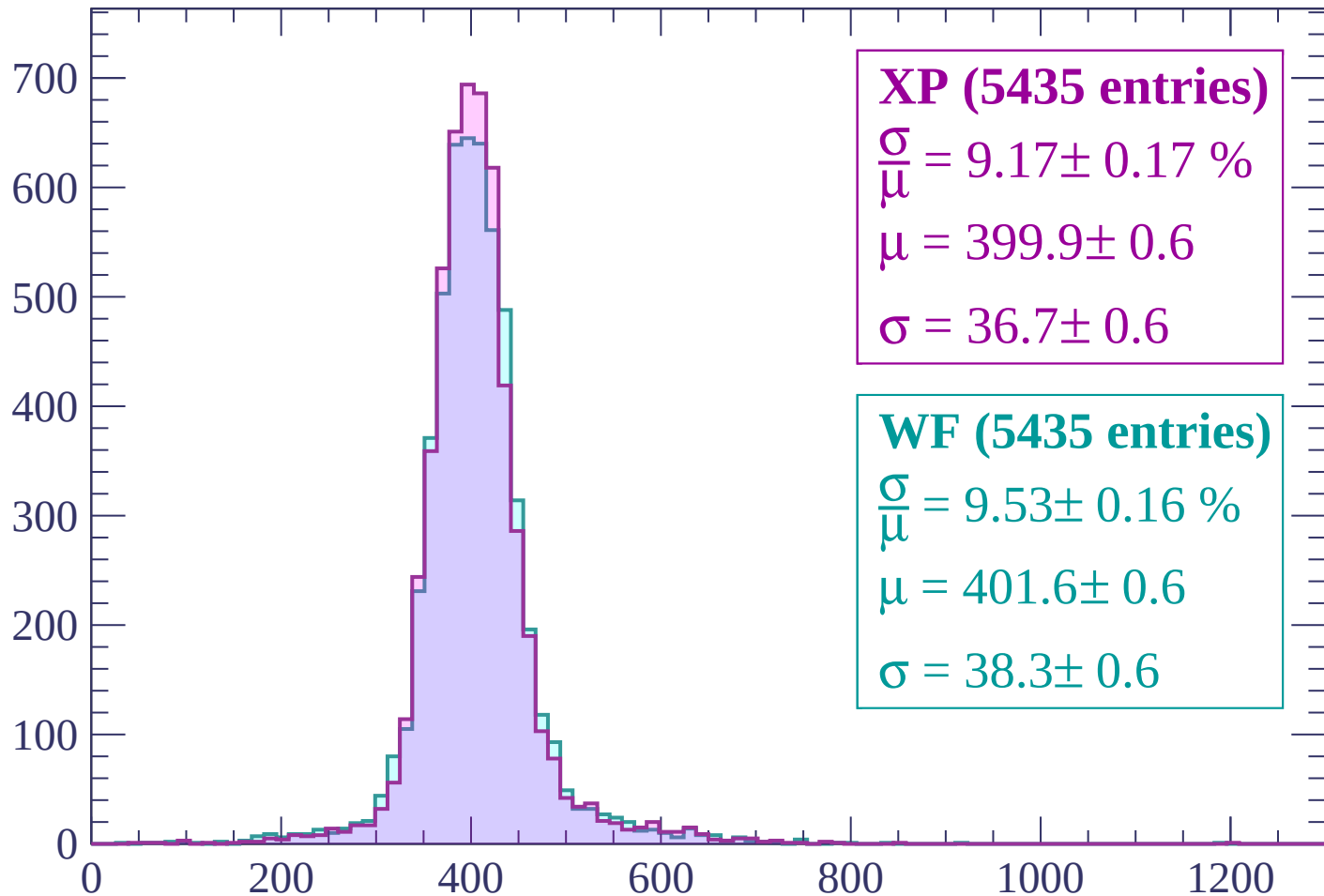
$$\mu = 411.3 \pm 0.9$$

$$\sigma = 41.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (5435 entries)

$$\frac{\sigma}{\mu} = 9.17 \pm 0.17 \%$$

$$\mu = 399.9 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

WF (5435 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$$

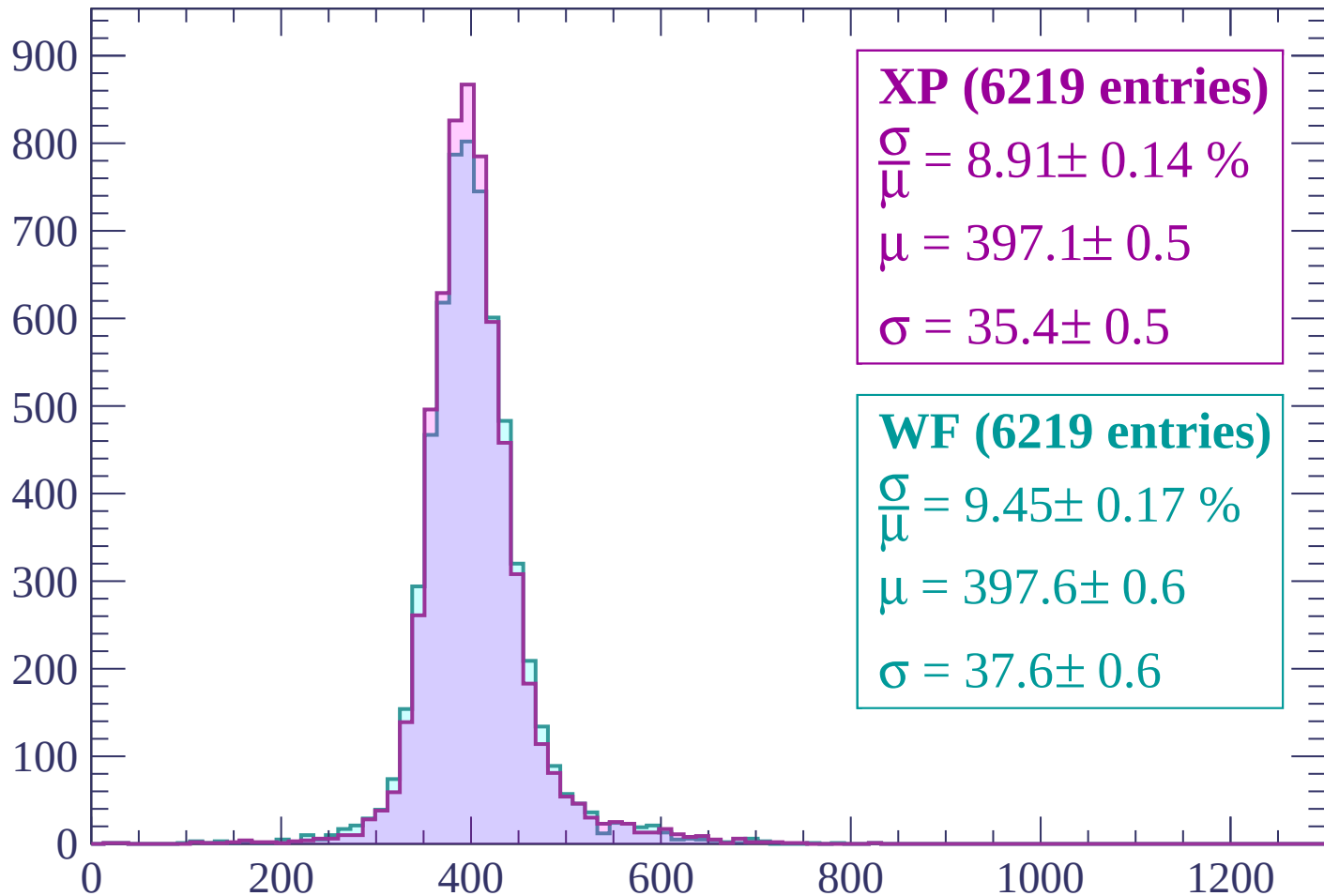
$$\mu = 401.6 \pm 0.6$$

$$\sigma = 38.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (6219 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.14 \%$$

$$\mu = 397.1 \pm 0.5$$

$$\sigma = 35.4 \pm 0.5$$

WF (6219 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.17 \%$$

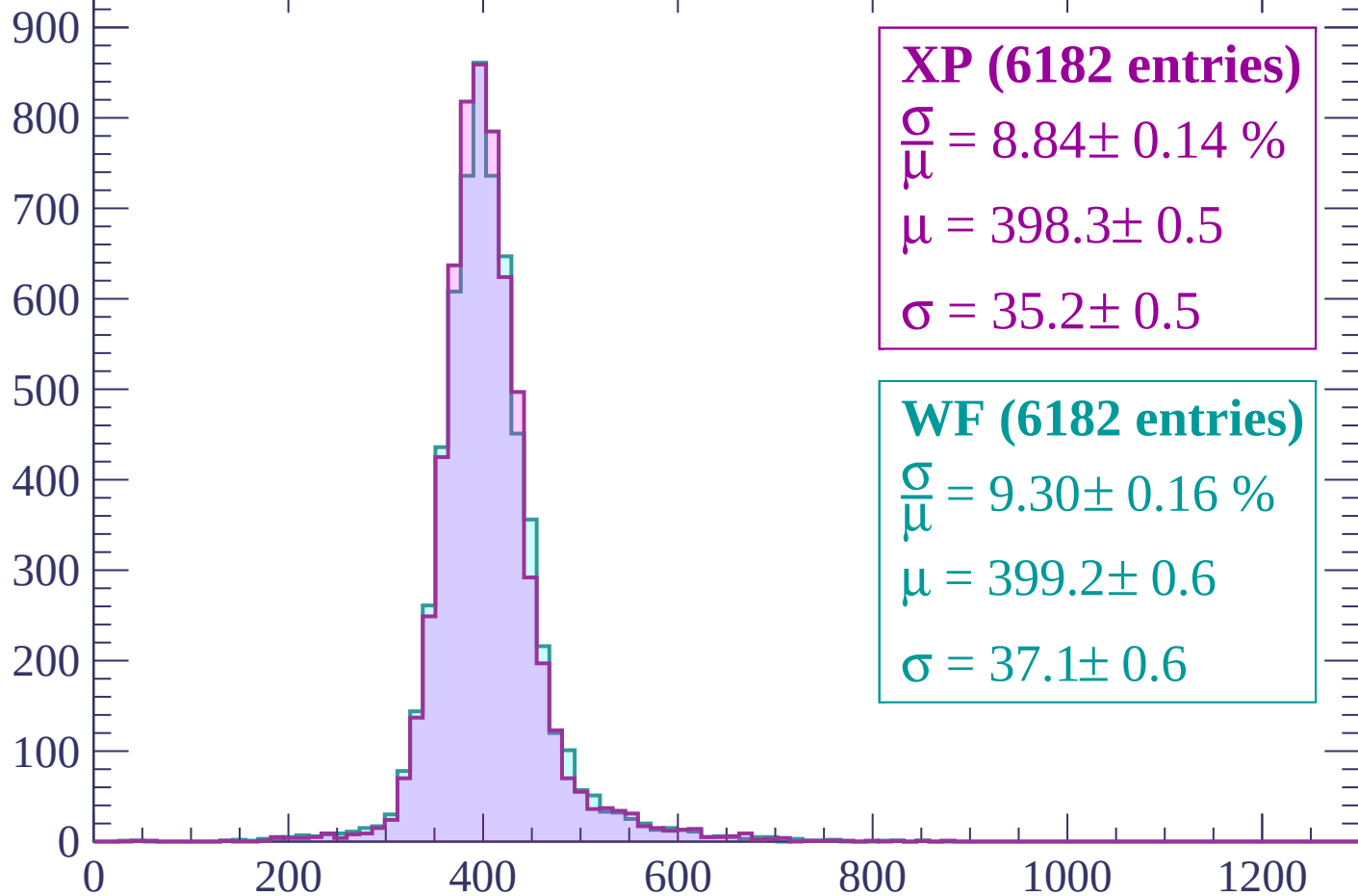
$$\mu = 397.6 \pm 0.6$$

$$\sigma = 37.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (6182 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.14 \%$$

$$\mu = 398.3 \pm 0.5$$

$$\sigma = 35.2 \pm 0.5$$

WF (6182 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.16 \%$$

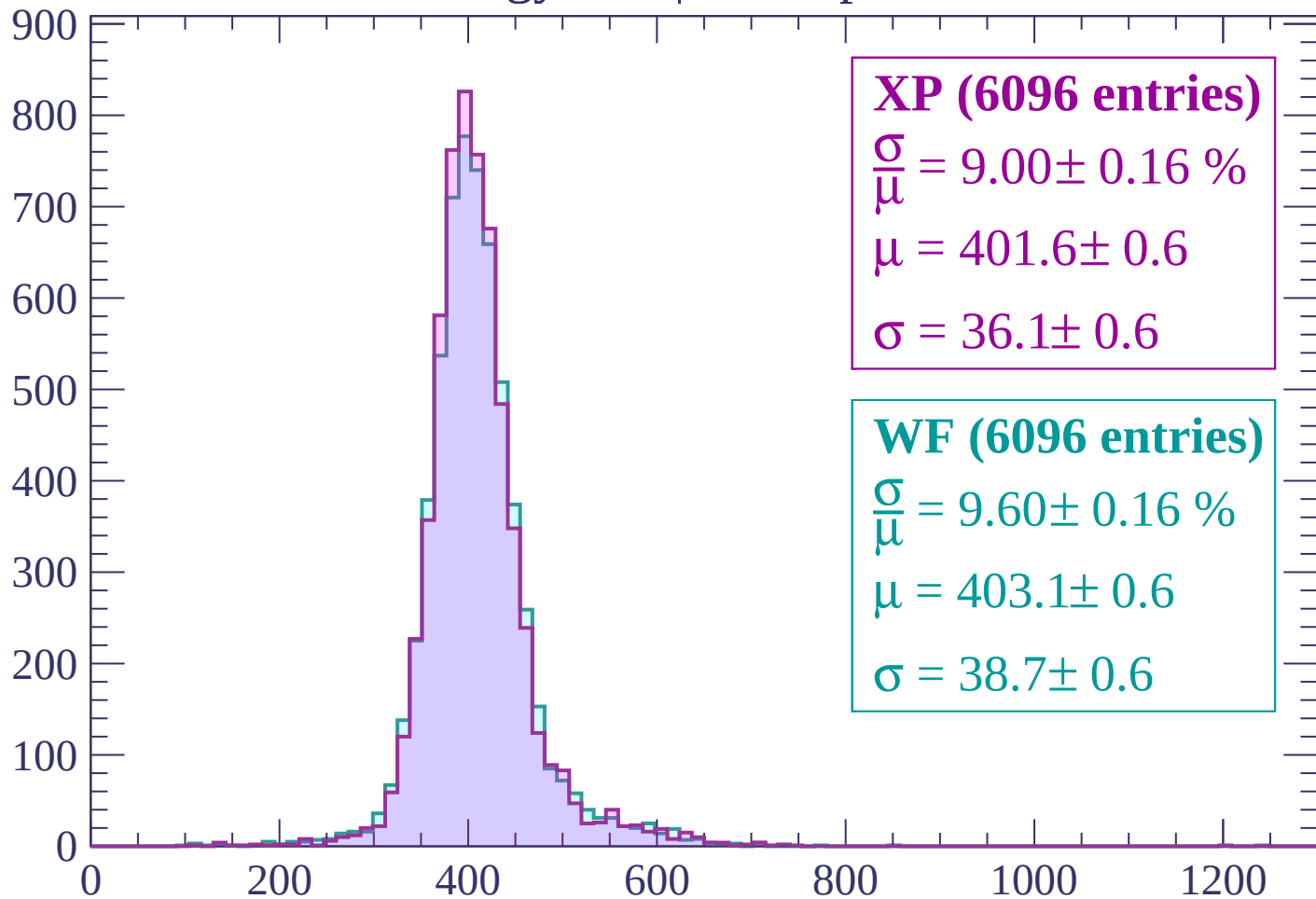
$$\mu = 399.2 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (6096 entries)

$$\frac{\sigma}{\mu} = 9.00 \pm 0.16 \%$$

$$\mu = 401.6 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

WF (6096 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$$

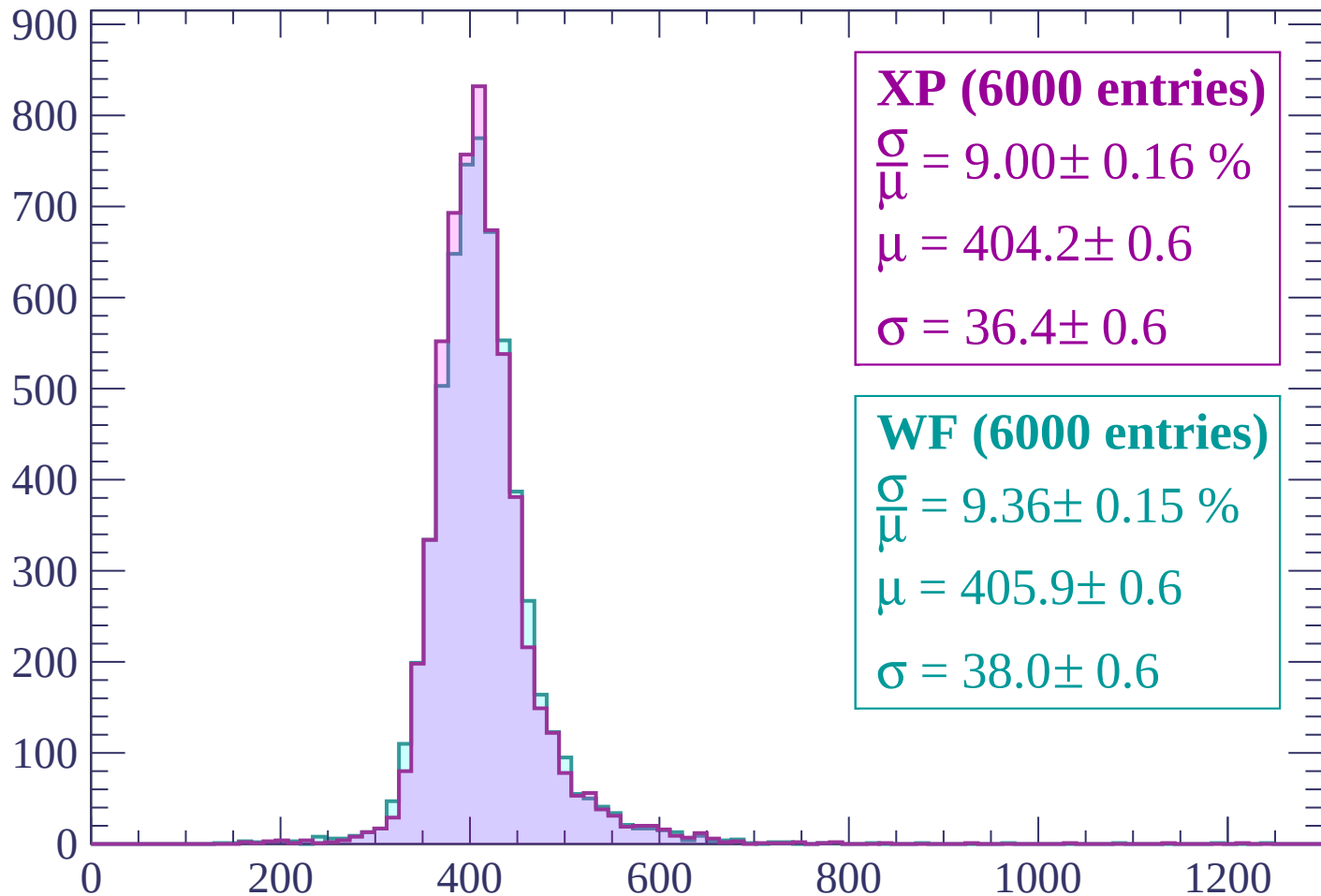
$$\mu = 403.1 \pm 0.6$$

$$\sigma = 38.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (6000 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.16 \%$

$\mu = 404.2 \pm 0.6$

$\sigma = 36.4 \pm 0.6$

WF (6000 entries)

$\frac{\sigma}{\mu} = 9.36 \pm 0.15 \%$

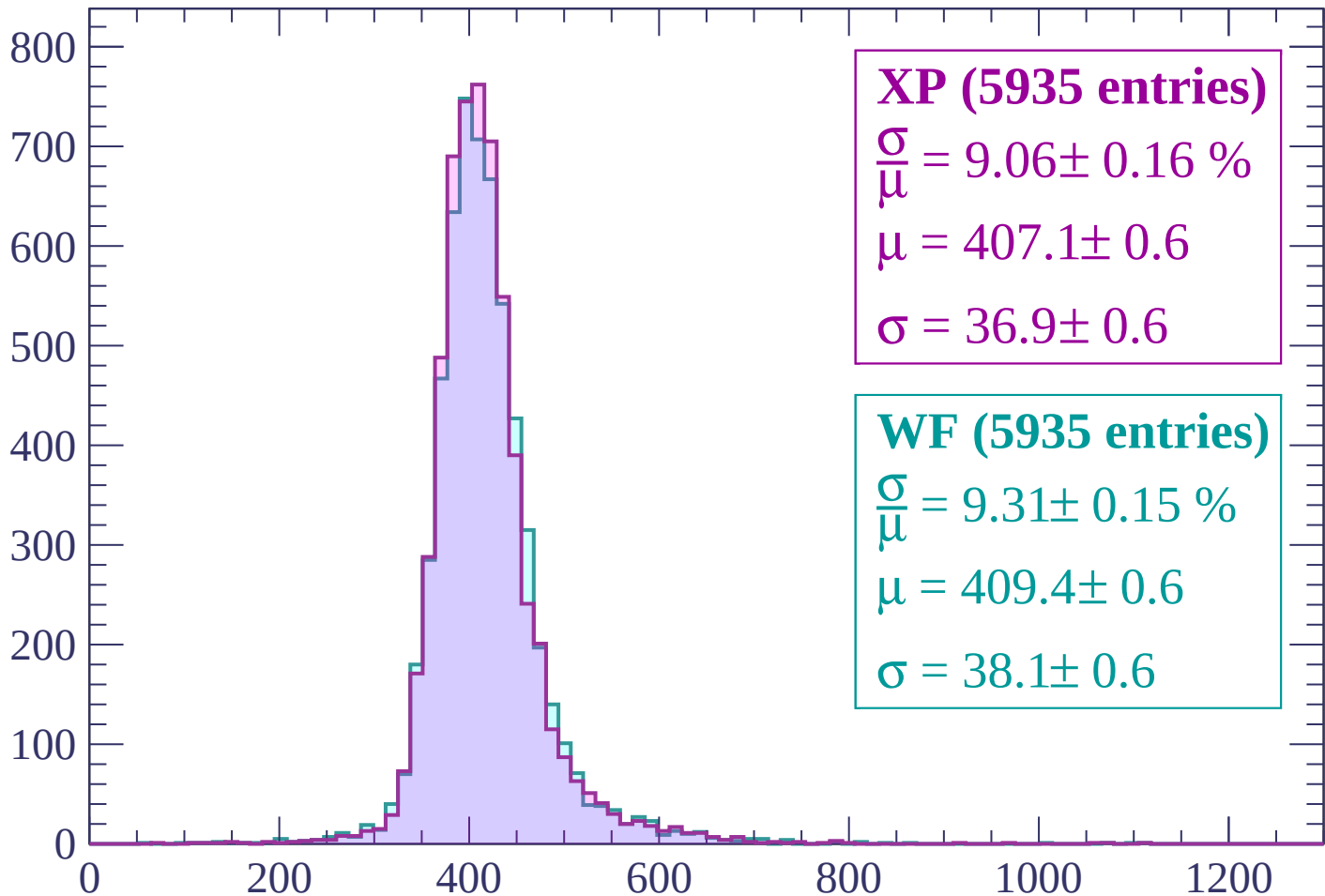
$\mu = 405.9 \pm 0.6$

$\sigma = 38.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (5935 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.16 \%$$

$$\mu = 407.1 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

WF (5935 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.15 \%$$

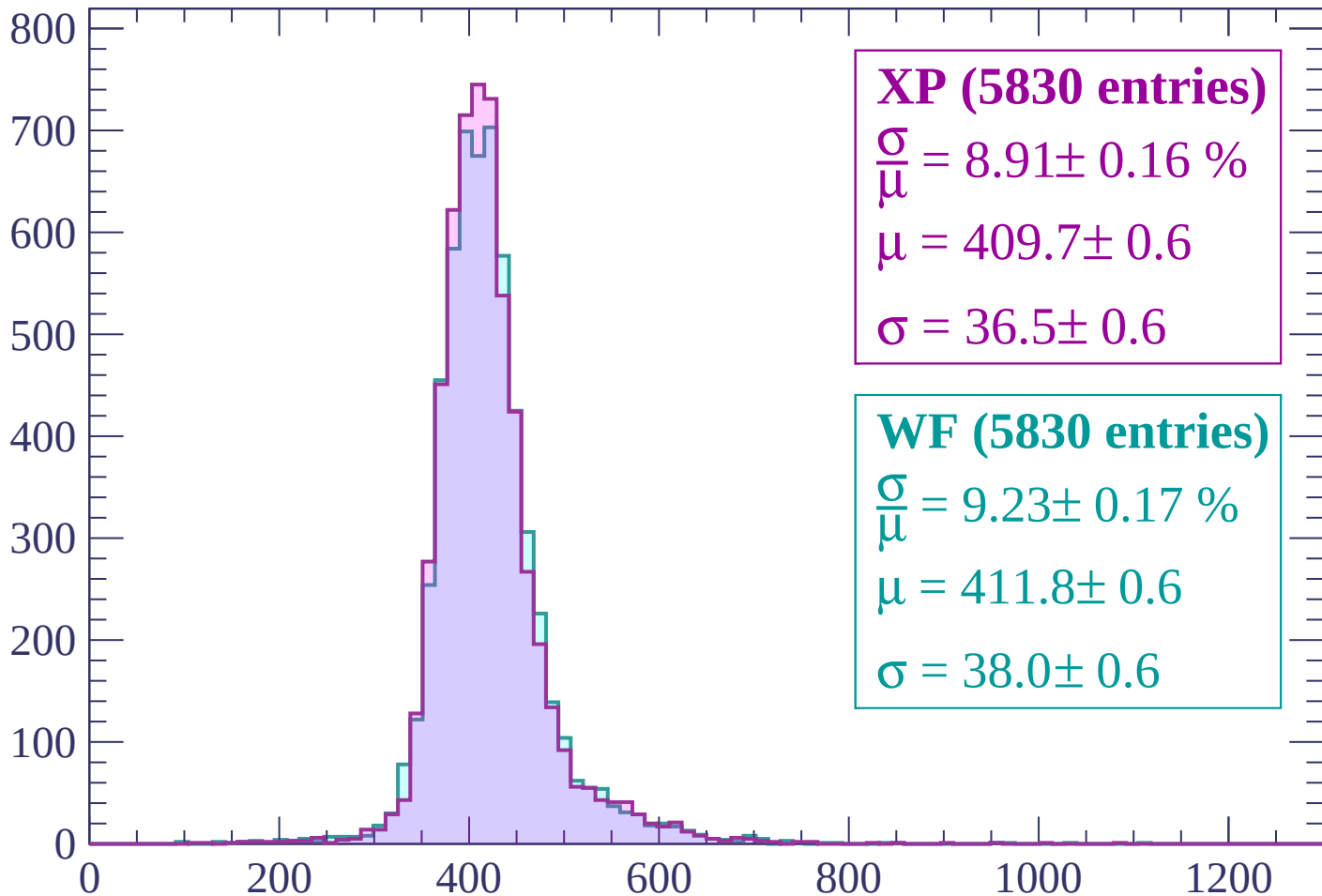
$$\mu = 409.4 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (5830 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.16 \%$$

$$\mu = 409.7 \pm 0.6$$

$$\sigma = 36.5 \pm 0.6$$

WF (5830 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.17 \%$$

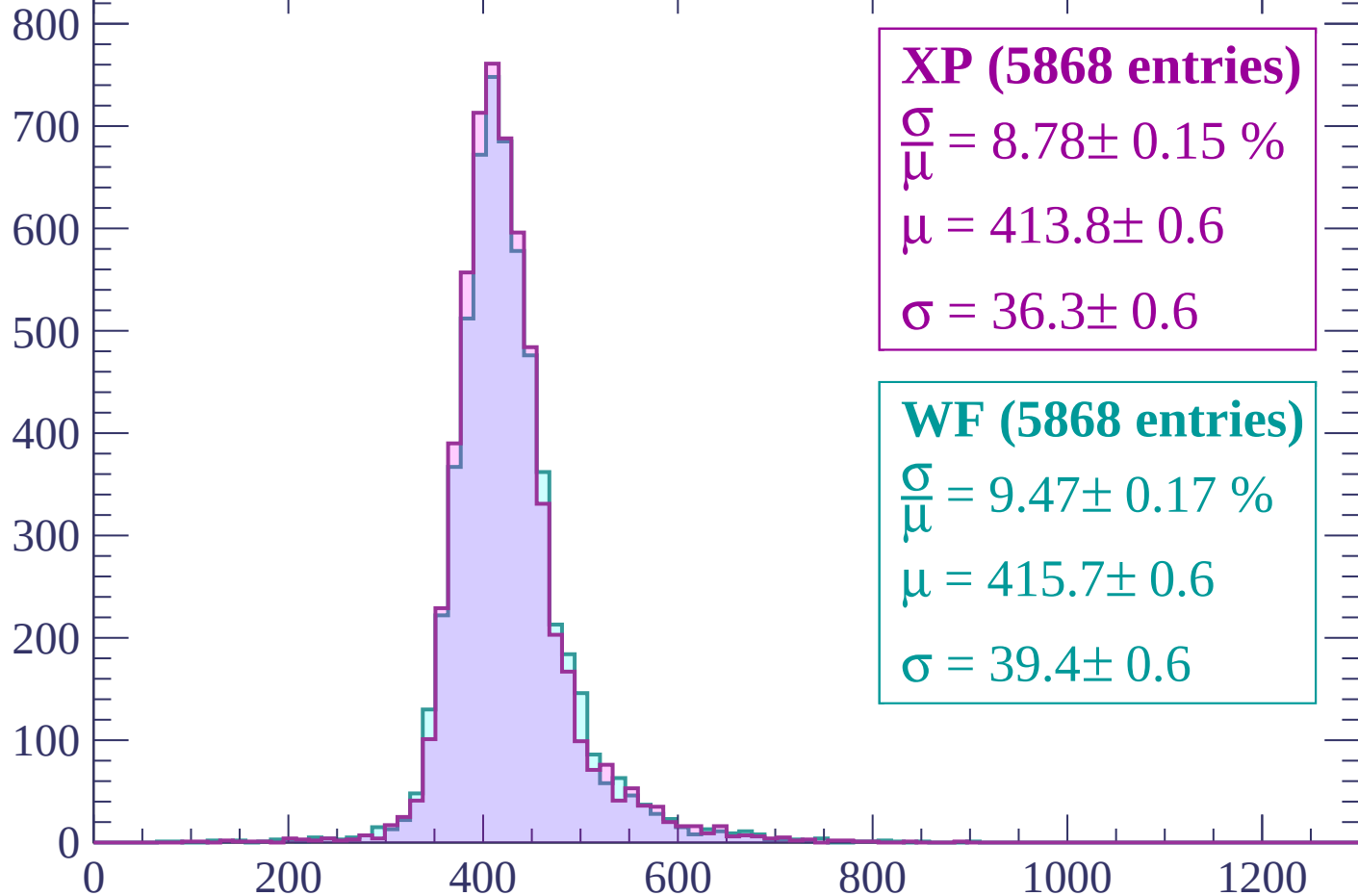
$$\mu = 411.8 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (5868 entries)

$$\frac{\sigma}{\mu} = 8.78 \pm 0.15 \%$$

$$\mu = 413.8 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF (5868 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.17 \%$$

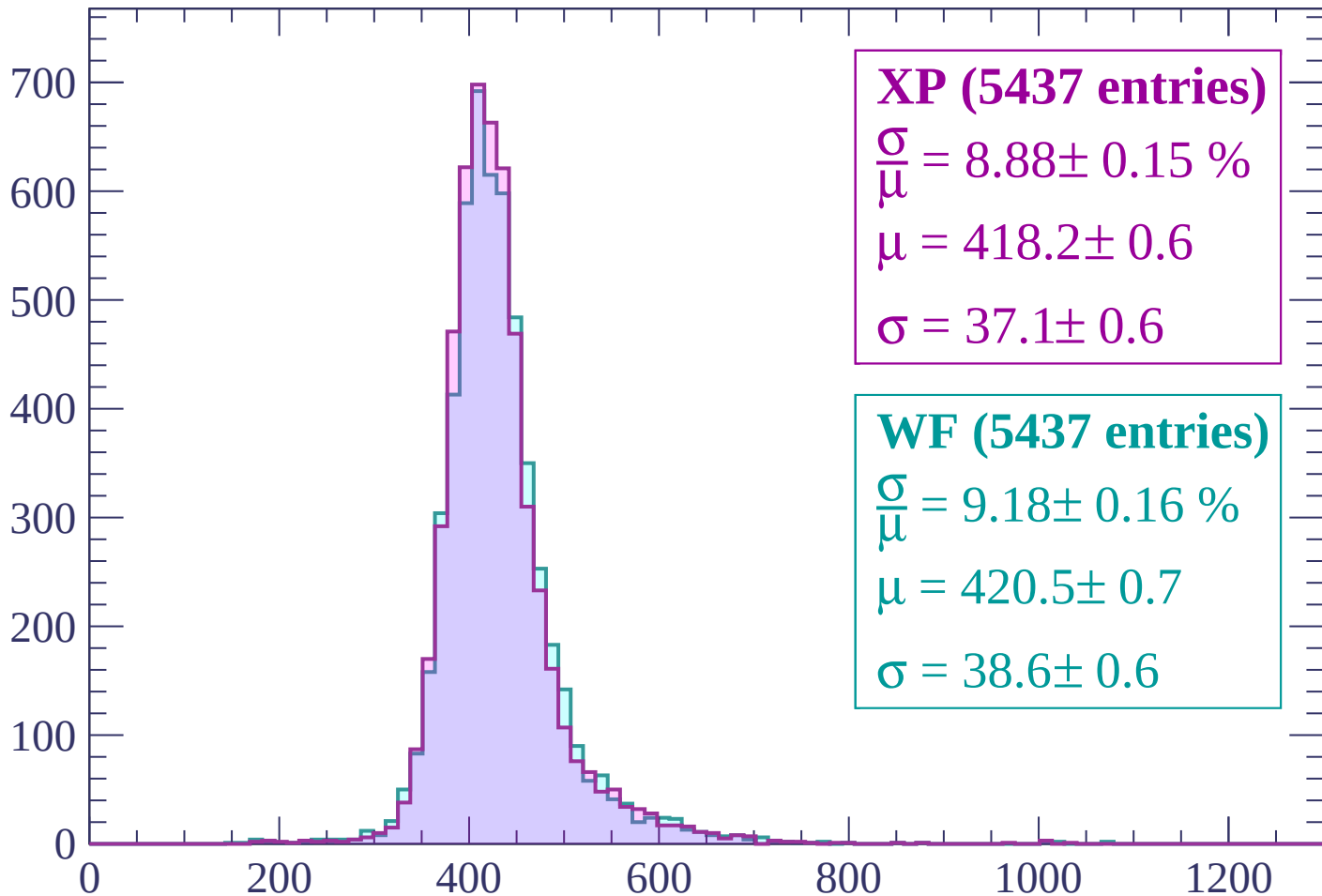
$$\mu = 415.7 \pm 0.6$$

$$\sigma = 39.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (5437 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.15 \%$

$\mu = 418.2 \pm 0.6$

$\sigma = 37.1 \pm 0.6$

WF (5437 entries)

$\frac{\sigma}{\mu} = 9.18 \pm 0.16 \%$

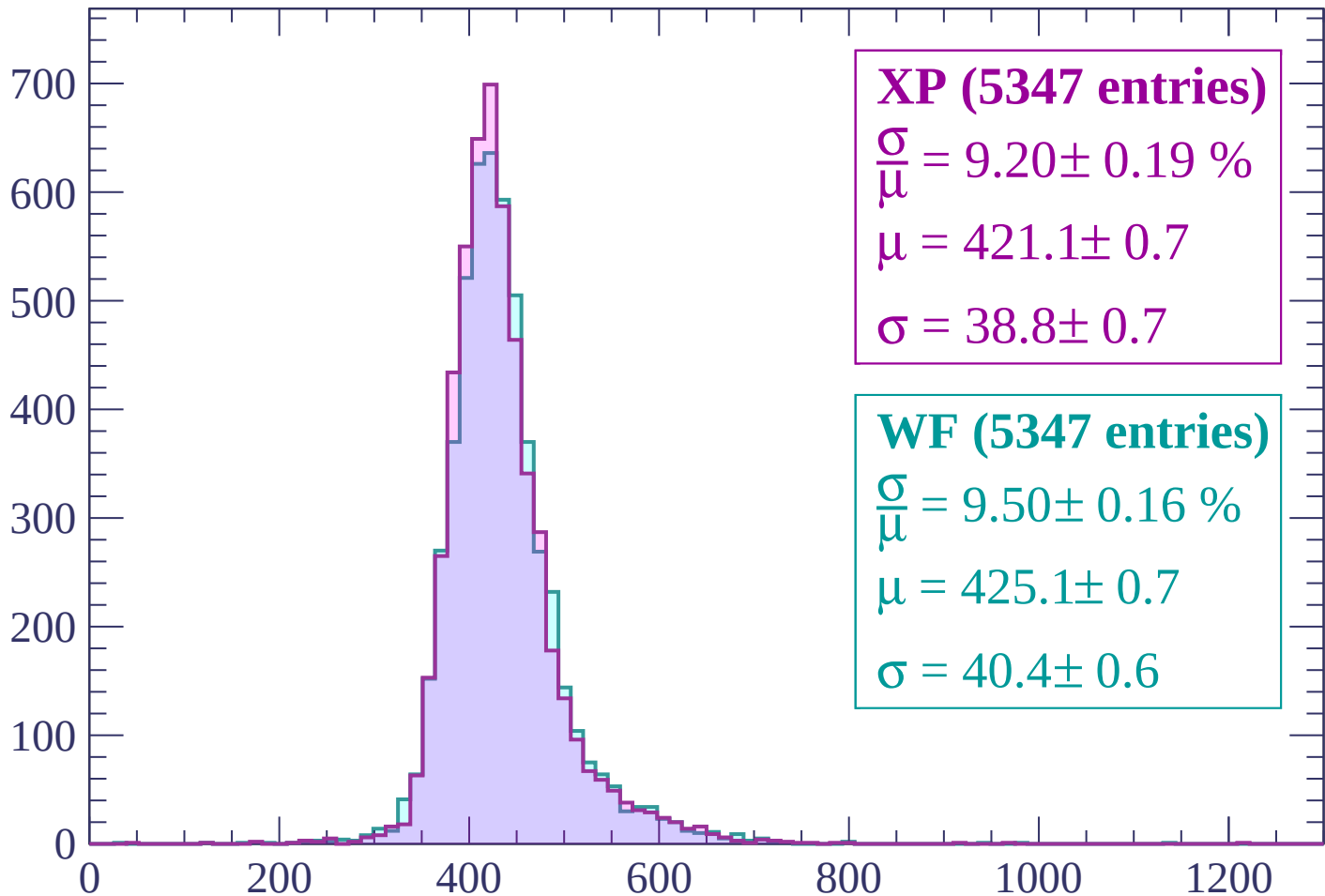
$\mu = 420.5 \pm 0.7$

$\sigma = 38.6 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (5347 entries)

$\frac{\sigma}{\mu} = 9.20 \pm 0.19 \%$

$\mu = 421.1 \pm 0.7$

$\sigma = 38.8 \pm 0.7$

WF (5347 entries)

$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$

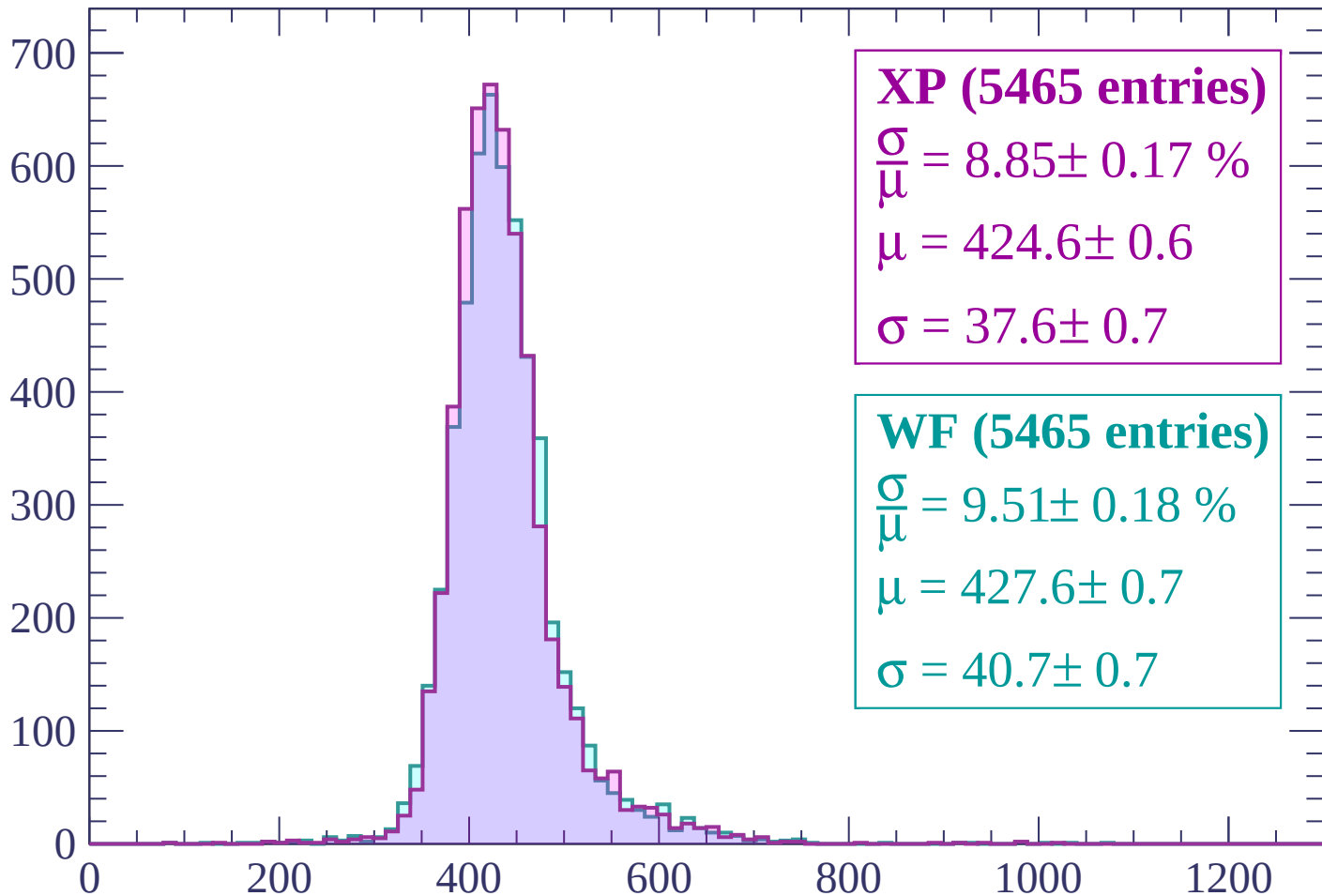
$\mu = 425.1 \pm 0.7$

$\sigma = 40.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (5465 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

$$\mu = 424.6 \pm 0.6$$

$$\sigma = 37.6 \pm 0.7$$

WF (5465 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.18 \%$$

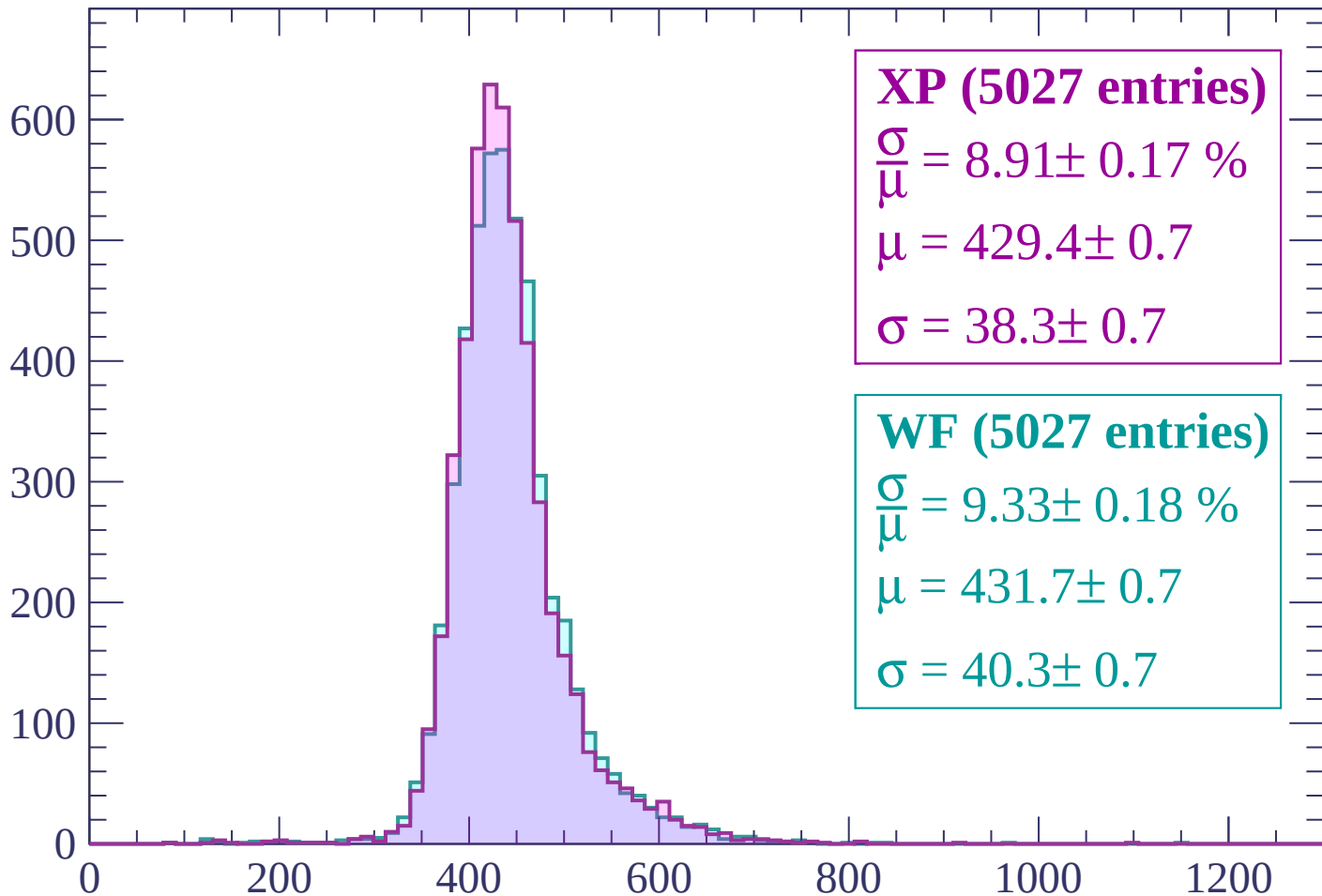
$$\mu = 427.6 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (5027 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.17 \%$$

$$\mu = 429.4 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

WF (5027 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.18 \%$$

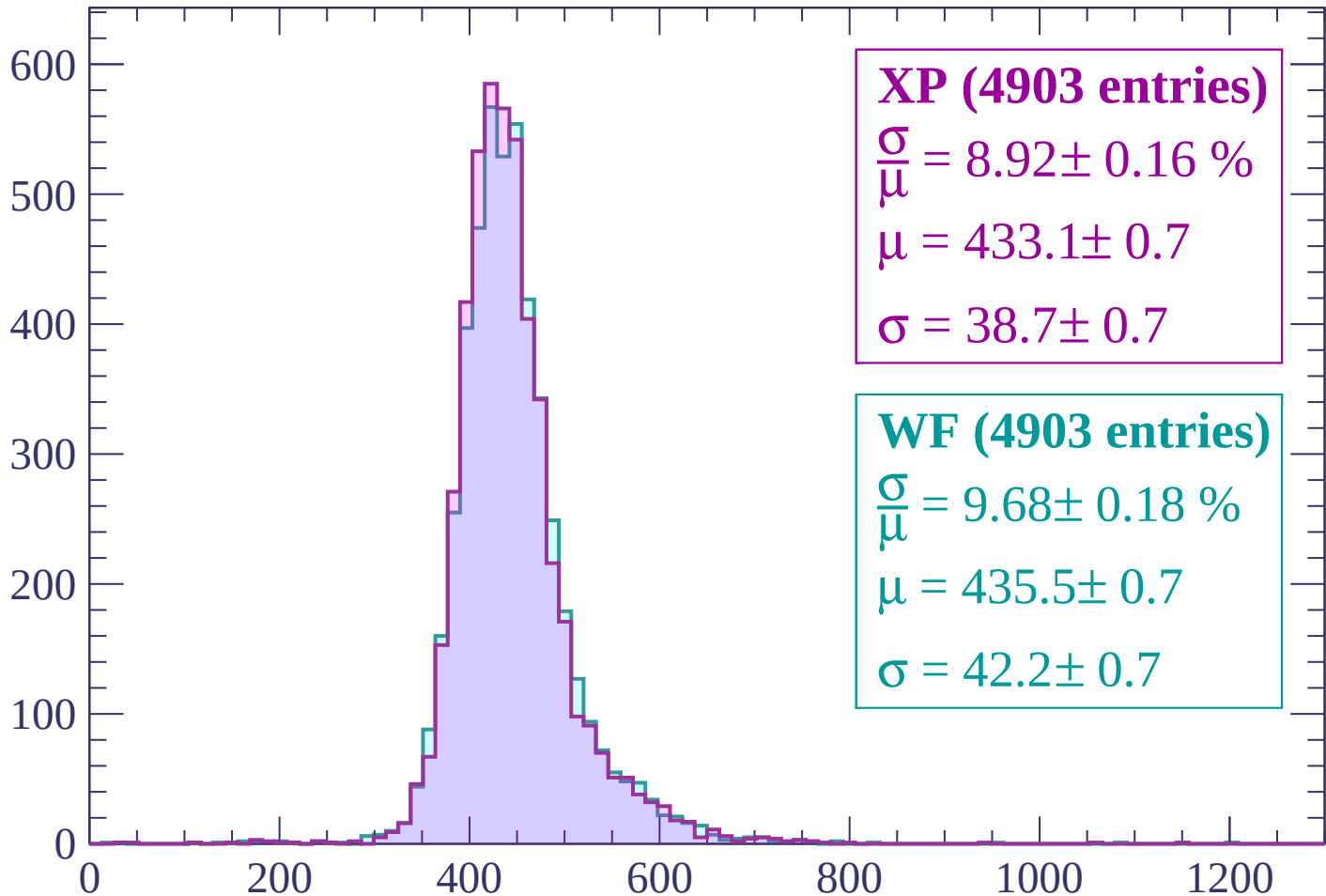
$$\mu = 431.7 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (4903 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

$$\mu = 433.1 \pm 0.7$$

$$\sigma = 38.7 \pm 0.7$$

WF (4903 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.18 \%$$

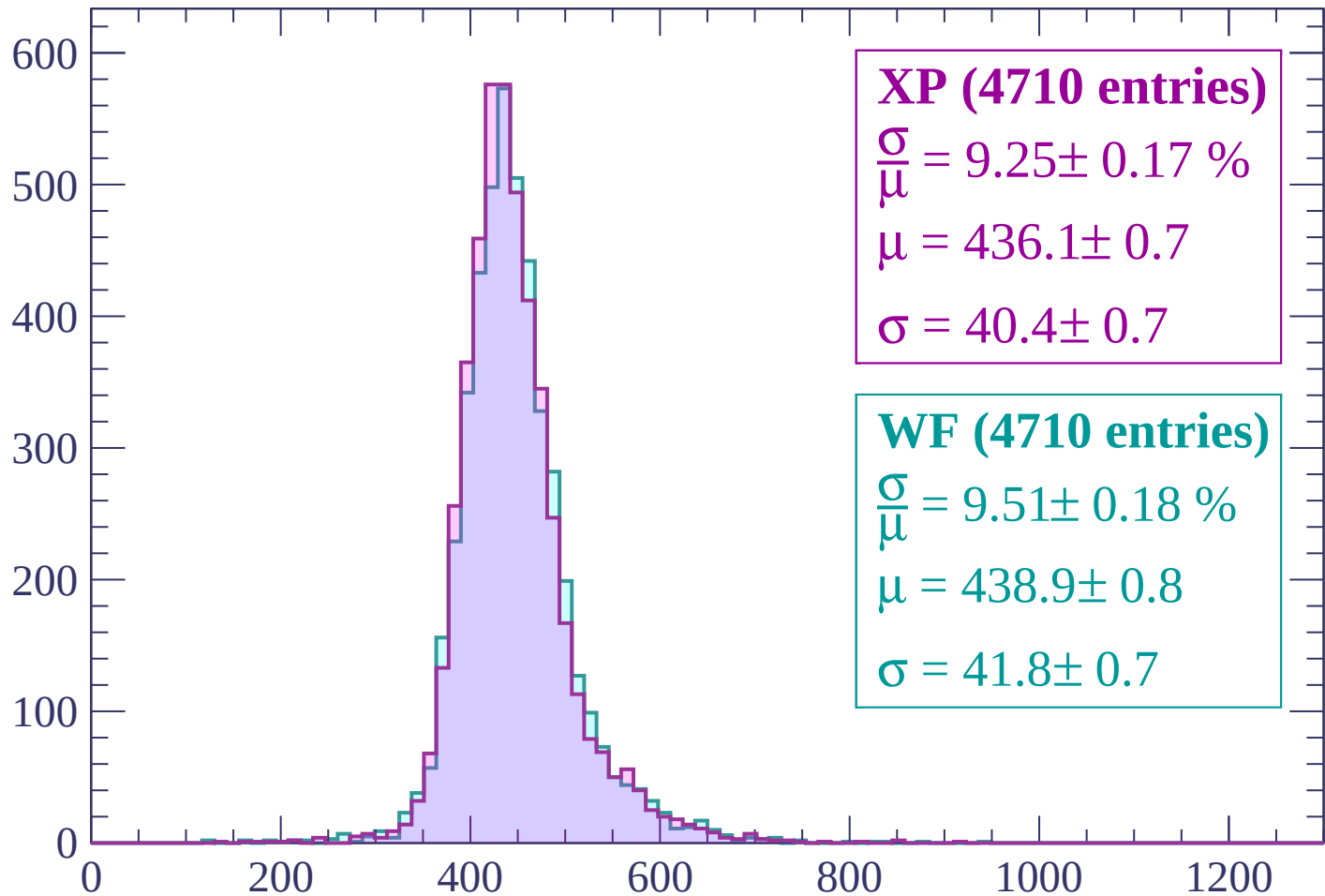
$$\mu = 435.5 \pm 0.7$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (4710 entries)

$\frac{\sigma}{\mu} = 9.25 \pm 0.17 \%$

$\mu = 436.1 \pm 0.7$

$\sigma = 40.4 \pm 0.7$

WF (4710 entries)

$\frac{\sigma}{\mu} = 9.51 \pm 0.18 \%$

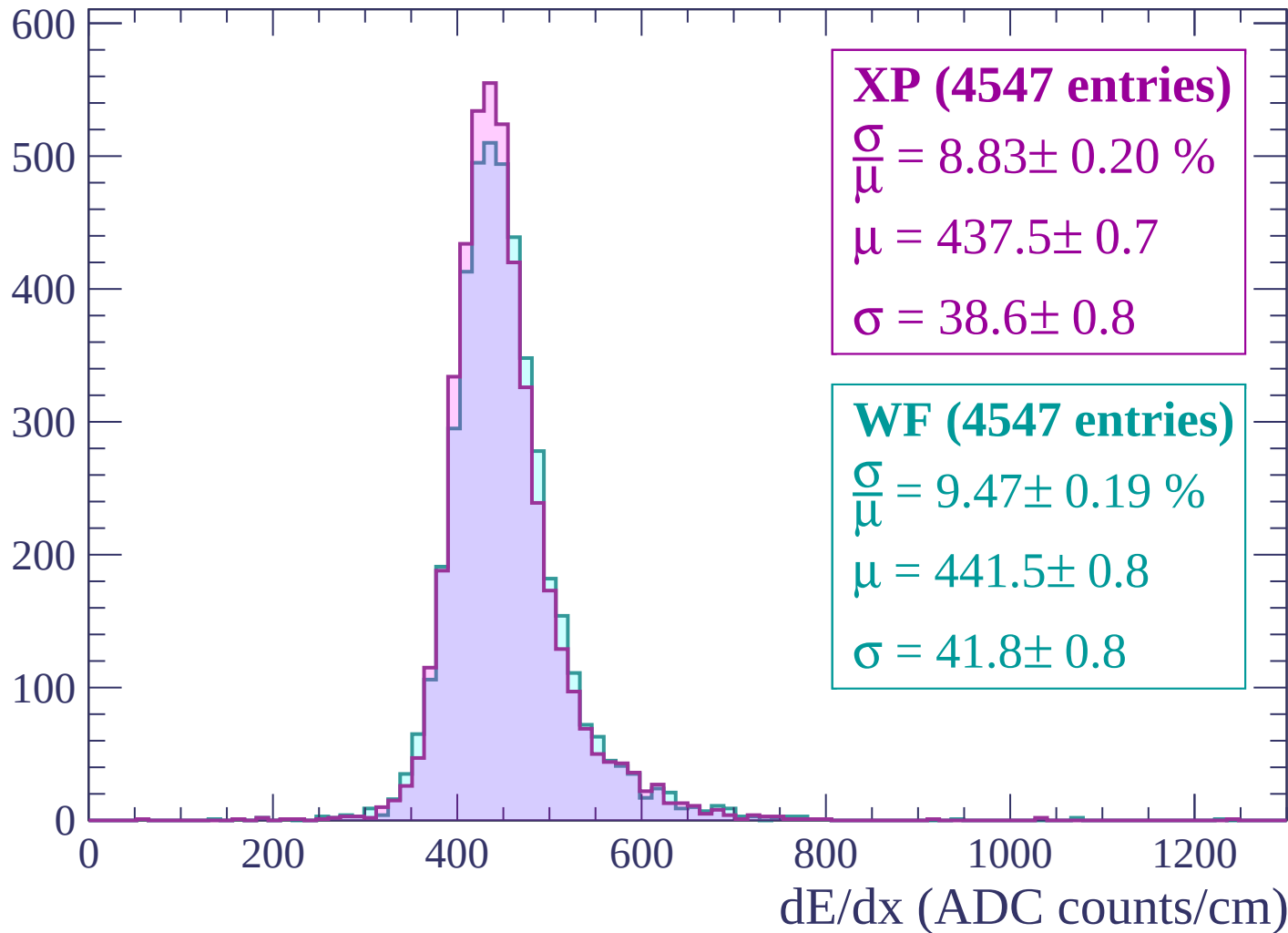
$\mu = 438.9 \pm 0.8$

$\sigma = 41.8 \pm 0.7$

dE/dx (ADC counts/cm)

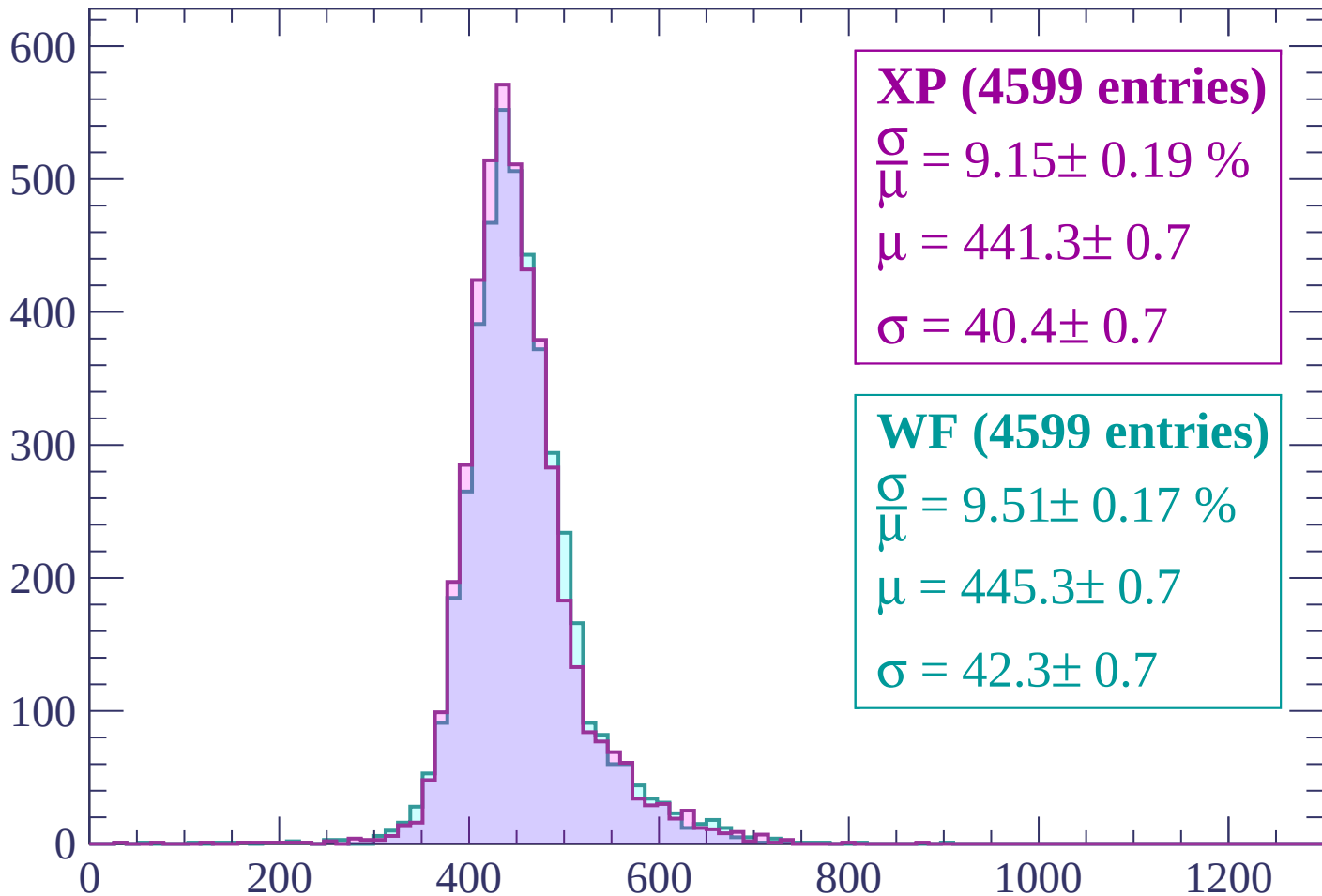
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (4599 entries)

$$\frac{\sigma}{\mu} = 9.15 \pm 0.19 \%$$

$$\mu = 441.3 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

WF (4599 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.17 \%$$

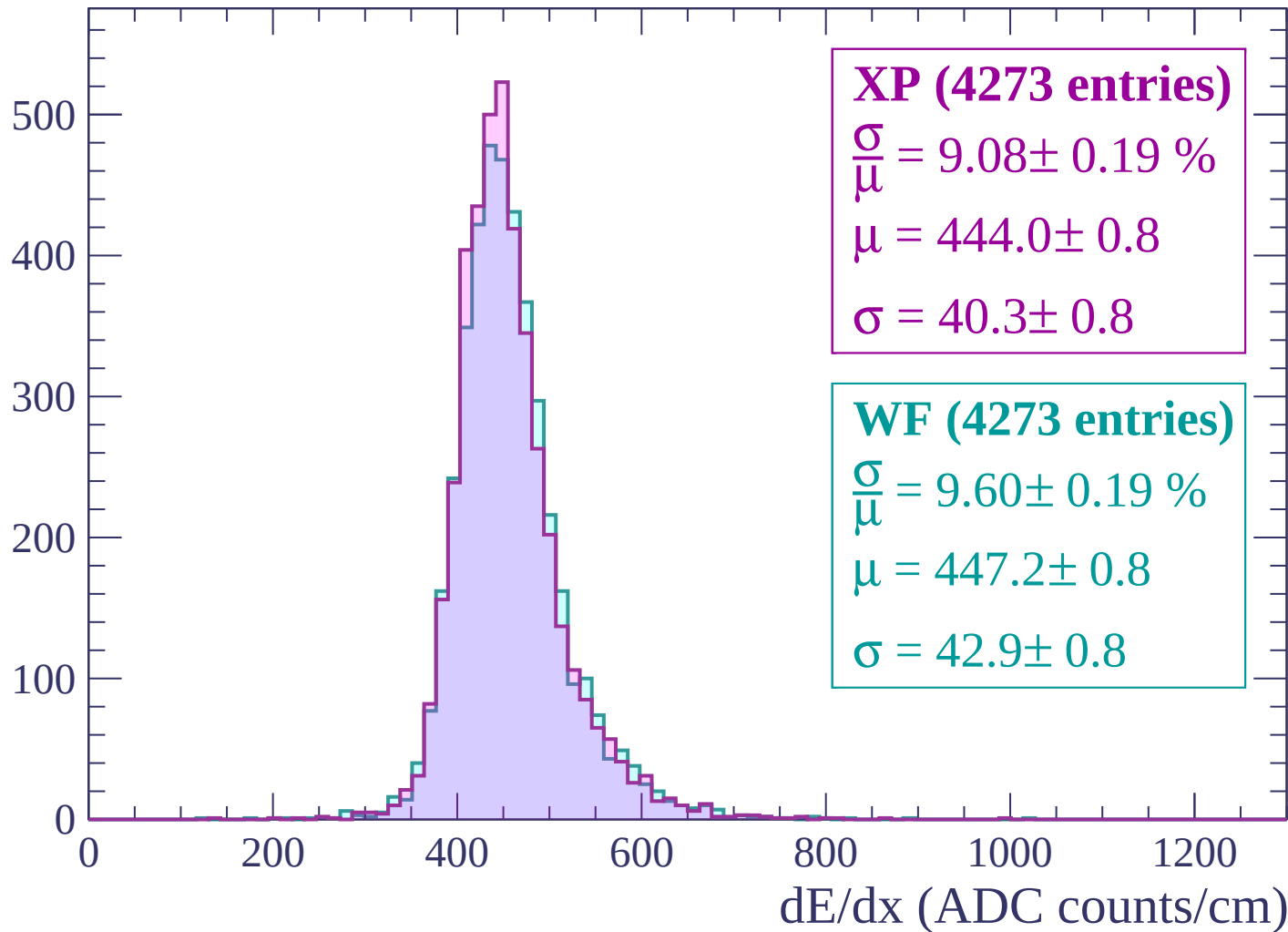
$$\mu = 445.3 \pm 0.7$$

$$\sigma = 42.3 \pm 0.7$$

dE/dx (ADC counts/cm)

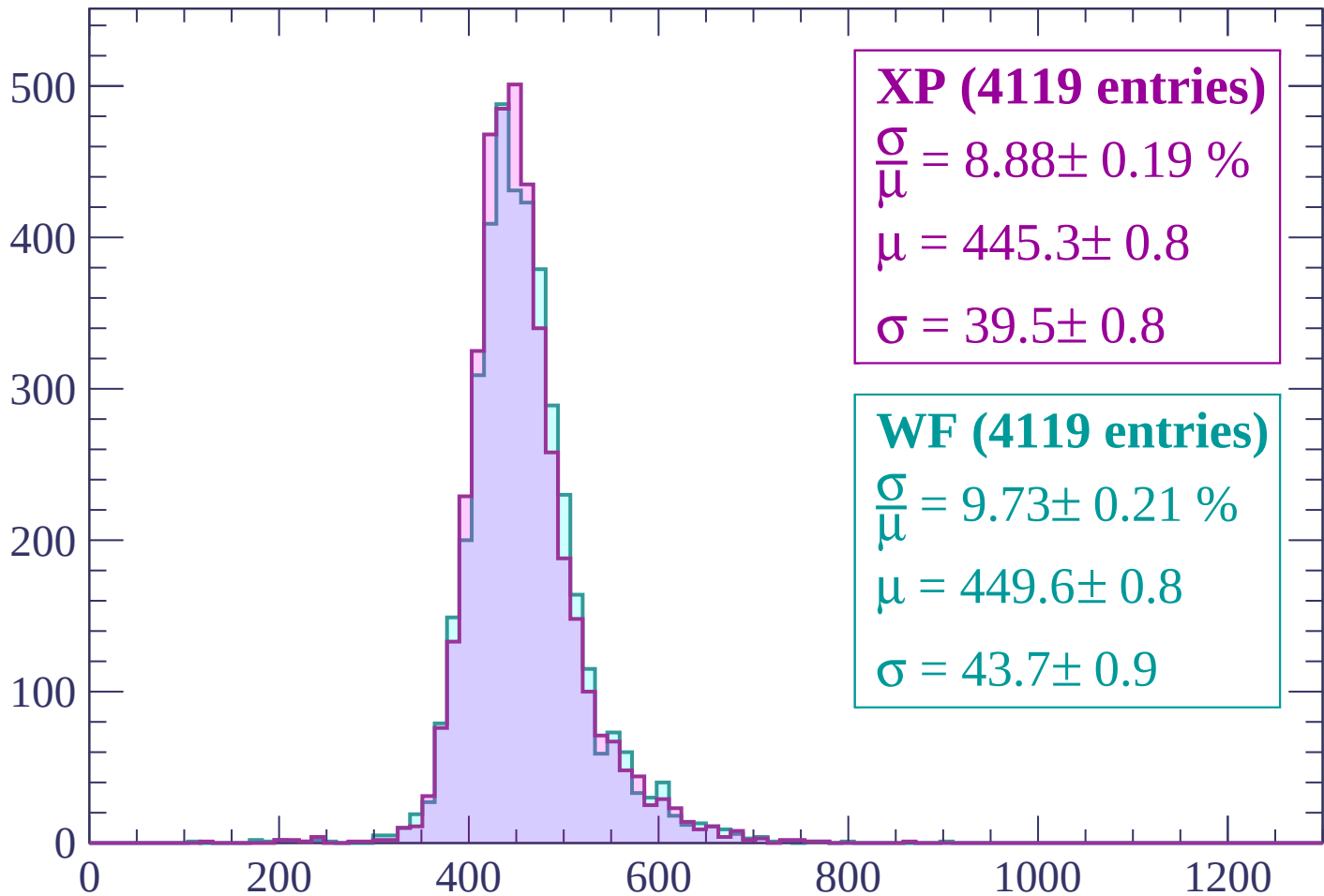
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP (4119 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.19 \%$$

$$\mu = 445.3 \pm 0.8$$

$$\sigma = 39.5 \pm 0.8$$

WF (4119 entries)

$$\frac{\sigma}{\mu} = 9.73 \pm 0.21 \%$$

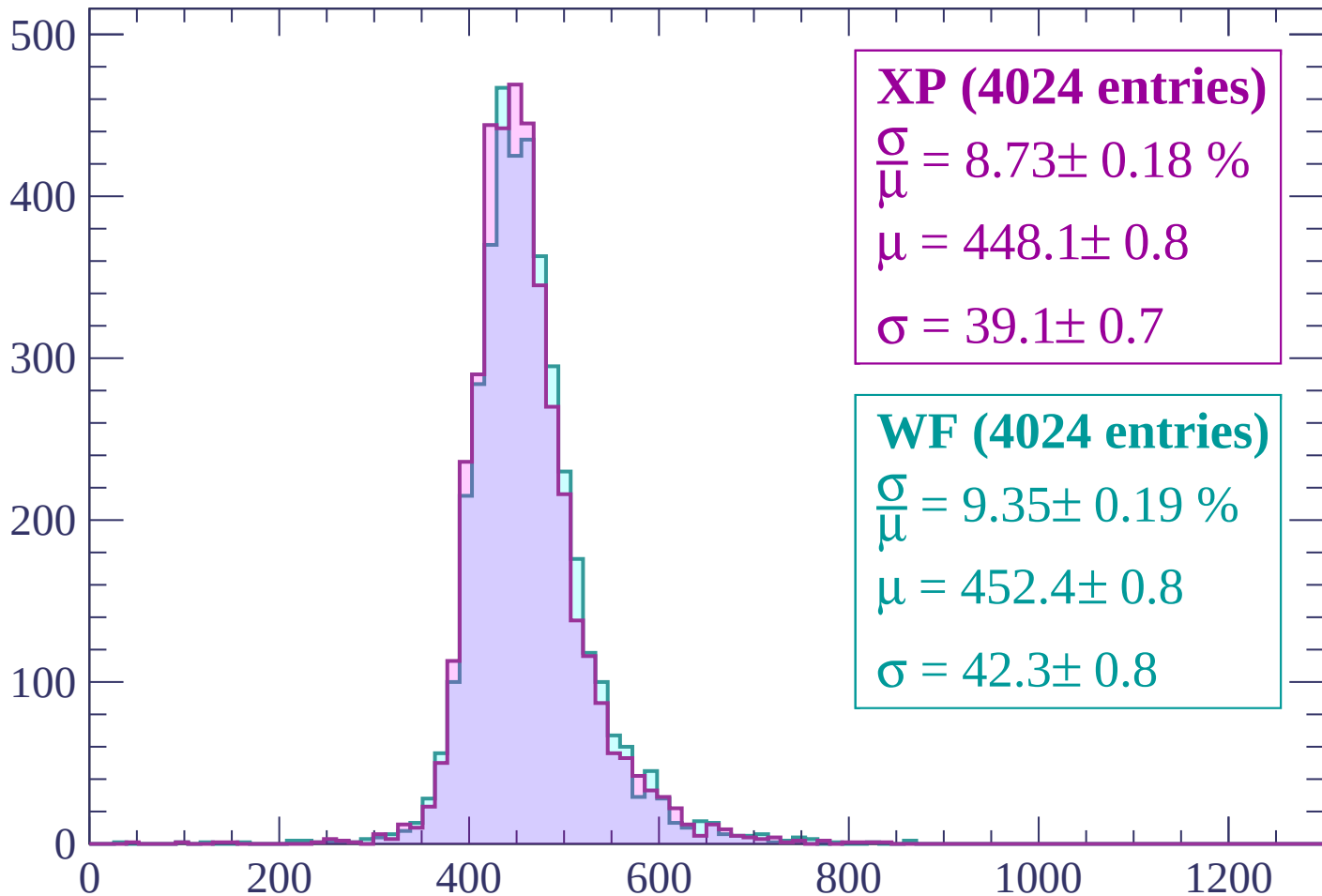
$$\mu = 449.6 \pm 0.8$$

$$\sigma = 43.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (4024 entries)

$$\frac{\sigma}{\mu} = 8.73 \pm 0.18 \%$$

$$\mu = 448.1 \pm 0.8$$

$$\sigma = 39.1 \pm 0.7$$

WF (4024 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.19 \%$$

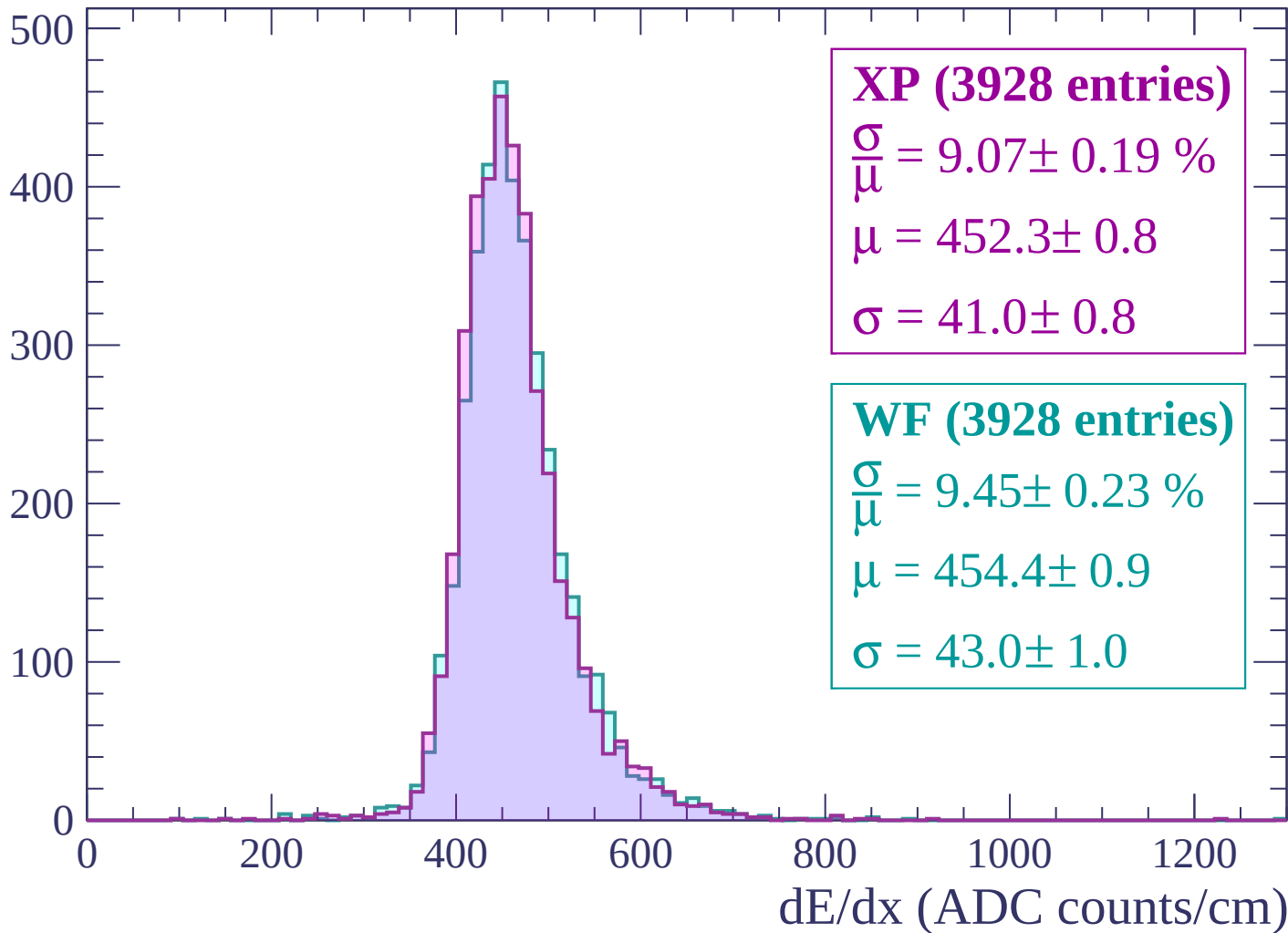
$$\mu = 452.4 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

dE/dx (ADC counts/cm)

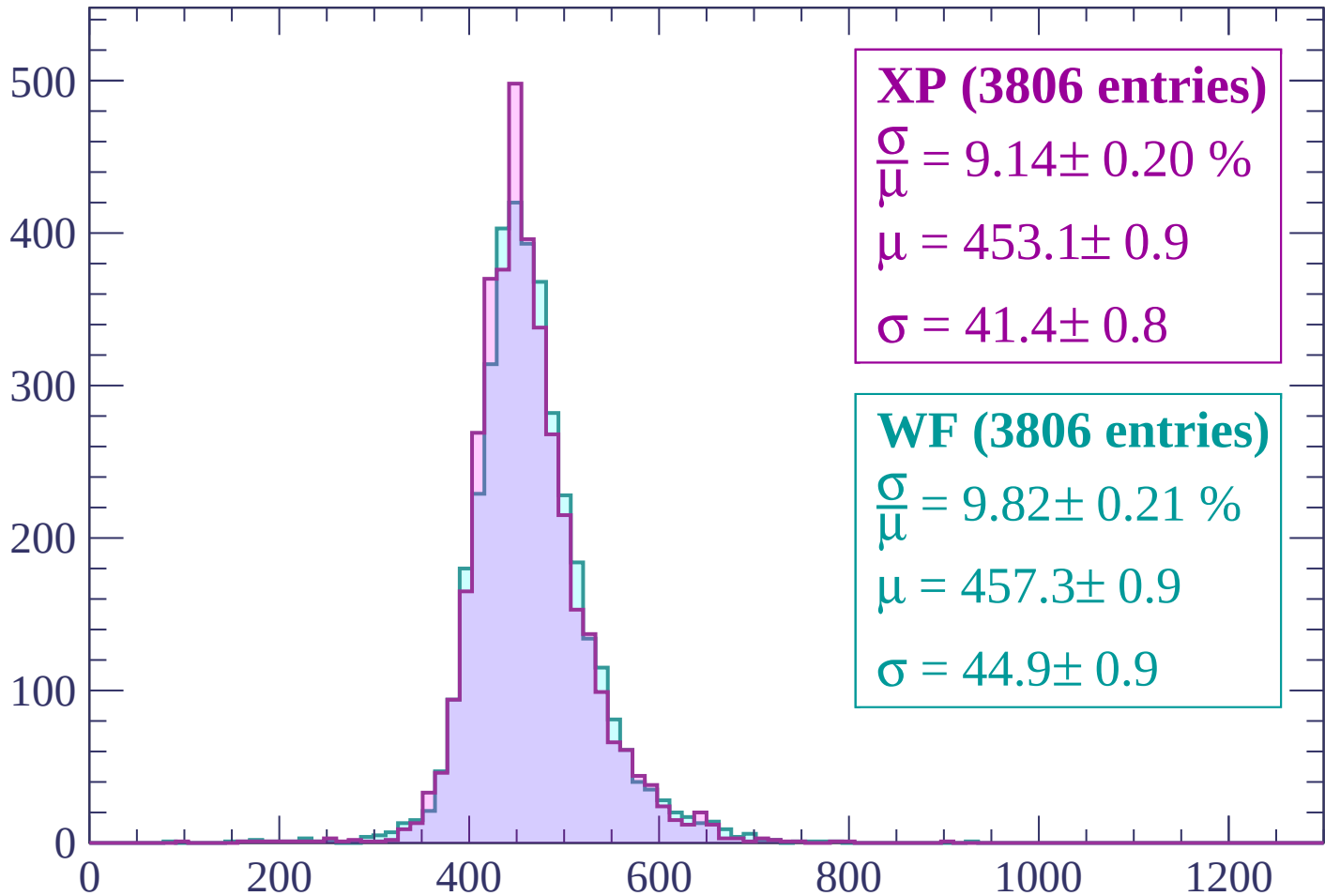
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (3806 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.20 \%$$

$$\mu = 453.1 \pm 0.9$$

$$\sigma = 41.4 \pm 0.8$$

WF (3806 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.21 \%$$

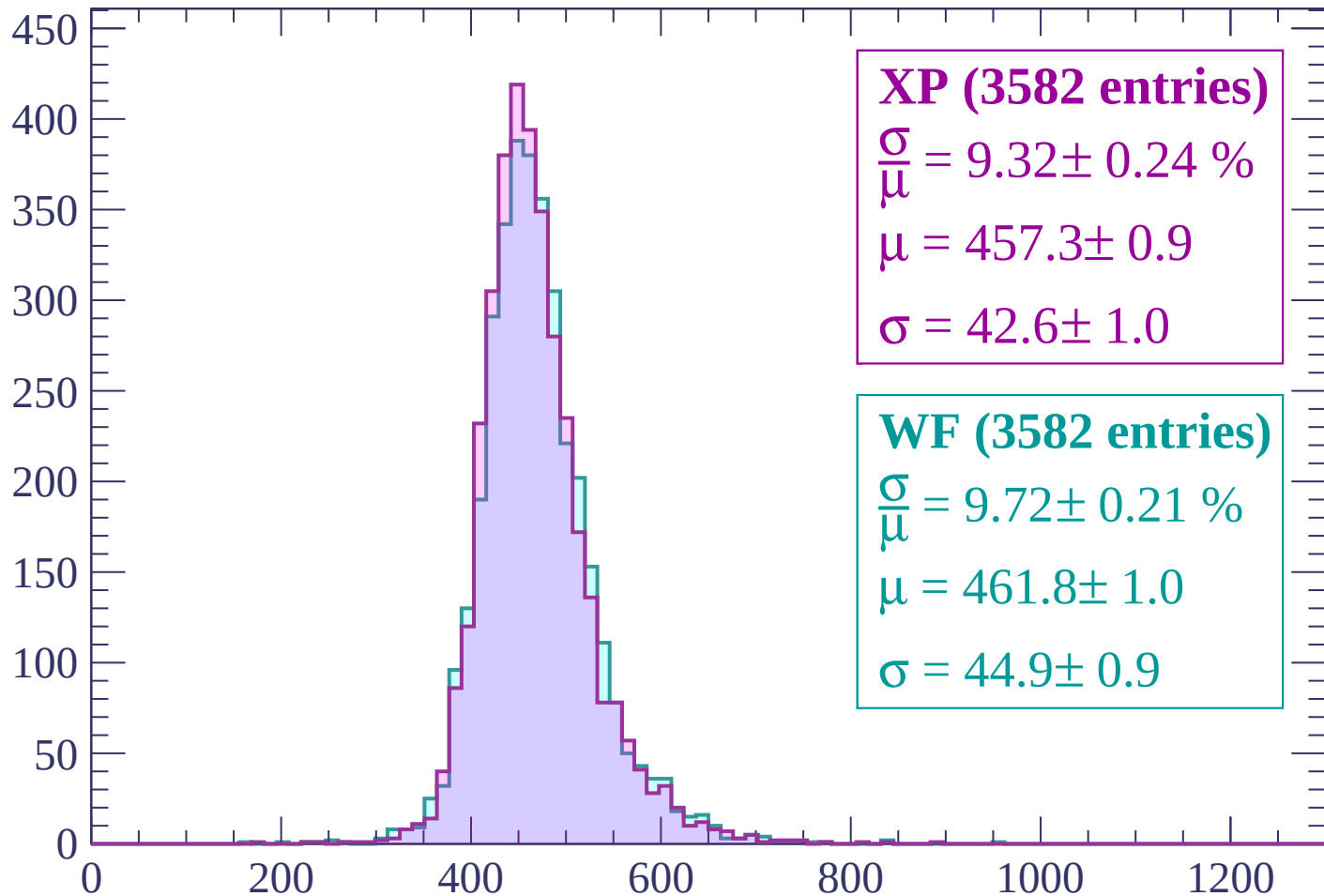
$$\mu = 457.3 \pm 0.9$$

$$\sigma = 44.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (3582 entries)

$$\frac{\sigma}{\mu} = 9.32 \pm 0.24 \%$$

$$\mu = 457.3 \pm 0.9$$

$$\sigma = 42.6 \pm 1.0$$

WF (3582 entries)

$$\frac{\sigma}{\mu} = 9.72 \pm 0.21 \%$$

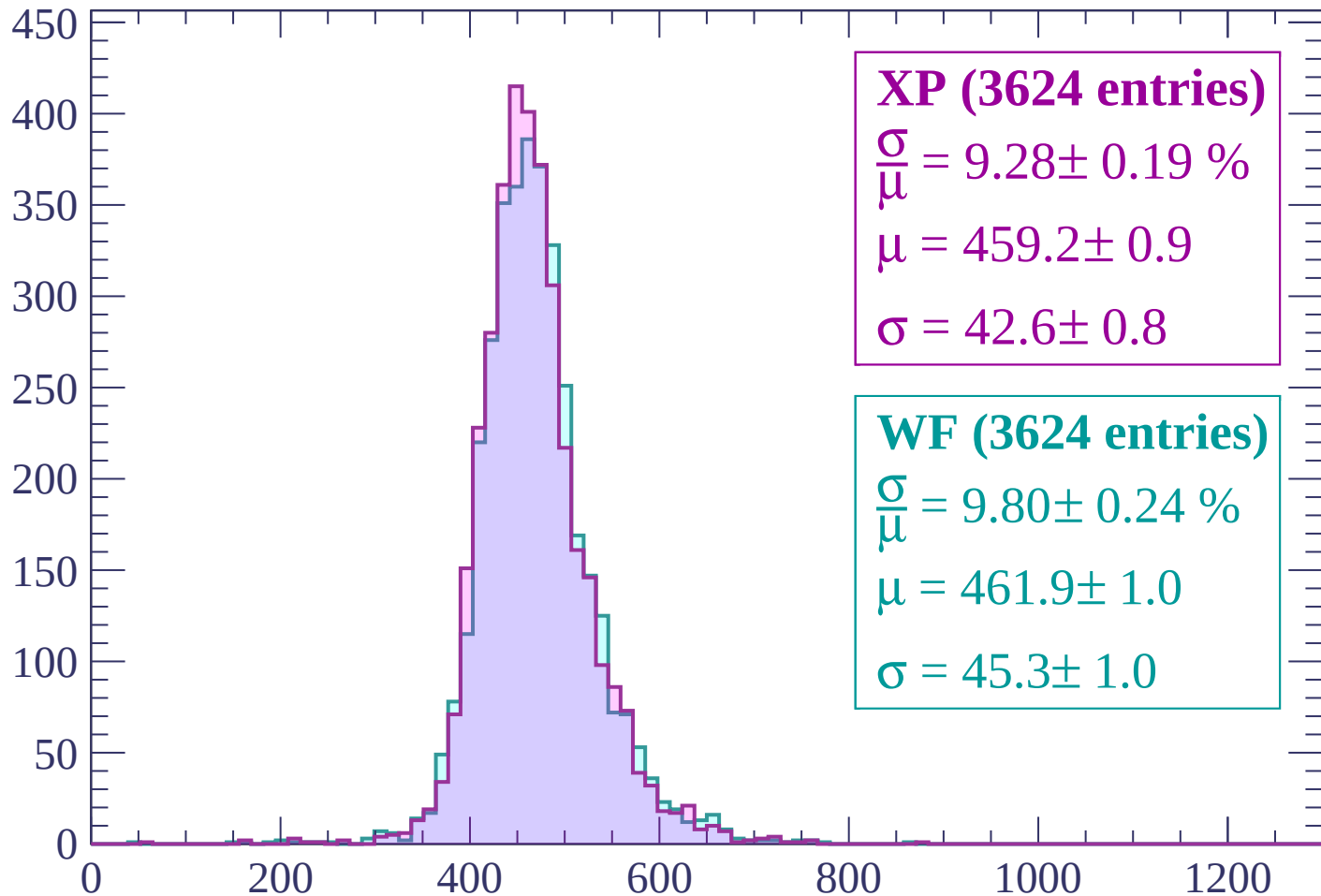
$$\mu = 461.8 \pm 1.0$$

$$\sigma = 44.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (3624 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.19 \%$$

$$\mu = 459.2 \pm 0.9$$

$$\sigma = 42.6 \pm 0.8$$

WF (3624 entries)

$$\frac{\sigma}{\mu} = 9.80 \pm 0.24 \%$$

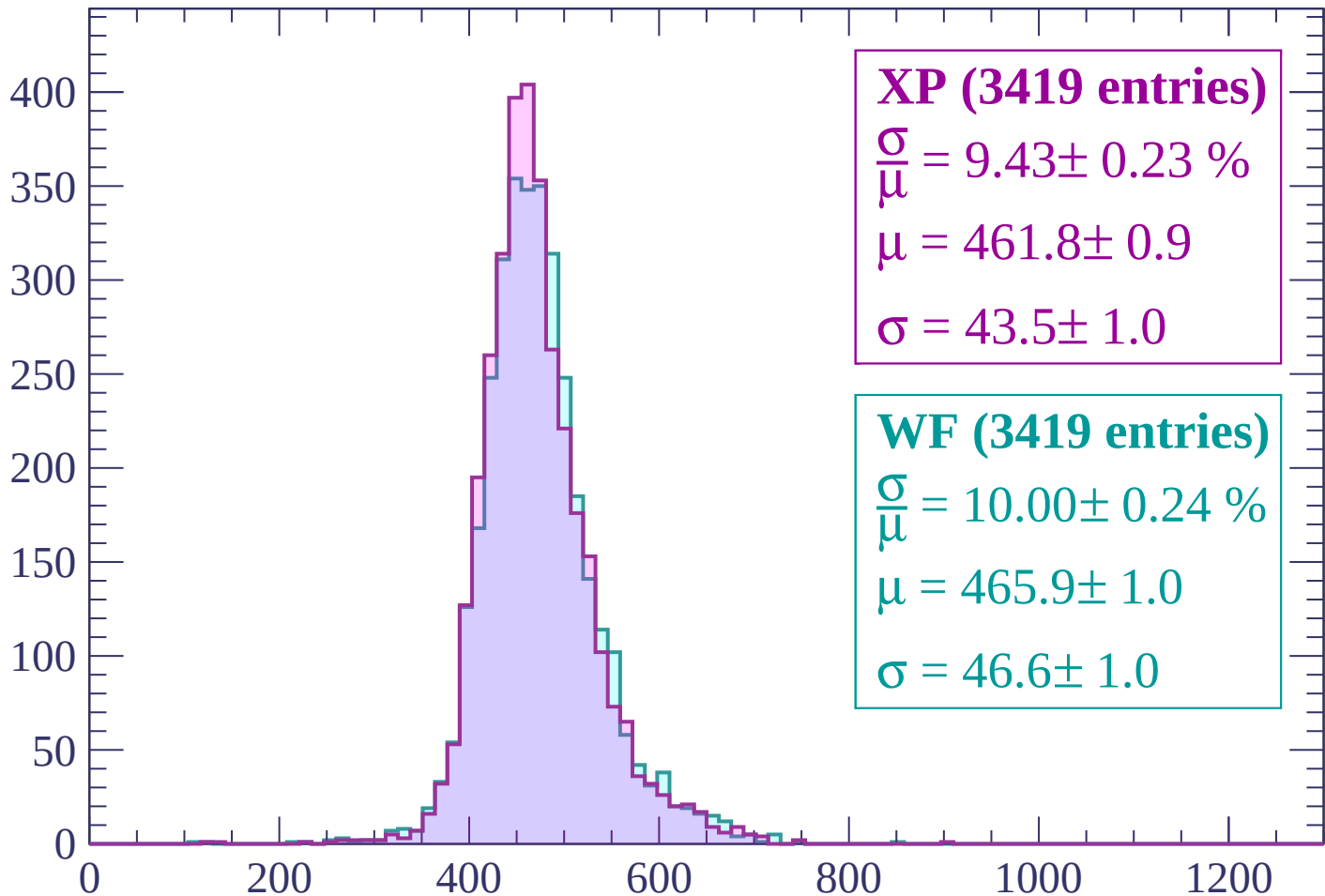
$$\mu = 461.9 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (3419 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.23 \%$$

$$\mu = 461.8 \pm 0.9$$

$$\sigma = 43.5 \pm 1.0$$

WF (3419 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.24 \%$$

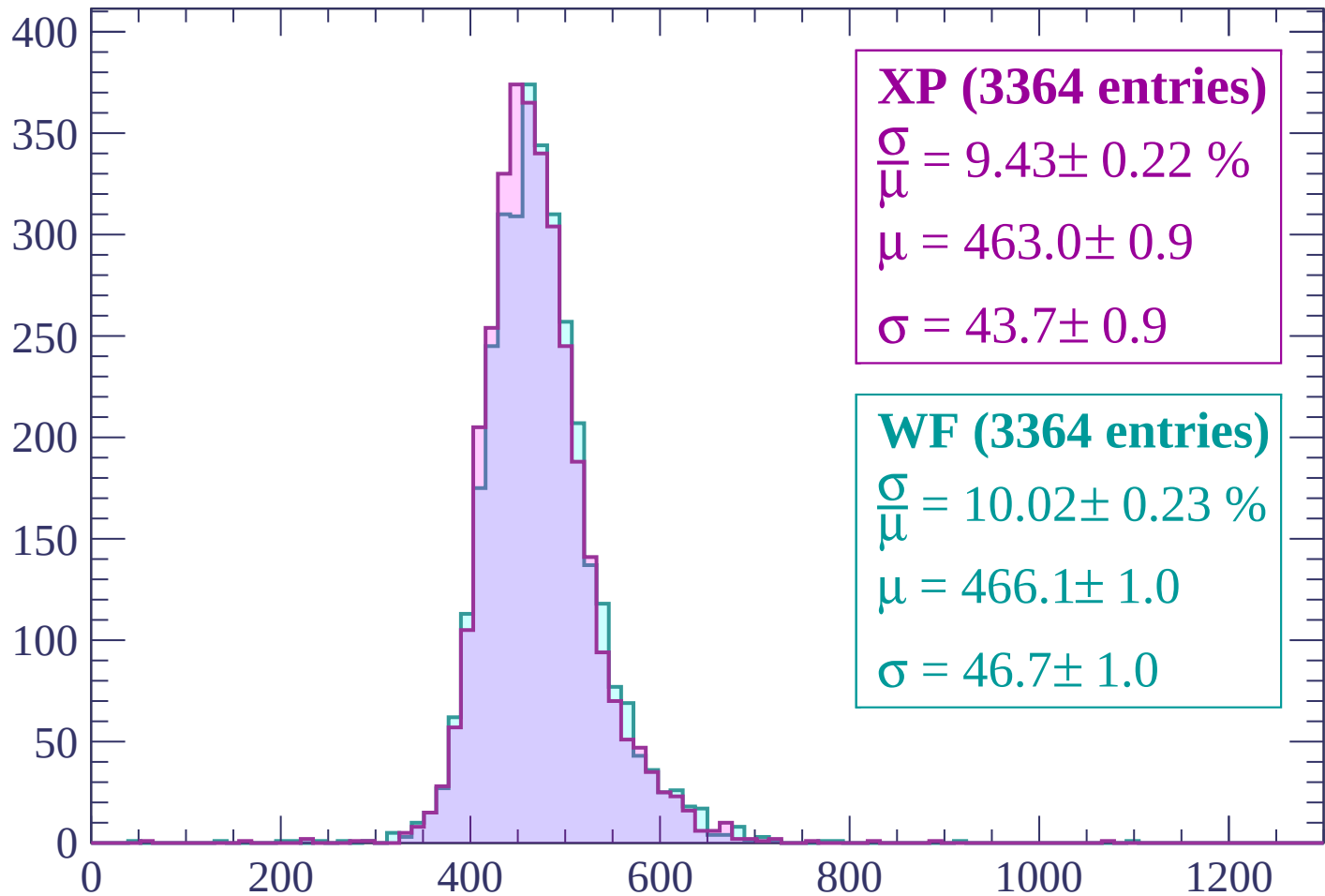
$$\mu = 465.9 \pm 1.0$$

$$\sigma = 46.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (3364 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.22 \%$$

$$\mu = 463.0 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

WF (3364 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.23 \%$$

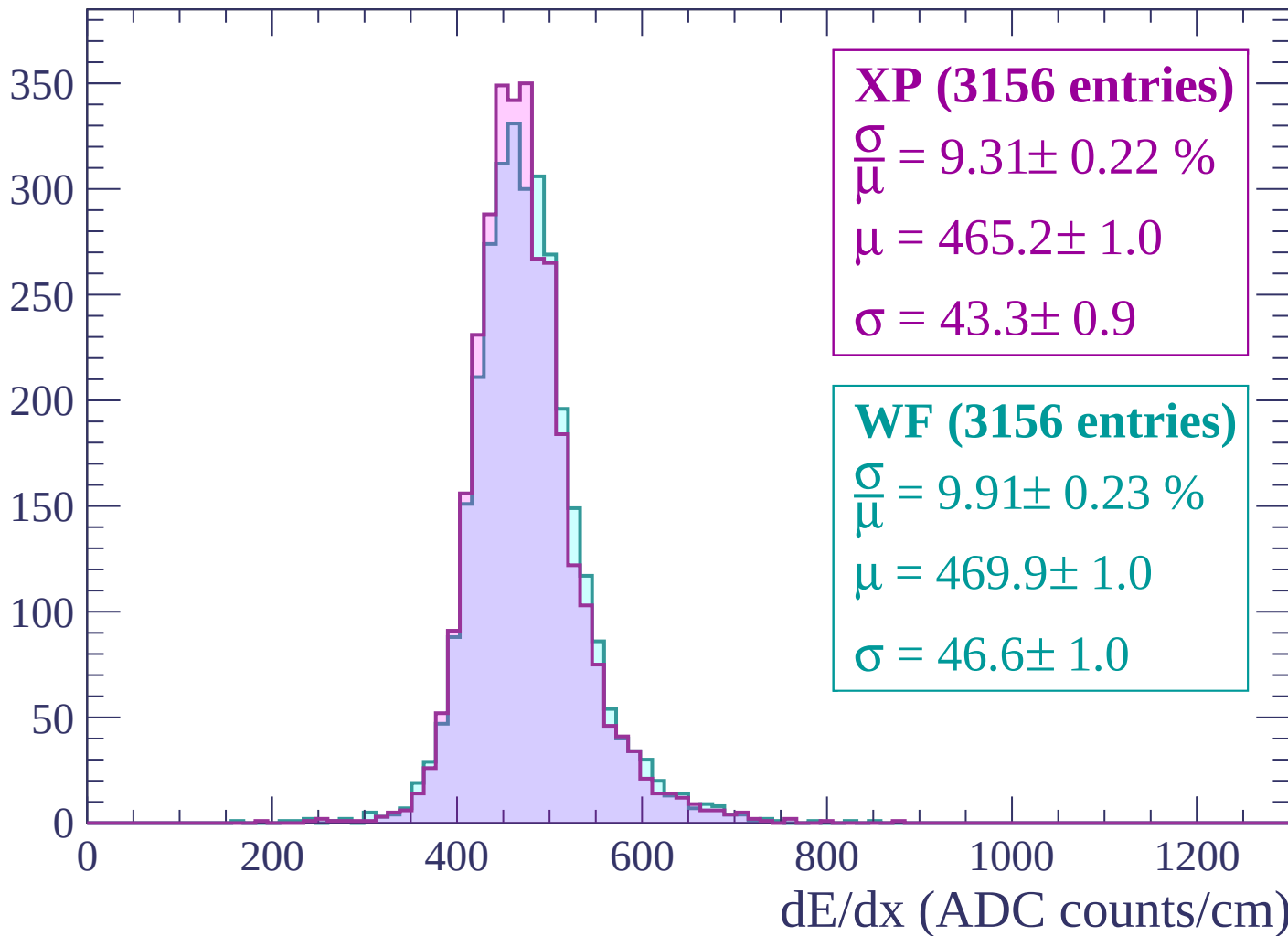
$$\mu = 466.1 \pm 1.0$$

$$\sigma = 46.7 \pm 1.0$$

dE/dx (ADC counts/cm)

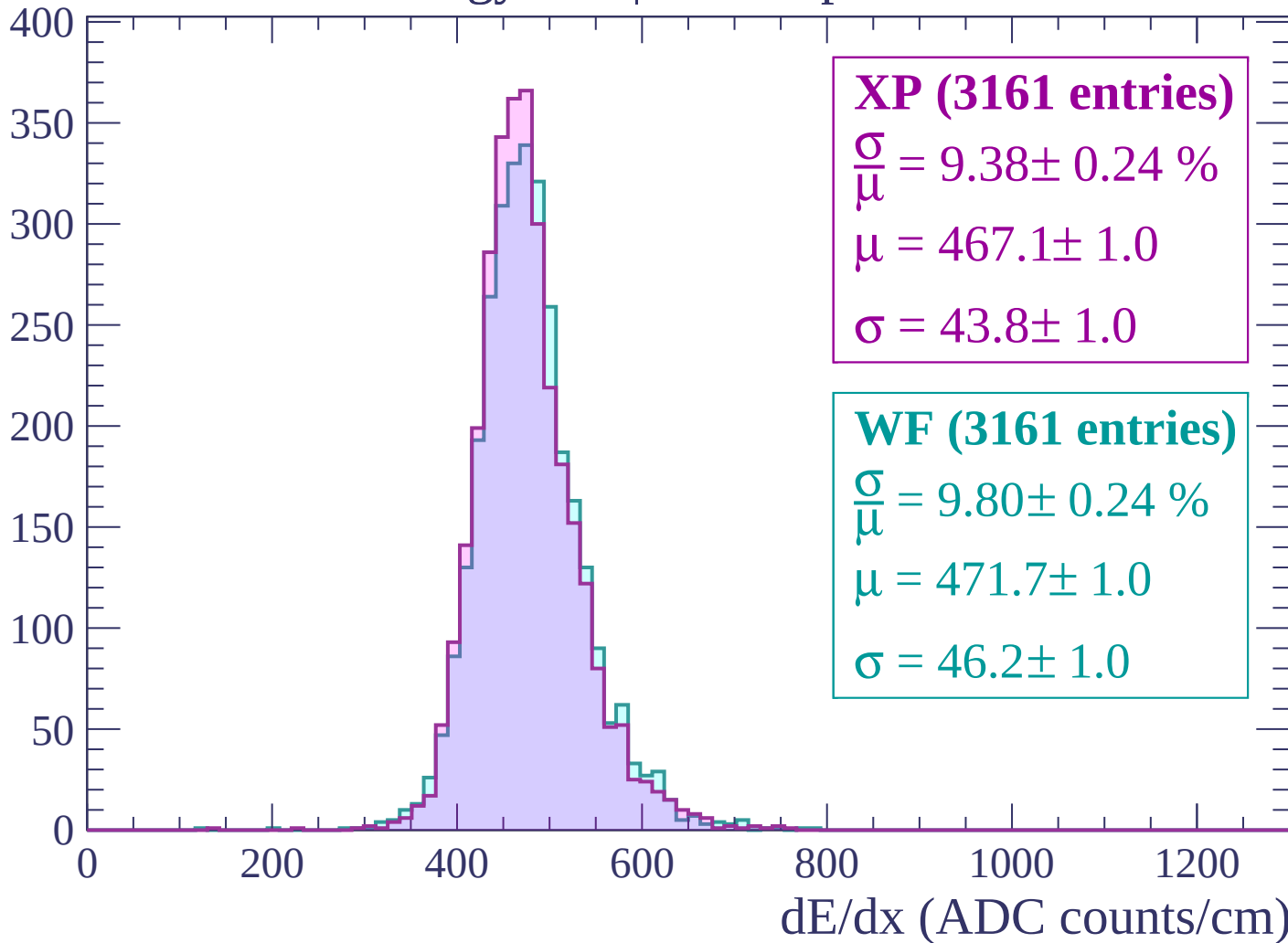
Energy loss | $1800 < p < 1860$

Count



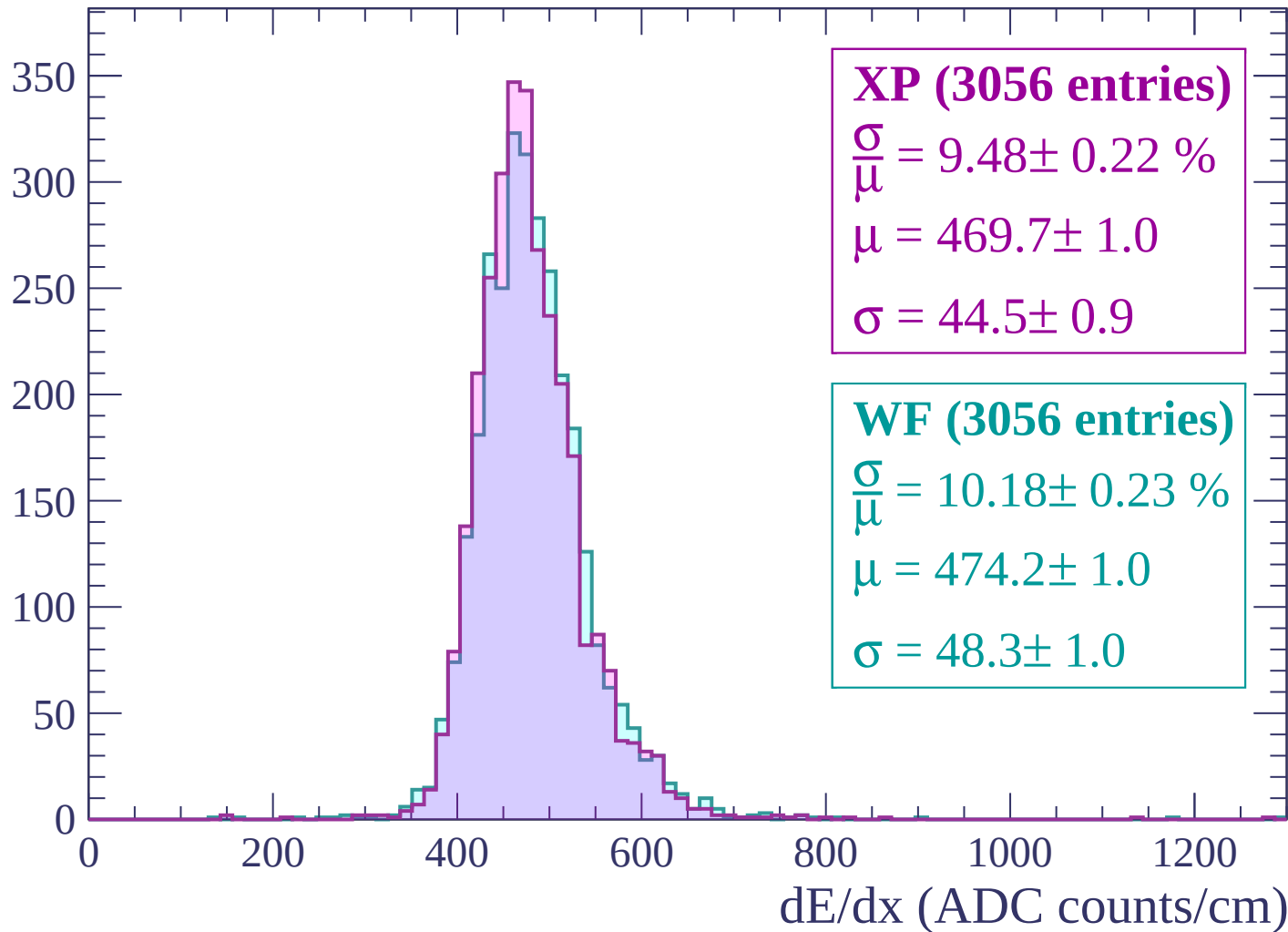
Energy loss | $1860 < p < 1920$

Count



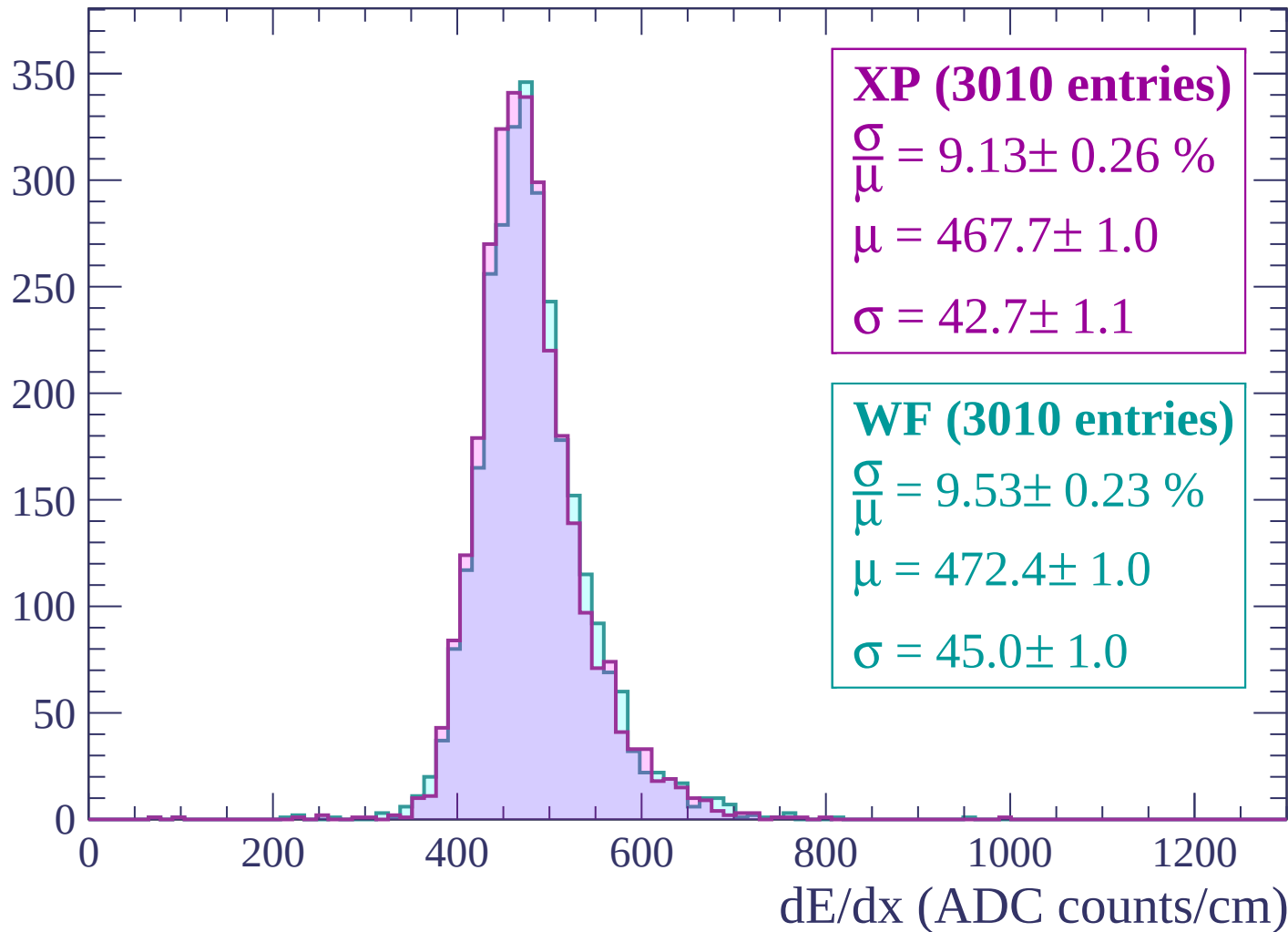
Energy loss | $1920 < p < 1980$

Count



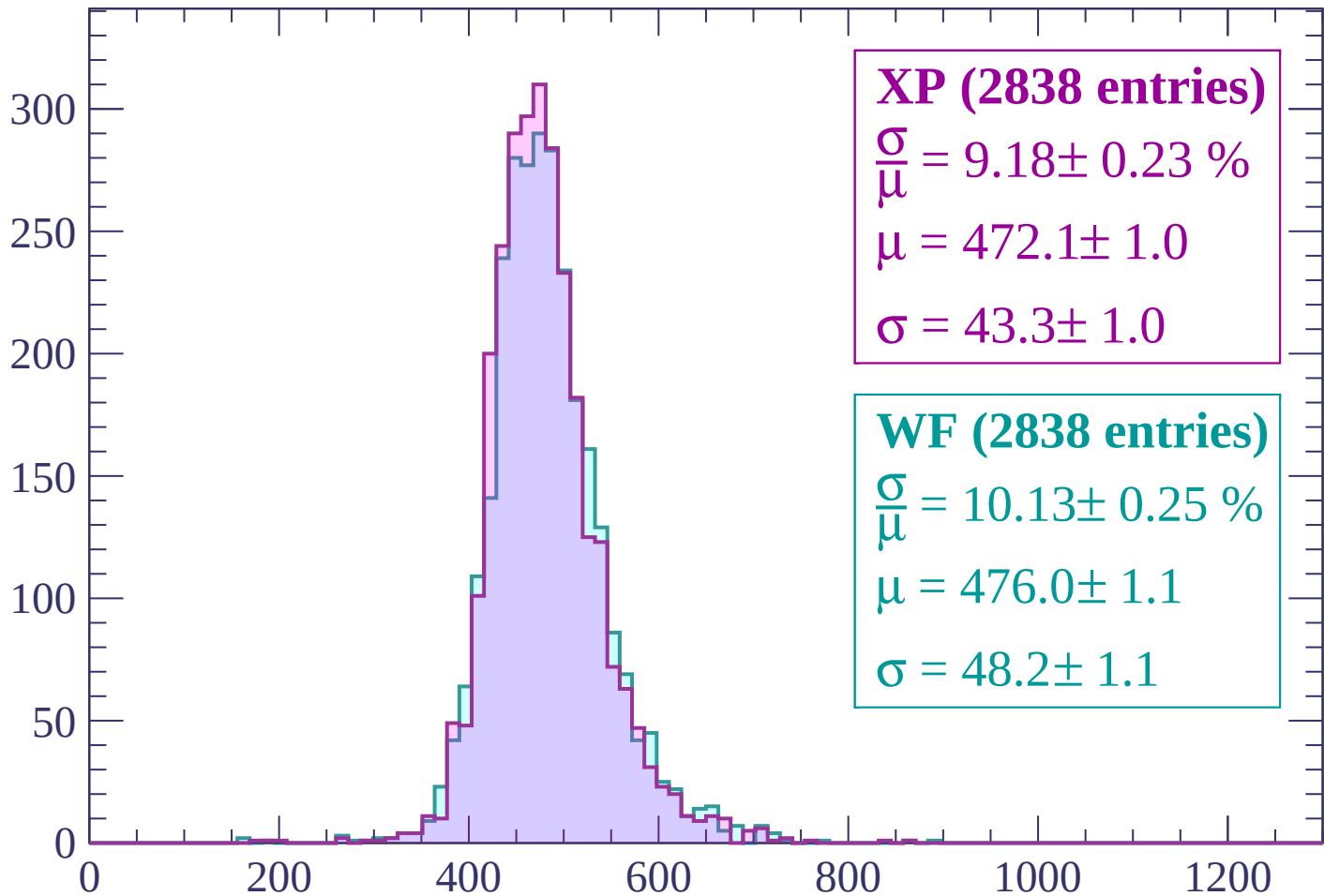
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (2838 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.23 \%$$

$$\mu = 472.1 \pm 1.0$$

$$\sigma = 43.3 \pm 1.0$$

WF (2838 entries)

$$\frac{\sigma}{\mu} = 10.13 \pm 0.25 \%$$

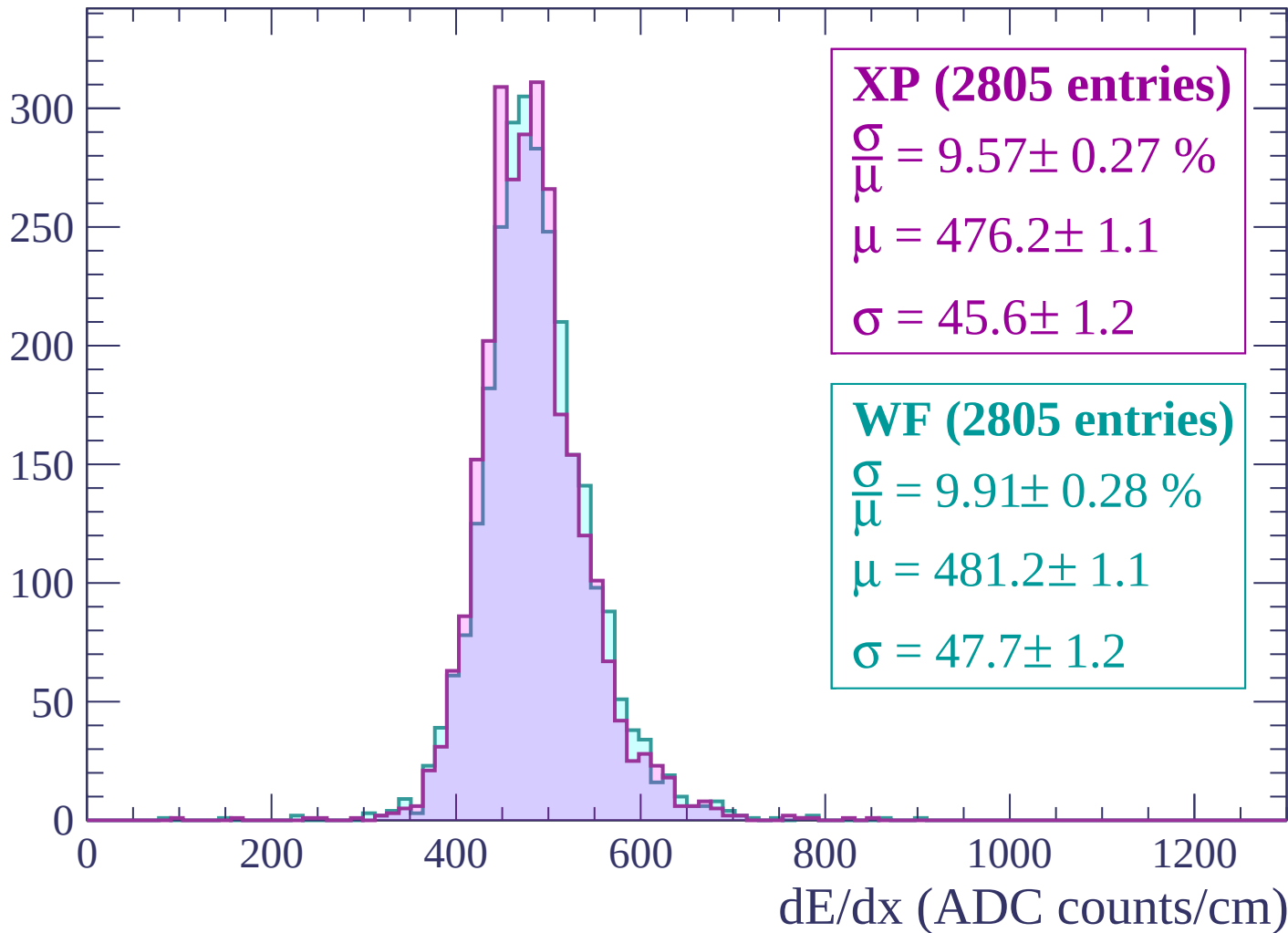
$$\mu = 476.0 \pm 1.1$$

$$\sigma = 48.2 \pm 1.1$$

dE/dx (ADC counts/cm)

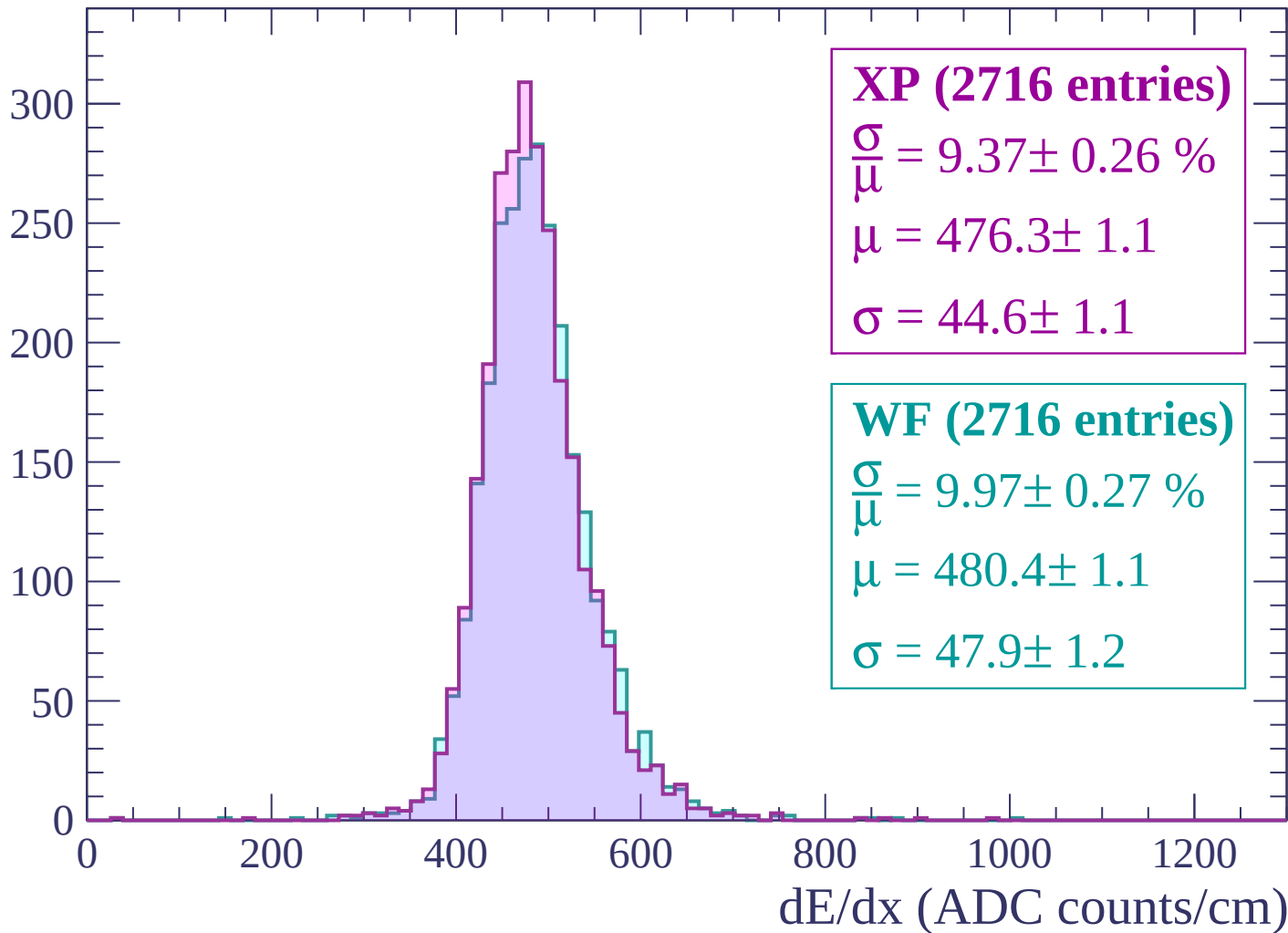
Energy loss | $2100 < p < 2160$

Count



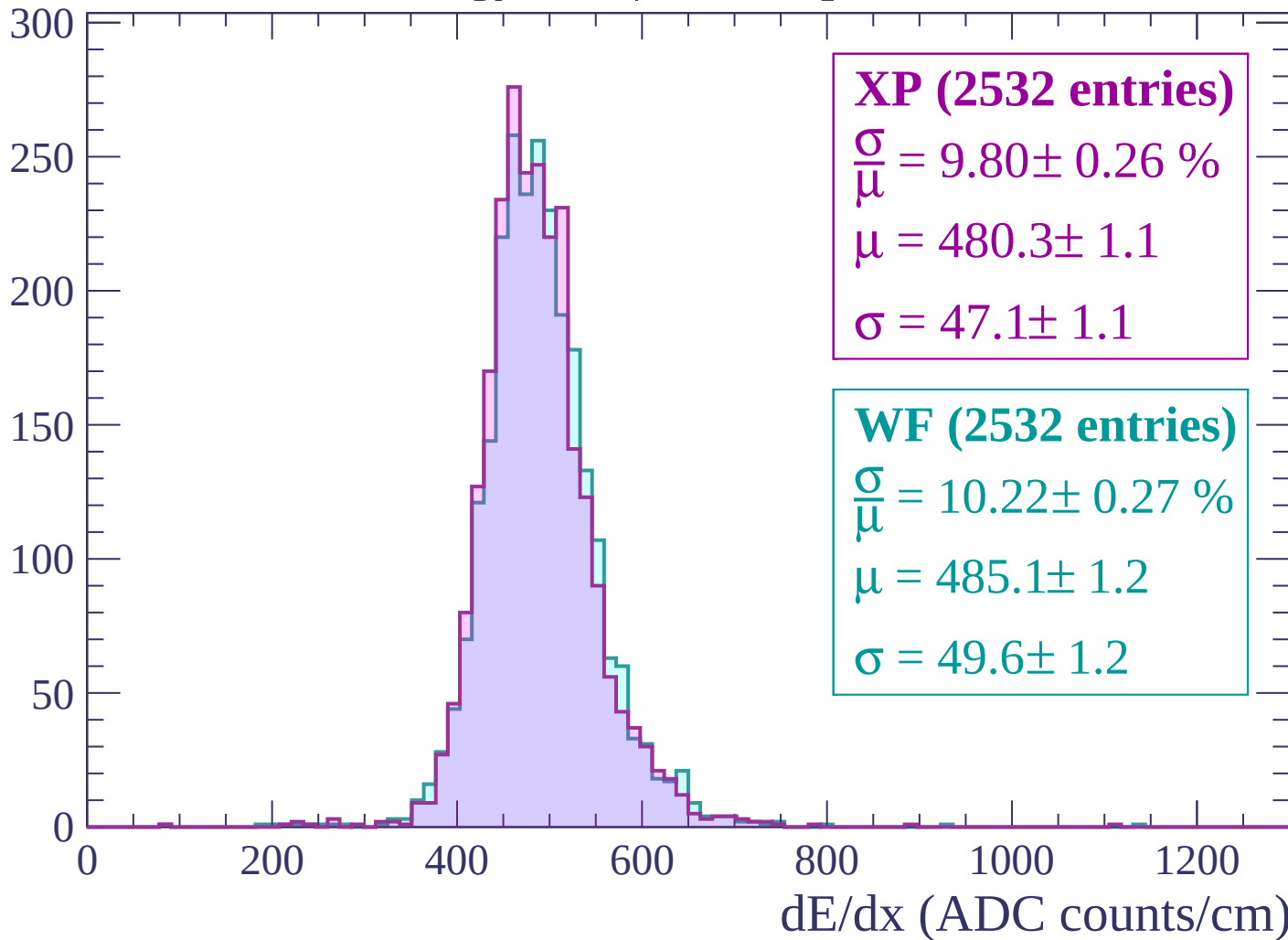
Energy loss | $2160 < p < 2220$

Count



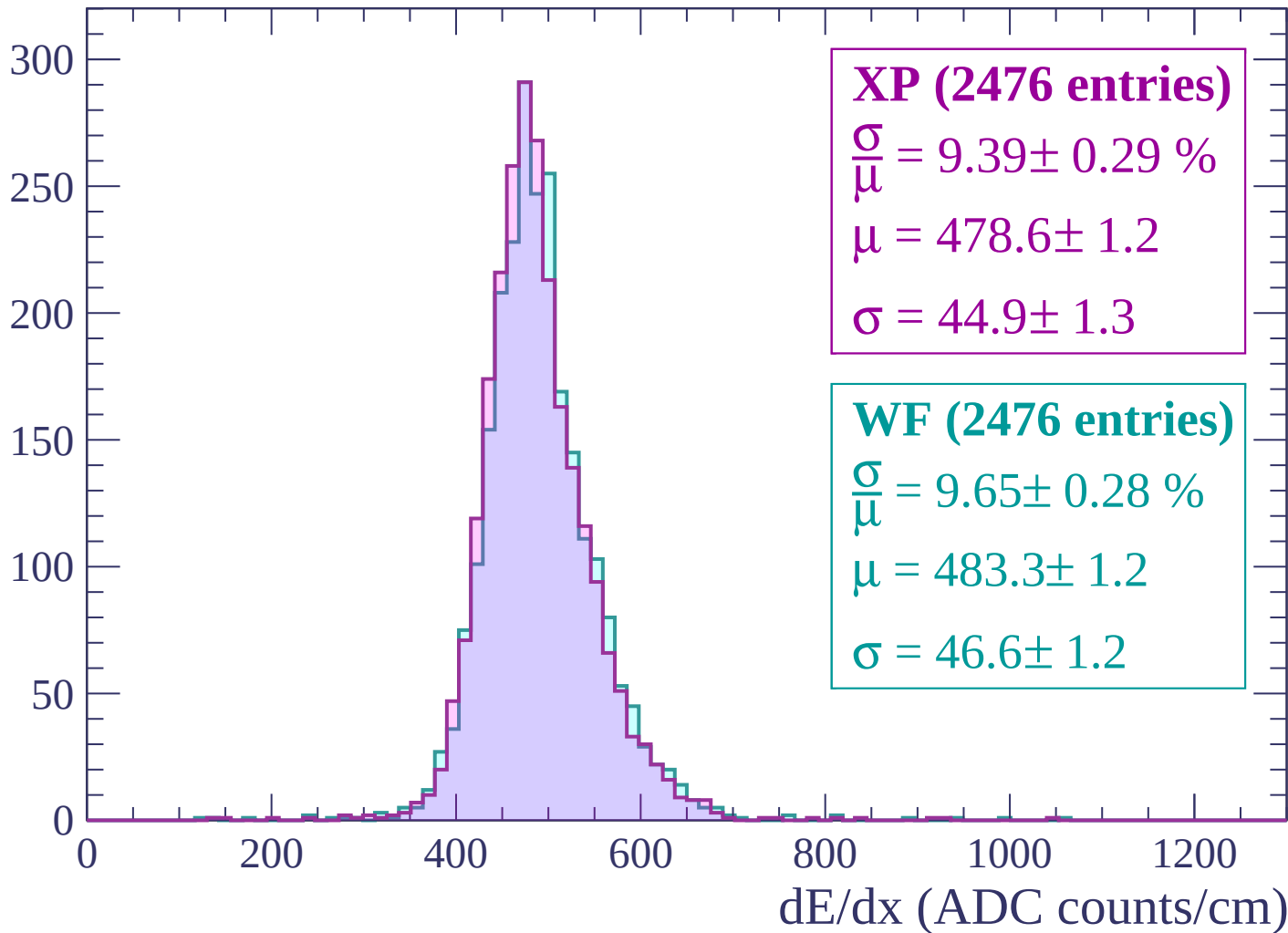
Energy loss | $2220 < p < 2280$

Count



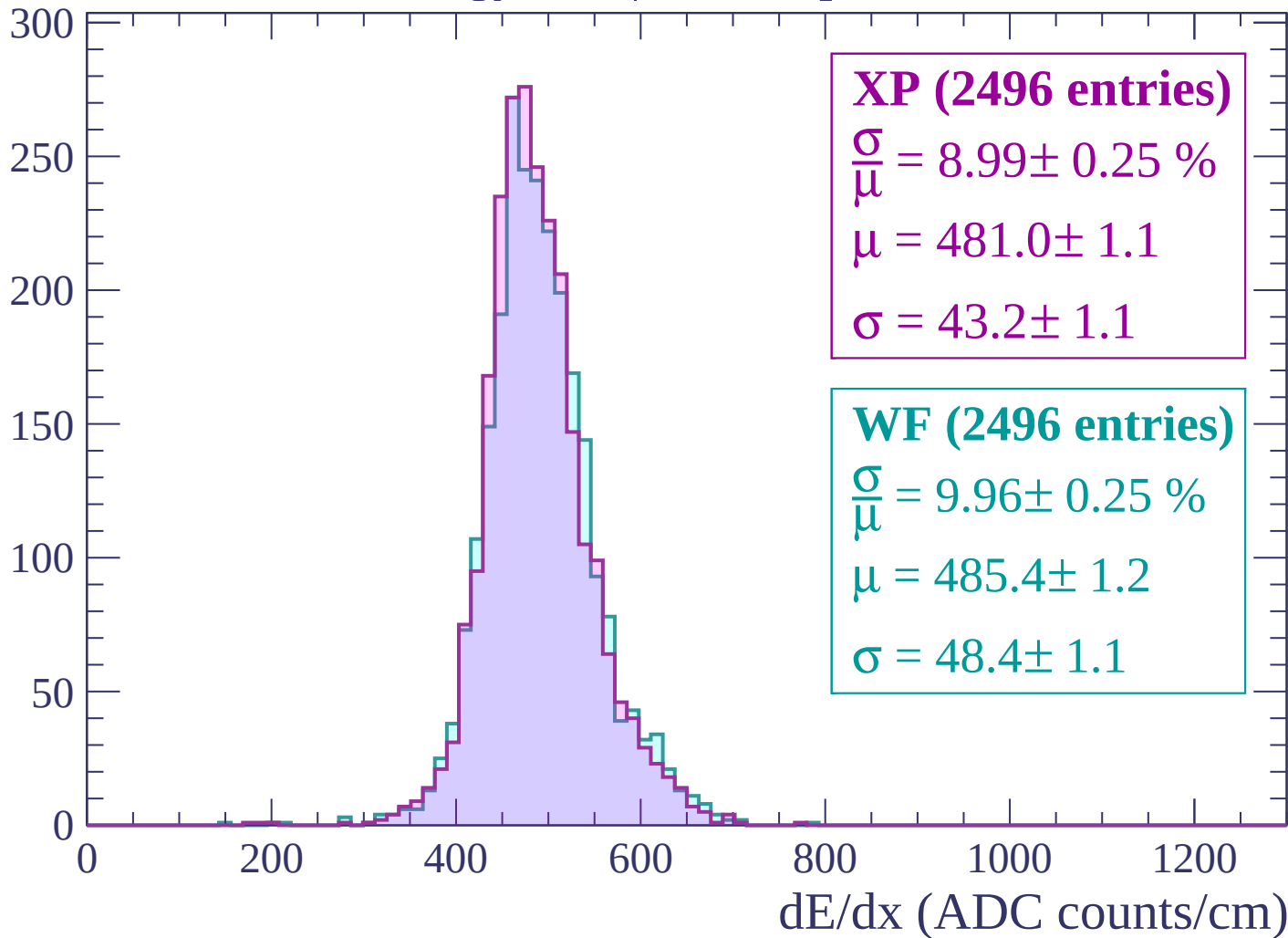
Energy loss | $2280 < p < 2340$

Count



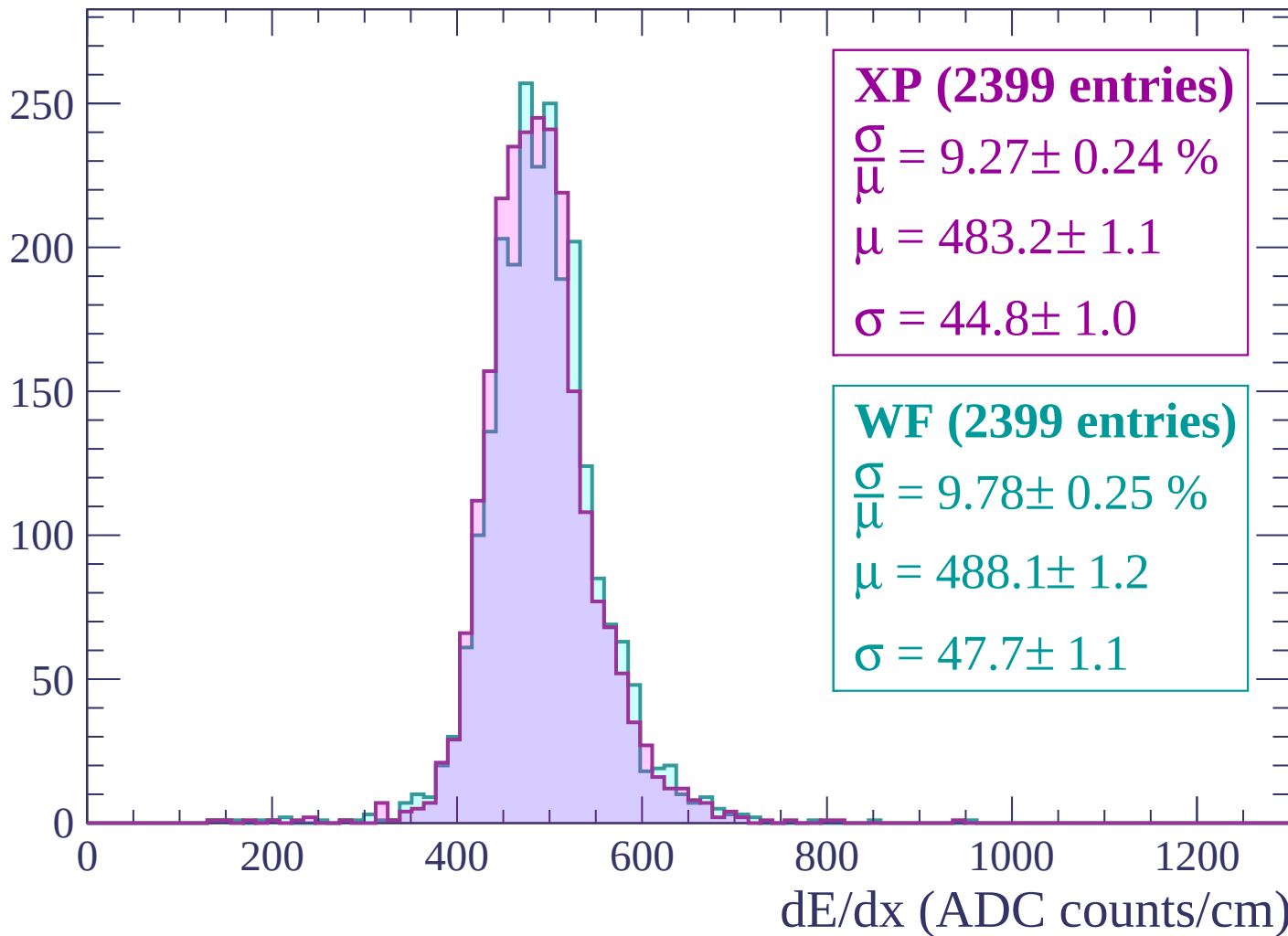
Energy loss | $2340 < p < 2400$

Count



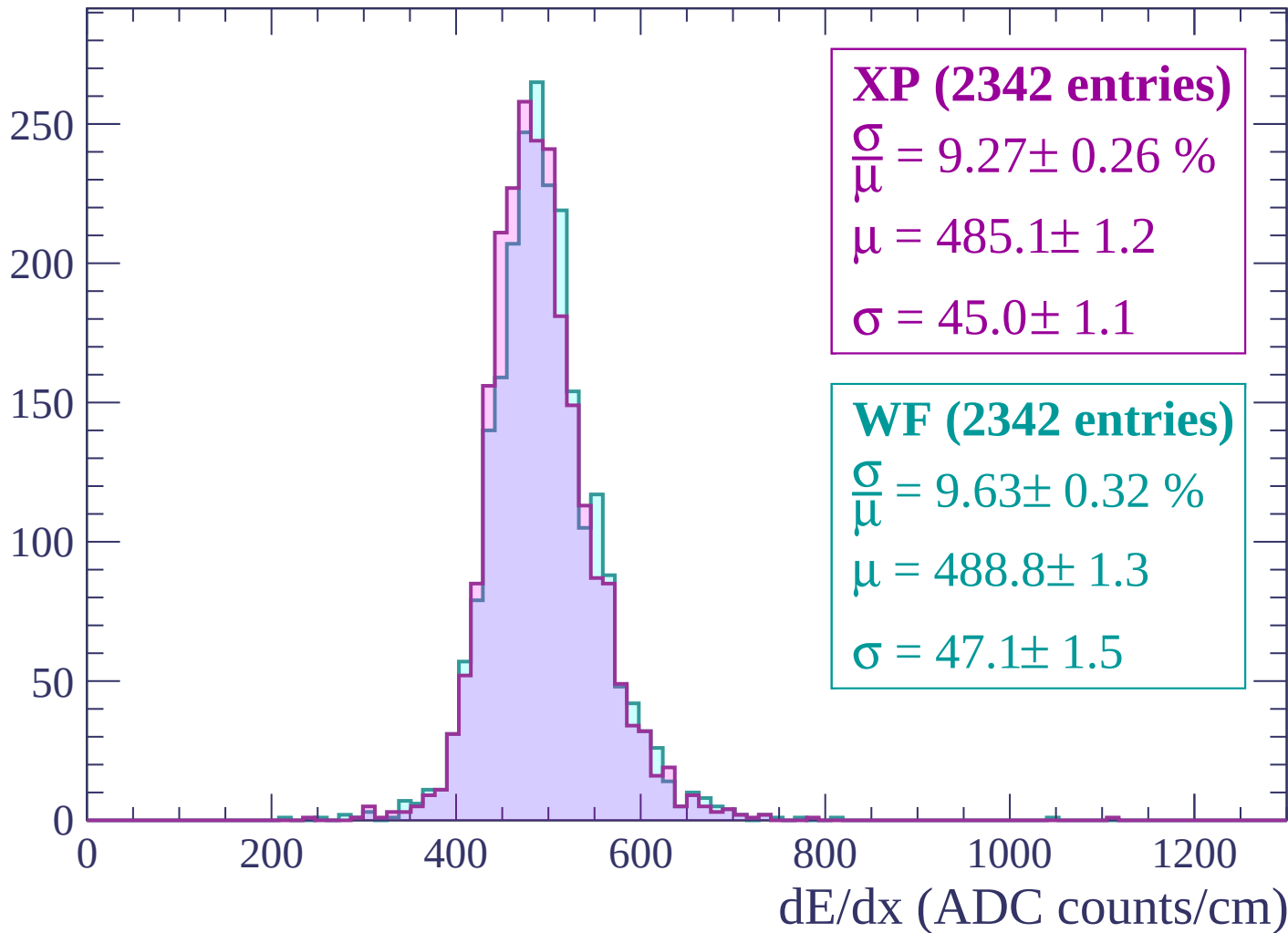
Energy loss | $2400 < p < 2460$

Count



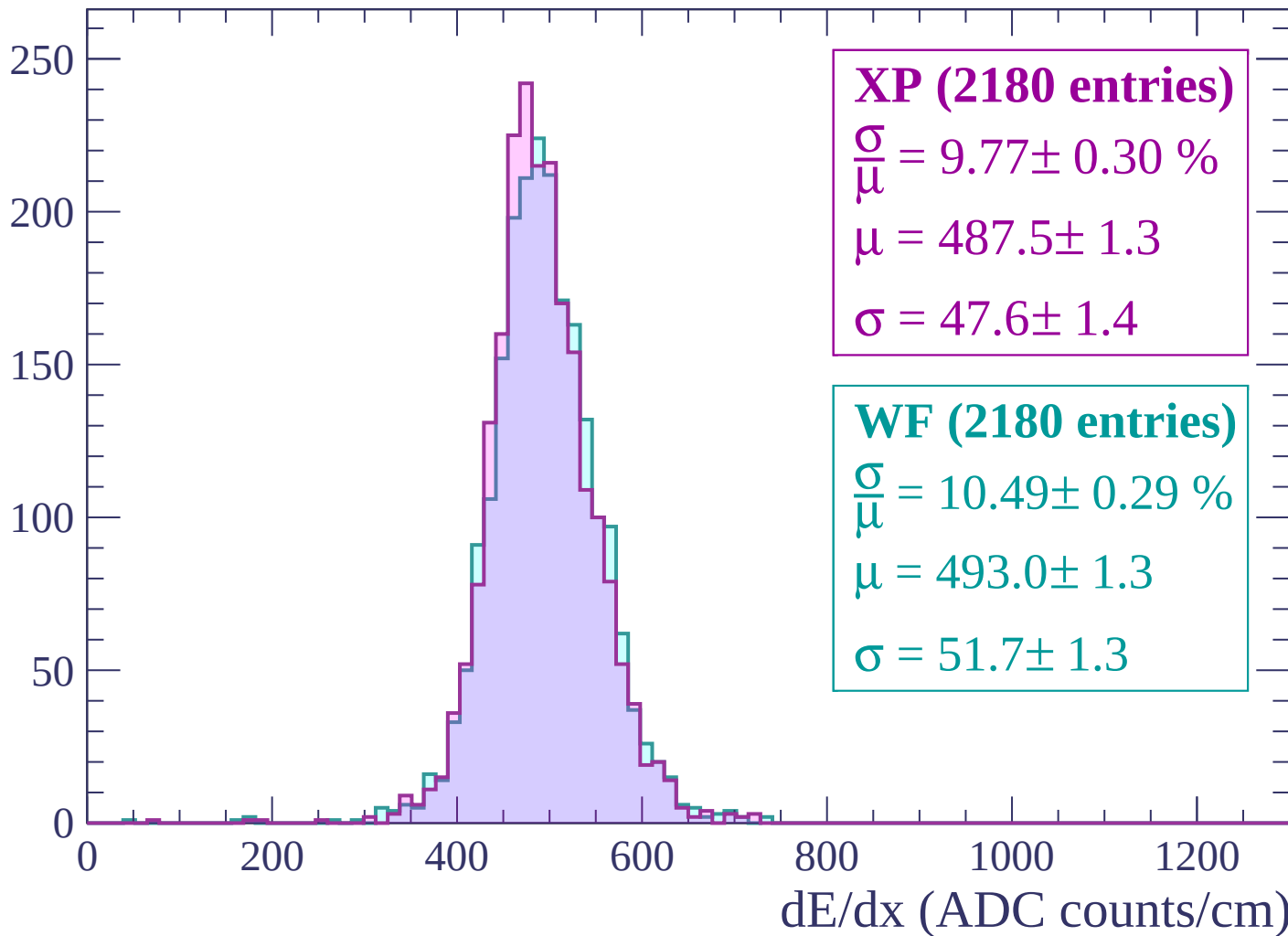
Energy loss | $2460 < p < 2520$

Count



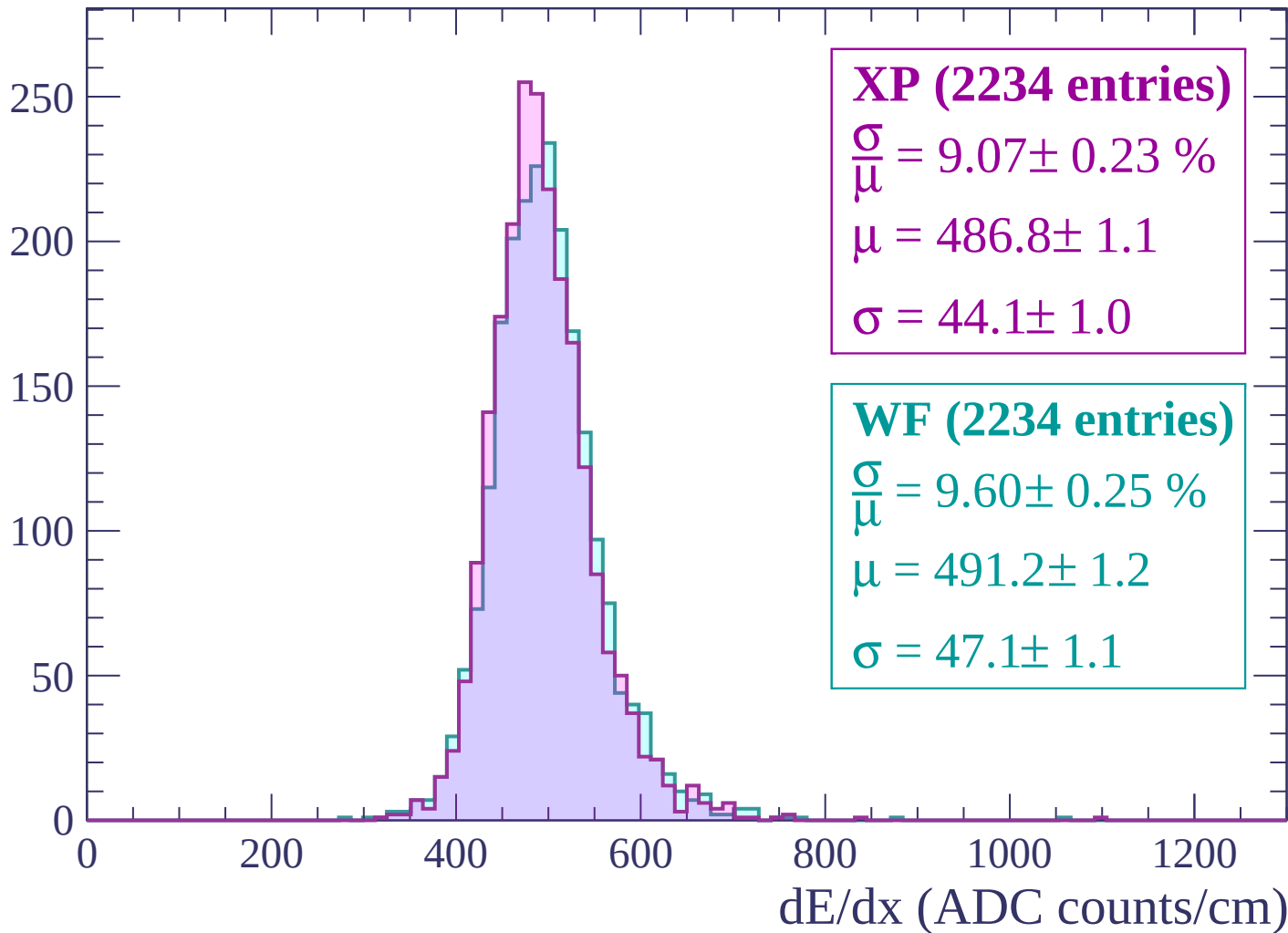
Energy loss | $2520 < p < 2580$

Count



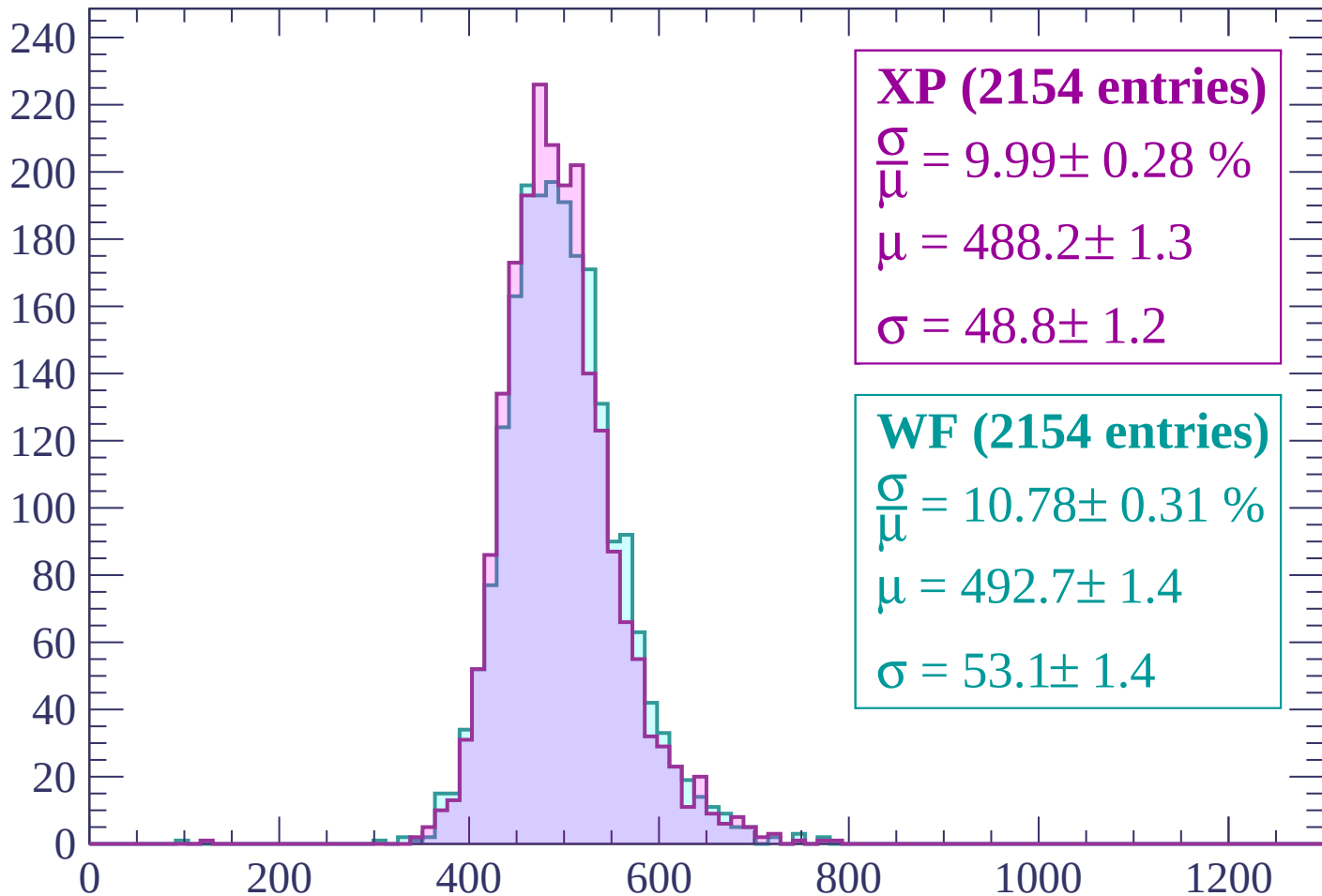
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP (2154 entries)

$\frac{\sigma}{\mu} = 9.99 \pm 0.28 \%$

$\mu = 488.2 \pm 1.3$

$\sigma = 48.8 \pm 1.2$

WF (2154 entries)

$\frac{\sigma}{\mu} = 10.78 \pm 0.31 \%$

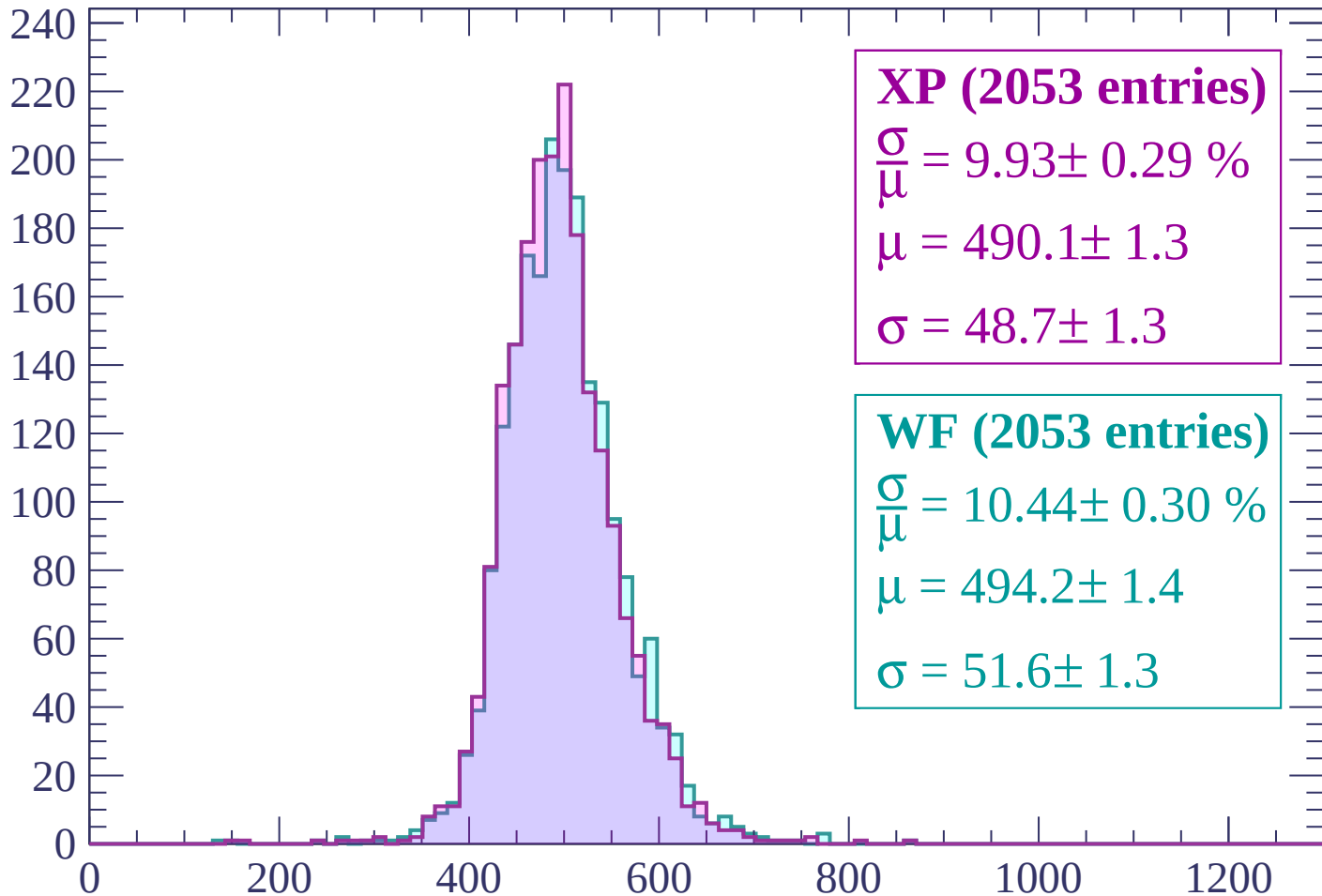
$\mu = 492.7 \pm 1.4$

$\sigma = 53.1 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (2053 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.29 \%$$

$$\mu = 490.1 \pm 1.3$$

$$\sigma = 48.7 \pm 1.3$$

WF (2053 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 0.30 \%$$

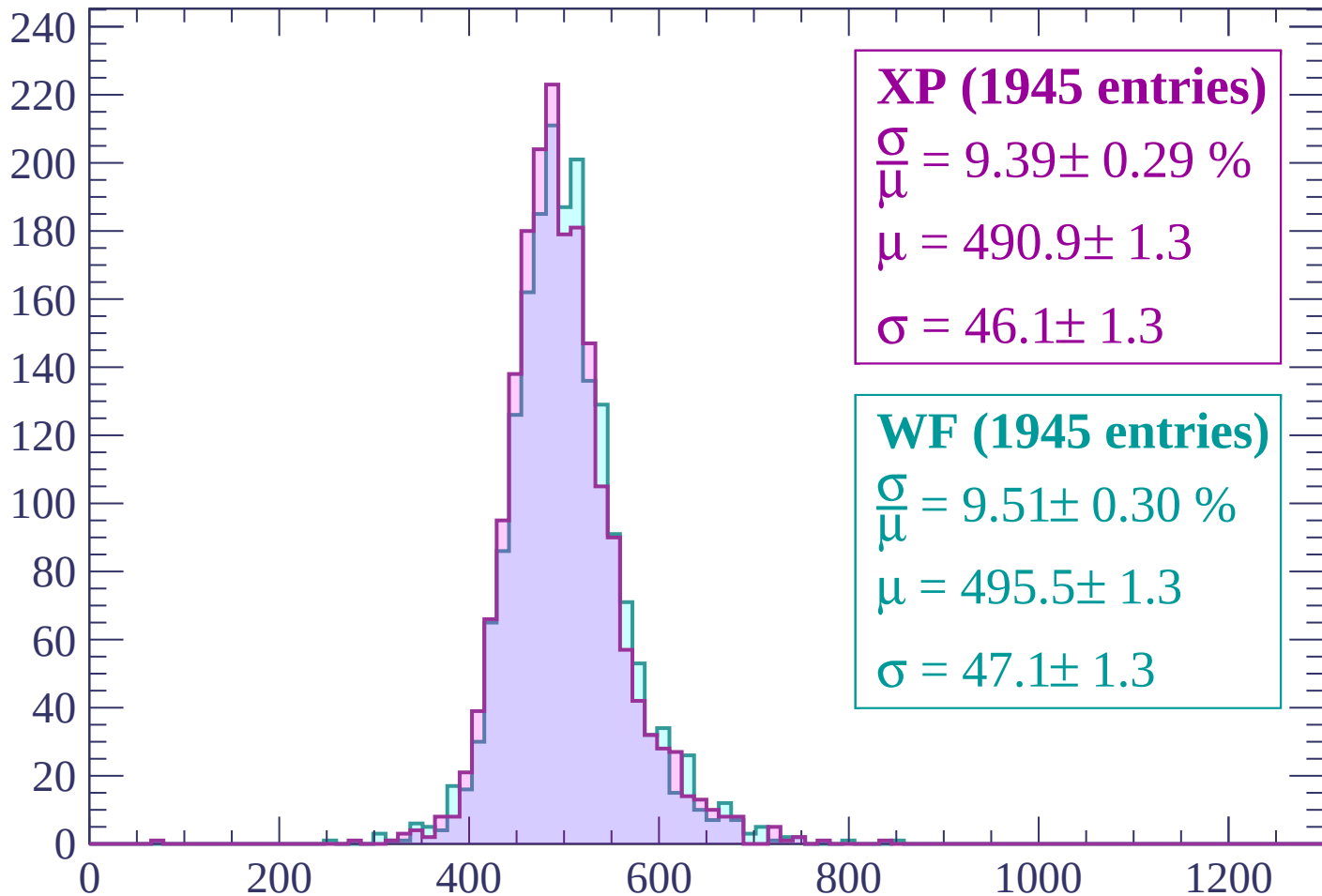
$$\mu = 494.2 \pm 1.4$$

$$\sigma = 51.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2760 < p < 2820$

Count



XP (1945 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.29 \%$$

$$\mu = 490.9 \pm 1.3$$

$$\sigma = 46.1 \pm 1.3$$

WF (1945 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.30 \%$$

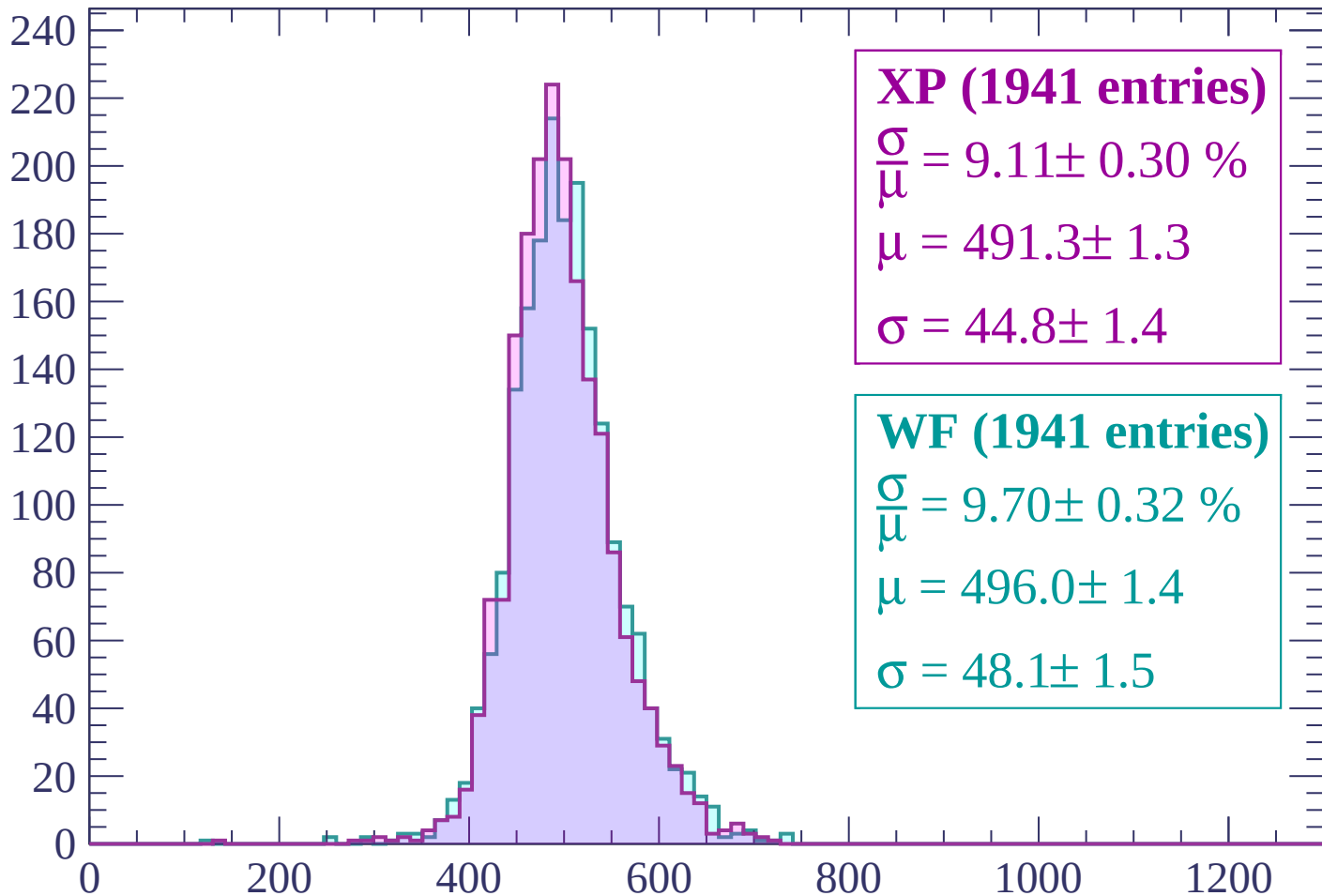
$$\mu = 495.5 \pm 1.3$$

$$\sigma = 47.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP (1941 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.30 \%$$

$$\mu = 491.3 \pm 1.3$$

$$\sigma = 44.8 \pm 1.4$$

WF (1941 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.32 \%$$

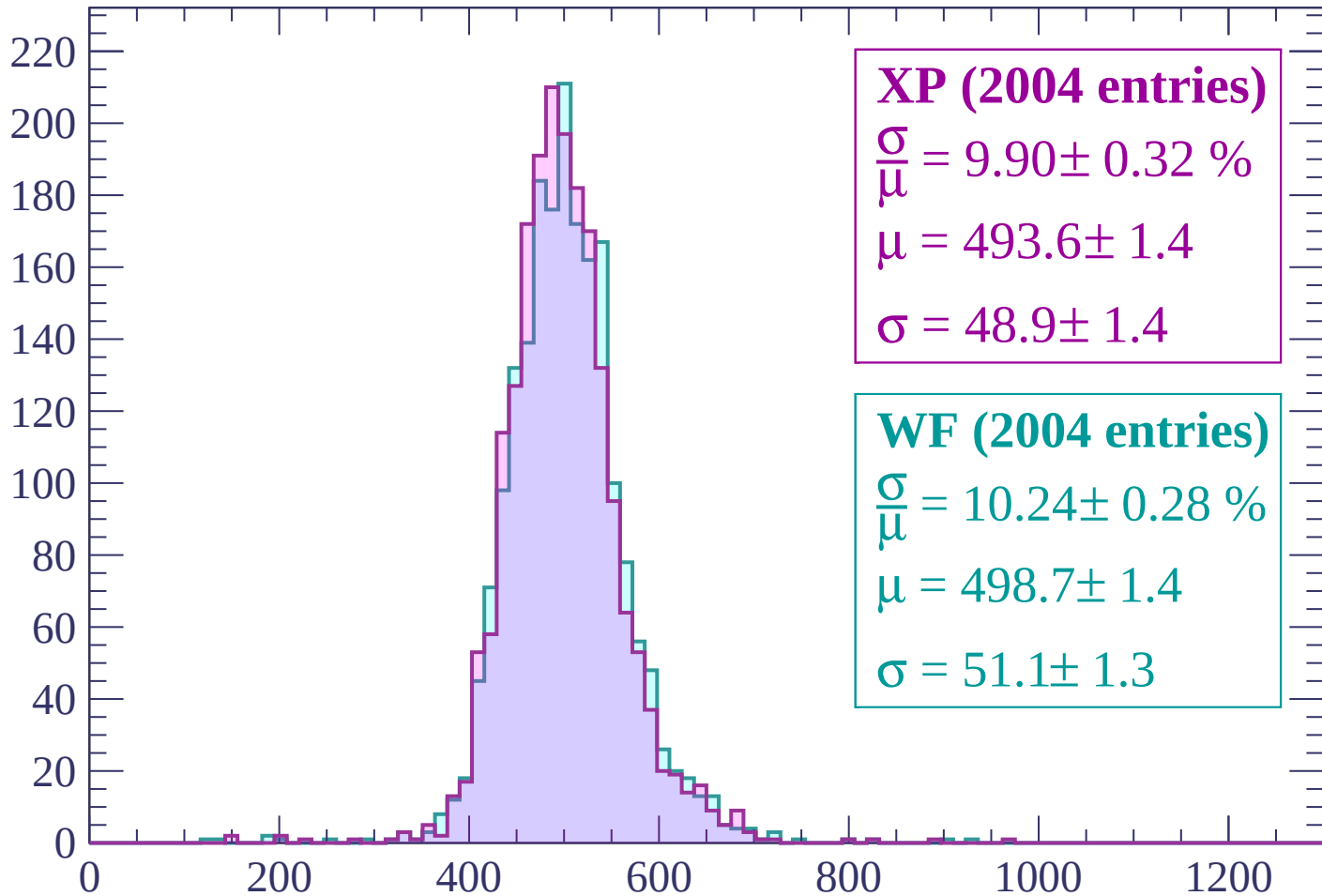
$$\mu = 496.0 \pm 1.4$$

$$\sigma = 48.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (2004 entries)

$$\frac{\sigma}{\mu} = 9.90 \pm 0.32 \%$$

$$\mu = 493.6 \pm 1.4$$

$$\sigma = 48.9 \pm 1.4$$

WF (2004 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.28 \%$$

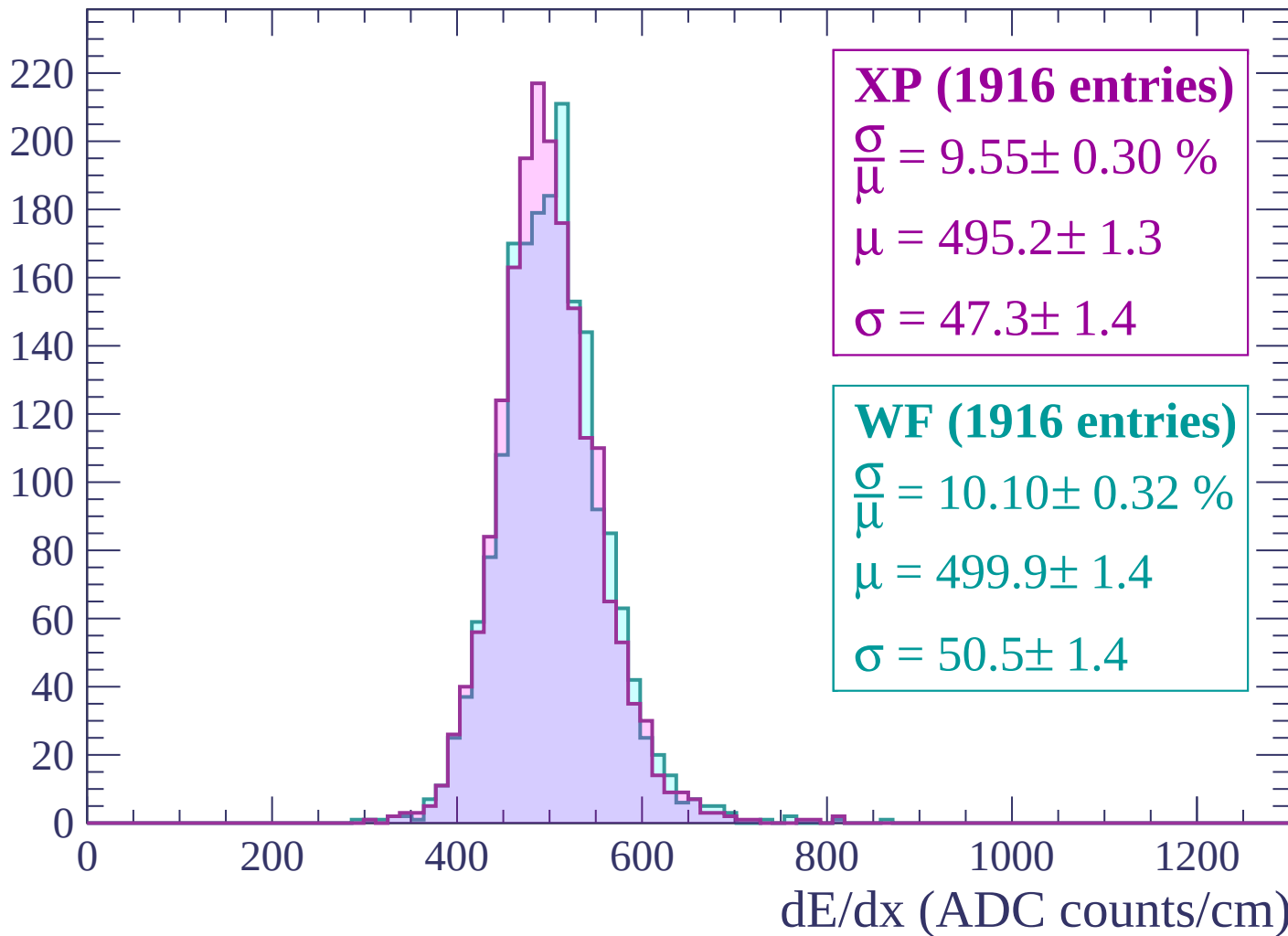
$$\mu = 498.7 \pm 1.4$$

$$\sigma = 51.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

